# Supplementary Material for "Model Averaging Approach for Analyzing Complex Survey Data"

This Supplementary Material is organized as follows. Section S1 explains the motivation for the proposed weight selection criterion. Additional theoretical results are presented in Section S2, and Section S3 provides proofs of all theoretical results. Section S4 discusses the relationship between model averaging and high-dimensional regression in survey sampling. Section S5 presents additional numerical analyses.

## S1 Motivation for Weight Selection Criterion

In frequentist model averaging, Mallows-type criteria are typically constructed from an unbiased estimator of the risk function and consist of a goodness-of-fit term and a penalty term (Hansen 2007, Wan et al. 2010). Based on this principle, we propose a Mallows-type criterion for complex survey data. Note that

$$\begin{aligned} E\left\|\boldsymbol{y}_s-\hat{\boldsymbol{\mu}}_s(\boldsymbol{\omega})\right\|_{\boldsymbol{\Phi}_s}^2 &= E\left\|\boldsymbol{\mu}_s-\tilde{\boldsymbol{\mu}}_s(\boldsymbol{\omega})\right\|_{\boldsymbol{\Phi}_s}^2 + E\left\|\tilde{\boldsymbol{\mu}}_s(\boldsymbol{\omega})-\hat{\boldsymbol{\mu}}_s(\boldsymbol{\omega})\right\|_{\boldsymbol{\Phi}_s}^2 \\ &\quad + 2E\left[\left\{\boldsymbol{\mu}_s-\tilde{\boldsymbol{\mu}}_s(\boldsymbol{\omega})\right\}'\boldsymbol{\Phi}_s\left\{\tilde{\boldsymbol{\mu}}_s(\boldsymbol{\omega})-\hat{\boldsymbol{\mu}}_s(\boldsymbol{\omega})\right\}\right] \\ &\quad + 2E\left[\boldsymbol{e}_s'\boldsymbol{\Phi}_s\left\{\boldsymbol{\mu}_s-\hat{\boldsymbol{\mu}}_s(\boldsymbol{\omega})\right\}\right] + E_\epsilon(\boldsymbol{e}_U'\boldsymbol{e}_U) \\ &= R_N(\boldsymbol{\omega}) + N\sigma^2 + 2E\left[\boldsymbol{e}_s'\boldsymbol{\Phi}_s\left\{\boldsymbol{\mu}_s-\hat{\boldsymbol{\mu}}_s(\boldsymbol{\omega})\right\}\right] \\ &\quad + \left\{\tilde{R}_N(\boldsymbol{\omega})-R_N(\boldsymbol{\omega})\right\} + E\left\|\tilde{\boldsymbol{\mu}}_s(\boldsymbol{\omega})-\hat{\boldsymbol{\mu}}_s(\boldsymbol{\omega})\right\|_{\boldsymbol{\Phi}_s}^2 \\ &\quad + 2E\left[\left\{\boldsymbol{\mu}_s-\tilde{\boldsymbol{\mu}}_s(\boldsymbol{\omega})\right\}'\boldsymbol{\Phi}_s\left\{\tilde{\boldsymbol{\mu}}_s(\boldsymbol{\omega})-\hat{\boldsymbol{\mu}}_s(\boldsymbol{\omega})\right\}\right]. \end{aligned} \tag{S.1}$$

According to (S.24)–(S.26) in the proof of Theorem 1, the last three terms in the above equation are asymptotically negligible relative to $R_N(\boldsymbol{\omega})$. Consequently,

$$E\left\|\boldsymbol{y}_s-\hat{\boldsymbol{\mu}}_s(\boldsymbol{\omega})\right\|_{\boldsymbol{\Phi}_s}^2 - 2E\left[\boldsymbol{e}_s'\boldsymbol{\Phi}_s\left\{\boldsymbol{\mu}_s-\hat{\boldsymbol{\mu}}_s(\boldsymbol{\omega})\right\}\right] \approx R_N(\boldsymbol{\omega}) + N\sigma^2.$$

Hence, a feasible penalty term can be constructed by estimating $2E\left[\boldsymbol{e}_s'\boldsymbol{\Phi}_s\left\{\boldsymbol{\mu}_s-\hat{\boldsymbol{\mu}}_s(\boldsymbol{\omega})\right\}\right]$. When $E_\epsilon E_d(\cdot)=E_d E_\epsilon(\cdot)$, which holds under a non-informative sampling design, a design unrelated to the study variable (Särndal et al. 1978, Raghunath 1990), we have

$$
\begin{aligned}
&2E\left[\boldsymbol{e}_s'\boldsymbol{\Phi}_s\left\{\boldsymbol{\mu}_s-\hat{\boldsymbol{\mu}}_s(\boldsymbol{\omega})\right\}\right]\\
&=2E_\epsilon E_d\left[\boldsymbol{e}_s'\boldsymbol{\Phi}_s\boldsymbol{\mu}_s-\sum_{m=1}^{M}\omega_m\boldsymbol{e}_s'\boldsymbol{\Phi}_s\boldsymbol{X}_{(m)s}\left(\boldsymbol{X}_{(m)s}'\boldsymbol{\Phi}_s\boldsymbol{X}_{(m)s}\right)^{-1}\boldsymbol{X}_{(m)s}'\boldsymbol{\Phi}_s\boldsymbol{y}_s\right]\\
&=2E_dE_\epsilon\left[\boldsymbol{e}_s'\boldsymbol{\Phi}_s\boldsymbol{\mu}_s-\sum_{m=1}^{M}\omega_m\boldsymbol{e}_s'\boldsymbol{\Phi}_s\boldsymbol{X}_{(m)s}\left(\boldsymbol{X}_{(m)s}'\boldsymbol{\Phi}_s\boldsymbol{X}_{(m)s}\right)^{-1}\boldsymbol{X}_{(m)s}'\boldsymbol{\Phi}_s\boldsymbol{\mu}_s\right]\\
&\quad-2E_dE_\epsilon\left[\sum_{m=1}^{M}\omega_m\boldsymbol{e}_s'\boldsymbol{\Phi}_s\boldsymbol{X}_{(m)s}\left(\boldsymbol{X}_{(m)s}'\boldsymbol{\Phi}_s\boldsymbol{X}_{(m)s}\right)^{-1}\boldsymbol{X}_{(m)s}'\boldsymbol{\Phi}_s\boldsymbol{e}_s\right]\\
&=-2\sigma^2E_d\left[\sum_{m=1}^{M}\omega_m\,\mathrm{tr}\left\{\boldsymbol{\Phi}_s\boldsymbol{X}_{(m)s}\left(\boldsymbol{X}_{(m)s}'\boldsymbol{\Phi}_s\boldsymbol{X}_{(m)s}\right)^{-1}\boldsymbol{X}_{(m)s}'\boldsymbol{\Phi}_s\right\}\right]\\
&\triangleq-2\sigma^2E_d\left[k_s(\boldsymbol{\omega})\right].
\end{aligned}
\tag{S.2}
$$

This indicates that $-2\sigma^2k_s(\boldsymbol{\omega})$ is a design-unbiased estimator of $2E\left[\boldsymbol{e}_s'\boldsymbol{\Phi}_s\left\{\boldsymbol{\mu}_s-\hat{\boldsymbol{\mu}}_s(\boldsymbol{\omega})\right\}\right]$. Therefore, the corresponding penalty term is $2\sigma^2k_s(\boldsymbol{\omega})$, where

$$
k_s(\boldsymbol{\omega})=\sum_{m=1}^{M}\omega_m\mathrm{tr}\{\boldsymbol{\Phi}_s\boldsymbol{X}_{(m)s}(\boldsymbol{X}_{(m)s}'\boldsymbol{\Phi}_s\boldsymbol{X}_{(m)s})^{-1}\boldsymbol{X}_{(m)s}'\boldsymbol{\Phi}_s\}.
$$

Noting that the trace term $\mathrm{tr}\{\boldsymbol{\Phi}_s\boldsymbol{X}_{(m)s}(\boldsymbol{X}_{(m)s}'\boldsymbol{\Phi}_s\boldsymbol{X}_{(m)s})^{-1}\boldsymbol{X}_{(m)s}'\boldsymbol{\Phi}_s\}$ in $k_s(\boldsymbol{\omega})$ is complicated, we replace each first-order inclusion probability $\pi_i$ in the first $\boldsymbol{\Phi}_s$ by their population mean, $\sum_U\pi_i/N=n/N=f$. A similar idea of replacing individual values by their average is used to construct the generalized cross-validation (GCV) criterion (Craven & Wahba 1979). This approximation yields the simplified penalty term $2f^{-1}\sigma^2k(\boldsymbol{\omega})$, where $k(\boldsymbol{\omega})=\sum_{m=1}^{M}\omega_mk_m$. The resulting weight selection criterion is

$$
C_n(\boldsymbol{\omega})=\left(\boldsymbol{y}_s-\hat{\boldsymbol{\mu}}_s(\boldsymbol{\omega})\right)'\boldsymbol{\Phi}_s\left(\boldsymbol{y}_s-\hat{\boldsymbol{\mu}}_s(\boldsymbol{\omega})\right)+2f^{-1}\sigma^2k(\boldsymbol{\omega}).
$$

# S2 Additional Theoretical Results

In this section, we present two additional theoretical results that follow as corollaries of Theorems 1 and 2.

**Corollary 1** *If Conditions 1–2, $E_\epsilon E_d(\cdot) = E_d E_\epsilon(\cdot)$, and $k_M(f\xi_N)^{-1} = o(1)$ hold, then*

$$E\left(C_n(\boldsymbol{\omega})\right) = R_N(\boldsymbol{\omega}) + N\sigma^2 + o\left(R_N(\boldsymbol{\omega})\right).$$

*Moreover, if $\boldsymbol{\mu}_U'\boldsymbol{\mu}_U = O(N)$ and $k_M^2 = O(N)$ also hold, then*

$$E\left(C_n^*(\boldsymbol{\omega})\right) = R_N(\boldsymbol{\omega}) + N\sigma^2 + o\left(R_N(\boldsymbol{\omega})\right).$$

**Proof.** See Supplementary Material S3.3.

Corollary 1 shows that, under the interchangeability condition $E_\epsilon E_d(\cdot) = E_d E_\epsilon(\cdot)$, the proposed criteria $C_n(\boldsymbol{\omega})$ and $C_n^*(\boldsymbol{\omega})$ remain asymptotically unbiased for $R_N(\boldsymbol{\omega}) + N\sigma^2$ even when Condition 3 is relaxed to the weaker assumption $k_M(f\xi_N)^{-1} = o(1)$.

**Corollary 2** *If Conditions 1–4 hold, then*

$$C_n(\boldsymbol{\omega}) = R_N(\boldsymbol{\omega}) + N\sigma^2 + o_p\left(R_N(\boldsymbol{\omega})\right).$$

*Moreover, if $\boldsymbol{\mu}_U'\boldsymbol{\mu}_U = O(N)$ and $k_M^2 = O(N)$ also hold, then*

$$C_n^*(\boldsymbol{\omega}) = R_N(\boldsymbol{\omega}) + N\sigma^2 + o_p\left(R_N(\boldsymbol{\omega})\right).$$

**Proof.** See Supplementary Material S3.5.

As shown in Corollary 2, the proposed criteria $C_n(\boldsymbol{\omega})$ and $C_n^*(\boldsymbol{\omega})$ are consistent for $R_N(\boldsymbol{\omega}) + N\sigma^2$ in a relative sense. Specifically, after normalization by $N$, we have $\bar{C}_n(\boldsymbol{\omega}) = \bar{R}_N(\boldsymbol{\omega}) + \sigma^2 + o_p(\bar{R}_N(\boldsymbol{\omega}))$ and $\bar{C}_n^*(\boldsymbol{\omega}) = \bar{R}_N(\boldsymbol{\omega}) + \sigma^2 + o_p(\bar{R}_N(\boldsymbol{\omega}))$, where $\bar{C}_n(\boldsymbol{\omega}) = N^{-1}C_n(\boldsymbol{\omega})$, $\bar{C}_n^*(\boldsymbol{\omega}) = N^{-1}C_n^*(\boldsymbol{\omega})$, and $\bar{R}_N(\boldsymbol{\omega}) = N^{-1}R_N(\boldsymbol{\omega})$ is typically bounded.

# S3 Proofs of Theoretical Results

In this section, we provide proofs of all theoretical results in the main manuscript and the Supplementary Material S2. Throughout the proofs, we use $C$ to denote a generic positive constant whose value may vary from place to place. For any $m \times 1$ vector $b$ and $m \times m$ symmetric matrix $A$, we write $\|b\|_A^2 \triangleq b'Ab$.

## S3.1 Lemmas for the proofs of Theorems 1–4

**Lemma 1** *If Condition 1 holds, then*

$$E_\epsilon \left\| \tilde{\boldsymbol{\beta}}_{(m)} - \boldsymbol{\beta}^*_{(m)} \right\|^4 \leq CN^{-2}k_m^2.$$

**Proof.** By the definition of $\boldsymbol{\beta}^*_{(m)}$, we have

$$\sum_{i \in U} \boldsymbol{x}_{i(m)} \left( \mu_i - \boldsymbol{x}'_{i(m)} \boldsymbol{\beta}^*_{(m)} \right) = \mathbf{0}.$$

Since $e^*_{i(m)} - e_i = \mu_i - \boldsymbol{x}'_{i(m)}\boldsymbol{\beta}^*_{(m)}$, it follows that

$$\sum_{i \in U} \boldsymbol{x}_{i(m)} e^*_{i(m)} = \sum_{i \in U} \boldsymbol{x}_{i(m)} e_i.$$

Consequently, for each $j = 1, \ldots, k_m$,

$$E_\epsilon \left( \sum_{i \in U} x_{ij(m)} e^*_{i(m)} \right) = 0. \tag{S.3}$$

Applying Loève's $c_r$ inequality, for any $q \geq 1$, yields

$$\begin{aligned} E_\epsilon \left\| \frac{1}{N} \sum_{i \in U} \boldsymbol{x}_{i(m)} e^*_{i(m)} \right\|^{2q} &= \frac{1}{N^{2q}} E_\epsilon \left| \sum_{j=1}^{k_m} \left( \sum_{i \in U} x_{ij(m)} e^*_{i(m)} \right)^2 \right|^q \\ &\leq \frac{C k_m^{q-1}}{N^{2q}} \sum_{j=1}^{k_m} E_\epsilon \left| \sum_{i \in U} x_{ij(m)} e^*_{i(m)} \right|^{2q} \\ &\leq \frac{C k_m^{q-1}}{N^{2q}} \sum_{j=1}^{k_m} \left\{ \sum_{i \in U} \left( E_\epsilon \left| x_{ij(m)} e^*_{i(m)} \right|^{2q} \right)^{\frac{1}{q}} \right\}^q, \end{aligned} \tag{S.4}$$

where the last inequality follows from (S.3) and Theorem 2 of Whittle (1960). Then, by Condition 1, we have

$$\begin{aligned} E_\epsilon \left\| \tilde{\boldsymbol{\beta}}_{(m)} - \boldsymbol{\beta}^*_{(m)} \right\|^4 =& E_\epsilon \left\| \left( \sum_{i \in U} \boldsymbol{x}_{i(m)} \boldsymbol{x}'_{i(m)} \right)^{-1} \sum_{i \in U} \boldsymbol{x}_{i(m)} e^*_{i(m)} \right\|^4 \\ \leq & E_\epsilon \left\{ \lambda_{\min}^{-4} \left( \frac{1}{N} \boldsymbol{X}'_{(m)U} \boldsymbol{X}_{(m)U} \right) \cdot \left\| \frac{1}{N} \sum_{i \in U} \boldsymbol{x}_{i(m)} e^*_{i(m)} \right\|^4 \right\} \\ \leq & C N^{-2} k_m^2. \end{aligned}$$

This completes the proof of Lemma 1.

**Lemma 2** *If Condition 2 holds, then*

$$E \left( \frac{1}{N} \sum_{i \in s} \frac{z_i}{\pi_i} - \frac{1}{N} \sum_{i \in U} z_i \right)^2 \leq \frac{C}{nN} \sum_{i \in U} E_\epsilon \left( z_i^2 \right).$$

**Proof.** Let $I_i = 1$ if $i \in s$ and $I_i = 0$ otherwise. By Condition 2, we have

$$\begin{aligned} E \left( \frac{1}{N} \sum_{i \in s} \frac{z_i}{\pi_i} - \frac{1}{N} \sum_{i \in U} z_i \right)^2 &= E_\epsilon \left[ E_d \left\{ \frac{1}{N} \sum_{i \in U} \frac{z_i (I_i - \pi_i)}{\pi_i} \right\}^2 \right] \\ &= E_\epsilon \left( \frac{1}{N^2} \sum_{i \in U} \frac{1 - \pi_i}{\pi_i} z_i^2 + \frac{1}{N^2} \sum_{i \neq j \in U} \frac{\pi_{ij} - \pi_i \pi_j}{\pi_i \pi_j} z_i z_j \right) \\ &\leq E_\epsilon \left( \frac{C}{N^2} \sum_{i \in U} z_i^2 \right) + E_\epsilon \left( \frac{C}{nN^2} \sum_{i \neq j \in U} |z_i z_j| \right) \\ &\leq \frac{C}{nN} \sum_{i \in U} E_\epsilon \left( z_i^2 \right). \end{aligned}$$

This completes the proof of Lemma 2.

**Lemma 3** *If Conditions 1–2 hold, then*

$$E \left\| \hat{\boldsymbol{\beta}}_{(m)} - \boldsymbol{\beta}^*_{(m)} \right\|^2 \leq C n^{-1} k_m.$$

**Proof.** Condition 1 implies

$$E_\epsilon \left( x_{ij(m)} e^*_{i(m)} \right)^2 \leq C.$$

Applying Lemma 2, we have

$$E\left(\frac{1}{N}\sum_{i\in s}\frac{x_{ij(m)}e^*_{i(m)}}{\pi_i}-\frac{1}{N}\sum_{i\in U}x_{ij(m)}e^*_{i(m)}\right)^2\leq Cn^{-1}.$$

Combining this bound with (S.4) gives

$$\begin{aligned}
&E\left\|\frac{1}{N}\sum_{i\in s}\frac{\boldsymbol{x}_{i(m)}e^*_{i(m)}}{\pi_i}\right\|^2\\
&=E\left\|\frac{1}{N}\sum_{i\in s}\frac{\boldsymbol{x}_{i(m)}e^*_{i(m)}}{\pi_i}-\frac{1}{N}\sum_{i\in U}\boldsymbol{x}_{i(m)}e^*_{i(m)}+\frac{1}{N}\sum_{i\in U}\boldsymbol{x}_{i(m)}e^*_{i(m)}\right\|^2\\
&\leq 2E\left\|\frac{1}{N}\sum_{i\in s}\frac{\boldsymbol{x}_{i(m)}e^*_{i(m)}}{\pi_i}-\frac{1}{N}\sum_{i\in U}\boldsymbol{x}_{i(m)}e^*_{i(m)}\right\|^2+2E\left\|\frac{1}{N}\sum_{i\in U}\boldsymbol{x}_{i(m)}e^*_{i(m)}\right\|^2\\
&=2\sum_{j=1}^{k_m}\left\{E\left(\frac{1}{N}\sum_{i\in s}\frac{x_{ij(m)}e^*_{i(m)}}{\pi_i}-\frac{1}{N}\sum_{i\in U}x_{ij(m)}e^*_{i(m)}\right)^2\right\}+2E\left\|\frac{1}{N}\sum_{i\in U}\boldsymbol{x}_{i(m)}e^*_{i(m)}\right\|^2\\
&\leq Cn^{-1}k_m. \qquad \text{(S.5)}
\end{aligned}$$

From Condition 1, it follows that

$$\begin{aligned}
E\left\|\hat{\boldsymbol{\beta}}_{(m)}-\boldsymbol{\beta}^*_{(m)}\right\|^2=&E\left\|\left(\sum_{i\in s}\frac{\boldsymbol{x}_{i(m)}\boldsymbol{x}'_{i(m)}}{\pi_i}\right)^{-1}\sum_{i\in s}\frac{\boldsymbol{x}_{i(m)}e^*_{i(m)}}{\pi_i}\right\|^2\\
\leq&E\left\{\lambda^{-2}_{\min}\left(\boldsymbol{X}'_{(m)s}\boldsymbol{\Phi}_s\boldsymbol{X}_{(m)s}\right)\cdot\left\|\sum_{i\in s}\frac{\boldsymbol{x}_{i(m)}e^*_{i(m)}}{\pi_i}\right\|^2\right\}\\
\leq&CE\left\|\frac{1}{N}\sum_{i\in s}\frac{\boldsymbol{x}_{i(m)}e^*_{i(m)}}{\pi_i}\right\|^2. \qquad \text{(S.6)}
\end{aligned}$$

Combining (S.5) with (S.6) yields the desired result.

**Lemma 4** *If Conditions 1–2 and $k_M(f\xi_N)^{-1}=o(1)$ hold, then*

$$\sup_{\boldsymbol{\omega}\in\mathcal{W}}\frac{\left|\tilde{R}_N(\boldsymbol{\omega})-R_N(\boldsymbol{\omega})\right|}{R_N(\boldsymbol{\omega})}=o(1).$$

**Proof.** Note that

$$\begin{aligned}
\tilde{R}_N(\boldsymbol{\omega})&=E\left\|\boldsymbol{\mu}_U-\hat{\boldsymbol{\mu}}_U(\boldsymbol{\omega})+\hat{\boldsymbol{\mu}}_U(\boldsymbol{\omega})-\tilde{\boldsymbol{\mu}}_U(\boldsymbol{\omega})\right\|^2\\
&=R_N(\boldsymbol{\omega})+E\left\|\hat{\boldsymbol{\mu}}_U(\boldsymbol{\omega})-\tilde{\boldsymbol{\mu}}_U(\boldsymbol{\omega})\right\|^2+2E\left\{\boldsymbol{\mu}_U-\hat{\boldsymbol{\mu}}_U(\boldsymbol{\omega})\right\}'\left\{\hat{\boldsymbol{\mu}}_U(\boldsymbol{\omega})-\tilde{\boldsymbol{\mu}}_U(\boldsymbol{\omega})\right\}.
\end{aligned}$$

Thus, it suffices to show that

$$\sup_{\boldsymbol{\omega}\in\mathcal{W}} \frac{E\left\|\hat{\boldsymbol{\mu}}_U(\boldsymbol{\omega}) - \tilde{\boldsymbol{\mu}}_U(\boldsymbol{\omega})\right\|^2}{R_N(\boldsymbol{\omega})} = o(1) \tag{S.7}$$

and

$$\sup_{\boldsymbol{\omega}\in\mathcal{W}} \frac{E\left|\{\boldsymbol{\mu}_U - \hat{\boldsymbol{\mu}}_U(\boldsymbol{\omega})\}'\{\hat{\boldsymbol{\mu}}_U(\boldsymbol{\omega}) - \tilde{\boldsymbol{\mu}}_U(\boldsymbol{\omega})\}\right|}{R_N(\boldsymbol{\omega})} = o(1). \tag{S.8}$$

Lemmas 1 and 3, together with Condition 1, imply that

$$\begin{aligned}
&E\left\|\hat{\boldsymbol{\mu}}_U(\boldsymbol{\omega}) - \tilde{\boldsymbol{\mu}}_U(\boldsymbol{\omega})\right\|^2 \\
&= E\left|\sum_{m=1}^{M}\sum_{l=1}^{M}\omega_m\omega_l\left(\boldsymbol{X}_{(m)U}\hat{\boldsymbol{\beta}}_{(m)} - \boldsymbol{X}_{(m)U}\tilde{\boldsymbol{\beta}}_{(m)}\right)'\left(\boldsymbol{X}_{(l)U}\hat{\boldsymbol{\beta}}_{(l)} - \boldsymbol{X}_{(l)U}\tilde{\boldsymbol{\beta}}_{(l)}\right)\right| \\
&\leq \sum_{m=1}^{M}\sum_{l=1}^{M}\omega_m\omega_l E\left|\left(\hat{\boldsymbol{\beta}}_{(m)} - \tilde{\boldsymbol{\beta}}_{(m)}\right)'\boldsymbol{X}'_{(m)U}\boldsymbol{X}_{(l)U}\left(\hat{\boldsymbol{\beta}}_{(l)} - \tilde{\boldsymbol{\beta}}_{(l)}\right)\right| \\
&\leq \left\{\sum_{m=1}^{M}\omega_m\cdot\lambda_{\max}^{1/2}\left(\boldsymbol{X}'_{(m)U}\boldsymbol{X}_{(m)U}\right)\cdot\left(E\left\|\tilde{\boldsymbol{\beta}}_{(m)} - \hat{\boldsymbol{\beta}}_{(m)}\right\|^2\right)^{1/2}\right\}^2 \\
&\leq Cf^{-1}k_M.
\end{aligned} \tag{S.9}$$

Thus, (S.7) follows from (S.9) and $k_M(f\xi_N)^{-1} = o(1)$.

We next verify (S.8). By the Cauchy–Schwarz inequality and (S.7),

$$\begin{aligned}
&\sup_{\boldsymbol{\omega}\in\mathcal{W}} \frac{E\left|\{\boldsymbol{\mu}_U - \hat{\boldsymbol{\mu}}_U(\boldsymbol{\omega})\}'\{\hat{\boldsymbol{\mu}}_U(\boldsymbol{\omega}) - \tilde{\boldsymbol{\mu}}_U(\boldsymbol{\omega})\}\right|}{R_N(\boldsymbol{\omega})} \\
&\leq \sup_{\boldsymbol{\omega}\in\mathcal{W}}\left\{\frac{R_N(\boldsymbol{\omega})\cdot E\left\|\hat{\boldsymbol{\mu}}_U(\boldsymbol{\omega}) - \tilde{\boldsymbol{\mu}}_U(\boldsymbol{\omega})\right\|^2}{R_N^2(\boldsymbol{\omega})}\right\}^{1/2} \\
&= o(1).
\end{aligned}$$

This completes the proof of Lemma 4.

**Lemma 5** *If Conditions 1–3 hold, then*

$$\sup_{\boldsymbol{\omega}\in\mathcal{W}} \frac{\left|\tilde{L}_N(\boldsymbol{\omega}) - L_N(\boldsymbol{\omega})\right|}{R_N(\boldsymbol{\omega})} = o_p(1).$$

**Proof.** Note that

$$\begin{aligned}\tilde{L}_N(\boldsymbol{\omega}) &= \|\boldsymbol{\mu}_U - \hat{\boldsymbol{\mu}}_U(\boldsymbol{\omega}) + \hat{\boldsymbol{\mu}}_U(\boldsymbol{\omega}) - \tilde{\boldsymbol{\mu}}_U(\boldsymbol{\omega})\|^2 \\ &= L_N(\boldsymbol{\omega}) + \|\hat{\boldsymbol{\mu}}_U(\boldsymbol{\omega}) - \tilde{\boldsymbol{\mu}}_U(\boldsymbol{\omega})\|^2 + 2\left\{\boldsymbol{\mu}_U - \hat{\boldsymbol{\mu}}_U(\boldsymbol{\omega})\right\}'\left\{\hat{\boldsymbol{\mu}}_U(\boldsymbol{\omega}) - \tilde{\boldsymbol{\mu}}_U(\boldsymbol{\omega})\right\}.\end{aligned}$$

Thus, it suffices to show that

$$\sup_{\boldsymbol{\omega}\in\mathcal{W}} \frac{\|\hat{\boldsymbol{\mu}}_U(\boldsymbol{\omega}) - \tilde{\boldsymbol{\mu}}_U(\boldsymbol{\omega})\|^2}{R_N(\boldsymbol{\omega})} = o_p(1) \tag{S.10}$$

and

$$\sup_{\boldsymbol{\omega}\in\mathcal{W}} \frac{\left|\left\{\boldsymbol{\mu}_U - \hat{\boldsymbol{\mu}}_U(\boldsymbol{\omega})\right\}'\left\{\hat{\boldsymbol{\mu}}_U(\boldsymbol{\omega}) - \tilde{\boldsymbol{\mu}}_U(\boldsymbol{\omega})\right\}\right|}{R_N(\boldsymbol{\omega})} = o_p(1). \tag{S.11}$$

Using Bonferroni's inequality and Markov's inequality, for any $\delta > 0$, we have

$$\begin{aligned}&P\left(\sup_{\boldsymbol{\omega}\in\mathcal{W}}\left\{R_N^{-1}(\boldsymbol{\omega})\|\hat{\boldsymbol{\mu}}_U(\boldsymbol{\omega}) - \tilde{\boldsymbol{\mu}}_U(\boldsymbol{\omega})\|^2\right\} > \delta\right) \\ &\leq P\left(\sup_{\boldsymbol{\omega}\in\mathcal{W}}\|\hat{\boldsymbol{\mu}}_U(\boldsymbol{\omega}) - \tilde{\boldsymbol{\mu}}_U(\boldsymbol{\omega})\|^2 > \delta\xi_N\right) \\ &= P\left(\sup_{\boldsymbol{\omega}\in\mathcal{W}}\left\{\sum_{m=1}^{M}\sum_{t=1}^{M}\omega_m\omega_t\left(\tilde{\boldsymbol{\beta}}_{(m)} - \hat{\boldsymbol{\beta}}_{(m)}\right)'\boldsymbol{X}'_{(m)U}\boldsymbol{X}_{(t)U}\left(\tilde{\boldsymbol{\beta}}_{(t)} - \hat{\boldsymbol{\beta}}_{(t)}\right)\right\} > \delta\xi_N\right) \\ &\leq \sum_{m=1}^{M}\sum_{t=1}^{M} P\left(\left|\left(\tilde{\boldsymbol{\beta}}_{(m)} - \hat{\boldsymbol{\beta}}_{(m)}\right)'\boldsymbol{X}'_{(m)U}\boldsymbol{X}_{(t)U}\left(\tilde{\boldsymbol{\beta}}_{(t)} - \hat{\boldsymbol{\beta}}_{(t)}\right)\right| > \delta\xi_N\right) \\ &\leq \delta^{-1}\xi_N^{-1}\sum_{m=1}^{M}\sum_{t=1}^{M} E\left|\left(\tilde{\boldsymbol{\beta}}_{(m)} - \hat{\boldsymbol{\beta}}_{(m)}\right)'\boldsymbol{X}'_{(m)U}\boldsymbol{X}_{(t)U}\left(\tilde{\boldsymbol{\beta}}_{(t)} - \hat{\boldsymbol{\beta}}_{(t)}\right)\right| \\ &\leq M\delta^{-1}\xi_N^{-1}\sum_{m=1}^{M}\left\{\lambda_{\max}\left(\boldsymbol{X}'_{(m)U}\boldsymbol{X}_{(m)U}\right)\cdot E\left\|\tilde{\boldsymbol{\beta}}_{(m)} - \hat{\boldsymbol{\beta}}_{(m)}\right\|^2\right\} \\ &= O\left(\frac{M^2 k_M}{f\xi_N}\right),\end{aligned}$$

where the last bound follows from Lemmas 1 and 3, together with Condition 1. Condition 3 then implies (S.10).

We next verify (S.11). By Condition 1 and Lemma 3,

$$E\left\|\boldsymbol{\mu}_U - \boldsymbol{X}_{(m)U}\hat{\boldsymbol{\beta}}_{(m)}\right\|^2 = E\left\|\boldsymbol{e}^*_{(m)U} - \boldsymbol{e}_U + \boldsymbol{X}_{(m)U}\boldsymbol{\beta}^*_{(m)} - \boldsymbol{X}_{(m)U}\hat{\boldsymbol{\beta}}_{(m)}\right\|^2$$

$$\leq 3E_\epsilon \left\| \boldsymbol{e}^*_{(m)U} \right\|^2 + 3E_\epsilon \left\| \boldsymbol{e}_U \right\|^2 + 3E \left\| \boldsymbol{X}_{(m)U}\boldsymbol{\beta}^*_{(m)} - \boldsymbol{X}_{(m)U}\hat{\boldsymbol{\beta}}_{(m)} \right\|^2$$

$$\leq CN, \tag{S.12}$$

where $\boldsymbol{e}^*_{(m)U} = [e^*_{i(m)}]_{i\in U}$ and $\boldsymbol{e}_U = [e_i]_{i\in U}$. Therefore, for any $\delta > 0$,

$$\begin{aligned}
&P\left( \sup_{\boldsymbol{\omega}\in\mathcal{W}} R_N^{-1}(\boldsymbol{\omega}) \left| \{\boldsymbol{\mu}_U - \hat{\boldsymbol{\mu}}_U(\boldsymbol{\omega})\}' \{\hat{\boldsymbol{\mu}}_U(\boldsymbol{\omega}) - \tilde{\boldsymbol{\mu}}_U(\boldsymbol{\omega})\} \right| > \delta \right) \\
&\leq P\left( \sup_{\boldsymbol{\omega}\in\mathcal{W}} \left| \{\boldsymbol{\mu}_U - \hat{\boldsymbol{\mu}}_U(\boldsymbol{\omega})\}' \{\hat{\boldsymbol{\mu}}_U(\boldsymbol{\omega}) - \tilde{\boldsymbol{\mu}}_U(\boldsymbol{\omega})\} \right| > \delta\xi_N \right) \\
&= P\left( \sup_{\boldsymbol{\omega}\in\mathcal{W}} \left\{ \sum_{m=1}^M \sum_{t=1}^M \omega_m\omega_t \left| \left(\boldsymbol{\mu}_U - \boldsymbol{X}_{(m)U}\hat{\boldsymbol{\beta}}_{(m)}\right)' \boldsymbol{X}_{(t)U} \left(\tilde{\boldsymbol{\beta}}_{(t)} - \hat{\boldsymbol{\beta}}_{(t)}\right) \right| \right\} > \delta\xi_N \right) \\
&\leq \sum_{m=1}^M \sum_{t=1}^M P\left( \left| \left(\boldsymbol{\mu}_U - \boldsymbol{X}_{(m)U}\hat{\boldsymbol{\beta}}_{(m)}\right)' \boldsymbol{X}_{(t)U} \left(\tilde{\boldsymbol{\beta}}_{(t)} - \hat{\boldsymbol{\beta}}_{(t)}\right) \right| > \delta\xi_N \right) \\
&\leq \delta^{-1}\xi_N^{-1} \sum_{m=1}^M \sum_{t=1}^M E \left| \left(\boldsymbol{\mu}_U - \boldsymbol{X}_{(m)U}\hat{\boldsymbol{\beta}}_{(m)}\right)' \boldsymbol{X}_{(t)U} \left(\tilde{\boldsymbol{\beta}}_{(t)} - \hat{\boldsymbol{\beta}}_{(t)}\right) \right| \\
&\leq \delta^{-1}\xi_N^{-1} \sum_{m=1}^M \sum_{t=1}^M \left\{ E \left\| \boldsymbol{\mu}_U - \boldsymbol{X}_{(m)U}\hat{\boldsymbol{\beta}}_{(m)} \right\|^2 \cdot \lambda_{\max}\left( \boldsymbol{X}_{(t)U}\boldsymbol{X}'_{(t)U} \right) \cdot E \left\| \tilde{\boldsymbol{\beta}}_{(t)} - \hat{\boldsymbol{\beta}}_{(t)} \right\|^2 \right\}^{1/2} \\
&= O\left( \frac{M^2\sqrt{nk_M}}{f\xi_N} \right),
\end{aligned}$$

where the last bound follows from Lemmas 1 and 3, (S.12), and Condition 1. Condition 3 then implies (S.11). This completes the proof of Lemma 5.

**Lemma 6** *If Conditions 1–2 and $k_M(f\xi_N)^{-1} = o(1)$ hold, then*

$$E \left\| \boldsymbol{\mu}_U - \hat{\boldsymbol{\mu}}_U(\boldsymbol{\omega}^*) \right\|^2 = E \left\| \boldsymbol{\mu}_s - \hat{\boldsymbol{\mu}}_s(\boldsymbol{\omega}^*) \right\|^2_{\boldsymbol{\Phi}_s} + o(\xi_N),$$

*where $\boldsymbol{\omega}^* = \arg\min_{\boldsymbol{\omega}\in\mathcal{W}} \{R_N(\boldsymbol{\omega})\}$.*

**Proof.** Note that

$$\begin{aligned}
E \left\| \boldsymbol{\mu}_U - \hat{\boldsymbol{\mu}}_U(\boldsymbol{\omega}^*) \right\|^2 &= E \left\| \boldsymbol{\mu}_s - \tilde{\boldsymbol{\mu}}_s(\boldsymbol{\omega}^*) \right\|^2_{\boldsymbol{\Phi}_s} + E \left\| \tilde{\boldsymbol{\mu}}_U(\boldsymbol{\omega}^*) - \hat{\boldsymbol{\mu}}_U(\boldsymbol{\omega}^*) \right\|^2 \\
&\quad + 2E \left\{ \boldsymbol{\mu}_U - \tilde{\boldsymbol{\mu}}_U(\boldsymbol{\omega}^*) \right\}' \left\{ \tilde{\boldsymbol{\mu}}_U(\boldsymbol{\omega}^*) - \hat{\boldsymbol{\mu}}_U(\boldsymbol{\omega}^*) \right\} \\
&= E \left\| \boldsymbol{\mu}_s - \hat{\boldsymbol{\mu}}_s(\boldsymbol{\omega}^*) \right\|^2_{\boldsymbol{\Phi}_s} + E \left\| \hat{\boldsymbol{\mu}}_s(\boldsymbol{\omega}^*) - \tilde{\boldsymbol{\mu}}_s(\boldsymbol{\omega}^*) \right\|^2_{\boldsymbol{\Phi}_s} \\
&\quad + 2E \left\{ \boldsymbol{\mu}_s - \hat{\boldsymbol{\mu}}_s(\boldsymbol{\omega}^*) \right\}' \boldsymbol{\Phi}_s \left\{ \hat{\boldsymbol{\mu}}_s(\boldsymbol{\omega}^*) - \tilde{\boldsymbol{\mu}}_s(\boldsymbol{\omega}^*) \right\}
\end{aligned}$$

$$+ 2E\left\{\boldsymbol{\mu}_U - \tilde{\boldsymbol{\mu}}_U(\boldsymbol{\omega}^*)\right\}'\left\{\tilde{\boldsymbol{\mu}}_U(\boldsymbol{\omega}^*) - \hat{\boldsymbol{\mu}}_U(\boldsymbol{\omega}^*)\right\}$$
$$+ E\left\|\tilde{\boldsymbol{\mu}}_U(\boldsymbol{\omega}^*) - \hat{\boldsymbol{\mu}}_U(\boldsymbol{\omega}^*)\right\|^2.$$

Thus, it suffices to show that

$$\xi_N^{-1} E\left\|\tilde{\boldsymbol{\mu}}_U(\boldsymbol{\omega}^*) - \hat{\boldsymbol{\mu}}_U(\boldsymbol{\omega}^*)\right\|^2 = o(1), \tag{S.13}$$

$$\xi_N^{-1} E\left\|\hat{\boldsymbol{\mu}}_s(\boldsymbol{\omega}^*) - \tilde{\boldsymbol{\mu}}_s(\boldsymbol{\omega}^*)\right\|_{\boldsymbol{\Phi}_s}^2 = o(1), \tag{S.14}$$

$$\xi_N^{-1}\left|E\left\{\boldsymbol{\mu}_U - \tilde{\boldsymbol{\mu}}_U(\boldsymbol{\omega}^*)\right\}'\left\{\tilde{\boldsymbol{\mu}}_U(\boldsymbol{\omega}^*) - \hat{\boldsymbol{\mu}}_U(\boldsymbol{\omega}^*)\right\}\right| = o(1) \tag{S.15}$$

and

$$\xi_N^{-1}\left|E\left\{\boldsymbol{\mu}_s - \hat{\boldsymbol{\mu}}_s(\boldsymbol{\omega}^*)\right\}'\boldsymbol{\Phi}_s\left\{\hat{\boldsymbol{\mu}}_s(\boldsymbol{\omega}^*) - \tilde{\boldsymbol{\mu}}_s(\boldsymbol{\omega}^*)\right\}\right| = o(1). \tag{S.16}$$

Lemmas 1 and 3, together with Condition 1, imply that

$$\begin{aligned}
E\left\|\hat{\boldsymbol{\mu}}_s(\boldsymbol{\omega}) - \tilde{\boldsymbol{\mu}}_s(\boldsymbol{\omega})\right\|_{\boldsymbol{\Phi}_s}^2 &= E\left|\sum_{m=1}^{M}\sum_{l=1}^{M}\omega_m\omega_l\left(\hat{\boldsymbol{\beta}}_{(m)} - \tilde{\boldsymbol{\beta}}_{(m)}\right)'\boldsymbol{X}'_{(m)s}\boldsymbol{\Phi}_s\boldsymbol{X}_{(l)s}\left(\hat{\boldsymbol{\beta}}_{(l)} - \tilde{\boldsymbol{\beta}}_{(l)}\right)\right| \\
&\leq \sum_{m=1}^{M}\sum_{l=1}^{M}\omega_m\omega_l E\left|\left(\hat{\boldsymbol{\beta}}_{(m)} - \tilde{\boldsymbol{\beta}}_{(m)}\right)'\boldsymbol{X}'_{(m)s}\boldsymbol{\Phi}_s\boldsymbol{X}_{(l)s}\left(\hat{\boldsymbol{\beta}}_{(l)} - \tilde{\boldsymbol{\beta}}_{(l)}\right)\right| \\
&\leq \left\{\sum_{m=1}^{M}\omega_m\cdot\lambda_{\max}^{1/2}\left(\boldsymbol{X}'_{(m)s}\boldsymbol{\Phi}_s\boldsymbol{X}_{(m)s}\right)\cdot\left(E\left\|\hat{\boldsymbol{\beta}}_{(m)} - \tilde{\boldsymbol{\beta}}_{(m)}\right\|^2\right)^{1/2}\right\}^2 \\
&\leq Cf^{-1}k_M.
\end{aligned} \tag{S.17}$$

Using (S.9), (S.17), and $k_M(f\xi_N)^{-1} = o(1)$, we obtain (S.13) and (S.14).

By the Cauchy–Schwarz inequality,

$$\begin{aligned}
&\left|E\left\{\boldsymbol{\mu}_U - \tilde{\boldsymbol{\mu}}_U(\boldsymbol{\omega}^*)\right\}'\left\{\tilde{\boldsymbol{\mu}}_U(\boldsymbol{\omega}^*) - \hat{\boldsymbol{\mu}}_U(\boldsymbol{\omega}^*)\right\}\right| \\
&\leq \left\{E\left\|\boldsymbol{\mu}_U - \tilde{\boldsymbol{\mu}}_U(\boldsymbol{\omega}^*)\right\|^2\cdot E\left\|\tilde{\boldsymbol{\mu}}_U(\boldsymbol{\omega}^*) - \hat{\boldsymbol{\mu}}_U(\boldsymbol{\omega}^*)\right\|^2\right\}^{1/2} \\
&= \left\{\tilde{R}_N(\boldsymbol{\omega}^*)\cdot E\left\|\tilde{\boldsymbol{\mu}}_U(\boldsymbol{\omega}^*) - \hat{\boldsymbol{\mu}}_U(\boldsymbol{\omega}^*)\right\|^2\right\}^{1/2}
\end{aligned}$$

$$= \left\{ \xi_N \cdot \frac{\tilde{R}_N(\boldsymbol{\omega}^*)}{R_N(\boldsymbol{\omega}^*)} \cdot E \left\| \tilde{\boldsymbol{\mu}}_U(\boldsymbol{\omega}^*) - \hat{\boldsymbol{\mu}}_U(\boldsymbol{\omega}^*) \right\|^2 \right\}^{1/2}.$$

Therefore, Lemma 4, (S.9), and $k_M(f\xi_N)^{-1} = o(1)$ imply (S.15).

We now prove (S.16). Note that

$$\begin{aligned} E \left\| \boldsymbol{\mu}_s - \hat{\boldsymbol{\mu}}_s(\boldsymbol{\omega}) \right\|_{\boldsymbol{\Phi}_s}^2 &= E \left\| \boldsymbol{\mu}_s - \tilde{\boldsymbol{\mu}}_s(\boldsymbol{\omega}) + \tilde{\boldsymbol{\mu}}_s(\boldsymbol{\omega}) - \hat{\boldsymbol{\mu}}_s(\boldsymbol{\omega}) \right\|_{\boldsymbol{\Phi}_s}^2 \\ &\leq 2E \left\| \boldsymbol{\mu}_s - \tilde{\boldsymbol{\mu}}_s(\boldsymbol{\omega}) \right\|_{\boldsymbol{\Phi}_s}^2 + 2E \left\| \tilde{\boldsymbol{\mu}}_s(\boldsymbol{\omega}) - \hat{\boldsymbol{\mu}}_s(\boldsymbol{\omega}) \right\|_{\boldsymbol{\Phi}_s}^2 \\ &= 2\tilde{R}_N(\boldsymbol{\omega}) + 2E \left\| \tilde{\boldsymbol{\mu}}_s(\boldsymbol{\omega}) - \hat{\boldsymbol{\mu}}_s(\boldsymbol{\omega}) \right\|_{\boldsymbol{\Phi}_s}^2. \end{aligned}$$

By the Cauchy–Schwarz inequality,

$$\begin{aligned} &\left| E \left\{ \boldsymbol{\mu}_s - \hat{\boldsymbol{\mu}}_s(\boldsymbol{\omega}^*) \right\}' \boldsymbol{\Phi}_s \left\{ \hat{\boldsymbol{\mu}}_s(\boldsymbol{\omega}^*) - \tilde{\boldsymbol{\mu}}_s(\boldsymbol{\omega}^*) \right\} \right| \\ &\leq \left( E \left\| \boldsymbol{\mu}_s - \hat{\boldsymbol{\mu}}_s(\boldsymbol{\omega}^*) \right\|_{\boldsymbol{\Phi}_s}^2 \cdot E \left\| \hat{\boldsymbol{\mu}}_s(\boldsymbol{\omega}^*) - \tilde{\boldsymbol{\mu}}_s(\boldsymbol{\omega}^*) \right\|_{\boldsymbol{\Phi}_s}^2 \right)^{1/2} \\ &\leq CE \left\| \hat{\boldsymbol{\mu}}_s(\boldsymbol{\omega}^*) - \tilde{\boldsymbol{\mu}}_s(\boldsymbol{\omega}^*) \right\|_{\boldsymbol{\Phi}_s}^2 + C \left\{ \tilde{R}_N(\boldsymbol{\omega}^*) \cdot E \left\| \hat{\boldsymbol{\mu}}_s(\boldsymbol{\omega}^*) - \tilde{\boldsymbol{\mu}}_s(\boldsymbol{\omega}^*) \right\|_{\boldsymbol{\Phi}_s}^2 \right\}^{1/2} \\ &= CE \left\| \hat{\boldsymbol{\mu}}_s(\boldsymbol{\omega}^*) - \tilde{\boldsymbol{\mu}}_s(\boldsymbol{\omega}^*) \right\|_{\boldsymbol{\Phi}_s}^2 + C \left\{ \xi_N \cdot \frac{\tilde{R}_N(\boldsymbol{\omega}^*)}{R_N(\boldsymbol{\omega}^*)} \cdot E \left\| \hat{\boldsymbol{\mu}}_s(\boldsymbol{\omega}^*) - \tilde{\boldsymbol{\mu}}_s(\boldsymbol{\omega}^*) \right\|_{\boldsymbol{\Phi}_s}^2 \right\}^{1/2}. \end{aligned}$$

Thus, (S.16) follows from (S.17), Lemma 4, and $k_M(f\xi_N)^{-1} = o(1)$. This completes the proof of Lemma 6.

**Lemma 7** *If Conditions 1–2 and $\boldsymbol{\mu}_U'\boldsymbol{\mu}_U = O(N)$ hold, then*

$$\sum_{m=1}^{M} \left( \boldsymbol{e}_s' \boldsymbol{\Phi}_s \hat{\boldsymbol{\mu}}_{(m)s} \right)^2 = O_p\left( f^{-1} N M k_M \right).$$

**Proof.** By Condition 1 and $\boldsymbol{\mu}_U'\boldsymbol{\mu}_U = O(N)$,

$$E\left( \boldsymbol{y}_s' \boldsymbol{\Phi}_s \boldsymbol{y}_s \right) = E_\epsilon \|\boldsymbol{y}_U\|^2 = E_\epsilon \left\| \boldsymbol{\mu}_U + \boldsymbol{e}_U \right\|^2 = O(N). \tag{S.18}$$

Note that

$$E \left\| \boldsymbol{X}_{(m)s}' \boldsymbol{\Phi}_s \boldsymbol{e}_s \right\|^2 = E \left\| \sum_{i \in s} \frac{\boldsymbol{x}_{i(m)} e_i}{\pi_i} - \sum_{i \in U} \boldsymbol{x}_{i(m)} e_i + \sum_{i \in U} \boldsymbol{x}_{i(m)} e_i \right\|^2$$

$$
\begin{aligned}
&\leq C\sum_{j=1}^{k_m} E\left\{\sum_{i\in U}\frac{(I_i-\pi_i)x_{ij(m)}e_i}{\pi_i}\right\}^2 + C\sum_{j=1}^{k_m} E\left(\sum_{i\in U} x_{ij(m)}e_i\right)^2 \\
&\leq \frac{CN}{n}\sum_{j=1}^{k_m}\sum_{i\in U} E_\epsilon\left(x_{ij(m)}^2 e_i^2\right) + C\sum_{j=1}^{k_m}\sum_{i\in U} E_\epsilon\left(x_{ij(m)}^2 e_i^2\right) \\
&\leq Cf^{-1}Nk_m, \qquad \text{(S.19)}
\end{aligned}
$$

where the third line follows from Condition 1, Lemma 2, and Theorem 2 of Whittle (1960). It follows from (S.18), (S.19), and Condition 2 that

$$
\begin{aligned}
\sum_{m=1}^{M}\left(\boldsymbol{e}_s'\boldsymbol{\Phi}_s\hat{\boldsymbol{\mu}}_{(m)s}\right)^2 &= \sum_{m=1}^{M}\left\{\boldsymbol{e}_s'\boldsymbol{\Phi}_s\boldsymbol{X}_{(m)s}\left(\boldsymbol{X}_{(m)s}'\boldsymbol{\Phi}_s\boldsymbol{X}_{(m)s}\right)^{-1}\boldsymbol{X}_{(m)s}'\boldsymbol{\Phi}_s\boldsymbol{y}_s\right\}^2 \\
&\leq \sum_{m=1}^{M}\left\{\boldsymbol{e}_s'\boldsymbol{\Phi}_s\boldsymbol{X}_{(m)s}\left(\boldsymbol{X}_{(m)s}'\boldsymbol{\Phi}_s\boldsymbol{X}_{(m)s}\right)^{-1}\boldsymbol{X}_{(m)s}'\boldsymbol{\Phi}_s\boldsymbol{e}_s\right\}\cdot(\boldsymbol{y}_s'\boldsymbol{\Phi}_s\boldsymbol{y}_s) \\
&\leq \frac{C}{N}\cdot\left(\sum_{m=1}^{M}\left\|\boldsymbol{X}_{(m)s}'\boldsymbol{\Phi}_s\boldsymbol{e}_s\right\|^2\right)\cdot(\boldsymbol{y}_s'\boldsymbol{\Phi}_s\boldsymbol{y}_s) \\
&= O_p\left(f^{-1}NMk_M\right).
\end{aligned}
$$

This completes the proof of Lemma 7.

## S3.2 Proof of Theorem 1

The proof of this theorem relies on Lemmas 1–4.

**Proof.** We begin by establishing the asymptotic unbiasedness of $C_n(\boldsymbol{\omega})$ for $R_N(\boldsymbol{\omega}) + N\sigma^2$. By direct calculation, we have

$$
\begin{aligned}
C_n(\boldsymbol{\omega}) &= \|\boldsymbol{y}_s - \hat{\boldsymbol{\mu}}_s(\boldsymbol{\omega})\|_{\boldsymbol{\Phi}_s}^2 + 2f^{-1}\sigma^2 k(\boldsymbol{\omega}) \\
&= \|\boldsymbol{\mu}_s - \hat{\boldsymbol{\mu}}_s(\boldsymbol{\omega})\|_{\boldsymbol{\Phi}_s}^2 + 2\boldsymbol{e}_s'\boldsymbol{\Phi}_s\left\{\boldsymbol{\mu}_s - \hat{\boldsymbol{\mu}}_s(\boldsymbol{\omega})\right\} + \boldsymbol{e}_s'\boldsymbol{\Phi}_s\boldsymbol{e}_s + 2f^{-1}\sigma^2 k(\boldsymbol{\omega}) \\
&= \|\boldsymbol{\mu}_s - \tilde{\boldsymbol{\mu}}_s(\boldsymbol{\omega})\|_{\boldsymbol{\Phi}_s}^2 + \|\tilde{\boldsymbol{\mu}}_s(\boldsymbol{\omega}) - \hat{\boldsymbol{\mu}}_s(\boldsymbol{\omega})\|_{\boldsymbol{\Phi}_s}^2 \\
&\quad + 2\left\{\boldsymbol{\mu}_s - \tilde{\boldsymbol{\mu}}_s(\boldsymbol{\omega})\right\}'\boldsymbol{\Phi}_s\left\{\tilde{\boldsymbol{\mu}}_s(\boldsymbol{\omega}) - \hat{\boldsymbol{\mu}}_s(\boldsymbol{\omega})\right\} + 2f^{-1}\sigma^2 k(\boldsymbol{\omega}) \\
&\quad + 2\boldsymbol{e}_s'\boldsymbol{\Phi}_s\left\{\boldsymbol{\mu}_s - \tilde{\boldsymbol{\mu}}_s(\boldsymbol{\omega})\right\} + 2\boldsymbol{e}_s'\boldsymbol{\Phi}_s\left\{\tilde{\boldsymbol{\mu}}_s(\boldsymbol{\omega}) - \hat{\boldsymbol{\mu}}_s(\boldsymbol{\omega})\right\} + \boldsymbol{e}_s'\boldsymbol{\Phi}_s\boldsymbol{e}_s. \qquad \text{(S.20)}
\end{aligned}
$$

Note that

$$E(\boldsymbol{e}_s'\boldsymbol{\Phi}_s\boldsymbol{e}_s) = E_\epsilon E_d\left(\sum_{i\in s}\frac{e_i^2}{\pi_i}\right) = E_\epsilon\left(\sum_{i\in U} e_i^2\right) = N\sigma^2, \tag{S.21}$$

$$\begin{aligned} E\left\|\boldsymbol{\mu}_s - \tilde{\boldsymbol{\mu}}_s(\boldsymbol{\omega})\right\|_{\boldsymbol{\Phi}_s}^2 &= E_\epsilon E_d\left[\sum_{i\in s}\frac{\left\{\mu_i - \tilde{\mu}_i(\boldsymbol{\omega})\right\}^2}{\pi_i}\right] \\ &= E_\epsilon\left[\sum_{i\in U}\left\{\mu_i - \tilde{\mu}_i(\boldsymbol{\omega})\right\}^2\right] \\ &= \tilde{R}_N(\boldsymbol{\omega}), \end{aligned} \tag{S.22}$$

and

$$\begin{aligned} E\left[2\boldsymbol{e}_s'\boldsymbol{\Phi}_s\left\{\boldsymbol{\mu}_s - \tilde{\boldsymbol{\mu}}_s(\boldsymbol{\omega})\right\}\right] &= 2E_\epsilon E_d\left(\sum_{i\in s}\frac{e_i\mu_i}{\pi_i}\right) - 2E_\epsilon E_d\left\{\sum_{i\in s}\frac{e_i\tilde{\mu}_i(\boldsymbol{\omega})}{\pi_i}\right\} \\ &= 2E_\epsilon(\boldsymbol{e}_U'\boldsymbol{\mu}_U) - 2E_\epsilon\left\{\boldsymbol{e}_U'\tilde{\boldsymbol{\mu}}_U(\boldsymbol{\omega})\right\} \\ &= -2\sum_{m=1}^{M}\left[\omega_m\cdot E_\epsilon\left\{\boldsymbol{e}_U'\boldsymbol{X}_{(m)U}\left(\boldsymbol{X}_{(m)U}'\boldsymbol{X}_{(m)U}\right)^{-1}\boldsymbol{X}_{(m)U}'\boldsymbol{e}_U\right\}\right] \\ &= -2\sigma^2 k(\boldsymbol{\omega}). \end{aligned} \tag{S.23}$$

Combining Equations (S.20)-(S.23), we have

$$\begin{aligned} E\left(C_n(\boldsymbol{\omega})\right) = \tilde{R}_N(\boldsymbol{\omega}) &+ E\left\|\tilde{\boldsymbol{\mu}}_s(\boldsymbol{\omega}) - \hat{\boldsymbol{\mu}}_s(\boldsymbol{\omega})\right\|_{\boldsymbol{\Phi}_s}^2 \\ &+ 2E\left\{\boldsymbol{\mu}_s - \tilde{\boldsymbol{\mu}}_s(\boldsymbol{\omega})\right\}'\boldsymbol{\Phi}_s\left\{\tilde{\boldsymbol{\mu}}_s(\boldsymbol{\omega}) - \hat{\boldsymbol{\mu}}_s(\boldsymbol{\omega})\right\} \\ &+ 2E\left[\boldsymbol{e}_s'\boldsymbol{\Phi}_s\left\{\tilde{\boldsymbol{\mu}}_s(\boldsymbol{\omega}) - \hat{\boldsymbol{\mu}}_s(\boldsymbol{\omega})\right\}\right] + N\sigma^2 \\ &+ 2(f^{-1} - 1)\sigma^2 k(\boldsymbol{\omega}). \end{aligned}$$

Therefore, it suffices to show that

$$\frac{\left|\tilde{R}_N(\boldsymbol{\omega}) - R_N(\boldsymbol{\omega})\right|}{R_N(\boldsymbol{\omega})} = o(1), \tag{S.24}$$

$$\frac{E\left\|\tilde{\boldsymbol{\mu}}_s(\boldsymbol{\omega}) - \hat{\boldsymbol{\mu}}_s(\boldsymbol{\omega})\right\|_{\boldsymbol{\Phi}_s}^2}{R_N(\boldsymbol{\omega})} = o(1), \tag{S.25}$$

$$\frac{E\left|\{\boldsymbol{\mu}_s - \tilde{\boldsymbol{\mu}}_s(\boldsymbol{\omega})\}' \boldsymbol{\Phi}_s \{\tilde{\boldsymbol{\mu}}_s(\boldsymbol{\omega}) - \hat{\boldsymbol{\mu}}_s(\boldsymbol{\omega})\}\right|}{R_N(\boldsymbol{\omega})} = o(1), \tag{S.26}$$

$$\frac{E\left|\boldsymbol{e}_s' \boldsymbol{\Phi}_s \{\tilde{\boldsymbol{\mu}}_s(\boldsymbol{\omega}) - \hat{\boldsymbol{\mu}}_s(\boldsymbol{\omega})\}\right|}{R_N(\boldsymbol{\omega})} = o(1), \tag{S.27}$$

and

$$\sup_{\boldsymbol{\omega}\in\mathcal{W}} \frac{2(1-f)\sigma^2 k(\boldsymbol{\omega})}{f R_N(\boldsymbol{\omega})} = o(1). \tag{S.28}$$

Condition 3 implies that $k_M(f\xi_N)^{-1} = o(1)$. Together with Lemma 4, this yields (S.24) and (S.28). For (S.25), Condition 1 and Lemmas 1 and 3 imply that

$$\begin{aligned}
E\left\|\hat{\boldsymbol{\mu}}_s(\boldsymbol{\omega}) - \tilde{\boldsymbol{\mu}}_s(\boldsymbol{\omega})\right\|_{\boldsymbol{\Phi}_s}^2 &= E\left|\sum_{m=1}^{M}\sum_{l=1}^{M} \omega_m\omega_l \left(\hat{\boldsymbol{\beta}}_{(m)} - \tilde{\boldsymbol{\beta}}_{(m)}\right)' \boldsymbol{X}_{(m)s}' \boldsymbol{\Phi}_s \boldsymbol{X}_{(l)s} \left(\hat{\boldsymbol{\beta}}_{(l)} - \tilde{\boldsymbol{\beta}}_{(l)}\right)\right| \\
&\le \sum_{m=1}^{M}\sum_{l=1}^{M} \omega_m\omega_l E\left|\left(\hat{\boldsymbol{\beta}}_{(m)} - \tilde{\boldsymbol{\beta}}_{(m)}\right)' \boldsymbol{X}_{(m)s}' \boldsymbol{\Phi}_s \boldsymbol{X}_{(l)s} \left(\hat{\boldsymbol{\beta}}_{(l)} - \tilde{\boldsymbol{\beta}}_{(l)}\right)\right| \\
&\le \left\{\sum_{m=1}^{M} \omega_m \cdot \lambda_{\max}^{1/2}\left(\boldsymbol{X}_{(m)s}' \boldsymbol{\Phi}_s \boldsymbol{X}_{(m)s}\right) \cdot \left(E\left\|\hat{\boldsymbol{\beta}}_{(m)} - \tilde{\boldsymbol{\beta}}_{(m)}\right\|^2\right)^{1/2}\right\}^2 \\
&\le C f^{-1} k_M.
\end{aligned} \tag{S.29}$$

Since $\xi_N \le R_N(\boldsymbol{\omega})$, it follows from $k_M(f\xi_N)^{-1} = o(1)$ that

$$\frac{E\left\|\tilde{\boldsymbol{\mu}}_s(\boldsymbol{\omega}) - \hat{\boldsymbol{\mu}}_s(\boldsymbol{\omega})\right\|_{\boldsymbol{\Phi}_s}^2}{R_N(\boldsymbol{\omega})} \le \frac{C k_M}{f\xi_N} = o(1).$$

Hence, (S.25) holds.

For (S.26), the Cauchy–Schwarz inequality implies

$$\begin{aligned}
&E\left|\left(\boldsymbol{\mu}_s - \tilde{\boldsymbol{\mu}}_s(\boldsymbol{\omega})\right)' \boldsymbol{\Phi}_s \left(\tilde{\boldsymbol{\mu}}_s(\boldsymbol{\omega}) - \hat{\boldsymbol{\mu}}_s(\boldsymbol{\omega})\right)\right| \\
&\le \left(E\left\|\boldsymbol{\mu}_s - \tilde{\boldsymbol{\mu}}_s(\boldsymbol{\omega})\right\|_{\boldsymbol{\Phi}_s}^2 \cdot E\left\|\tilde{\boldsymbol{\mu}}_s(\boldsymbol{\omega}) - \hat{\boldsymbol{\mu}}_s(\boldsymbol{\omega})\right\|_{\boldsymbol{\Phi}_s}^2\right)^{1/2} \\
&= \left(E_\epsilon\left\|\boldsymbol{\mu}_U - \tilde{\boldsymbol{\mu}}_U(\boldsymbol{\omega})\right\|^2 \cdot E\left\|\tilde{\boldsymbol{\mu}}_s(\boldsymbol{\omega}) - \hat{\boldsymbol{\mu}}_s(\boldsymbol{\omega})\right\|_{\boldsymbol{\Phi}_s}^2\right)^{1/2} \\
&= \left(\tilde{R}_N(\boldsymbol{\omega}) \cdot E\left\|\tilde{\boldsymbol{\mu}}_s(\boldsymbol{\omega}) - \hat{\boldsymbol{\mu}}_s(\boldsymbol{\omega})\right\|_{\boldsymbol{\Phi}_s}^2\right)^{1/2}.
\end{aligned}$$

Dividing both sides by $R_N(\boldsymbol{\omega})$ and using (S.24) and (S.25), we obtain (S.26).

For (S.27), applying the Cauchy–Schwarz inequality together with (S.29) yields

$$\begin{aligned} E\left|\boldsymbol{e}_s'\boldsymbol{\Phi}_s\left\{\tilde{\boldsymbol{\mu}}_s(\boldsymbol{\omega})-\hat{\boldsymbol{\mu}}_s(\boldsymbol{\omega})\right\}\right| &\leq \left(E\|\boldsymbol{e}_s\|_{\boldsymbol{\Phi}_s}^2\cdot E\left\|\tilde{\boldsymbol{\mu}}_s(\boldsymbol{\omega})-\hat{\boldsymbol{\mu}}_s(\boldsymbol{\omega})\right\|_{\boldsymbol{\Phi}_s}^2\right)^{1/2} \\ &= \left(N\sigma^2\cdot E\left\|\tilde{\boldsymbol{\mu}}_s(\boldsymbol{\omega})-\hat{\boldsymbol{\mu}}_s(\boldsymbol{\omega})\right\|_{\boldsymbol{\Phi}_s}^2\right)^{1/2} \\ &\leq Cf^{-1}(nk_M)^{1/2}. \end{aligned}$$

Dividing both sides by $R_N(\boldsymbol{\omega})$ and using Condition 3, we obtain (S.27). This completes the proof of the asymptotic unbiasedness of $C_n(\boldsymbol{\omega})$.

We next verify that $C_n^*(\boldsymbol{\omega})$ is asymptotically unbiased for $R_N(\boldsymbol{\omega})+N\sigma^2$. Comparing the definitions of $C_n^*(\boldsymbol{\omega})$ and $C_n(\boldsymbol{\omega})$, and using (S.20), it suffices to show that

$$\frac{k(\boldsymbol{\omega})\cdot E\left|\hat{\sigma}_{(M)}^2-\sigma^2\right|}{fR_N(\boldsymbol{\omega})}=o(1). \tag{S.30}$$

Note that

$$\left|\hat{\sigma}_{(M)}^2-\sigma^2\right|\leq\left|\hat{\sigma}_{(M)}^2-\tilde{\sigma}_{(M)}^2\right|+\left|\tilde{\sigma}_{(M)}^2-\sigma^2\right|,$$

where

$$\tilde{\sigma}_{(M)}^2=\frac{1}{N-k_M}\left\|\boldsymbol{y}_U-\boldsymbol{X}_{(M)U}\tilde{\boldsymbol{\beta}}_{(M)}\right\|^2.$$

Therefore, the proof of (S.30) reduces to verifying the following two relations:

$$\frac{k_M\cdot E\left|\hat{\sigma}_{(M)}^2-\tilde{\sigma}_{(M)}^2\right|}{f\xi_N}=o(1), \tag{S.31}$$

and

$$\frac{k_M\cdot E\left|\tilde{\sigma}_{(M)}^2-\sigma^2\right|}{f\xi_N}=o(1). \tag{S.32}$$

We first prove (S.31). Condition 1, Lemma 1 and $k_M^2=O(N)$ imply that

$$\begin{aligned} E_\epsilon\left(y_i-\tilde{\mu}_{i(m)}\right)^4 &= E_\epsilon\left(e_{i(m)}^*+\boldsymbol{x}_{i(m)}'\boldsymbol{\beta}_{(m)}^*-\boldsymbol{x}_{i(m)}'\tilde{\boldsymbol{\beta}}_{(m)}\right)^4 \\ &\leq CE_\epsilon\left(e_{i(m)}^{*4}\right)+Ck_m^2\cdot E_\epsilon\left\|\tilde{\boldsymbol{\beta}}_{(m)}-\boldsymbol{\beta}_{(m)}^*\right\|^4 \end{aligned}$$

$$= O(1) \tag{S.33}$$

uniformly in $i$ and $m$. Then, by Condition 2, we have

$$\begin{aligned}
&E\left|\frac{1}{n-k_M}\left\|\boldsymbol{y}_s-\boldsymbol{X}_{(M)s}\tilde{\boldsymbol{\beta}}_{(M)}\right\|^2-\frac{1}{N-k_M}\left\|\boldsymbol{y}_U-\boldsymbol{X}_{(M)U}\tilde{\boldsymbol{\beta}}_{(M)}\right\|^2\right| \\
&= E\left|\sum_{i\in U}\left(\frac{I_i}{n-k_M}-\frac{1}{N-k_M}\right)\left(y_i-\tilde{\mu}_{i(M)}\right)^2\right| \\
&\leq \frac{C}{n-k_M}\cdot\sum_{i\in U}E_\epsilon\left(y_i-\tilde{\mu}_{i(M)}\right)^2 \\
&= O(1),
\end{aligned}$$

where the last equality follows from $f=\sum_U \pi_i/N \geq c_5$ and $k_M^2 = O(N)$. Moreover, Condition 1, Lemma 3, and (S.33) yield

$$E\left\|\boldsymbol{X}_{(M)s}\tilde{\boldsymbol{\beta}}_{(M)}-\boldsymbol{X}_{(M)s}\hat{\boldsymbol{\beta}}_{(M)}\right\|^2 \leq E\left\{\lambda_{\max}\left(\boldsymbol{X}'_{(M)s}\boldsymbol{X}_{(M)s}\right)\left\|\tilde{\boldsymbol{\beta}}_{(M)}-\hat{\boldsymbol{\beta}}_{(M)}\right\|^2\right\} \leq Ck_M^2$$

and

$$\begin{aligned}
&E\left|\left(\boldsymbol{y}_s-\boldsymbol{X}_{(M)s}\tilde{\boldsymbol{\beta}}_{(M)}\right)'\left(\boldsymbol{X}_{(M)s}\tilde{\boldsymbol{\beta}}_{(M)}-\boldsymbol{X}_{(M)s}\hat{\boldsymbol{\beta}}_{(M)}\right)\right| \\
&\leq \left(E\left\|\boldsymbol{y}_s-\boldsymbol{X}_{(M)s}\tilde{\boldsymbol{\beta}}_{(M)}\right\|^2\cdot E\left\|\boldsymbol{X}_{(M)s}\tilde{\boldsymbol{\beta}}_{(M)}-\boldsymbol{X}_{(M)s}\hat{\boldsymbol{\beta}}_{(M)}\right\|^2\right)^{1/2} \\
&\leq C\left\{\sum_{i\in U}\pi_i E_\epsilon\left(y_i-\tilde{\mu}_{i(M)}\right)^2\cdot k_M^2\right\}^{1/2} \\
&= O(n^{1/2}k_M).
\end{aligned}$$

Using the preceding bounds and the triangle inequality, we obtain that

$$\begin{aligned}
&\frac{k_M\cdot E\left|\hat{\sigma}^2_{(M)}-\tilde{\sigma}^2_{(M)}\right|}{f\xi_N} \\
&\leq \frac{k_M}{f\xi_N}\cdot E\left|\frac{1}{n-k_M}\left\|\boldsymbol{y}_s-\boldsymbol{X}_{(M)s}\hat{\boldsymbol{\beta}}_{(M)}\right\|^2-\frac{1}{N-k_M}\left\|\boldsymbol{y}_U-\boldsymbol{X}_{(M)U}\tilde{\boldsymbol{\beta}}_{(M)}\right\|^2\right| \\
&= \frac{k_M}{f\xi_N}\cdot E\left|\frac{1}{n-k_M}\left\|\boldsymbol{y}_s-\boldsymbol{X}_{(M)s}\tilde{\boldsymbol{\beta}}_{(M)}\right\|^2-\frac{1}{N-k_M}\left\|\boldsymbol{y}_U-\boldsymbol{X}_{(M)U}\tilde{\boldsymbol{\beta}}_{(M)}\right\|^2\right. \\
&\quad+\frac{2}{n-k_M}\left(\boldsymbol{y}_s-\boldsymbol{X}_{(M)s}\tilde{\boldsymbol{\beta}}_{(M)}\right)'\left(\boldsymbol{X}_{(M)s}\tilde{\boldsymbol{\beta}}_{(M)}-\boldsymbol{X}_{(M)s}\hat{\boldsymbol{\beta}}_{(M)}\right) \\
&\quad\left.+\frac{1}{n-k_M}\left\|\boldsymbol{X}_{(M)s}\tilde{\boldsymbol{\beta}}_{(M)}-\boldsymbol{X}_{(M)s}\hat{\boldsymbol{\beta}}_{(M)}\right\|^2\right|
\end{aligned}$$

$$= o(1),$$

where the last step follows from the conditions $k_M(f\xi_N)^{-1} = o(1)$ and $k_M^2 = O(N)$. This proves (S.31).

We now verify (S.32). Note that

$$\tilde{\sigma}^2_{(M)} = \frac{1}{N-k_M}\left\{\boldsymbol{\mu}_U'\left(\boldsymbol{I}_N - \boldsymbol{P}_{(M)U}\right)\boldsymbol{\mu}_U + \boldsymbol{e}_U'\left(\boldsymbol{I}_N - \boldsymbol{P}_{(M)U}\right)\boldsymbol{e}_U \right.$$
$$\left. +2\boldsymbol{\mu}_U'\left(\boldsymbol{I}_N - \boldsymbol{P}_{(M)U}\right)\boldsymbol{e}_U\right\},$$

where $\boldsymbol{P}_{(M)U} = \boldsymbol{X}_{(M)U}\left(\boldsymbol{X}_{(M)U}'\boldsymbol{X}_{(M)U}\right)^{-1}\boldsymbol{X}_{(M)U}'$. Since

$$E\left\{\boldsymbol{\mu}_U'\left(\boldsymbol{I}_N - \boldsymbol{P}_{(M)U}\right)\boldsymbol{e}_U\right\} = 0$$

and

$$E\left\{\boldsymbol{e}_U'\left(\boldsymbol{I}_N - \boldsymbol{P}_{(M)U}\right)\boldsymbol{e}_U\right\} = (N-k_M)\sigma^2,$$

it follows from Theorem 2 of Whittle (1960) that

$$E\left\{\boldsymbol{\mu}_U'\left(\boldsymbol{I}_N - \boldsymbol{P}_{(M)U}\right)\boldsymbol{e}_U\right\}^2 \leq C\left\|\boldsymbol{\mu}_U'\left(\boldsymbol{I}_N - \boldsymbol{P}_{(M)U}\right)\right\|^2 = O(\|\boldsymbol{\mu}_U\|^2)$$

and

$$E\left\{\boldsymbol{e}_U'\left(\boldsymbol{I}_N - \boldsymbol{P}_{(M)U}\right)\boldsymbol{e}_U - (N-k_M)\sigma^2\right\}^2 \leq C\,\mathrm{tr}\left\{\left(\boldsymbol{I}_N - \boldsymbol{P}_{(M)U}\right)^2\right\} = O(N-k_M).$$

Then, using $\|\boldsymbol{\mu}_U\|^2 = O(N)$ and the fact that $\boldsymbol{I}_N - \boldsymbol{P}_{(M)U}$ is symmetric and idempotent, we have

$$\begin{aligned}
&\frac{k_M \cdot E\left|\tilde{\sigma}^2_{(M)} - \sigma^2\right|}{f\xi_N} \\
&\leq \frac{k_M}{(N-k_M)f\xi_N}\left|\boldsymbol{\mu}_U'\left(\boldsymbol{I}_N - \boldsymbol{P}_{(M)U}\right)\boldsymbol{\mu}_U\right| + \frac{2k_M}{(N-k_M)f\xi_N}E\left|\boldsymbol{\mu}_U'\left(\boldsymbol{I}_N - \boldsymbol{P}_{(M)U}\right)\boldsymbol{e}_U\right| \\
&\quad + \frac{k_M}{(N-k_M)f\xi_N}E\left|\boldsymbol{e}_U'\left(\boldsymbol{I}_N - \boldsymbol{P}_{(M)U}\right)\boldsymbol{e}_U - (N-k_M)\sigma^2\right| \\
&= O\left(\frac{Nk_M}{(N-k_M)f\xi_N}\right)
\end{aligned}$$

$$= o(1),$$

where the last step follows from the conditions $k_M(f\xi_N)^{-1} = o(1)$ and $k_M^2 = O(N)$. This proves (S.32) and completes the proof of Theorem 1.

## S3.3 Proof of Corollary 1

The proof of this corollary relies on Lemmas 1–4.

**Proof.** By Equations (S.1) and (S.2), we have

$$\begin{aligned} E\left(C_n(\boldsymbol{\omega})\right) = R_N(\boldsymbol{\omega}) + N\sigma^2 + 2f^{-1}\sigma^2 k(\boldsymbol{\omega}) - 2\sigma^2 E_d\left[k_s(\boldsymbol{\omega})\right] \\ + \left\{\tilde{R}_N(\boldsymbol{\omega}) - R_N(\boldsymbol{\omega})\right\} + E\left\|\tilde{\boldsymbol{\mu}}_s(\boldsymbol{\omega}) - \hat{\boldsymbol{\mu}}_s(\boldsymbol{\omega})\right\|_{\boldsymbol{\Phi}_s}^2 \\ + 2E\left[\left\{\boldsymbol{\mu}_s - \tilde{\boldsymbol{\mu}}_s(\boldsymbol{\omega})\right\}' \boldsymbol{\Phi}_s \left\{\tilde{\boldsymbol{\mu}}_s(\boldsymbol{\omega}) - \hat{\boldsymbol{\mu}}_s(\boldsymbol{\omega})\right\}\right]. \end{aligned}$$

Following Equations (S.24)–(S.26) in the proof of Theorem 1, the absolute values of the last three terms on the right-hand side are all of order $o(R_N(\boldsymbol{\omega}))$ under the condition $k_M(f\xi_N)^{-1} = o(1)$. Moreover, Conditions 1 and 2 imply that

$$\begin{aligned} E_d\left[k_s(\boldsymbol{\omega})\right] &= E_d\left[\sum_{m=1}^{M} \omega_m \operatorname{tr}\left\{\boldsymbol{\Phi}_s \boldsymbol{X}_{(m)s}\left(\boldsymbol{X}'_{(m)s}\boldsymbol{\Phi}_s\boldsymbol{X}_{(m)s}\right)^{-1}\boldsymbol{X}'_{(m)s}\boldsymbol{\Phi}_s\right\}\right] \\ &= E_d\left[\sum_{m=1}^{M} \omega_m \operatorname{tr}\left\{\left(\boldsymbol{X}'_{(m)s}\boldsymbol{\Phi}_s\boldsymbol{X}_{(m)s}\right)^{-1}\boldsymbol{X}'_{(m)s}\boldsymbol{\Phi}_s^2\boldsymbol{X}_{(m)s}\right\}\right] \\ &= O(k_M). \end{aligned}$$

Since $\xi_N \leq R_N(\boldsymbol{\omega})$, it follows from $k_M(f\xi_N)^{-1} = o(1)$ that

$$\frac{\left|2f^{-1}\sigma^2 k(\boldsymbol{\omega}) - 2\sigma^2 E_d\left(k_s(\boldsymbol{\omega})\right)\right|}{R_N(\boldsymbol{\omega})} \leq \frac{Ck_M}{f\xi_N} = o(1).$$

Therefore, $E\left(C_n(\boldsymbol{\omega})\right) = R_N(\boldsymbol{\omega}) + N\sigma^2 + o\left(R_N(\boldsymbol{\omega})\right)$.

The result for $C_n^*(\boldsymbol{\omega})$ follows from Equation (S.30) in the proof of Theorem 1, together with the conditions $\boldsymbol{\mu}_U'\boldsymbol{\mu}_U = O(N)$ and $k_M^2 = O(N)$. This completes the proof of Corollary 1.

## S3.4 Proof of Theorem 2

The proof of this theorem relies on Lemmas 1–5.

**Proof.** We first prove Equation (12) in Theorem 2. From Equation (S.20), we have

$$\begin{aligned} C_n(\boldsymbol{\omega}) = L_N(\boldsymbol{\omega}) &+ \left\{ \|\boldsymbol{\mu}_s - \tilde{\boldsymbol{\mu}}_s(\boldsymbol{\omega})\|^2_{\boldsymbol{\Phi}_s} - L_N(\boldsymbol{\omega}) \right\} + \|\tilde{\boldsymbol{\mu}}_s(\boldsymbol{\omega}) - \hat{\boldsymbol{\mu}}_s(\boldsymbol{\omega})\|^2_{\boldsymbol{\Phi}_s} \\ &+ 2\boldsymbol{e}_s'\boldsymbol{\Phi}_s \left\{ \tilde{\boldsymbol{\mu}}_s(\boldsymbol{\omega}) - \hat{\boldsymbol{\mu}}_s(\boldsymbol{\omega}) \right\} + 2\left[ \boldsymbol{e}_s'\boldsymbol{\Phi}_s \left\{ \boldsymbol{\mu}_s - \tilde{\boldsymbol{\mu}}_s(\boldsymbol{\omega}) \right\} + 2\sigma^2 k(\boldsymbol{\omega}) \right] \\ &+ 2\left\{ \boldsymbol{\mu}_s - \tilde{\boldsymbol{\mu}}_s(\boldsymbol{\omega}) \right\}' \boldsymbol{\Phi}_s \left\{ \tilde{\boldsymbol{\mu}}_s(\boldsymbol{\omega}) - \hat{\boldsymbol{\mu}}_s(\boldsymbol{\omega}) \right\} + \boldsymbol{e}_s'\boldsymbol{\Phi}_s\boldsymbol{e}_s + 2(f^{-1} - 1)\sigma^2 k(\boldsymbol{\omega}). \end{aligned} \tag{S.34}$$

By (S.28) and arguments similar to those in Li (1987) and Wan et al. (2010), it suffices to show that

$$\sup_{\boldsymbol{\omega}\in\mathcal{W}} \frac{\|\tilde{\boldsymbol{\mu}}_s(\boldsymbol{\omega}) - \hat{\boldsymbol{\mu}}_s(\boldsymbol{\omega})\|^2_{\boldsymbol{\Phi}_s}}{R_N(\boldsymbol{\omega})} = o_p(1), \tag{S.35}$$

$$\sup_{\boldsymbol{\omega}\in\mathcal{W}} \frac{|\boldsymbol{e}_s'\boldsymbol{\Phi}_s \left\{ \tilde{\boldsymbol{\mu}}_s(\boldsymbol{\omega}) - \hat{\boldsymbol{\mu}}_s(\boldsymbol{\omega}) \right\}|}{R_N(\boldsymbol{\omega})} = o_p(1), \tag{S.36}$$

$$\sup_{\boldsymbol{\omega}\in\mathcal{W}} \frac{\left|\left\{ \boldsymbol{\mu}_s - \tilde{\boldsymbol{\mu}}_s(\boldsymbol{\omega}) \right\}' \boldsymbol{\Phi}_s \left\{ \tilde{\boldsymbol{\mu}}_s(\boldsymbol{\omega}) - \hat{\boldsymbol{\mu}}_s(\boldsymbol{\omega}) \right\}\right|}{R_N(\boldsymbol{\omega})} = o_p(1), \tag{S.37}$$

$$\sup_{\boldsymbol{\omega}\in\mathcal{W}} \frac{|\boldsymbol{e}_s'\boldsymbol{\Phi}_s \left\{ \boldsymbol{\mu}_s - \tilde{\boldsymbol{\mu}}_s(\boldsymbol{\omega}) \right\} + \sigma^2 k(\boldsymbol{\omega})|}{R_N(\boldsymbol{\omega})} = o_p(1), \tag{S.38}$$

$$\sup_{\boldsymbol{\omega}\in\mathcal{W}} \frac{\left| \|\boldsymbol{\mu}_s - \tilde{\boldsymbol{\mu}}_s(\boldsymbol{\omega})\|^2_{\boldsymbol{\Phi}_s} - L_N(\boldsymbol{\omega}) \right|}{R_N(\boldsymbol{\omega})} = o_p(1), \tag{S.39}$$

and

$$\sup_{\boldsymbol{\omega}\in\mathcal{W}} \frac{|L_N(\boldsymbol{\omega}) - R_N(\boldsymbol{\omega})|}{R_N(\boldsymbol{\omega})} = o_p(1). \tag{S.40}$$

Using Bonferroni's inequality and Markov's inequality, for any $\delta > 0$, we have

$$P\left( \sup_{\boldsymbol{\omega}\in\mathcal{W}} \left\{ R_N^{-1}(\boldsymbol{\omega}) \|\tilde{\boldsymbol{\mu}}_s(\boldsymbol{\omega}) - \hat{\boldsymbol{\mu}}_s(\boldsymbol{\omega})\|^2_{\boldsymbol{\Phi}_s} \right\} > \delta \right)$$

$$
\begin{aligned}
&\leq P\left(\sup_{\boldsymbol{\omega}\in\mathcal{W}}\|\tilde{\boldsymbol{\mu}}_s(\boldsymbol{\omega})-\hat{\boldsymbol{\mu}}_s(\boldsymbol{\omega})\|_{\boldsymbol{\Phi}_s}^2>\delta\xi_N\right)\\
&=P\left(\sup_{\boldsymbol{\omega}\in\mathcal{W}}\left\{\sum_{m=1}^{M}\sum_{t=1}^{M}\omega_m\omega_t\left(\tilde{\boldsymbol{\beta}}_{(m)}-\hat{\boldsymbol{\beta}}_{(m)}\right)'\boldsymbol{X}_{(m)s}'\boldsymbol{\Phi}_s\boldsymbol{X}_{(t)s}\left(\tilde{\boldsymbol{\beta}}_{(t)}-\hat{\boldsymbol{\beta}}_{(t)}\right)\right\}>\delta\xi_N\right)\\
&\leq\sum_{m=1}^{M}\sum_{t=1}^{M}P\left(\left|\left(\tilde{\boldsymbol{\beta}}_{(m)}-\hat{\boldsymbol{\beta}}_{(m)}\right)'\boldsymbol{X}_{(m)s}'\boldsymbol{\Phi}_s\boldsymbol{X}_{(t)s}\left(\tilde{\boldsymbol{\beta}}_{(t)}-\hat{\boldsymbol{\beta}}_{(t)}\right)\right|>\delta\xi_N\right)\\
&\leq\delta^{-1}\xi_N^{-1}\sum_{m=1}^{M}\sum_{t=1}^{M}E\left|\left(\tilde{\boldsymbol{\beta}}_{(m)}-\hat{\boldsymbol{\beta}}_{(m)}\right)'\boldsymbol{X}_{(m)s}'\boldsymbol{\Phi}_s\boldsymbol{X}_{(t)s}\left(\tilde{\boldsymbol{\beta}}_{(t)}-\hat{\boldsymbol{\beta}}_{(t)}\right)\right|\\
&\leq\delta^{-1}\xi_N^{-1}E\left[\sum_{m=1}^{M}\left\{\lambda_{\max}^{1/2}\left(\boldsymbol{X}_{(m)s}'\boldsymbol{\Phi}_s\boldsymbol{X}_{(m)s}\right)\cdot\left\|\tilde{\boldsymbol{\beta}}_{(m)}-\hat{\boldsymbol{\beta}}_{(m)}\right\|\right\}\right]^2\\
&\leq M\delta^{-1}\xi_N^{-1}\sum_{m=1}^{M}\left\{\lambda_{\max}\left(\boldsymbol{X}_{(m)s}'\boldsymbol{\Phi}_s\boldsymbol{X}_{(m)s}\right)\cdot E\left\|\tilde{\boldsymbol{\beta}}_{(m)}-\hat{\boldsymbol{\beta}}_{(m)}\right\|^2\right\}\\
&=O\left(\frac{M^2k_M}{f\xi_N}\right),
\end{aligned}
$$

where the last bound follows from Lemmas 1 and 3, together with Condition 1. Hence, (S.35) follows from Condition 3.

Similarly, to prove (S.36), for any $\delta>0$, we have

$$
\begin{aligned}
&P\left(\sup_{\boldsymbol{\omega}\in\mathcal{W}}\left[R_N^{-1}(\boldsymbol{\omega})\left|\boldsymbol{e}_s'\boldsymbol{\Phi}_s\left\{\tilde{\boldsymbol{\mu}}_s(\boldsymbol{\omega})-\hat{\boldsymbol{\mu}}_s(\boldsymbol{\omega})\right\}\right|\right]>\delta\right)\\
&\leq P\left(\sup_{\boldsymbol{\omega}\in\mathcal{W}}\left|\boldsymbol{e}_s'\boldsymbol{\Phi}_s\left\{\tilde{\boldsymbol{\mu}}_s(\boldsymbol{\omega})-\hat{\boldsymbol{\mu}}_s(\boldsymbol{\omega})\right\}\right|>\delta\xi_N\right)\\
&\leq P\left(\sup_{\boldsymbol{\omega}\in\mathcal{W}}\left\{\sum_{m=1}^{M}\omega_m\left|\boldsymbol{e}_s'\boldsymbol{\Phi}_s\boldsymbol{X}_{(m)s}\left(\tilde{\boldsymbol{\beta}}_{(m)}-\hat{\boldsymbol{\beta}}_{(m)}\right)\right|\right\}>\delta\xi_N\right)\\
&\leq\sum_{m=1}^{M}P\left(\left|\boldsymbol{e}_s'\boldsymbol{\Phi}_s\boldsymbol{X}_{(m)s}\left(\tilde{\boldsymbol{\beta}}_{(m)}-\hat{\boldsymbol{\beta}}_{(m)}\right)\right|>\delta\xi_N\right)\\
&\leq\delta^{-1}\xi_N^{-1}\sum_{m=1}^{M}E\left|\boldsymbol{e}_s'\boldsymbol{\Phi}_s\boldsymbol{X}_{(m)s}\left(\tilde{\boldsymbol{\beta}}_{(m)}-\hat{\boldsymbol{\beta}}_{(m)}\right)\right|\\
&\leq\delta^{-1}\xi_N^{-1}\sum_{m=1}^{M}\left\{\lambda_{\max}\left(\boldsymbol{X}_{(m)s}\boldsymbol{\Phi}_s\boldsymbol{X}_{(m)s}'\right)\cdot E\left\|\tilde{\boldsymbol{\beta}}_{(m)}-\hat{\boldsymbol{\beta}}_{(m)}\right\|^2\cdot E\|\boldsymbol{e}_s\|_{\boldsymbol{\Phi}_s}^2\right\}^{1/2}\\
&=O\left(\frac{M\sqrt{nk_M}}{f\xi_N}\right),
\end{aligned}
$$

where the last bound follows from Lemmas 1 and 3, Condition 1, and the fact that $E\|\boldsymbol{e}_s\|_{\boldsymbol{\Phi}_s}^2=N\sigma^2$. Condition 3 then implies (S.36).

We now prove (S.37). By Lemma 1 and Conditions 1 and 4, we have

$$\begin{aligned} E_\epsilon\left(\mu_i - \tilde{\mu}_{i(m)}\right)^4 &= E_\epsilon\left(e^*_{i(m)} - e_i + \boldsymbol{x}'_{i(m)}\boldsymbol{\beta}^*_{(m)} - \boldsymbol{x}'_{i(m)}\tilde{\boldsymbol{\beta}}_{(m)}\right)^4 \\ &\leq 27E_\epsilon\left(e^{*4}_{i(m)}\right) + 27E_\epsilon\left(e_i^4\right) + 27k_m^2 \cdot E_\epsilon\left\|\tilde{\boldsymbol{\beta}}_{(m)} - \boldsymbol{\beta}^*_{(m)}\right\|^4 \\ &\leq C(1 + N^{-2}k_m^4) \end{aligned} \tag{S.41}$$

uniformly in $i$ and $m$. It follows that

$$E\left\|\boldsymbol{\mu}_s - \boldsymbol{X}_{(m)s}\tilde{\boldsymbol{\beta}}_{(m)}\right\|^2_{\boldsymbol{\Phi}_s} = \sum_{i\in U}\left\{E_\epsilon\left(\mu_i - \tilde{\mu}_{i(m)}\right)^2\right\} \leq C(N + k_m^2). \tag{S.42}$$

Therefore, for any $\delta > 0$,

$$\begin{aligned} &P\left(\sup_{\boldsymbol{\omega}\in\mathcal{W}} R_N^{-1}(\boldsymbol{\omega})\left|\left\{\boldsymbol{\mu}_s - \tilde{\boldsymbol{\mu}}_s(\boldsymbol{\omega})\right\}'\boldsymbol{\Phi}_s\left\{\tilde{\boldsymbol{\mu}}_s(\boldsymbol{\omega}) - \hat{\boldsymbol{\mu}}_s(\boldsymbol{\omega})\right\}\right| > \delta\right) \\ &\leq P\left(\sup_{\boldsymbol{\omega}\in\mathcal{W}}\left|\left\{\boldsymbol{\mu}_s - \tilde{\boldsymbol{\mu}}_s(\boldsymbol{\omega})\right\}'\boldsymbol{\Phi}_s\left\{\tilde{\boldsymbol{\mu}}_s(\boldsymbol{\omega}) - \hat{\boldsymbol{\mu}}_s(\boldsymbol{\omega})\right\}\right| > \delta\xi_N\right) \\ &= P\left(\sup_{\boldsymbol{\omega}\in\mathcal{W}}\left\{\sum_{m=1}^M\sum_{t=1}^M \omega_m\omega_t\left|\left(\boldsymbol{\mu}_s - \boldsymbol{X}_{(m)s}\tilde{\boldsymbol{\beta}}_{(m)}\right)'\boldsymbol{\Phi}_s\boldsymbol{X}_{(t)s}\left(\tilde{\boldsymbol{\beta}}_{(t)} - \hat{\boldsymbol{\beta}}_{(t)}\right)\right|\right\} > \delta\xi_N\right) \\ &\leq \sum_{m=1}^M\sum_{t=1}^M P\left(\left|\left(\boldsymbol{\mu}_s - \boldsymbol{X}_{(m)s}\tilde{\boldsymbol{\beta}}_{(m)}\right)'\boldsymbol{\Phi}_s\boldsymbol{X}_{(t)s}\left(\tilde{\boldsymbol{\beta}}_{(t)} - \hat{\boldsymbol{\beta}}_{(t)}\right)\right| > \delta\xi_N\right) \\ &\leq \delta^{-1}\xi_N^{-1}\sum_{m=1}^M\sum_{t=1}^M E\left|\left(\boldsymbol{\mu}_s - \boldsymbol{X}_{(m)s}\tilde{\boldsymbol{\beta}}_{(m)}\right)'\boldsymbol{\Phi}_s\boldsymbol{X}_{(t)s}\left(\tilde{\boldsymbol{\beta}}_{(t)} - \hat{\boldsymbol{\beta}}_{(t)}\right)\right| \\ &\leq \delta^{-1}\xi_N^{-1}\sum_{m=1}^M\sum_{t=1}^M\left\{\lambda_{\max}\left(\boldsymbol{X}'_{(t)s}\boldsymbol{\Phi}_s\boldsymbol{X}_{(t)s}\right)\cdot E\left\|\boldsymbol{\mu}_s - \boldsymbol{X}_{(m)s}\tilde{\boldsymbol{\beta}}_{(m)}\right\|^2_{\boldsymbol{\Phi}_s}\cdot E\left\|\tilde{\boldsymbol{\beta}}_{(t)} - \hat{\boldsymbol{\beta}}_{(t)}\right\|^2\right\}^{1/2} \\ &= O\left(\frac{M^2\sqrt{nk_M}}{f\xi_N}\right), \end{aligned}$$

where the last bound follows from (S.42), Lemmas 1 and 3, and Condition 1. Condition 3 then implies that the last term is $o(1)$, which proves (S.37).

To prove (S.38), it suffices to show that

$$\sup_{\boldsymbol{\omega}\in\mathcal{W}}\frac{\left|\boldsymbol{e}'_U\left\{\boldsymbol{\mu}_U - \tilde{\boldsymbol{\mu}}_U(\boldsymbol{\omega})\right\} + \sigma^2 k(\boldsymbol{\omega})\right|}{R_N(\boldsymbol{\omega})} = o_p(1) \tag{S.43}$$

and

$$\sup_{\boldsymbol{\omega}\in\mathcal{W}}\frac{\left|\boldsymbol{e}'_s\boldsymbol{\Phi}_s\left\{\boldsymbol{\mu}_s - \tilde{\boldsymbol{\mu}}_s(\boldsymbol{\omega})\right\} - \boldsymbol{e}'_U\left\{\boldsymbol{\mu}_U - \tilde{\boldsymbol{\mu}}_U(\boldsymbol{\omega})\right\}\right|}{R_N(\boldsymbol{\omega})} = o_p(1). \tag{S.44}$$

Note that

$$\boldsymbol{e}_U' \left\{\boldsymbol{\mu}_U - \tilde{\boldsymbol{\mu}}_U(\boldsymbol{\omega})\right\} + \sigma^2 k(\boldsymbol{\omega}) = \boldsymbol{e}_U' \left\{\boldsymbol{I}_N - \boldsymbol{P}_U(\boldsymbol{\omega})\right\} \boldsymbol{\mu}_U + \left\{\sigma^2 k(\boldsymbol{\omega}) - \boldsymbol{e}_U' \boldsymbol{P}_U(\boldsymbol{\omega}) \boldsymbol{e}_U\right\},$$

where $\boldsymbol{P}_U(\boldsymbol{\omega}) = \sum_{m=1}^{M} \omega_m \boldsymbol{P}_{(m)U}$ and $\boldsymbol{P}_{(m)U} = \boldsymbol{X}_{(m)U}(\boldsymbol{X}_{(m)U}'\boldsymbol{X}_{(m)U})^{-1}\boldsymbol{X}_{(m)U}'$. Then, by Equations (A.1) and (A.2) of Wan et al. (2010), together with Condition 4, we have

$$\sup_{\boldsymbol{\omega}\in\mathcal{W}} \frac{\left|\boldsymbol{e}_U' \left\{\boldsymbol{\mu}_U - \tilde{\boldsymbol{\mu}}_U(\boldsymbol{\omega})\right\} + \sigma^2 k(\boldsymbol{\omega})\right|}{\tilde{R}_N(\boldsymbol{\omega})} = o_p(1).$$

Furthermore, Lemma 4 and Condition 3 imply (S.43). Using Bonferroni's inequality, Markov's inequality, and Lemma 2, for any $\delta > 0$, we have

$$\begin{aligned}
&P\left(\sup_{\boldsymbol{\omega}\in\mathcal{W}} R_N^{-1}(\boldsymbol{\omega}) \left|\boldsymbol{e}_s' \boldsymbol{\Phi}_s \left\{\boldsymbol{\mu}_s - \tilde{\boldsymbol{\mu}}_s(\boldsymbol{\omega})\right\} - \boldsymbol{e}_U' \left\{\boldsymbol{\mu}_U - \tilde{\boldsymbol{\mu}}_U(\boldsymbol{\omega})\right\}\right| > \delta\right) \\
&\le P\left(\sup_{\boldsymbol{\omega}\in\mathcal{W}} \left|\sum_{i\in U} \frac{(I_i - \pi_i)\left\{\mu_i - \tilde{\mu}_i(\boldsymbol{\omega})\right\} e_i}{\pi_i}\right| > \delta\xi_N\right) \\
&\le P\left(\sup_{\boldsymbol{\omega}\in\mathcal{W}} \sum_{m=1}^{M} \omega_m \left|\sum_{i\in U} \frac{(I_i - \pi_i)\{\mu_i - \tilde{\mu}_{i(m)}\} e_i}{\pi_i}\right| > \delta\xi_N\right) \\
&\le \sum_{m=1}^{M} P\left(\left|\sum_{i\in U} \frac{(I_i - \pi_i)\{\mu_i - \tilde{\mu}_{i(m)}\} e_i}{\pi_i}\right| > \delta\xi_N\right) \\
&\le \delta^{-2}\xi_N^{-2} \sum_{m=1}^{M} E\left[\sum_{i\in U} \frac{(I_i - \pi_i)\{\mu_i - \tilde{\mu}_{i(m)}\} e_i}{\pi_i}\right]^2 \\
&\le f^{-1}\delta^{-2}\xi_N^{-2} \sum_{m=1}^{M} \sum_{i\in U} E_\epsilon\left[\{\mu_i - \tilde{\mu}_{i(m)}\}^2 e_i^2\right] \\
&= O\left(\frac{nM}{f^2\xi_N^2}\right),
\end{aligned}$$

where the last bound follows from the Cauchy–Schwarz inequality, (S.41), and Condition 4. Condition 3 then implies (S.44), and hence (S.38) follows.

Next, we establish (S.39) by verifying the following two relations:

$$\sup_{\boldsymbol{\omega}\in\mathcal{W}} \frac{\left|\tilde{L}_N(\boldsymbol{\omega}) - L_N(\boldsymbol{\omega})\right|}{R_N(\boldsymbol{\omega})} = o_p(1) \tag{S.45}$$

and

$$\sup_{\boldsymbol{\omega}\in\mathcal{W}} \frac{\left|\|\boldsymbol{\mu}_s - \tilde{\boldsymbol{\mu}}_s(\boldsymbol{\omega})\|_{\boldsymbol{\Phi}_s}^2 - \tilde{L}_N(\boldsymbol{\omega})\right|}{R_N(\boldsymbol{\omega})} = o_p(1). \tag{S.46}$$

Lemma 5 directly implies (S.45). Using Bonferroni's inequality, Markov's inequality, and Lemma 2, for any $\delta > 0$, we have

$$
\begin{aligned}
& P\left(\sup_{\omega\in\mathcal{W}} R_N^{-1}(\boldsymbol{\omega})\left\{\|\boldsymbol{\mu}_s - \tilde{\boldsymbol{\mu}}_s(\boldsymbol{\omega})\|_{\boldsymbol{\Phi}_s}^2 - \tilde{L}_N(\boldsymbol{\omega})\right\} > \delta\right) \\
& \le P\left(\sup_{\omega\in\mathcal{W}}\left|\sum_{i\in U}\frac{(I_i-\pi_i)\left\{\mu_i-\tilde{\mu}_i(\boldsymbol{\omega})\right\}^2}{\pi_i}\right| > \delta\xi_N\right) \\
& \le P\left(\sup_{\omega\in\mathcal{W}}\sum_{m=1}^{M}\sum_{t=1}^{M}\omega_m\omega_t\left|\sum_{i\in U}\frac{(I_i-\pi_i)\{\mu_i-\tilde{\mu}_{i(m)}\}\{\mu_i-\tilde{\mu}_{i(t)}\}}{\pi_i}\right| > \delta\xi_N\right) \\
& \le \sum_{m=1}^{M}\sum_{t=1}^{M} P\left(\left|\sum_{i\in U}\frac{(I_i-\pi_i)\{\mu_i-\tilde{\mu}_{i(m)}\}\{\mu_i-\tilde{\mu}_{i(t)}\}}{\pi_i}\right| > \delta\xi_N\right) \\
& \le \delta^{-2}\xi_N^{-2}\sum_{m=1}^{M}\sum_{t=1}^{M} E\left[\sum_{i\in U}\frac{(I_i-\pi_i)\{\mu_i-\tilde{\mu}_{i(m)}\}\{\mu_i-\tilde{\mu}_{i(t)}\}}{\pi_i}\right]^2 \\
& \le f^{-1}\delta^{-2}\xi_N^{-2}\sum_{m=1}^{M}\sum_{t=1}^{M}\sum_{i\in U} E_\epsilon\left[\{\mu_i-\tilde{\mu}_{i(m)}\}^2\{\mu_i-\tilde{\mu}_{i(t)}\}^2\right] \\
& = O\left(\frac{nM^2}{f^2\xi_N^2}\right),
\end{aligned}
$$

where the last bound follows from the Cauchy–Schwarz inequality and (S.41). By Condition 3, (S.46) holds, which further implies (S.39).

It remains to verify (S.40). Note that

$$
L_N(\boldsymbol{\omega}) - R_N(\boldsymbol{\omega}) = L_N(\boldsymbol{\omega}) - \tilde{L}_N(\boldsymbol{\omega}) + \tilde{L}_N(\boldsymbol{\omega}) - \tilde{R}_N(\boldsymbol{\omega}) + \tilde{R}_N(\boldsymbol{\omega}) - R_N(\boldsymbol{\omega}).
$$

By Lemmas 4 and 5, the verification of (S.40) reduces to showing that

$$
\sup_{\omega\in\mathcal{W}}\frac{\left|\tilde{L}_N(\boldsymbol{\omega}) - \tilde{R}_N(\boldsymbol{\omega})\right|}{R_N(\boldsymbol{\omega})} = o_p(1). \tag{S.47}
$$

By Condition 4 and Equation (A.3) of Wan et al. (2010), we have

$$
\sup_{\omega\in\mathcal{W}}\frac{\left|\tilde{L}_N(\boldsymbol{\omega}) - \tilde{R}_N(\boldsymbol{\omega})\right|}{\tilde{R}_N(\boldsymbol{\omega})} = o_p(1).
$$

Together with Lemma 4 and Condition 3, this establishes (S.47). Hence, (S.40) follows. This completes the proof of Equation (12) in Theorem 2.

We now turn to Equation (13) in Theorem 2. By (S.28) and (S.34), it suffices to establish

$$\sup_{\boldsymbol{\omega}\in\mathcal{W}} \frac{k(\boldsymbol{\omega}) \cdot \left|\hat{\sigma}^2_{(M)} - \sigma^2\right|}{R_N(\boldsymbol{\omega})} = o_p(1). \tag{S.48}$$

Note that

$$\sup_{\boldsymbol{\omega}\in\mathcal{W}} \frac{k(\boldsymbol{\omega}) \cdot \left|\hat{\sigma}^2_{(M)} - \sigma^2\right|}{R_N(\boldsymbol{\omega})} \leq \frac{k_M \cdot \left|\hat{\sigma}^2_{(M)} - \sigma^2\right|}{\xi_N}.$$

By (S.31), (S.32), and Markov's inequality, (S.48) follows. This completes the proof of Equation (13) in Theorem 2.

## S3.5 Proof of Corollary 2

The proof of this corollary relies on Lemmas 1–5.

**Proof.** By Equation (S.34), we have

$$\begin{aligned}
&C_n(\boldsymbol{\omega}) - R_N(\boldsymbol{\omega}) - N\sigma^2 \\
&= \{L_N(\boldsymbol{\omega}) - R_N(\boldsymbol{\omega})\} + \left\{\|\boldsymbol{\mu}_s - \tilde{\boldsymbol{\mu}}_s(\boldsymbol{\omega})\|^2_{\boldsymbol{\Phi}_s} - L_N(\boldsymbol{\omega})\right\} + \|\tilde{\boldsymbol{\mu}}_s(\boldsymbol{\omega}) - \hat{\boldsymbol{\mu}}_s(\boldsymbol{\omega})\|^2_{\boldsymbol{\Phi}_s} \\
&\quad + 2\boldsymbol{e}_s'\boldsymbol{\Phi}_s\{\tilde{\boldsymbol{\mu}}_s(\boldsymbol{\omega}) - \hat{\boldsymbol{\mu}}_s(\boldsymbol{\omega})\} + 2\left[\boldsymbol{e}_s'\boldsymbol{\Phi}_s\{\boldsymbol{\mu}_s - \tilde{\boldsymbol{\mu}}_s(\boldsymbol{\omega})\} + 2\sigma^2 k(\boldsymbol{\omega})\right] \\
&\quad + 2\{\boldsymbol{\mu}_s - \tilde{\boldsymbol{\mu}}_s(\boldsymbol{\omega})\}'\boldsymbol{\Phi}_s\{\tilde{\boldsymbol{\mu}}_s(\boldsymbol{\omega}) - \hat{\boldsymbol{\mu}}_s(\boldsymbol{\omega})\} + \left(\boldsymbol{e}_s'\boldsymbol{\Phi}_s\boldsymbol{e}_s - N\sigma^2\right) + 2(f^{-1} - 1)\sigma^2 k(\boldsymbol{\omega}).
\end{aligned}$$

Following Equations (S.28), (S.35)–(S.40), and (S.48) in the proof of Theorem 2, it remains to show that

$$\sup_{\boldsymbol{\omega}\in\mathcal{W}} \frac{|\boldsymbol{e}_s'\boldsymbol{\Phi}_s\boldsymbol{e}_s - N\sigma^2|}{R_N(\boldsymbol{\omega})} = o_p(1). \tag{S.49}$$

By Lemma 2, we have

$$E\left(\boldsymbol{e}_s'\boldsymbol{\Phi}_s\boldsymbol{e}_s - N\sigma^2\right)^2 \leq 2E\left(\boldsymbol{e}_s'\boldsymbol{\Phi}_s\boldsymbol{e}_s - \boldsymbol{e}_U'\boldsymbol{e}_U\right)^2 + 2E\left(\boldsymbol{e}_U'\boldsymbol{e}_U - N\sigma^2\right)^2$$

$$\leq \frac{CN}{n}\sum_{i\in U} Ee_i^4 + C\sum_{i\in U}\left(Ee_i^4 - \sigma^4\right)$$
$$\leq CNf^{-1}.$$

It follows from Markov's inequality that

$$\left|\boldsymbol{e}_s'\boldsymbol{\Phi}_s\boldsymbol{e}_s - N\sigma^2\right| = O_p\left(n^{1/2}f^{-1}\right).$$

Therefore,

$$\sup_{\boldsymbol{\omega}\in\mathcal{W}}\frac{|\boldsymbol{e}_s'\boldsymbol{\Phi}_s\boldsymbol{e}_s - N\sigma^2|}{R_N(\boldsymbol{\omega})} \leq \frac{|\boldsymbol{e}_s'\boldsymbol{\Phi}_s\boldsymbol{e}_s - N\sigma^2|}{\xi_N} = o_p(1),$$

where the last equality follows from Condition 3. This completes the proof of Corollary 2.

## S3.6 Proof of Theorem 3

The proof of this theorem relies on Lemmas 1–4 and 6–7.

**Proof.** Let $\epsilon = \xi_N^{1/2}N^{-1/2+\delta}$. To verify Equation (14) in Theorem 3, following Fan & Peng (2004) and Chen et al. (2018), it suffices to show that there exists a constant $C_0 > 0$ such that, for any $M$-dimensional vector $\boldsymbol{v} = (v_1, \ldots, v_M)'$,

$$\lim_{N\to\infty} P\left\{\inf_{\|\boldsymbol{v}\|=C_0,\, \boldsymbol{\omega}^*+\epsilon\boldsymbol{v}\in\mathcal{W}} C_n(\boldsymbol{\omega}^*+\epsilon\boldsymbol{v}) > C_n(\boldsymbol{\omega}^*)\right\} = 1.$$

This implies that there exists a minimizer $\breve{\boldsymbol{\omega}}$ in the set $\{\boldsymbol{\omega}^* + \epsilon\boldsymbol{v} \in \mathcal{W} : \|\boldsymbol{v}\| \leq C_0\}$ such that $\breve{\boldsymbol{\omega}} - \boldsymbol{\omega}^* = O_p(\epsilon)$.

Note that

$$\begin{aligned} C_n(\boldsymbol{\omega}^*+\epsilon\boldsymbol{v}) - C_n(\boldsymbol{\omega}^*) &= \|\boldsymbol{y}_s - \hat{\boldsymbol{\mu}}_s(\boldsymbol{\omega}^*+\epsilon\boldsymbol{v})\|_{\boldsymbol{\Phi}_s}^2 - \|\boldsymbol{y}_s - \hat{\boldsymbol{\mu}}_s(\boldsymbol{\omega}^*)\|_{\boldsymbol{\Phi}_s}^2 \\ &\quad + 2f^{-1}\sigma^2 k(\boldsymbol{\omega}^*+\epsilon\boldsymbol{v}) - 2f^{-1}\sigma^2 k(\boldsymbol{\omega}) \\ &= \|\hat{\boldsymbol{\mu}}_s(\epsilon\boldsymbol{v})\|_{\boldsymbol{\Phi}_s}^2 - 2\left\{\boldsymbol{\mu}_s - \hat{\boldsymbol{\mu}}_s(\boldsymbol{\omega}^*)\right\}'\boldsymbol{\Phi}_s\hat{\boldsymbol{\mu}}_s(\epsilon\boldsymbol{v}) \end{aligned}$$

$$
\begin{aligned}
&\quad - 2\boldsymbol{e}_s'\boldsymbol{\Phi}_s\hat{\boldsymbol{\mu}}_s(\epsilon\boldsymbol{v}) + 2f^{-1}\sigma^2 k(\epsilon\boldsymbol{v}) \\
&\triangleq \Delta_1 + \Delta_2 + \Delta_3 + \Delta_4.
\end{aligned}
$$

Let $\hat{\boldsymbol{\mu}}_{(m)s} = [\hat{\mu}_{i(m)}]_{i\in s}$ and $\boldsymbol{V}_1 = (\hat{\boldsymbol{\mu}}_{(1)s}, \ldots, \hat{\boldsymbol{\mu}}_{(M)s})$. By Condition 5, we have

$$
\begin{aligned}
\Delta_1 &= \left(\sum_{m=1}^{M} \epsilon v_m \hat{\boldsymbol{\mu}}_{(m)s}\right)' \boldsymbol{\Phi}_s \left(\sum_{m=1}^{M} \epsilon v_m \hat{\boldsymbol{\mu}}_{(m)s}\right) \\
&= \epsilon^2 \boldsymbol{v}'\boldsymbol{V}_1'\boldsymbol{\Phi}_s\boldsymbol{V}_1\boldsymbol{v} \\
&\geq \epsilon^2 \lambda_{\min}\left(\boldsymbol{V}_1'\boldsymbol{\Phi}_s\boldsymbol{V}_1\right)\|\boldsymbol{v}\|^2 \\
&\geq CN\epsilon^2 \cdot \|\boldsymbol{v}\|^2 > 0
\end{aligned}
$$

in probability tending to one. Thus, to prove Equation (14) in Theorem 3, it remains to show that $\Delta_2$, $\Delta_3$, and $\Delta_4$ are asymptotically negligible relative to $\Delta_1$.

By Lemma 6 and Condition 6, we have

$$
\xi_N = E\left\|\boldsymbol{\mu}_U - \hat{\boldsymbol{\mu}}_U(\boldsymbol{\omega}^*)\right\|^2 = E\left\|\boldsymbol{\mu}_s - \hat{\boldsymbol{\mu}}_s(\boldsymbol{\omega}^*)\right\|_{\boldsymbol{\Phi}_s}^2 + o(\xi_N).
$$

Hence, $\left\|\boldsymbol{\mu}_s - \hat{\boldsymbol{\mu}}_s(\boldsymbol{\omega}^*)\right\|_{\boldsymbol{\Phi}_s} = O_p(\xi_N^{1/2})$. Using Condition 5, we obtain that

$$
\begin{aligned}
|\Delta_2| &= 2\epsilon\left|\left\{\boldsymbol{\mu}_s - \hat{\boldsymbol{\mu}}_s(\boldsymbol{\omega}^*)\right\}'\boldsymbol{\Phi}_s\boldsymbol{V}_1\boldsymbol{v}\right| \\
&\leq 2\epsilon\left\|\boldsymbol{\mu}_s - \hat{\boldsymbol{\mu}}_s(\boldsymbol{\omega}^*)\right\|_{\boldsymbol{\Phi}_s} \lambda_{\max}^{1/2}(\boldsymbol{V}_1'\boldsymbol{\Phi}_s\boldsymbol{V}_1)\|\boldsymbol{v}\| \\
&= O_p\left(\xi_N^{1/2}N^{1/2}\epsilon\right)\cdot\|\boldsymbol{v}\|.
\end{aligned}
$$

Since $\epsilon = \xi_N^{1/2}N^{-1/2+\delta}$, we have

$$
\frac{|\Delta_2|}{\Delta_1} = O_p\left(\frac{\xi_N^{1/2}N^{1/2}\epsilon\cdot\|\boldsymbol{v}\|}{N\epsilon^2\cdot\|\boldsymbol{v}\|^2}\right) = O_p(N^{-\delta}) = o_p(1).
$$

Thus, $\Delta_2$ is asymptotically negligible relative to $\Delta_1$.

Next, we consider $\Delta_3$. By Lemma 7, we have

$$
|\Delta_3| = 2\epsilon\left|\sum_{m=1}^{M}\left(v_m\boldsymbol{e}_s'\boldsymbol{\Phi}_s\hat{\boldsymbol{\mu}}_{(m)s}\right)\right|
$$

$$\leq 2\epsilon \left\{ \sum_{m=1}^{M} \left( \boldsymbol{e}_s' \boldsymbol{\Phi}_s \hat{\boldsymbol{\mu}}_{(m)s} \right)^2 \right\}^{1/2} \|\boldsymbol{v}\|$$
$$= O_p\left( f^{-1/2} (NMk_M)^{1/2} \epsilon \right) \cdot \|\boldsymbol{v}\|.$$

For $\Delta_4$, it follows that

$$|\Delta_4| = 2f^{-1}\sigma^2 \epsilon \left| \sum_{m=1}^{M} k_m v_m \right| \tag{S.50}$$
$$= O_p\left( f^{-1} M^{1/2} k_M \epsilon \right) \cdot \|\boldsymbol{v}\|.$$

Thus, by Condition 6, $\Delta_3$ and $\Delta_4$ are asymptotically negligible relative to $\Delta_1$.

We next prove Equation (15) in Theorem 3. Comparing $C_n(\boldsymbol{\omega})$ with $C_n^*(\boldsymbol{\omega})$, it suffices to replace $\sigma^2$ in (S.50) by $\hat{\sigma}^2_{(M)}$ and show that the resulting term is asymptotically negligible relative to $\Delta_1$. By Conditions 1–2 and Lemma 3, we have

$$\begin{aligned}
\hat{\sigma}^2_{(M)} &= \frac{1}{n - k_M} \sum_{i \in s} \left( y_i - \boldsymbol{x}_{i(M)}' \hat{\boldsymbol{\beta}}_{(M)} \right)^2 \\
&\leq \frac{C}{n - k_M} \sum_{i \in s} \left\{ e^{*2}_{i(M)} + \left( \boldsymbol{x}_{i(M)} \boldsymbol{\beta}^*_{(M)} - \boldsymbol{x}_{i(M)}' \hat{\boldsymbol{\beta}}_{(M)} \right)^2 \right\} \\
&= O_p\left( \frac{n + k_M^2}{n - k_M} \right).
\end{aligned}$$

Since $k_M^2 = O(N)$ and $f \geq c_5$, it follows that

$$|\Delta_4'| \triangleq 2f^{-1}\epsilon \hat{\sigma}^2_{(M)} \left| \sum_{m=1}^{M} k_m v_m \right|$$
$$= O_p\left( f^{-1} M^{1/2} k_M \epsilon \right) \cdot \|\boldsymbol{v}\|.$$

Thus, by Condition 6, $\Delta_4'$ is asymptotically negligible relative to $\Delta_1$. This completes the proof of Theorem 3.

## S3.7 Proof of Theorem 4

**Proof.** We first prove Equation (16) in Theorem 4. Following Fan & Peng (2004) and Chen et al. (2018), it is sufficient to show that

$$\lim_{N\to\infty} P\left(\sup_{\check{\boldsymbol{\omega}}\in\mathcal{W}_1,\tilde{\boldsymbol{\omega}}\in\mathcal{W}_2}\{C_n(\check{\boldsymbol{\omega}})-C_n(\tilde{\boldsymbol{\omega}})\}<0\right)=1.$$

For any $\check{\boldsymbol{\omega}}\in\mathcal{W}_1$ and $\tilde{\boldsymbol{\omega}}\in\mathcal{W}_2$, we have

$$\begin{aligned}
C_n(\check{\boldsymbol{\omega}})-C_n(\tilde{\boldsymbol{\omega}}) &= \|\boldsymbol{y}_s-\hat{\boldsymbol{\mu}}_s(\check{\boldsymbol{\omega}})\|^2_{\boldsymbol{\Phi}_s}-\|\boldsymbol{y}_s-\hat{\boldsymbol{\mu}}_s(\tilde{\boldsymbol{\omega}})\|^2_{\boldsymbol{\Phi}_s}\\
&\quad+2f^{-1}\sigma^2k(\check{\boldsymbol{\omega}})-2f^{-1}\sigma^2k(\tilde{\boldsymbol{\omega}})\\
&=-\tilde{R}_N(\tilde{\boldsymbol{\omega}})+\|\boldsymbol{\mu}_s-\hat{\boldsymbol{\mu}}_s(\check{\boldsymbol{\omega}})\|^2_{\boldsymbol{\Phi}_s}\\
&\quad+\left\{\tilde{R}_N(\tilde{\boldsymbol{\omega}})-\|\boldsymbol{\mu}_s-\hat{\boldsymbol{\mu}}_s(\tilde{\boldsymbol{\omega}})\|^2_{\boldsymbol{\Phi}_s}\right\}\\
&\quad+2\left[\{\boldsymbol{\mu}_s-\hat{\boldsymbol{\mu}}_s(\check{\boldsymbol{\omega}})\}'\boldsymbol{\Phi}_s\boldsymbol{e}_s+f^{-1}\sigma^2k(\check{\boldsymbol{\omega}})\right]\\
&\quad-2\left[\{\boldsymbol{\mu}_s-\hat{\boldsymbol{\mu}}_s(\tilde{\boldsymbol{\omega}})\}'\boldsymbol{\Phi}_s\boldsymbol{e}_s+f^{-1}\sigma^2k(\tilde{\boldsymbol{\omega}})\right]\\
&\triangleq-\tilde{R}_N(\tilde{\boldsymbol{\omega}})+\tilde{\Delta}_1+\tilde{\Delta}_2+\tilde{\Delta}_3+\tilde{\Delta}_4.
\end{aligned}$$

Note that, for each correct model $m$, $\boldsymbol{\mu}_U=\boldsymbol{X}_{(m)U}\boldsymbol{\beta}_{(m)}$ and $\boldsymbol{\mu}_s=\boldsymbol{X}_{(m)s}\boldsymbol{\beta}_{(m)}$. Therefore, for any $\check{\boldsymbol{\omega}}\in\mathcal{W}_1$, whose weights are assigned only to correct models, we have

$$\begin{aligned}
\boldsymbol{\mu}_s-\tilde{\boldsymbol{\mu}}_s(\check{\boldsymbol{\omega}}) &= \boldsymbol{\mu}_s-\sum_{m=1}^{M}\check{\omega}_m\boldsymbol{X}_{(m)s}\left(\boldsymbol{X}'_{(m)U}\boldsymbol{X}_{(m)U}\right)^{-1}\boldsymbol{X}'_{(m)U}\boldsymbol{y}_U\\
&=\boldsymbol{\mu}_s-\sum_{m=1}^{M}\check{\omega}_m\boldsymbol{X}_{(m)s}\left(\boldsymbol{X}'_{(m)U}\boldsymbol{X}_{(m)U}\right)^{-1}\boldsymbol{X}'_{(m)U}\left(\boldsymbol{X}_{(m)U}\boldsymbol{\beta}_{(m)}+\boldsymbol{e}_U\right)\\
&=-\sum_{m=1}^{M}\check{\omega}_m\boldsymbol{X}_{(m)s}\left(\boldsymbol{X}'_{(m)U}\boldsymbol{X}_{(m)U}\right)^{-1}\boldsymbol{X}'_{(m)U}\boldsymbol{e}_U\\
&\triangleq-\tilde{\boldsymbol{P}}_s(\check{\boldsymbol{\omega}})\boldsymbol{e}_U,
\end{aligned}$$

where $\tilde{\boldsymbol{P}}_s(\check{\boldsymbol{\omega}})=\sum_{m=1}^{M}\check{\omega}_m\boldsymbol{X}_{(m)s}\left(\boldsymbol{X}'_{(m)U}\boldsymbol{X}_{(m)U}\right)^{-1}\boldsymbol{X}'_{(m)U}$. It follows that

$$|\tilde{\Delta}_1|=\|\boldsymbol{\mu}_s-\hat{\boldsymbol{\mu}}_s(\check{\boldsymbol{\omega}})\|^2_{\boldsymbol{\Phi}_s}$$

$$
\begin{aligned}
&= \|\boldsymbol{\mu}_s - \tilde{\boldsymbol{\mu}}_s(\breve{\boldsymbol{\omega}}) + \tilde{\boldsymbol{\mu}}_s(\breve{\boldsymbol{\omega}}) - \hat{\boldsymbol{\mu}}_s(\breve{\boldsymbol{\omega}})\|^2_{\boldsymbol{\Phi}_s} \\
&\leq 2\|\boldsymbol{\mu}_s - \tilde{\boldsymbol{\mu}}_s(\breve{\boldsymbol{\omega}})\|^2_{\boldsymbol{\Phi}_s} + 2\|\tilde{\boldsymbol{\mu}}_s(\breve{\boldsymbol{\omega}}) - \hat{\boldsymbol{\mu}}_s(\breve{\boldsymbol{\omega}})\|^2_{\boldsymbol{\Phi}_s} \\
&= 2\left\|\tilde{\boldsymbol{P}}_s(\breve{\boldsymbol{\omega}})\boldsymbol{e}_U\right\|^2_{\boldsymbol{\Phi}_s} + 2\|\tilde{\boldsymbol{\mu}}_s(\breve{\boldsymbol{\omega}}) - \hat{\boldsymbol{\mu}}_s(\breve{\boldsymbol{\omega}})\|^2_{\boldsymbol{\Phi}_s}
\end{aligned}
$$

and

$$
\begin{aligned}
|\tilde{\Delta}_3| &= 2\left|\{\boldsymbol{\mu}_s - \hat{\boldsymbol{\mu}}_s(\breve{\boldsymbol{\omega}})\}'\boldsymbol{\Phi}_s\boldsymbol{e}_s + f^{-1}\sigma^2 k(\breve{\boldsymbol{\omega}})\right| \\
&= 2\left|\{\boldsymbol{\mu}_s - \tilde{\boldsymbol{\mu}}_s(\breve{\boldsymbol{\omega}}) + \tilde{\boldsymbol{\mu}}_s(\breve{\boldsymbol{\omega}}) - \hat{\boldsymbol{\mu}}_s(\breve{\boldsymbol{\omega}})\}'\boldsymbol{\Phi}_s\boldsymbol{e}_s + f^{-1}\sigma^2 k(\breve{\boldsymbol{\omega}})\right| \\
&\leq 2\left|\{\boldsymbol{\mu}_s - \tilde{\boldsymbol{\mu}}_s(\breve{\boldsymbol{\omega}})\}'\boldsymbol{\Phi}_s\boldsymbol{e}_s + f^{-1}\sigma^2 k(\breve{\boldsymbol{\omega}})\right| + 2\left|\{\tilde{\boldsymbol{\mu}}_s(\breve{\boldsymbol{\omega}}) - \hat{\boldsymbol{\mu}}_s(\breve{\boldsymbol{\omega}})\}'\boldsymbol{\Phi}_s\boldsymbol{e}_s\right|.
\end{aligned}
$$

For any $\tilde{\boldsymbol{\omega}} \in \mathcal{W}_2$, we have

$$
\begin{aligned}
|\tilde{\Delta}_2| &= \left|\|\boldsymbol{\mu}_s - \hat{\boldsymbol{\mu}}_s(\tilde{\boldsymbol{\omega}})\|^2_{\boldsymbol{\Phi}_s} - \tilde{R}_N(\tilde{\boldsymbol{\omega}})\right| \\
&= \Big|\|\tilde{\boldsymbol{\mu}}_s(\tilde{\boldsymbol{\omega}}) - \hat{\boldsymbol{\mu}}_s(\tilde{\boldsymbol{\omega}})\|^2_{\boldsymbol{\Phi}_s} + 2\{\boldsymbol{\mu}_s - \tilde{\boldsymbol{\mu}}_s(\tilde{\boldsymbol{\omega}})\}'\boldsymbol{\Phi}_s\{\tilde{\boldsymbol{\mu}}_s(\tilde{\boldsymbol{\omega}}) - \hat{\boldsymbol{\mu}}_s(\tilde{\boldsymbol{\omega}})\} \\
&\quad + \|\boldsymbol{\mu}_s - \tilde{\boldsymbol{\mu}}_s(\tilde{\boldsymbol{\omega}})\|^2_{\boldsymbol{\Phi}_s} - \tilde{R}_N(\tilde{\boldsymbol{\omega}})\Big| \\
&\leq \|\tilde{\boldsymbol{\mu}}_s(\tilde{\boldsymbol{\omega}}) - \hat{\boldsymbol{\mu}}_s(\tilde{\boldsymbol{\omega}})\|^2_{\boldsymbol{\Phi}_s} + \left|\|\boldsymbol{\mu}_s - \tilde{\boldsymbol{\mu}}_s(\tilde{\boldsymbol{\omega}})\|^2_{\boldsymbol{\Phi}_s} - \tilde{R}_N(\tilde{\boldsymbol{\omega}})\right| \\
&\quad + 2\left|\{\boldsymbol{\mu}_s - \tilde{\boldsymbol{\mu}}_s(\tilde{\boldsymbol{\omega}})\}'\boldsymbol{\Phi}_s\{\tilde{\boldsymbol{\mu}}_s(\tilde{\boldsymbol{\omega}}) - \hat{\boldsymbol{\mu}}_s(\tilde{\boldsymbol{\omega}})\}\right|.
\end{aligned}
$$

Similarly,

$$
\begin{aligned}
|\tilde{\Delta}_4| &= 2\left|\{\boldsymbol{\mu}_s - \hat{\boldsymbol{\mu}}_s(\tilde{\boldsymbol{\omega}})\}'\boldsymbol{\Phi}_s\boldsymbol{e}_s + f^{-1}\sigma^2 k(\tilde{\boldsymbol{\omega}})\right| \\
&= 2\left|\{\boldsymbol{\mu}_s - \tilde{\boldsymbol{\mu}}_s(\tilde{\boldsymbol{\omega}}) + \tilde{\boldsymbol{\mu}}_s(\tilde{\boldsymbol{\omega}}) - \hat{\boldsymbol{\mu}}_s(\tilde{\boldsymbol{\omega}})\}'\boldsymbol{\Phi}_s\boldsymbol{e}_s + f^{-1}\sigma^2 k(\tilde{\boldsymbol{\omega}})\right| \\
&\leq 2\left|\{\boldsymbol{\mu}_s - \tilde{\boldsymbol{\mu}}_s(\tilde{\boldsymbol{\omega}})\}'\boldsymbol{\Phi}_s\boldsymbol{e}_s + f^{-1}\sigma^2 k(\tilde{\boldsymbol{\omega}})\right| + 2\left|\{\tilde{\boldsymbol{\mu}}_s(\tilde{\boldsymbol{\omega}}) - \hat{\boldsymbol{\mu}}_s(\tilde{\boldsymbol{\omega}})\}'\boldsymbol{\Phi}_s\boldsymbol{e}_s\right|.
\end{aligned}
$$

Therefore, the proof of Equation (16) in Theorem 4 reduces to showing that

$$
\sup_{\breve{\boldsymbol{\omega}}\in\mathcal{W}_1, \tilde{\boldsymbol{\omega}}\in\mathcal{W}_2} \tilde{R}_N^{-1}(\tilde{\boldsymbol{\omega}})\|\tilde{\boldsymbol{\mu}}_s(\breve{\boldsymbol{\omega}}) - \hat{\boldsymbol{\mu}}_s(\breve{\boldsymbol{\omega}})\|^2_{\boldsymbol{\Phi}_s} = o_p(1), \tag{S.51}
$$

$$
\sup_{\tilde{\boldsymbol{\omega}}\in\mathcal{W}_2} \tilde{R}_N^{-1}(\tilde{\boldsymbol{\omega}})\|\tilde{\boldsymbol{\mu}}_s(\tilde{\boldsymbol{\omega}}) - \hat{\boldsymbol{\mu}}_s(\tilde{\boldsymbol{\omega}})\|^2_{\boldsymbol{\Phi}_s} = o_p(1), \tag{S.52}
$$

$$\sup_{\breve{\omega}\in\mathcal{W}_1,\tilde{\omega}\in\mathcal{W}_2} \tilde{R}_N^{-1}(\tilde{\boldsymbol{\omega}}) \left|\{\tilde{\boldsymbol{\mu}}_s(\breve{\boldsymbol{\omega}}) - \hat{\boldsymbol{\mu}}_s(\breve{\boldsymbol{\omega}})\}' \boldsymbol{\Phi}_s \boldsymbol{e}_s\right| = o_p(1), \tag{S.53}$$

$$\sup_{\tilde{\omega}\in\mathcal{W}_2} \tilde{R}_N^{-1}(\tilde{\boldsymbol{\omega}}) \left|\{\tilde{\boldsymbol{\mu}}_s(\tilde{\boldsymbol{\omega}}) - \hat{\boldsymbol{\mu}}_s(\tilde{\boldsymbol{\omega}})\}' \boldsymbol{\Phi}_s \boldsymbol{e}_s\right| = o_p(1), \tag{S.54}$$

$$\sup_{\tilde{\omega}\in\mathcal{W}_2} \tilde{R}_N^{-1}(\tilde{\boldsymbol{\omega}}) \left|\{\boldsymbol{\mu}_s - \tilde{\boldsymbol{\mu}}_s(\tilde{\boldsymbol{\omega}})\}' \boldsymbol{\Phi}_s \{\tilde{\boldsymbol{\mu}}_s(\tilde{\boldsymbol{\omega}}) - \hat{\boldsymbol{\mu}}_s(\tilde{\boldsymbol{\omega}})\}\right| = o_p(1), \tag{S.55}$$

$$\sup_{\breve{\omega}\in\mathcal{W}_1,\tilde{\omega}\in\mathcal{W}_2} \tilde{R}_N^{-1}(\tilde{\boldsymbol{\omega}}) \left|\{\boldsymbol{\mu}_s - \tilde{\boldsymbol{\mu}}_s(\breve{\boldsymbol{\omega}})\}' \boldsymbol{\Phi}_s \boldsymbol{e}_s + f^{-1}\sigma^2 k(\breve{\boldsymbol{\omega}})\right| = o_p(1), \tag{S.56}$$

$$\sup_{\tilde{\omega}\in\mathcal{W}_2} \tilde{R}_N^{-1}(\tilde{\boldsymbol{\omega}}) \left|\{\boldsymbol{\mu}_s - \tilde{\boldsymbol{\mu}}_s(\tilde{\boldsymbol{\omega}})\}' \boldsymbol{\Phi}_s \boldsymbol{e}_s + f^{-1}\sigma^2 k(\tilde{\boldsymbol{\omega}})\right| = o_p(1), \tag{S.57}$$

$$\sup_{\tilde{\omega}\in\mathcal{W}_2} \tilde{R}_N^{-1}(\tilde{\boldsymbol{\omega}}) \left|\|\boldsymbol{\mu}_s - \tilde{\boldsymbol{\mu}}_s(\tilde{\boldsymbol{\omega}})\|_{\boldsymbol{\Phi}_s}^2 - \tilde{R}_N(\tilde{\boldsymbol{\omega}})\right| = o_p(1), \tag{S.58}$$

and

$$\sup_{\breve{\omega}\in\mathcal{W}_1,\tilde{\omega}\in\mathcal{W}_2} \tilde{R}_N^{-1}(\tilde{\boldsymbol{\omega}}) \left\|\tilde{\boldsymbol{P}}_s(\breve{\boldsymbol{\omega}})\boldsymbol{e}_U\right\|_{\boldsymbol{\Phi}_s}^2 = o_p(1). \tag{S.59}$$

Using the same argument as in the proof of Equation (S.35), for any $\delta > 0$, we have

$$\begin{aligned}
&P\left(\sup_{\breve{\omega}\in\mathcal{W}_1,\tilde{\omega}\in\mathcal{W}_2} \tilde{R}_N^{-1}(\tilde{\boldsymbol{\omega}}) \|\tilde{\boldsymbol{\mu}}_s(\breve{\boldsymbol{\omega}}) - \hat{\boldsymbol{\mu}}_s(\breve{\boldsymbol{\omega}})\|_{\boldsymbol{\Phi}_s}^2 > \delta\right) \\
&\leq P\left(\sup_{\breve{\omega}\in\mathcal{W}_1} \|\tilde{\boldsymbol{\mu}}_s(\breve{\boldsymbol{\omega}}) - \hat{\boldsymbol{\mu}}_s(\breve{\boldsymbol{\omega}})\|_{\boldsymbol{\Phi}_s}^2 > \delta\tilde{\xi}_N^*\right) \\
&= O\left(\frac{s_0^2 k_M^2}{f\tilde{\xi}_N^*}\right)
\end{aligned}$$

and

$$\begin{aligned}
&P\left(\sup_{\tilde{\omega}\in\mathcal{W}_2} \tilde{R}_N^{-1}(\tilde{\boldsymbol{\omega}}) \|\tilde{\boldsymbol{\mu}}_s(\tilde{\boldsymbol{\omega}}) - \hat{\boldsymbol{\mu}}_s(\tilde{\boldsymbol{\omega}})\|_{\boldsymbol{\Phi}_s}^2 > \delta\right) \\
&\leq P\left(\sup_{\tilde{\omega}\in\mathcal{W}_2} \|\tilde{\boldsymbol{\mu}}_s(\tilde{\boldsymbol{\omega}}) - \hat{\boldsymbol{\mu}}_s(\tilde{\boldsymbol{\omega}})\|_{\boldsymbol{\Phi}_s}^2 > \delta\tilde{\xi}_N^*\right) \\
&= O\left(\frac{(M-s_0)^2 k_M^2}{f\tilde{\xi}_N^*}\right).
\end{aligned}$$

By Condition 7, both upper bounds are $o(1)$. This establishes Equations (S.51) and (S.52).

Similarly, using the same argument as in the proof of Equation (S.36), for any $\delta > 0$, we have

$$\begin{aligned}
&P\left(\sup_{\check{\boldsymbol{\omega}}\in\mathcal{W}_1,\tilde{\boldsymbol{\omega}}\in\mathcal{W}_2} \tilde{R}_N^{-1}(\tilde{\boldsymbol{\omega}})\left|\{\tilde{\boldsymbol{\mu}}_s(\check{\boldsymbol{\omega}})-\hat{\boldsymbol{\mu}}_s(\check{\boldsymbol{\omega}})\}'\boldsymbol{\Phi}_s\boldsymbol{e}_s\right|>\delta\right)\\
&\leq P\left(\sup_{\check{\boldsymbol{\omega}}\in\mathcal{W}_1}\left|\{\tilde{\boldsymbol{\mu}}_s(\check{\boldsymbol{\omega}})-\hat{\boldsymbol{\mu}}_s(\check{\boldsymbol{\omega}})\}'\boldsymbol{\Phi}_s\boldsymbol{e}_s\right|>\delta\tilde{\xi}_N^*\right)\\
&=O\left(\frac{s_0\sqrt{nk_M}}{f\tilde{\xi}_N^*}\right)
\end{aligned}$$

and

$$\begin{aligned}
&P\left(\sup_{\tilde{\boldsymbol{\omega}}\in\mathcal{W}_2} \tilde{R}_N^{-1}(\tilde{\boldsymbol{\omega}})\left|\{\tilde{\boldsymbol{\mu}}_s(\tilde{\boldsymbol{\omega}})-\hat{\boldsymbol{\mu}}_s(\tilde{\boldsymbol{\omega}})\}'\boldsymbol{\Phi}_s\boldsymbol{e}_s\right|>\delta\right)\\
&\leq P\left(\sup_{\tilde{\boldsymbol{\omega}}\in\mathcal{W}_2}\left|\{\tilde{\boldsymbol{\mu}}_s(\tilde{\boldsymbol{\omega}})-\hat{\boldsymbol{\mu}}_s(\tilde{\boldsymbol{\omega}})\}'\boldsymbol{\Phi}_s\boldsymbol{e}_s\right|>\delta\tilde{\xi}_N^*\right)\\
&=O\left(\frac{(M-s_0)\sqrt{nk_M}}{f\tilde{\xi}_N^*}\right).
\end{aligned}$$

By Condition 7, both upper bounds are $o(1)$. This establishes Equations (S.53) and (S.54).

By the argument used to prove Equation (S.37), for any $\delta > 0$, we have

$$\begin{aligned}
&P\left(\sup_{\tilde{\boldsymbol{\omega}}\in\mathcal{W}_2} \tilde{R}_N^{-1}(\tilde{\boldsymbol{\omega}})\left|\{\boldsymbol{\mu}_s-\tilde{\boldsymbol{\mu}}_s(\tilde{\boldsymbol{\omega}})\}'\boldsymbol{\Phi}_s\{\tilde{\boldsymbol{\mu}}_s(\tilde{\boldsymbol{\omega}})-\hat{\boldsymbol{\mu}}_s(\tilde{\boldsymbol{\omega}})\}\right|>\delta\right)\\
&\leq P\left(\sup_{\tilde{\boldsymbol{\omega}}\in\mathcal{W}_2}\left|\{\boldsymbol{\mu}_s-\tilde{\boldsymbol{\mu}}_s(\tilde{\boldsymbol{\omega}})\}'\boldsymbol{\Phi}_s\{\tilde{\boldsymbol{\mu}}_s(\tilde{\boldsymbol{\omega}})-\hat{\boldsymbol{\mu}}_s(\tilde{\boldsymbol{\omega}})\}\right|>\delta\tilde{\xi}_N^*\right)\\
&=O\left(\frac{(M-s_0)^2\sqrt{nk_M}}{f\tilde{\xi}_N^*}\right).
\end{aligned}$$

By Condition 7, the upper bound is $o(1)$, and hence Equation (S.55) holds.

Following the proofs of Equations (S.28) and (S.38), and using Conditions 7 and 8, Equations (S.56) and (S.57) follow by replacing $\tilde{R}_N(\boldsymbol{\omega})$ and $\tilde{\xi}_N$ with $\tilde{R}_N(\tilde{\boldsymbol{\omega}})$ and $\tilde{\xi}_N^*$, respectively. Similarly, Equation (S.58) follows from the same arguments as those used in the proofs of Equations (S.39) and (S.40).

It remains to prove Equation (S.59). By Markov's inequality and the Cauchy–Schwarz inequality, we have

$$
\begin{aligned}
&P\left(\sup_{\check{\boldsymbol{\omega}}\in\mathcal{W}_1,\tilde{\boldsymbol{\omega}}\in\mathcal{W}_2}\tilde{R}_N^{-1}(\tilde{\boldsymbol{\omega}})\left\|\tilde{\boldsymbol{P}}_s(\check{\boldsymbol{\omega}})\boldsymbol{e}_U\right\|_{\boldsymbol{\Phi}_s}^2>\delta\right)\\
&\leq P\left(\sup_{\check{\boldsymbol{\omega}}\in\mathcal{W}_1}\left|\boldsymbol{e}_U'\tilde{\boldsymbol{P}}_s(\check{\boldsymbol{\omega}})'\boldsymbol{\Phi}_s\tilde{\boldsymbol{P}}_s(\check{\boldsymbol{\omega}})\boldsymbol{e}_U\right|>\delta\tilde{\xi}_N^*\right)\\
&\leq \sum_{m=1}^{s_0}\sum_{t=1}^{s_0}P\left(\left|\boldsymbol{e}_U'\boldsymbol{X}_{(m)U}\left(\boldsymbol{X}_{(m)U}'\boldsymbol{X}_{(m)U}\right)^{-1}\boldsymbol{X}_{(m)s}'\boldsymbol{\Phi}_s\boldsymbol{X}_{(t)s}\left(\boldsymbol{X}_{(t)U}'\boldsymbol{X}_{(t)U}\right)^{-1}\boldsymbol{X}_{(t)U}'\boldsymbol{e}_U\right|>\delta\tilde{\xi}_N^*\right)\\
&\leq \delta^{-1}\tilde{\xi}_N^{*-1}\cdot\sum_{m=1}^{s_0}\sum_{t=1}^{s_0}E\left|\boldsymbol{e}_U'\boldsymbol{X}_{(m)U}\left(\boldsymbol{X}_{(m)U}'\boldsymbol{X}_{(m)U}\right)^{-1}\boldsymbol{X}_{(m)s}'\boldsymbol{\Phi}_s\boldsymbol{X}_{(t)s}\left(\boldsymbol{X}_{(t)U}'\boldsymbol{X}_{(t)U}\right)^{-1}\boldsymbol{X}_{(t)U}'\boldsymbol{e}_U\right|\\
&\leq \delta^{-1}\tilde{\xi}_N^{*-1}\cdot\left[\sum_{m=1}^{s_0}\left\{E\left\|\boldsymbol{e}_U'\boldsymbol{X}_{(m)U}\left(\boldsymbol{X}_{(m)U}'\boldsymbol{X}_{(m)U}\right)^{-1}\boldsymbol{X}_{(m)s}'\boldsymbol{\Phi}_s^{1/2}\right\|^2\right\}^{1/2}\right]^2\\
&= \delta^{-1}\tilde{\xi}_N^{*-1}\cdot\left[\sum_{m=1}^{s_0}\left\{E_m\left(\boldsymbol{e}_U'\boldsymbol{P}_{(m)U}\boldsymbol{e}_U\right)\right\}^{1/2}\right]^2\\
&= \sigma^2\delta^{-1}\tilde{\xi}_N^{*-1}\cdot\left[\sum_{m=1}^{s_0}\left\{\operatorname{tr}\left(\boldsymbol{P}_{(m)U}\right)\right\}^{1/2}\right]^2\\
&\leq \sigma^2\delta^{-1}s_0^2k_M\tilde{\xi}_N^{*-1},
\end{aligned}
$$

where $\boldsymbol{P}_{(m)U}=\boldsymbol{X}_{(m)U}\left(\boldsymbol{X}_{(m)U}'\boldsymbol{X}_{(m)U}\right)^{-1}\boldsymbol{X}_{(m)U}'$. By Condition 7, the last bound is $o(1)$, and hence Equation (S.59) holds. This completes the proof of Equation (16) in Theorem 4.

We now prove Equation (17) in Theorem 4. By the same argument as that used in the proof of Equation (S.48), together with Conditions 7 and 8, Equation (17) follows after replacing $\tilde{R}_N(\boldsymbol{\omega})$ and $\tilde{\xi}_N$ with $\tilde{R}_N(\tilde{\boldsymbol{\omega}})$ and $\tilde{\xi}_N^*$, respectively. This completes the proof of Theorem 4.

## S3.8 Lemmas for the proofs of Theorems 5–7

**Lemma 8** *If Conditions 2 and 9 hold, then*

$$
E\left(\frac{1}{N}\sum_{i\in s}\frac{z_i}{\pi_i}-\frac{1}{N}\sum_{i\in U}z_i\right)^4\leq Cn^{-2}\cdot\frac{1}{N}\sum_{i\in U}E_\epsilon(z_i^4).
$$

**Proof.** By Conditions 2 and 9, Loève's $c_r$ inequality, and the Cauchy–Schwarz inequality,

$$
\begin{aligned}
& E\left(\frac{1}{N}\sum_{i\in s}\frac{z_i}{\pi_i}-\frac{1}{N}\sum_{i\in U}z_i\right)^4 \\
&= \left|E_\epsilon E_d\left[\frac{1}{N}\sum_{i\in U}\frac{z_i(I_i-\pi_i)}{\pi_i}\right\}^4\right| \\
&\le \frac{C}{N^4}E_\epsilon\left\{\sum_{i,j,k,l\in \boldsymbol{D}_4}|z_iz_jz_kz_l|\cdot\left|E_d\left\{(I_i-\pi_i)(I_j-\pi_j)(I_k-\pi_k)(I_l-\pi_l)\right\}\right|\right. \\
&\quad +\sum_{i,j,k\in \boldsymbol{D}_3}|z_i^2z_jz_k|\cdot\left|E_d\left\{(I_i-\pi_i)^2(I_j-\pi_j)(I_k-\pi_k)\right\}\right| \\
&\quad +\sum_{i\neq j\in U}|z_i^2z_j^2|\cdot\left|E_d\left\{(I_i-\pi_i)^2(I_j-\pi_j)^2+(I_i-\pi_i)^3(I_j-\pi_j)\right\}\right| \\
&\quad \left.+\sum_{i\in U}|z_i^4|\cdot E_d\left\{(I_i-\pi_i)^4\right\}\right] \\
&\le Cn^{-2}\cdot\frac{1}{N}\sum_{i\in U}E_\epsilon(z_i^4).
\end{aligned}
$$

This completes the proof of Lemma 8.

**Lemma 9** *If Conditions 1–2 and 9 hold, then*

$$E\left\|\hat{\boldsymbol{\beta}}_{(m)}-\boldsymbol{\beta}^*_{(m)}\right\|^4\le Cn^{-2}k_m^2.$$

**Proof.** By Condition 1, we have

$$
\begin{aligned}
E\left\|\hat{\boldsymbol{\beta}}_{(m)}-\boldsymbol{\beta}^*_{(m)}\right\|^4 =& E\left\|\left(\sum_{i\in s}\frac{\boldsymbol{x}_{i(m)}\boldsymbol{x}'_{i(m)}}{\pi_i}\right)^{-1}\sum_{i\in s}\frac{\boldsymbol{x}_{i(m)}e^*_{i(m)}}{\pi_i}\right\|^4 \\
\le& E\left\{\lambda_{\min}^{-1}\left(\boldsymbol{X}'_{(m)s}\boldsymbol{\Phi}_s\boldsymbol{X}_{(m)s}\right)\cdot\left\|\sum_{i\in s}\frac{\boldsymbol{x}_{i(m)}e^*_{i(m)}}{\pi_i}\right\|\right\}^4 \\
\le& CE\left\|\frac{1}{N}\sum_{i\in s}\frac{\boldsymbol{x}_{i(m)}e^*_{i(m)}}{\pi_i}\right\|^4. \qquad \text{(S.60)}
\end{aligned}
$$

Moreover, Condition 1 implies that

$$E_\epsilon(x_{ij(m)}e^*_{i(m)})^4\le C.$$

Applying Lemma 8 yields

$$E\left(\frac{1}{N}\sum_{i\in s}\frac{x_{ij(m)}e^*_{i(m)}}{\pi_i}-\frac{1}{N}\sum_{i\in U}x_{ij(m)}e^*_{i(m)}\right)^4\le Cn^{-2}.$$

Therefore,

$$
\begin{aligned}
& E\left\|\frac{1}{N}\sum_{i\in s}\frac{\boldsymbol{x}_{i(m)}e^*_{i(m)}}{\pi_i}\right\|^4 \\
&= E\left\|\frac{1}{N}\sum_{i\in s}\frac{\boldsymbol{x}_{i(m)}e^*_{i(m)}}{\pi_i}-\frac{1}{N}\sum_{i\in U}\boldsymbol{x}_{i(m)}e^*_{i(m)}+\frac{1}{N}\sum_{i\in U}\boldsymbol{x}_{i(m)}e^*_{i(m)}\right\|^4 \\
&\le CE\left\|\frac{1}{N}\sum_{i\in s}\frac{\boldsymbol{x}_{i(m)}e^*_{i(m)}}{\pi_i}-\frac{1}{N}\sum_{i\in U}\boldsymbol{x}_{i(m)}e^*_{i(m)}\right\|^4+CE_\epsilon\left\|\frac{1}{N}\sum_{i\in U}\boldsymbol{x}_{i(m)}e^*_{i(m)}\right\|^4 \\
&\le CE\left\{\sum_{j=1}^{k_m}\left(\frac{1}{N}\sum_{i\in s}\frac{x_{ij(m)}e^*_{i(m)}}{\pi_i}-\frac{1}{N}\sum_{i\in U}x_{ij}e^*_{i(m)}\right)^2\right\}^2+CE_\epsilon\left\|\frac{1}{N}\sum_{i\in U}\boldsymbol{x}_{i(m)}e^*_{i(m)}\right\|^4 \\
&\le Ck_m\sum_{j=1}^{k_m}\left[E\left(\frac{1}{N}\sum_{i\in s}\frac{x_{ij(m)}e^*_{i(m)}}{\pi_i}-\frac{1}{N}\sum_{i\in U}x_{ij}e^*_{i(m)}\right)^4+\frac{1}{N^4}\left\{\sum_{i\in U}\sqrt{E_\epsilon\left(x_{ij}e^*_{i(m)}\right)^4}\right\}^2\right] \\
&\le Cn^{-2}k_m^2, \qquad \text{(S.61)}
\end{aligned}
$$

where the third inequality follows from Equation (S.4) with $q = 2$. Combining (S.60) and (S.61), we obtain the desired result.

**Lemma 10** *Let* $\bar{\boldsymbol{T}}_{(m)} = N^{-1}\boldsymbol{X}'_{(m)U}\boldsymbol{X}_{(m)U}$, $\bar{\boldsymbol{t}}_{(m)} = N^{-1}\boldsymbol{X}'_{(m)U}\boldsymbol{y}_U$, $\hat{\bar{\boldsymbol{T}}}_{(m)} = N^{-1}\boldsymbol{X}'_{(m)s}\boldsymbol{\Phi}_s\boldsymbol{X}_{(m)s}$, *and* $\hat{\bar{\boldsymbol{t}}}_{(m)} = N^{-1}\boldsymbol{X}'_{(m)s}\boldsymbol{\Phi}_s\boldsymbol{y}_s$. *By a Taylor expansion, we have*

$$\hat{\boldsymbol{\beta}}_{(m)} = \tilde{\boldsymbol{\beta}}_{(m)} + \bar{\boldsymbol{T}}^{-1}_{(m)}\left(\hat{\bar{\boldsymbol{t}}}_{(m)} - \hat{\bar{\boldsymbol{T}}}_{(m)}\tilde{\boldsymbol{\beta}}_{(m)}\right) + \boldsymbol{r}_{(m)},$$

*where* $\boldsymbol{r}_{(m)}$ *denotes the remainder term. If Conditions 1–2 and 9–10 hold, then*

$$E\left(r^2_{j(m)}\right) = O(n^{-2}),$$

*where* $r_{j(m)}$ *is the* $j$*th element of* $\boldsymbol{r}_{(m)}$.

**Proof.** Let $\bar{t}_{jj'(m)}$, $\bar{t}_{j(m)}$, $\hat{\bar{t}}_{jj'(m)}$, and $\hat{\bar{t}}_{j(m)}$ be the corresponding elements of $\bar{\boldsymbol{T}}_{(m)}$, $\bar{\boldsymbol{t}}_{(m)}$, $\hat{\bar{\boldsymbol{T}}}_{(m)}$, and $\hat{\bar{\boldsymbol{t}}}_{(m)}$, respectively. A Taylor linearization of $\hat{\boldsymbol{\beta}}_{(m)}$ around $(\bar{\boldsymbol{T}}_{(m)}, \bar{\boldsymbol{t}}_{(m)})$ yields

$$\hat{\boldsymbol{\beta}}_{(m)} = \tilde{\boldsymbol{\beta}}_{(m)} + \sum_{j=1}^{k_m}\sum_{j'\le j} a_{jj'(m)}\left(\hat{\bar{t}}_{jj'(m)} - \bar{t}_{jj'(m)}\right) + \sum_{j=1}^{k_m} a_{j(m)}\left(\hat{\bar{t}}_{j(m)} - \bar{t}_{j(m)}\right) + \boldsymbol{r}_{(m)},$$

where $a_{jj'(m)} = \left.\dfrac{\partial\hat{\boldsymbol{\beta}}_{(m)}}{\partial\hat{\bar{t}}_{jj'(m)}}\right|_{\hat{\bar{\boldsymbol{T}}}_{(m)}=\bar{\boldsymbol{T}}_{(m)},\hat{\bar{\boldsymbol{t}}}_{(m)}=\bar{\boldsymbol{t}}_{(m)}}$, $a_{j(m)} = \left.\dfrac{\partial\hat{\boldsymbol{\beta}}_{(m)}}{\partial\hat{\bar{t}}_{j(m)}}\right|_{\hat{\bar{\boldsymbol{T}}}_{(m)}=\bar{\boldsymbol{T}}_{(m)},\hat{\bar{\boldsymbol{t}}}_{(m)}=\bar{\boldsymbol{t}}_{(m)}}$ and $\boldsymbol{r}_{(m)}$ is the remainder term. By standard matrix differentiation results (see, e.g., Graybill 1983), we

have

$$\frac{\partial \hat{\boldsymbol{\beta}}_{(m)}}{\partial \hat{\bar{t}}_{jj'(m)}} = \left( \frac{\partial \hat{\bar{\boldsymbol{T}}}_{(m)}^{-1}}{\partial \hat{\bar{t}}_{jj'(m)}} \right) \hat{\bar{\boldsymbol{t}}}_{(m)} = \left( -\hat{\bar{\boldsymbol{T}}}_{(m)}^{-1} \boldsymbol{\Lambda}_{jj'(m)} \hat{\bar{\boldsymbol{T}}}_{(m)}^{-1} \right) \hat{\bar{\boldsymbol{t}}}_{(m)} = -\hat{\bar{\boldsymbol{T}}}_{(m)}^{-1} \boldsymbol{\Lambda}_{jj'(m)} \hat{\boldsymbol{\beta}}_{(m)},$$

where $\boldsymbol{\Lambda}_{jj'(m)}$ is a $k_m \times k_m$ matrix with one at positions $(j, j')$ and $(j', j)$ and zeros elsewhere. Similarly,

$$\frac{\partial \hat{\boldsymbol{\beta}}_{(m)}}{\partial \hat{\bar{t}}_{j(m)}} = \hat{\bar{\boldsymbol{T}}}_{(m)}^{-1} \left( \frac{\partial \hat{\bar{\boldsymbol{t}}}_{(m)}}{\partial \hat{\bar{t}}_{j(m)}} \right) = \hat{\bar{\boldsymbol{T}}}_{(m)}^{-1} \boldsymbol{\lambda}_{j(m)},$$

where $\boldsymbol{\lambda}_{j(m)}$ is a $k_m$-vector with the $j$th component equal to one and zeros elsewhere. Therefore,

$$\begin{aligned}
\hat{\boldsymbol{\beta}}_{(m)} &= \tilde{\boldsymbol{\beta}}_{(m)} - \sum_{j=1}^{k_m} \sum_{j' \leq j} \bar{\boldsymbol{T}}_{(m)}^{-1} \boldsymbol{\Lambda}_{jj'(m)} \tilde{\boldsymbol{\beta}}_{(m)} \left( \hat{\bar{t}}_{jj'(m)} - \bar{t}_{jj'(m)} \right) \\
&\quad + \sum_{j=1}^{k_m} \bar{\boldsymbol{T}}_{(m)}^{-1} \boldsymbol{\lambda}_{j(m)} \left( \hat{\bar{t}}_{j(m)} - \bar{t}_{j(m)} \right) + \boldsymbol{r}_{(m)} \\
&= \tilde{\boldsymbol{\beta}}_{(m)} - \bar{\boldsymbol{T}}_{(m)}^{-1} \left( \hat{\bar{\boldsymbol{T}}}_{(m)} - \bar{\boldsymbol{T}}_{(m)} \right) \tilde{\boldsymbol{\beta}}_{(m)} + \bar{\boldsymbol{T}}_{(m)}^{-1} \left( \hat{\bar{\boldsymbol{t}}}_{(m)} - \bar{\boldsymbol{t}}_{(m)} \right) + \boldsymbol{r}_{(m)} \\
&= \tilde{\boldsymbol{\beta}}_{(m)} + \bar{\boldsymbol{T}}_{(m)}^{-1} \left( \hat{\bar{\boldsymbol{t}}}_{(m)} - \hat{\bar{\boldsymbol{T}}}_{(m)} \tilde{\boldsymbol{\beta}}_{(m)} \right) + \boldsymbol{r}_{(m)}.
\end{aligned}$$

It follows from the above equation that

$$r_{j(m)} = \boldsymbol{\lambda}_{j(m)}' \boldsymbol{r}_{(m)} = \boldsymbol{\lambda}_{j(m)}' \hat{\bar{\boldsymbol{T}}}_{(m)}^{-1} \hat{\bar{\boldsymbol{t}}}_{(m)} - \boldsymbol{\lambda}_{j(m)}' \tilde{\boldsymbol{\beta}}_{(m)} - \boldsymbol{\lambda}_{j(m)}' \bar{\boldsymbol{T}}_{(m)}^{-1} \left( \hat{\bar{\boldsymbol{t}}}_{(m)} - \hat{\bar{\boldsymbol{T}}}_{(m)} \tilde{\boldsymbol{\beta}}_{(m)} \right).$$

Since $\hat{\bar{\boldsymbol{T}}}_{(m)}$ and $\hat{\bar{\boldsymbol{t}}}_{(m)}$ consist of HT estimators, $r_{j(m)}$ is a nonlinear function of these HT estimators. By Lemma 8 and Conditions 1 and 10, the fourth-order moment of each HT estimator is $O(n^{-2})$. Moreover, $r_{j(m)}^2$ and its first three derivatives are equal to zero at the point $(\bar{\boldsymbol{T}}_{(m)}, \bar{\boldsymbol{t}}_{(m)})$. Therefore, applying Theorem 5.4.3 of Fuller (1996) with $\alpha = 1$, $s = 4$, and $a_N^4 = O(n^{-2})$, we conclude that

$$E\left( r_{j(m)}^2 \right) = O(n^{-2}).$$

This completes the proof of Lemma 10.

**Lemma 11** *If Conditions 1–2 and 9–10 and $k_M^2 = O(N)$ hold, then*

$$E(b_N^2) = E\left[\frac{1}{N}\sum_{i\in s}\frac{\{\tilde{\mu}_i(\boldsymbol{\omega}^*) - \hat{\mu}_i(\boldsymbol{\omega}^*)\}}{\pi_i} - \frac{1}{N}\sum_{i\in U}\{\tilde{\mu}_i(\boldsymbol{\omega}^*) - \hat{\mu}_i(\boldsymbol{\omega}^*)\}\right]^2 = O(n^{-2}k_M^2),$$

*where $\boldsymbol{\omega}^*$ is the optimal weight vector.*

**Proof.** By Lemma 10, we have

$$\begin{aligned}
\hat{\mu}_i(\boldsymbol{\omega}^*) - \tilde{\mu}_i(\boldsymbol{\omega}^*) &= \sum_{m=1}^{M}\omega_m^*\boldsymbol{x}'_{i(m)}\left\{\bar{\boldsymbol{T}}_{(m)}^{-1}\left(\hat{\bar{\boldsymbol{t}}}_{(m)} - \hat{\bar{\boldsymbol{T}}}_{(m)}\tilde{\boldsymbol{\beta}}_{(m)}\right) + \boldsymbol{r}_{(m)}\right\} \\
&= \frac{1}{N}\sum_{j\in U}\frac{I_j - \pi_j}{\pi_j}\sum_{m=1}^{M}\left\{\omega_m^*\boldsymbol{x}'_{i(m)}\bar{\boldsymbol{T}}_{(m)}^{-1}\boldsymbol{x}_{j(m)}\left(y_j - \boldsymbol{x}'_{j(m)}\tilde{\boldsymbol{\beta}}_{(m)}\right)\right\} \\
&\quad + \sum_{m=1}^{M}\left(\omega_m^*\boldsymbol{x}'_{i(m)}\boldsymbol{r}_{(m)}\right) \\
&\triangleq \frac{1}{N}\sum_{j\in U}\frac{I_j - \pi_j}{\pi_j}z_{ij} + r_i(\boldsymbol{\omega}^*).
\end{aligned}$$

Then

$$\begin{aligned}
b_N^2 &= \frac{1}{N^2}\sum_{i,j\in U}\frac{(\pi_i - I_i)(\pi_j - I_j)\{\hat{\mu}_i(\boldsymbol{\omega}^*) - \tilde{\mu}_i(\boldsymbol{\omega}^*)\}\{\hat{\mu}_j(\boldsymbol{\omega}^*) - \tilde{\mu}_j(\boldsymbol{\omega}^*)\}}{\pi_i\pi_j} \\
&= \frac{1}{N^4}\sum_{i,j,k,l\in U}\frac{(\pi_i - I_i)(\pi_j - I_j)(\pi_k - I_k)(\pi_l - I_l)z_{ik}z_{jl}}{\pi_i\pi_j\pi_k\pi_l} \\
&\quad + \frac{2}{N^3}\sum_{i,j,k\in U}\frac{(\pi_i - I_i)(\pi_j - I_j)(\pi_k - I_k)z_{ik}r_j(\boldsymbol{\omega}^*)}{\pi_i\pi_j\pi_k} \\
&\quad + \frac{1}{N^2}\sum_{i,j\in U}\frac{(\pi_i - I_i)(\pi_j - I_j)r_i(\boldsymbol{\omega}^*)r_j(\boldsymbol{\omega}^*)}{\pi_i\pi_j} \\
&\triangleq b_{1N} + b_{2N} + b_{3N}.
\end{aligned}$$

First, we prove that $|E(b_{1N})| = O(n^{-2}k_M^2)$. Condition 1 and Lemma 1 imply that

$$\left|\boldsymbol{x}'_{i(m)}\bar{\boldsymbol{T}}_{(m)}^{-1}\boldsymbol{x}_{j(m)}\right| \le \left\|\boldsymbol{x}_{i(m)}\right\|^2 \cdot \lambda_{\min}^{-1}\left(\bar{\boldsymbol{T}}_{(m)}\right) \le Ck_m$$

and

$$\begin{aligned}
&E_\epsilon\left|\left(y_j - \boldsymbol{x}'_{j(m)}\tilde{\boldsymbol{\beta}}_{(m)}\right)\left(y_j - \boldsymbol{x}'_{j(l)}\tilde{\boldsymbol{\beta}}_{(l)}\right)\right| \\
&\le \left(E_\epsilon\left|e^*_{j(m)} + \boldsymbol{x}'_{j(m)}\boldsymbol{\beta}^*_{(m)} - \boldsymbol{x}'_{j(m)}\tilde{\boldsymbol{\beta}}_{(m)}\right|^2 \cdot E_\epsilon\left|e^*_{j(l)} + \boldsymbol{x}'_{j(l)}\beta^*_{(l)} - \boldsymbol{x}'_{j(l)}\tilde{\boldsymbol{\beta}}_{(l)}\right|^2\right)^{1/2}
\end{aligned}$$

$$\leq C \left\{ \left(1 + \frac{k_m^2}{N}\right) \left(1 + \frac{k_l^2}{N}\right) \right\}^{1/2}.$$

It follows from the condition $k_M^2 = O(N)$ that

$$E_\epsilon(z_{ij}^2) \leq C \sum_{m=1}^{M} \sum_{l=1}^{M} \omega_m^* \omega_l^* k_m k_l \left\{ \left(1 + \frac{k_m^2}{N}\right) \left(1 + \frac{k_l^2}{N}\right) \right\}^{1/2} = O(k_M^2)$$

uniformly in $i$ and $j$. Thus, using Conditions 2 and 9, we obtain that

$$\begin{aligned}
|E(b_{1N})| &= \left| E \left\{ \frac{1}{N^4} \sum_{i,j,k,l \in U} \frac{(\pi_i - I_i)(\pi_j - I_j)(\pi_k - I_k)(\pi_l - I_l) z_{ik} z_{jl}}{\pi_i \pi_j \pi_k \pi_l} \right\} \right| \\
&= \left| \frac{1}{N^4} \sum_{i,j,k,l \in U} E_\epsilon(z_{ik} z_{jl}) \cdot E_d \left\{ \frac{(\pi_i - I_i)(\pi_j - I_j)(\pi_k - I_k)(\pi_l - I_l)}{\pi_i \pi_j \pi_k \pi_l} \right\} \right| \\
&\leq \frac{C k_M^2}{N^4} \sum_{i,j,k,l \in U} \left| E_d \left\{ (\pi_i - I_i)(\pi_j - I_j)(\pi_k - I_k)(\pi_l - I_l) \right\} \right| \\
&\leq \frac{C k_M^2}{N^4} \left[ \sum_{i,j,k,l \in \boldsymbol{D}_4} \left| E_d \left\{ (I_i - \pi_i)(I_j - \pi_j)(I_k - \pi_k)(I_l - \pi_l) \right\} \right| \right. \\
&\quad + \sum_{i,j,k \in \boldsymbol{D}_3} \left| E_d \left\{ (I_i - \pi_i)^2 (I_j - \pi_j)(I_k - \pi_k) \right\} \right| \\
&\quad + \sum_{i \neq j \in U} \left| E_d \left\{ (I_i - \pi_i)^2 (I_j - \pi_j)^2 + (I_i - \pi_i)(I_j - \pi_j)^3 \right\} \right| \\
&\quad \left. + \sum_{i \in U} \left| E_d \left\{ (I_i - \pi_i)^4 \right\} \right| \right] \\
&= O\left( n^{-2} k_M^2 \right).
\end{aligned}$$

Next, we consider $|E(b_{3N})|$. By Lemma 10, we have

$$E \left\| \boldsymbol{r}_{(m)} \right\|^2 = \sum_{j=1}^{k_m} E\left( r_{j(m)}^2 \right) = O(n^{-2} k_m)$$

uniformly in $m$. It follows from Condition 1 that

$$\begin{aligned}
\sum_{i \in U} E\left\{ r_i^2(\boldsymbol{\omega}^*) \right\} &= E \left\{ \sum_{m=1}^{M} \sum_{l=1}^{M} \omega_m^* \omega_l^* \boldsymbol{r}'_{(m)} \left( \sum_{i \in U} \boldsymbol{x}_{i(m)} \boldsymbol{x}'_{i(l)} \right) \boldsymbol{r}_{(l)} \right\} \\
&= E \left( \sum_{m=1}^{M} \sum_{l=1}^{M} \omega_m^* \omega_l^* \boldsymbol{r}'_{(m)} \boldsymbol{X}'_{(m)U} \boldsymbol{X}_{(l)U} \boldsymbol{r}_{(l)} \right) \\
&\leq C \sum_{m=1}^{M} \sum_{l=1}^{M} \omega_m^* \omega_l^* \left( E \left\| \boldsymbol{r}_{(m)} \right\|^2 E \left\| \boldsymbol{r}_{(l)} \right\|^2 \right)^{1/2}
\end{aligned}$$

$$= O(Nn^{-2}k_M^2). \tag{S.62}$$

Applying Loève's $c_r$ inequality, we obtain that

$$\begin{aligned}
|E(b_{3N})| &= \left| E\left\{ \frac{1}{N^2} \sum_{i,j\in U} \frac{(\pi_i - I_i)(\pi_j - I_j) r_i(\boldsymbol{\omega}^*) r_j(\boldsymbol{\omega}^*)}{\pi_i \pi_j} \right\} \right| \\
&= E\left\{ \frac{1}{N} \sum_{i\in U} \frac{r_i(\boldsymbol{\omega}^*)(\pi_i - I_i)}{\pi_i} \right\}^2 \\
&\leq \frac{1}{N} \sum_{i\in U} E\left\{ r_i^2(\boldsymbol{\omega}^*) \right\} \\
&= O\left( n^{-2} k_M^2 \right).
\end{aligned}$$

Finally, we prove that $|E(b_{2N})| = O\left(n^{-2}k_M^2\right)$. By the Cauchy–Schwarz inequality,

$$\begin{aligned}
|E(b_{2N})| &= \left| E\left\{ \frac{2}{N^3} \sum_{i,j,k\in U} \frac{(\pi_i - I_i)(\pi_j - I_j)(\pi_k - I_k) z_{ik} r_j(\boldsymbol{\omega}^*)}{\pi_i \pi_j \pi_k} \right\} \right| \\
&= \frac{2}{N^3} \left| E\left\{ \sum_{i,k\in U} \frac{(\pi_i - I_i)(\pi_k - I_k) z_{ik}}{\pi_i \pi_k} \cdot \sum_{j\in U} \frac{(\pi_j - I_j) r_j(\boldsymbol{\omega}^*)}{\pi_j} \right\} \right| \\
&\leq \frac{2}{N^3} \left[ E\left\{ \sum_{i,k\in U} \frac{(\pi_i - I_i)(\pi_k - I_k) z_{ik}}{\pi_i \pi_k} \right\}^2 \cdot E\left\{ \sum_{j\in U} \frac{(\pi_j - I_j) r_j(\boldsymbol{\omega}^*)}{\pi_j} \right\}^2 \right]^{1/2} \\
&= \frac{2}{N^3} \cdot \sqrt{N^4 |E(b_{1N})| \cdot N^2 |E(b_{3N})|} \\
&= O\left( n^{-2} k_M^2 \right).
\end{aligned}$$

This completes the proof of Lemma 11.

**Lemma 12** *If Conditions 1–2 and 10 hold, then*

$$\frac{1}{N} \sum_{i\in U} E\left\{ \hat{\mu}_i(\hat{\boldsymbol{\omega}}_0) - \tilde{\mu}_i(\boldsymbol{\omega}^*) \right\}^2 = O\left( n^{-1} M k_M + M \zeta_n^{1/2} \right),$$

*where* $\hat{\boldsymbol{\omega}}_0$ *is an estimator of* $\boldsymbol{\omega}^*$ *and* $\zeta_n = E\|\hat{\boldsymbol{\omega}}_0 - \boldsymbol{\omega}^*\|^4$.

**Proof.** Conditions 1–2 and 10 imply that

$$E\|\boldsymbol{y}_U\|^4 \leq N \sum_{i\in U} E_\epsilon (\mu_i + e_i)^4$$

$$\leq CN \sum_{i \in U} \mu_i^4 + CN \sum_{i \in U} E_\epsilon \left( e_i^4 \right)$$

$$= O(N^2).$$

Then, by Condition 1, Lemma 3, Loève's $c_r$ inequality, and the Cauchy–Schwarz inequality, we have

$$\begin{aligned}
&\sum_{i \in U} E \left\{ \hat{\mu}_i(\hat{\boldsymbol{\omega}}_0) - \tilde{\mu}_i(\boldsymbol{\omega}^*) \right\}^2 \\
&= \sum_{i \in U} E \left\{ \hat{\mu}_i(\hat{\boldsymbol{\omega}}_0) - \tilde{\mu}_i(\hat{\boldsymbol{\omega}}_0) + \tilde{\mu}_i(\hat{\boldsymbol{\omega}}_0) - \tilde{\mu}_i(\boldsymbol{\omega}^*) \right\}^2 \\
&\leq 2 \sum_{i \in U} E \left[ \sum_{m=1}^{M} \left\{ \hat{\omega}_{0m} \left( \boldsymbol{x}'_{i(m)} \hat{\boldsymbol{\beta}}_{(m)} - \boldsymbol{x}'_{i(m)} \tilde{\boldsymbol{\beta}}_{(m)} \right) \right\} \right]^2 \\
&\quad + 2 \sum_{i \in U} E \left[ \sum_{m=1}^{M} \left\{ (\hat{\omega}_{0m} - \omega_m^*) \boldsymbol{x}'_{i(m)} \tilde{\boldsymbol{\beta}}_{(m)} \right\} \right]^2 \\
&\leq 2E \left\{ \sum_{m=1}^{M} \left( \hat{\boldsymbol{\beta}}_{(m)} - \tilde{\boldsymbol{\beta}}_{(m)} \right)' \boldsymbol{X}'_{(m)U} \boldsymbol{X}_{(m)U} \left( \hat{\boldsymbol{\beta}}_{(m)} - \tilde{\boldsymbol{\beta}}_{(m)} \right) \right\} \\
&\quad + 2E \left\{ \| \hat{\boldsymbol{\omega}}_0 - \boldsymbol{\omega}^* \|^2 \cdot \sum_{m=1}^{M} \boldsymbol{y}'_U \boldsymbol{X}_{(m)U} \left( \boldsymbol{X}'_{(m)U} \boldsymbol{X}_{(m)U} \right)^{-1} \boldsymbol{X}'_{(m)U} \boldsymbol{y}_U \right\} \\
&\leq C f^{-1} M k_M + CM \left( E \| \boldsymbol{y}_U \|^4 \cdot E \| \hat{\boldsymbol{\omega}}_0 - \boldsymbol{\omega}^* \|^4 \right)^{1/2} \\
&= O \left( f^{-1} M k_M + MN \zeta_n^{1/2} \right),
\end{aligned}$$

where $\hat{\omega}_{0m}$ and $\omega_m^*$ denote the $m$th elements of $\hat{\boldsymbol{\omega}}_0$ and $\boldsymbol{\omega}^*$, respectively. Therefore,

$$\frac{1}{N} \sum_{i \in U} E \left\{ \hat{\mu}_i(\hat{\boldsymbol{\omega}}_0) - \tilde{\mu}_i(\boldsymbol{\omega}^*) \right\}^2 = O \left( n^{-1} M k_M + M \zeta_n^{1/2} \right).$$

This completes the proof of Lemma 12.

## S3.9 Proof of Theorem 5

The proof of this theorem relies on Lemmas 1–2 and 8–11.

**Proof.** Note that

$$\hat{\bar{t}}_y(\hat{\boldsymbol{\omega}}_0) - \bar{t}_y = \left( \hat{\bar{t}}_y(\boldsymbol{\omega}^*) - \bar{t}_y \right) + \left( \hat{\bar{t}}_y(\hat{\boldsymbol{\omega}}_0) - \hat{\bar{t}}_y(\boldsymbol{\omega}^*) \right)$$

$$
\begin{aligned}
&= \frac{1}{N}\left\{\sum_{i\in s}\frac{y_i-\hat{\mu}_i(\boldsymbol{\omega}^*)}{\pi_i}+\sum_{i\in U}\hat{\mu}_i(\boldsymbol{\omega}^*)\right\}-\bar{t}_y \\
&\quad+\frac{1}{N}\sum_{m=1}^{M}\left\{(\hat{\omega}_{0m}-\omega_m^*)\left(\sum_{i\in s}\frac{\hat{\mu}_{i(m)}}{\pi_i}-\sum_{i\in U}\hat{\mu}_{i(m)}\right)\right\} \\
&= \frac{1}{N}\sum_{i\in U}\frac{(I_i-\pi_i)\left\{y_i-\tilde{\mu}_i(\boldsymbol{\omega}^*)\right\}}{\pi_i} \\
&\quad+\frac{1}{N}\sum_{i\in U}\frac{(\pi_i-I_i)\left\{\hat{\mu}_i(\boldsymbol{\omega}^*)-\tilde{\mu}_i(\boldsymbol{\omega}^*)\right\}}{\pi_i} \\
&\quad+\frac{1}{N}\sum_{m=1}^{M}\left\{(\hat{\omega}_{0m}-\omega_m^*)\left(\sum_{i\in s}\frac{\hat{\mu}_{i(m)}}{\pi_i}-\sum_{i\in U}\hat{\mu}_{i(m)}\right)\right\} \\
&\triangleq a_N+b_N+c_N. \qquad \text{(S.63)}
\end{aligned}
$$

By Condition 1, Lemma 1, Loève's $c_r$ inequality, and the Cauchy–Schwarz inequality,

$$
\begin{aligned}
\sum_{i\in U}E_\epsilon\left\{y_i-\tilde{\mu}_i(\boldsymbol{\omega}^*)\right\}^2 &= \sum_{i\in U}E_\epsilon\left\{y_i-\mu_i^*(\boldsymbol{\omega}^*)+\mu_i^*(\boldsymbol{\omega}^*)-\tilde{\mu}_i(\boldsymbol{\omega}^*)\right\}^2 \\
&\le 2\sum_{i\in U}E_\epsilon\left(\sum_{m=1}^{M}\omega_m^*e_{i(m)}^*\right)^2+2\sum_{i\in U}E_\epsilon\left\{\sum_{m=1}^{M}\omega_m^*\boldsymbol{x}_{i(m)}'\left(\boldsymbol{\beta}_{(m)}^*-\tilde{\boldsymbol{\beta}}_{(m)}\right)\right\}^2 \\
&\le 2\sum_{i\in U}\sum_{m=1}^{M}\sum_{l=1}^{M}\omega_m^*\omega_l^*E_\epsilon\left(e_{i(m)}^*e_{i(l)}^*\right) \\
&\quad+2\sum_{m=1}^{M}\sum_{l=1}^{M}\omega_m^*\omega_l^*E\left|\left(\boldsymbol{\beta}_{(m)}^*-\tilde{\boldsymbol{\beta}}_{(m)}\right)'\boldsymbol{X}_{(m)U}'\boldsymbol{X}_{(l)U}\left(\boldsymbol{\beta}_{(m)}^*-\tilde{\boldsymbol{\beta}}_{(m)}\right)\right| \\
&\le 2\sum_{i\in U}\left[\sum_{m=1}^{M}\omega_m^*\left\{E_\epsilon\left(e_{i(m)}^{*2}\right)\right\}^{1/2}\right]^2 \\
&\quad+2\left\{\sum_{m=1}^{M}\omega_m^*\cdot\lambda_{\max}^{1/2}\left(\boldsymbol{X}_{(m)U}'\boldsymbol{X}_{(m)U}\right)\cdot\left(E\left\|\boldsymbol{\beta}_{(m)}^*-\tilde{\boldsymbol{\beta}}_{(m)}\right\|^2\right)^{1/2}\right\}^2 \\
&= O(N),
\end{aligned}
$$

where $\mu_i^*(\boldsymbol{\omega}^*)=\sum_{m=1}^{M}\omega_m^*\boldsymbol{x}_{i(m)}'\boldsymbol{\beta}_{(m)}^*$. Then, by Lemma 2, we have

$$
\begin{aligned}
E(a_N^2) &= E\left[\frac{1}{N}\sum_{i\in U}\frac{(I_i-\pi_i)\left\{y_i-\tilde{\mu}_i(\boldsymbol{\omega}^*)\right\}}{\pi_i}\right]^2 \\
&= \frac{1}{N^2}\sum_{i,j\in U}\frac{\pi_{ij}-\pi_i\pi_j}{\pi_i\pi_j}E_\epsilon\left[\left\{y_i-\tilde{\mu}_i(\boldsymbol{\omega}^*)\right\}\left\{y_j-\tilde{\mu}_j(\boldsymbol{\omega}^*)\right\}\right] \\
&= O\left(n^{-1}\right). \qquad \text{(S.64)}
\end{aligned}
$$

For $b_N$, Lemma 11 and Condition 11 imply that

$$E(b_N^2) = O\left(n^{-2}k_M^2\right) = o\left(n^{-1}\right). \tag{S.65}$$

Combining (S.63), (S.64), and (S.65), we obtain that

$$\begin{aligned}\mathrm{MSE}\left(\hat{\bar{t}}_y(\boldsymbol{\omega}^*)\right) &= E(a_N + b_N)^2 \\ &= \frac{1}{N^2}\sum_{i,j\in U}\frac{\pi_{ij}-\pi_i\pi_j}{\pi_i\pi_j}E_\epsilon\left[\left\{y_i - \tilde{\mu}_i(\boldsymbol{\omega}^*)\right\}\left\{y_j - \tilde{\mu}_j(\boldsymbol{\omega}^*)\right\}\right] + O\left(\frac{k_M}{n^{3/2}}\right). \end{aligned} \tag{S.66}$$

Next, we consider $c_N$. By Conditions 1 and 10,

$$\begin{aligned}\sum_{i\in U}E_\epsilon\left(\mu_{i(m)}^{*4}\right) &= \sum_{i\in U}E_\epsilon\left(\mu_i + e_i - e_{i(m)}^*\right)^4 \\ &\le 27\sum_{i\in U}\mu_i^4 + 27\sum_{i\in U}E_\epsilon\left(e_i^4\right) + 27\sum_{i\in U}E_\epsilon\left(e_{i(m)}^{*4}\right) \\ &= O(N).\end{aligned}$$

Then, Lemmas 8–9 and Loève's $c_r$ inequality yield

$$\begin{aligned}&E\left(\sum_{i\in s}\frac{\hat{\mu}_{i(m)}}{\pi_i} - \sum_{i\in U}\hat{\mu}_{i(m)}\right)^4 \\ &= E\left\{\sum_{i\in U}\frac{(I_i-\pi_i)\left(\hat{\mu}_{i(m)} - \mu_{i(m)}^*\right)}{\pi_i} + \sum_{i\in U}\frac{(I_i-\pi_i)\mu_{i(m)}^*}{\pi_i}\right\}^4 \\ &\le CN^3\sum_{i\in U}E\left\{\boldsymbol{x}_{i(m)}'\left(\hat{\boldsymbol{\beta}}_{(m)} - \boldsymbol{\beta}_{(m)}^*\right)\right\}^4 + Cn^{-2}N^3\sum_{i=1}^N E_\epsilon\left(\mu_{i(m)}^{*4}\right) \\ &\le Cn^{-2}N^4k_M^4.\end{aligned}$$

Hence, using the Cauchy–Schwarz inequality and Condition 11, we have

$$\begin{aligned}E\left(c_N^2\right) &= E\left\{\frac{1}{N}\sum_{m=1}^M(\hat{\omega}_{0m} - \omega_m^*)\left(\sum_{i\in s}\frac{\hat{\mu}_{i(m)}}{\pi_i} - \sum_{i\in U}\hat{\mu}_{i(m)}\right)\right\}^2 \\ &\le \frac{1}{N^2}E\left\{\sum_{m=1}^M(\hat{\omega}_{0m} - \omega_m^*)^2\sum_{m=1}^M\left(\sum_{i\in s}\frac{\hat{\mu}_{i(m)}}{\pi_i} - \sum_{i\in U}\hat{\mu}_{i(m)}\right)^2\right\} \\ &\le \frac{1}{N^2}\left[E\left\{\sum_{m=1}^M(\hat{\omega}_{0m} - \omega_m^*)^2\right\}^2 E\left\{\sum_{m=1}^M\left(\sum_{i\in s}\frac{\hat{\mu}_{i(m)}}{\pi_i} - \sum_{i\in U}\hat{\mu}_{i(m)}\right)^2\right\}^2\right]^{1/2}\end{aligned}$$

$$
\begin{aligned}
&\leq \frac{1}{N^2}\left[E\left\|\hat{\boldsymbol{\omega}}_0-\boldsymbol{\omega}^*\right\|^4\cdot M\left\{\sum_{m=1}^{M}E\left(\sum_{i\in s}\frac{\hat{\mu}_{i(m)}}{\pi_i}-\sum_{i\in U}\hat{\mu}_{i(m)}\right)^4\right\}\right]^{1/2}\\
&=O\left(n^{-1}\zeta_n^{1/2}Mk_M^2\right)\\
&=o\left(n^{-1}\right),
\end{aligned}
\tag{S.67}
$$

where $\hat{\omega}_{0m}$ and $\omega_m^*$ are the $m$th elements of $\hat{\boldsymbol{\omega}}_0$ and $\boldsymbol{\omega}^*$, respectively. Combining (S.63), (S.66), (S.67) and Condition 11, we obtain that

$$
\begin{aligned}
\mathrm{MSE}\left(\hat{\bar{t}}_y(\hat{\boldsymbol{\omega}}_0)\right)&=\frac{1}{N^2}\sum_{i,j\in U}\frac{\pi_{ij}-\pi_i\pi_j}{\pi_i\pi_j}E_\epsilon\left[\left\{y_i-\tilde{\mu}_i(\boldsymbol{\omega}^*)\right\}\left\{y_j-\tilde{\mu}_j(\boldsymbol{\omega}^*)\right\}\right]\\
&\quad+O\left(\frac{1}{n}\sqrt{\zeta_n^{1/2}Mk_M^2\left(1+\frac{k_M}{\sqrt{n}}\right)}\right)\\
&=\frac{1}{N^2}\sum_{i,j\in U}\frac{\pi_{ij}-\pi_i\pi_j}{\pi_i\pi_j}E_\epsilon\left[\left\{y_i-\tilde{\mu}_i(\boldsymbol{\omega}^*)\right\}\left\{y_j-\tilde{\mu}_j(\boldsymbol{\omega}^*)\right\}\right]+o\left(n^{-1}\right).
\end{aligned}
$$

This completes the proof of Theorem 5.

## S3.10 Proof of Theorem 6

The proof of this theorem relies on Lemmas 1–3 and 8–12.

**Proof.** By Theorem 5, it suffices to show that

$$
\lim_{n\to\infty}nE\left|\widehat{\mathrm{MSE}}\left(\hat{\bar{t}}_y(\hat{\boldsymbol{\omega}}_0)\right)-\mathrm{AMSE}\left(\hat{\bar{t}}_y(\boldsymbol{\omega}^*)\right)\right|=o\left(1\right),
$$

where

$$
\mathrm{AMSE}\left(\hat{\bar{t}}_y(\boldsymbol{\omega}^*)\right)=\frac{1}{N^2}\sum_{i,j\in U}\frac{\pi_{ij}-\pi_i\pi_j}{\pi_i\pi_j}E_\epsilon\left[\left\{y_i-\tilde{\mu}_i(\boldsymbol{\omega}^*)\right\}\left\{y_j-\tilde{\mu}_j(\boldsymbol{\omega}^*)\right\}\right].
$$

Let

$$
A_N=E\left|\frac{n}{N^2}\sum_{i,j\in U}\left\{y_i-\tilde{\mu}_i(\boldsymbol{\omega}^*)\right\}\left\{y_j-\tilde{\mu}_j(\boldsymbol{\omega}^*)\right\}\frac{\pi_{ij}-\pi_i\pi_j}{\pi_i\pi_j}\frac{I_iI_j-\pi_{ij}}{\pi_{ij}}\right|
$$

and

$$
B_N=E\left|\frac{n}{N^2}\sum_{i,j\in U}\left[2\left\{y_i-\tilde{\mu}_i(\boldsymbol{\omega}^*)\right\}\left\{\tilde{\mu}_j(\boldsymbol{\omega}^*)-\hat{\mu}_j(\hat{\boldsymbol{\omega}}_0)\right\}\right.
$$

$$+ \left\{\tilde{\mu}_i(\boldsymbol{\omega}^*) - \hat{\mu}_i(\hat{\boldsymbol{\omega}}_0)\right\}\left\{\tilde{\mu}_j(\boldsymbol{\omega}^*) - \hat{\mu}_j(\hat{\boldsymbol{\omega}}_0)\right\}\Big] \frac{\pi_{ij} - \pi_i\pi_j}{\pi_i\pi_j}\frac{I_iI_j}{\pi_{ij}}\Bigg|.$$

Then

$$nE\left|\widehat{\mathrm{MSE}}\left(\hat{\bar{t}}_y(\hat{\boldsymbol{\omega}}_0)\right) - \mathrm{AMSE}\left(\hat{\bar{t}}_y(\boldsymbol{\omega}^*)\right)\right| \leq A_N + B_N.$$

Hence, the proof is completed once we establish that $A_N = o(1)$ and $B_N = o(1)$.

First, we prove that $A_N = o(1)$. Note that

$$\begin{aligned}
&E\left[\frac{n}{N^2}\sum_{i,j\in U}\left\{y_i - \tilde{\mu}_i(\boldsymbol{\omega}^*)\right\}\left\{y_j - \tilde{\mu}_j(\boldsymbol{\omega}^*)\right\}\frac{\pi_{ij}-\pi_i\pi_j}{\pi_i\pi_j}\frac{I_iI_j - \pi_{ij}}{\pi_{ij}}\right]^2 \\
&= n^2E_\epsilon\left[\sum_{i\neq j\in U}\sum_{k\neq l\in U}\frac{(\pi_{ij}-\pi_i\pi_j)(\pi_{kl}-\pi_k\pi_l)}{\pi_i\pi_j\pi_k\pi_l}\frac{E_d\left\{(I_iI_j-\pi_{ij})(I_kI_l-\pi_{kl})\right\}}{\pi_{ij}\pi_{kl}}\right. \\
&\quad\left.\times\frac{\left\{y_i - \tilde{\mu}_i(\boldsymbol{\omega}^*)\right\}\left\{y_j - \tilde{\mu}_j(\boldsymbol{\omega}^*)\right\}\left\{y_k - \tilde{\mu}_k(\boldsymbol{\omega}^*)\right\}\left\{y_l - \tilde{\mu}_l(\boldsymbol{\omega}^*)\right\}}{N^4}\right] \\
&\quad + n^2E_\epsilon\left[\sum_{i\in U}\sum_{k\in U}\frac{(1-\pi_i)(1-\pi_k)}{\pi_i\pi_k}\frac{E_d\left\{(I_i-\pi_i)(I_k-\pi_k)\right\}}{\pi_i\pi_k}\right. \\
&\quad\left.\times\frac{\left\{y_i - \tilde{\mu}_i(\boldsymbol{\omega}^*)\right\}^2\left\{y_k - \tilde{\mu}_k(\boldsymbol{\omega}^*)\right\}^2}{N^4}\right] \\
&\quad + 2n^2E_\epsilon\left[\sum_{i\in U}\sum_{k\neq l\in U}\frac{(1-\pi_i)(\pi_{kl}-\pi_k\pi_l)}{\pi_i\pi_k\pi_l}\frac{E_d\left\{(I_i-\pi_i)(I_kI_l-\pi_{kl})\right\}}{\pi_i\pi_{kl}}\right. \\
&\quad\left.\times\frac{\left\{y_i - \tilde{\mu}_i(\boldsymbol{\omega}^*)\right\}^2\left\{y_k - \tilde{\mu}_k(\boldsymbol{\omega}^*)\right\}\left\{y_l - \tilde{\mu}_l(\boldsymbol{\omega}^*)\right\}}{N^4}\right] \\
&\triangleq A_{1N} + A_{2N} + A_{3N}.
\end{aligned}$$

By Lemma 1 and Conditions 1 and 11, we have

$$\begin{aligned}
E_\epsilon\left\{y_i - \tilde{\mu}_i(\boldsymbol{\omega}^*)\right\}^4 &= E_\epsilon\left\{y_i - \mu_i^*(\boldsymbol{\omega}^*) + \mu_i^*(\boldsymbol{\omega}^*) - \tilde{\mu}_i(\boldsymbol{\omega}^*)\right\}^4 \\
&\leq CE_\epsilon\left(\sum_{m=1}^M \omega_m^* e_{i(m)}^*\right)^4 + CE_\epsilon\left[\sum_{m=1}^M \omega_m^*\left\{\boldsymbol{x}_{i(m)}'\left(\boldsymbol{\beta}_{(m)}^* - \tilde{\boldsymbol{\beta}}_{(m)}\right)\right\}\right]^4 \\
&\leq C\left[\sum_{m=1}^M \omega_m^*\left\{E_\epsilon\left(e_{i(m)}^{*4}\right)\right\}^{1/4}\right]^4 \\
&\quad + C\left[\sum_{m=1}^M \omega_m^*\left\{E_\epsilon\left(\boldsymbol{x}_{i(m)}'\boldsymbol{\beta}_{(m)}^* - \boldsymbol{x}_{i(m)}'\tilde{\boldsymbol{\beta}}_{(m)}\right)^4\right\}^{1/4}\right]^4 \\
&= O(1) \qquad \text{(S.68)}
\end{aligned}$$

uniformly in $i$. Using Conditions 2 and 9, we obtain that

$$
\begin{aligned}
|A_{1N}| &= n^2 E_\epsilon \left[ \sum_{i\neq j\in U} \sum_{k\neq l\in U} \frac{(\pi_{ij}-\pi_i\pi_j)(\pi_{kl}-\pi_k\pi_l)}{\pi_i\pi_j\pi_k\pi_l} \frac{E_d\left\{(I_iI_j-\pi_{ij})(I_kI_l-\pi_{kl})\right\}}{\pi_{ij}\pi_{kl}} \right. \\
&\quad \left. \times \frac{\left\{y_i-\tilde{\mu}_i(\boldsymbol{\omega}^*)\right\}\left\{y_j-\tilde{\mu}_j(\boldsymbol{\omega}^*)\right\}\left\{y_k-\tilde{\mu}_k(\boldsymbol{\omega}^*)\right\}\left\{y_l-\tilde{\mu}_l(\boldsymbol{\omega}^*)\right\}}{N^4} \right] \\
&\leq C\left(n\max_{i\neq j\in U}|\pi_{ij}-\pi_i\pi_j|\right)^2 E_\epsilon \left[ \sum_{i\neq j\in U} \sum_{k\neq l\in U} \left|E_d\left\{(I_iI_j-\pi_{ij})(I_kI_l-\pi_{kl})\right\}\right| \right. \\
&\quad \left. \times \frac{\left|\left\{y_i-\tilde{\mu}_i(\boldsymbol{\omega}^*)\right\}\left\{y_j-\tilde{\mu}_j(\boldsymbol{\omega}^*)\right\}\left\{y_k-\tilde{\mu}_k(\boldsymbol{\omega}^*)\right\}\left\{y_l-\tilde{\mu}_l(\boldsymbol{\omega}^*)\right\}\right|}{N^4} \right] \\
&\leq O\left(\frac{1}{N}\right) + \frac{C}{N}\sum_{i\in U} E_\epsilon\{y_i-\tilde{\mu}_i(\boldsymbol{\omega}^*)\}^4 \times \max_{i,j,k,l\in \boldsymbol{D}_4} \left|E_d\left\{(I_iI_j-\pi_{ij})(I_kI_l-\pi_{kl})\right\}\right| \\
&= o(1).
\end{aligned}
$$

For $A_{2N}$, Conditions 2 and (S.68) imply that

$$
\begin{aligned}
|A_{2N}| &\leq \frac{Cn^2}{N^4}\sum_{i\in U} E_\epsilon\left\{y_i-\tilde{\mu}_i(\boldsymbol{\omega}^*)\right\}^4 + \frac{Cn^2}{N}\sum_{i\in U} E_\epsilon\left\{y_i-\tilde{\mu}_i(\boldsymbol{\omega}^*)\right\}^4 \times \max_{i\neq k\in U}|\pi_{ik}-\pi_i\pi_k| \\
&= o(1).
\end{aligned}
$$

Similarly, combining Conditions 2 and 9 with (S.68), we obtain $|A_{3N}| = o(1)$. Consequently, $A_N = o(1)$.

Next, we prove that $B_N = o(1)$. By (S.68), Lemma 12, and Condition 11, we have

$$
\begin{aligned}
B_N &= E\left| \frac{n}{N^2}\sum_{i,j\in U} \left[2\left\{y_i-\tilde{\mu}_i(\boldsymbol{\omega}^*)\right\}\left\{\tilde{\mu}_j(\boldsymbol{\omega}^*)-\hat{\mu}_j(\hat{\boldsymbol{\omega}}_0)\right\} \right.\right. \\
&\quad \left.\left. + \left\{\tilde{\mu}_i(\boldsymbol{\omega}^*)-\hat{\mu}_i(\hat{\boldsymbol{\omega}}_0)\right\}\left\{\tilde{\mu}_j(\boldsymbol{\omega}^*)-\hat{\mu}_j(\hat{\boldsymbol{\omega}}_0)\right\}\right] \frac{\pi_{ij}-\pi_i\pi_j}{\pi_i\pi_j}\frac{I_iI_j}{\pi_{ij}} \right| \\
&\leq \frac{C}{N}\sum_{j\in U} E\left|\tilde{\mu}_j(\boldsymbol{\omega}^*)-\hat{\mu}_j(\hat{\boldsymbol{\omega}}_0)\right| \times \left(f + n\max_{i\neq j\in U}|\pi_{ij}-\pi_i\pi_j|\right) \\
&\quad + \frac{C}{N}\sum_{j\in U} E\left\{\tilde{\mu}_j(\boldsymbol{\omega}^*)-\hat{\mu}_j(\hat{\boldsymbol{\omega}}_0)\right\}^2 \times \left(f + n\max_{i\neq j\in U}|\pi_{ij}-\pi_i\pi_j|\right) \\
&= O\left(n^{-1}Mk_M + M\zeta_n^{1/2}\right) \\
&= o(1).
\end{aligned}
$$

This completes the proof of Theorem 6.

## S3.11 Proof of Theorem 7

The proof of this theorem relies on Lemmas 1–3 and 8–12.

**Proof.** It follows from (S.63) that

$$\begin{aligned}
\hat{\bar{t}}_y(\hat{\boldsymbol{\omega}}_0) - \bar{t}_y &= a_N + b_N + c_N \\
&= \frac{1}{N}\sum_{i\in U}\frac{(I_i-\pi_i)\left\{y_i-\tilde{\mu}_i(\boldsymbol{\omega}^*)\right\}}{\pi_i} + b_N + c_N \\
&= \frac{1}{N}\sum_{i\in U}\frac{(I_i-\pi_i)\left\{y_i-\mu_i+\mu_i-\tilde{\mu}_i(\boldsymbol{\omega}^*)\right\}}{\pi_i} + b_N + c_N \\
&= \frac{1}{N}\left(\sum_{i\in s}\frac{y_i-\mu_i}{\pi_i}+\sum_{i\in U}\mu_i\right) - \bar{t}_y \\
&\quad + \frac{1}{N}\left[\sum_{i\in s}\frac{\mu_i-\tilde{\mu}_i(\boldsymbol{\omega}^*)}{\pi_i} - \sum_{i\in U}\left\{\mu_i-\tilde{\mu}_i(\boldsymbol{\omega}^*)\right\}\right] + b_N + c_N \\
&= \left(\hat{\bar{t}}_y - \bar{t}_y\right) + a_{2N} + b_N + c_N. \qquad \text{(S.69)}
\end{aligned}$$

Using Conditions 2 and 11, we obtain that

$$\begin{aligned}
E(a_{2N}^2) &= \left|\frac{1}{N^2}\sum_{i,j\in U}\frac{\pi_{ij}-\pi_i\pi_j}{\pi_i\pi_j}E_\epsilon\left[\left\{\mu_i-\tilde{\mu}_i(\boldsymbol{\omega}^*)\right\}\left\{\mu_j-\tilde{\mu}_j(\boldsymbol{\omega}^*)\right\}\right]\right| \\
&\le E_\epsilon\left|\frac{1}{N^2}\sum_{i\in U}\frac{1-\pi_i}{\pi_i}\left\{\mu_i-\tilde{\mu}_i(\boldsymbol{\omega}^*)\right\}^2\right| \\
&\quad + E_\epsilon\left|\frac{1}{N^2}\sum_{i\neq j\in U}\frac{\pi_{ij}-\pi_i\pi_j}{\pi_i\pi_j}\left[\left\{\mu_i-\tilde{\mu}_i(\boldsymbol{\omega}^*)\right\}\left\{\mu_j-\tilde{\mu}_j(\boldsymbol{\omega}^*)\right\}\right]\right| \\
&\le \frac{C}{N^2}E_\epsilon\left[\sum_{i\in U}\left\{\mu_i-\tilde{\mu}_i(\boldsymbol{\omega}^*)\right\}^2\right] + \frac{C}{nN^2}E_\epsilon\left|\sum_{i\neq j\in U}\left[\left\{\mu_i-\tilde{\mu}_i(\boldsymbol{\omega}^*)\right\}\left\{\mu_j-\tilde{\mu}_j(\boldsymbol{\omega}^*)\right\}\right]\right| \\
&\le \frac{C}{N^2}\sum_{i\in U}E_\epsilon\left\{\mu_i-\tilde{\mu}_i(\boldsymbol{\omega}^*)\right\}^2 + \frac{C}{nN}\sum_{i\in U}E_\epsilon\left\{\mu_i-\tilde{\mu}_i(\boldsymbol{\omega}^*)\right\}^2 \\
&= O\left(\frac{\tilde{R}_N(\boldsymbol{\omega}^*)}{nN}\right) \\
&= o\left(n^{-1}\right).
\end{aligned}$$

Lemma 11 and Condition 11 imply that $E(b_N^2) = o\left(n^{-1}\right)$. By (S.67), $E(c_N^2) = o\left(n^{-1}\right)$. Hence, combining (S.69), Markov's inequality, and the Cauchy–Schwarz inequality, we have

$$\hat{\bar{t}}_y(\hat{\boldsymbol{\omega}}_0) - \bar{t}_y = \left(\hat{\bar{t}}_y - \bar{t}_y\right) + o_p\left(n^{-1/2}\right) \qquad \text{(S.70)}$$

and

$$\begin{aligned}\mathrm{MSE}\left(\hat{\bar{t}}_y(\hat{\boldsymbol{\omega}}_0)\right) &= E\left(\hat{\bar{t}}_y(\hat{\boldsymbol{\omega}}_0) - \bar{t}_y\right)^2 \\ &= E\left(\hat{\bar{t}}_y - \bar{t}_y\right)^2 + o\left(n^{-1}\right) \\ &= \mathrm{MSE}(\hat{\bar{t}}_y) + o\left(n^{-1}\right). \end{aligned} \tag{S.71}$$

It follows from (S.70) that

$$\begin{aligned}&\frac{\hat{\bar{t}}_y(\hat{\boldsymbol{\omega}}_0) - \bar{t}_y}{\sqrt{\widehat{\mathrm{MSE}}\left(\hat{\bar{t}}_y(\hat{\boldsymbol{\omega}}_0)\right)}} \\ &= \frac{\hat{\bar{t}}_y - \bar{t}_y}{\sqrt{\widehat{\mathrm{MSE}}\left(\hat{\bar{t}}_y(\hat{\boldsymbol{\omega}}_0)\right)}} + o_p\left(\frac{1}{\sqrt{n\widehat{\mathrm{MSE}}\left(\hat{\bar{t}}_y(\hat{\boldsymbol{\omega}}_0)\right)}}\right) \\ &= \frac{\hat{\bar{t}}_y - \bar{t}_y}{\sqrt{\mathrm{MSE}(\hat{\bar{t}}_y)}}\sqrt{\frac{\mathrm{MSE}(\hat{\bar{t}}_y)}{\widehat{\mathrm{MSE}}\left(\hat{\bar{t}}_y(\hat{\boldsymbol{\omega}}_0)\right)}} + o_p\left(\frac{1}{\sqrt{n\mathrm{MSE}(\hat{\bar{t}}_y)}}\sqrt{\frac{\mathrm{MSE}(\hat{\bar{t}}_y)}{\widehat{\mathrm{MSE}}\left(\hat{\bar{t}}_y(\hat{\boldsymbol{\omega}}_0)\right)}}\right). \end{aligned} \tag{S.72}$$

To complete the proof of this theorem, since $n\mathrm{MSE}(\hat{\bar{t}}_y) > c_9$, it suffices to show that

$$\left|\widehat{\mathrm{MSE}}\left(\hat{\bar{t}}_y(\hat{\boldsymbol{\omega}}_0)\right) - \mathrm{MSE}(\hat{\bar{t}}_y)\right| = o_p(n^{-1}).$$

By Theorem 6 and Markov's inequality, we have

$$\left|\widehat{\mathrm{MSE}}\left(\hat{\bar{t}}_y(\hat{\boldsymbol{\omega}}_0)\right) - \mathrm{MSE}\left(\hat{\bar{t}}_y(\hat{\boldsymbol{\omega}}_0)\right)\right| = o_p(n^{-1}).$$

Together with (S.71), it follows that

$$\left|\widehat{\mathrm{MSE}}\left(\hat{\bar{t}}_y(\hat{\boldsymbol{\omega}}_0)\right) - \mathrm{MSE}(\hat{\bar{t}}_y)\right| = o_p(n^{-1}).$$

This completes the proof of Theorem 7.

## S3.12 Lemmas for the proof of Theorem 8

Note that the following theoretical results are established with respect to the sampling design randomness only.

**Lemma 13** *If Conditions 2 and 9 hold, then*

$$E_d\left(\frac{1}{N}\sum_{i\in s}\frac{z_i}{\pi_i}-\frac{1}{N}\sum_{i\in U}z_i\right)^2 \le Cn^{-1}\cdot\frac{1}{N}\sum_{i\in U}z_i^2$$

*and*

$$E_d\left(\frac{1}{N}\sum_{i\in s}\frac{z_i}{\pi_i}-\frac{1}{N}\sum_{i\in U}z_i\right)^4 \le Cn^{-2}\cdot\frac{1}{N}\sum_{i\in U}z_i^4.$$

**Proof.** By Condition 2 and the Cauchy–Schwarz inequality, we have

$$\begin{aligned}
E_d\left(\frac{1}{N}\sum_{i\in s}\frac{z_i}{\pi_i}-\frac{1}{N}\sum_{i\in U}z_i\right)^2 &= E_d\left\{\frac{1}{N}\sum_{i\in U}\frac{z_i(I_i-\pi_i)}{\pi_i}\right\}^2 \\
&= \frac{1}{N^2}\left\{\sum_{i\in U}(1-\pi_i)z_i^2+\sum_{i\neq j\in U}\frac{\pi_{ij}-\pi_i\pi_j}{\pi_i\pi_j}z_iz_j\right\} \\
&\le \frac{C}{N^2}\sum_{i\in U}z_i^2+\frac{C}{nN^2}\left(\sum_{i\in U}z_i\right)^2 \\
&\le Cn^{-1}\cdot\frac{1}{N}\sum_{i\in U}z_i^2.
\end{aligned}$$

Furthermore, Conditions 2 and 9, together with Loève's $c_r$ inequality and the Cauchy–Schwarz inequality, imply that

$$\begin{aligned}
&E_d\left(\frac{1}{N}\sum_{i\in s}\frac{z_i}{\pi_i}-\frac{1}{N}\sum_{i\in U}z_i\right)^4 \\
&\le \frac{C}{N^4}\Bigg[\sum_{i,j,k,l\in\boldsymbol{D}_4}z_iz_jz_kz_lE_d\left\{(I_i-\pi_i)(I_j-\pi_j)(I_k-\pi_k)(I_l-\pi_l)\right\} \\
&\quad+\sum_{i,j,k\in\boldsymbol{D}_3}z_i^2z_jz_kE_d\left\{(I_i-\pi_i)^2(I_j-\pi_j)(I_k-\pi_k)\right\} \\
&\quad+\sum_{i\neq j\in U}z_i^2z_j^2E_d\left\{(I_i-\pi_i)^2(I_j-\pi_j)^2+(I_i-\pi_i)^3(I_j-\pi_j)\right\} \\
&\quad+\sum_{i\in U}z_i^4E_d\left\{(I_i-\pi_i)^4\right\}\Bigg] \\
&\le Cn^{-2}\cdot\frac{1}{N}\sum_{i\in U}z_i^4.
\end{aligned}$$

This completes the proof of Lemma 13.

**Lemma 14** *Using a Taylor approximation, we have*

$$\hat{\boldsymbol{\beta}}_{(m)}=\tilde{\boldsymbol{\beta}}_{(m)}+\bar{\boldsymbol{T}}_{(m)}^{-1}\left(\hat{\bar{\boldsymbol{t}}}_{(m)}-\hat{\bar{\boldsymbol{T}}}_{(m)}\tilde{\boldsymbol{\beta}}_{(m)}\right)+\boldsymbol{r}_{(m)},$$

*where* $\boldsymbol{r}_{(m)}$ *is the remainder term of* $\hat{\boldsymbol{\beta}}_{(m)}$*. If Conditions 1, 2, 9, and 13 hold, then*

$$E_d\left(r_{j(m)}^2\right) = O(n^{-2}),$$

*where* $r_{j(m)}$ *is the* $j$*th element of* $\boldsymbol{r}_{(m)}$*.*

**Proof.** Similar to the proof of Lemma 10, by applying Taylor linearization to $\hat{\boldsymbol{\beta}}_{(m)}$ at the point $(\bar{\boldsymbol{T}}_{(m)}, \bar{\boldsymbol{t}}_{(m)})$, we have

$$\hat{\boldsymbol{\beta}}_{(m)} = \tilde{\boldsymbol{\beta}}_{(m)} + \bar{\boldsymbol{T}}_{(m)}^{-1}\left(\hat{\bar{\boldsymbol{t}}}_{(m)} - \hat{\bar{\boldsymbol{T}}}_{(m)}\tilde{\boldsymbol{\beta}}_{(m)}\right) + \boldsymbol{r}_{(m)}.$$

Thus

$$r_{j(m)} = \boldsymbol{\lambda}_{j(m)}'\boldsymbol{r}_{(m)} = \boldsymbol{\lambda}_{j(m)}'\hat{\bar{\boldsymbol{T}}}_{(m)}^{-1}\hat{\bar{\boldsymbol{t}}}_{(m)} - \boldsymbol{\lambda}_{j(m)}'\tilde{\boldsymbol{\beta}}_{(m)} - \boldsymbol{\lambda}_{j(m)}'\bar{\boldsymbol{T}}_{(m)}^{-1}\left(\hat{\bar{\boldsymbol{t}}}_{(m)} - \hat{\bar{\boldsymbol{T}}}_{(m)}\tilde{\boldsymbol{\beta}}_{(m)}\right).$$

Note that the elements of $\hat{\bar{\boldsymbol{T}}}_{(m)}^{-1}$ and $\hat{\bar{\boldsymbol{t}}}_{(m)}$ are HT estimators, so $r_{j(m)}$ is a nonlinear function of the HT estimators. From Lemma 13 and Conditions 1, 2 and 13, we can see that the fourth-order moment of each HT estimator is $O(n^{-2})$. Additionally, $r_{j(m)}^2$ and its first three derivatives are equal to zero at the point $(\bar{\boldsymbol{T}}_{(m)}, \bar{\boldsymbol{t}}_{(m)})$. Therefore, using Theorem 5.4.3 of Fuller (1996), we conclude that

$$E_d\left(r_{j(m)}^2\right) = O(n^{-2}).$$

This completes the proof of Lemma 14.

**Lemma 15** *If Conditions 1, 2, 9, and 13 hold, then*

$$E_d(b_N^2) = E_d\left[\frac{1}{N}\sum_{i\in U}\frac{I_i - \pi_i}{\pi_i}\left\{\tilde{\mu}_i(\boldsymbol{\omega}^*) - \hat{\mu}_i(\boldsymbol{\omega}^*)\right\}\right]^2 = O(n^{-2}k_M^2)$$

*and*

$$E_d\left[\frac{1}{N}\sum_{i\in U}\frac{I_i - \pi_i}{\pi_i}\left\{\tilde{\mu}_{i(m)} - \hat{\mu}_{i(m)}\right\}\right]^2 \leq Cn^{-2}k_m^2.$$

**Proof.** By Lemma 14, we have

$$\hat{\mu}_i(\boldsymbol{\omega}^*) - \tilde{\mu}_i(\boldsymbol{\omega}^*) = \sum_{m=1}^{M}\omega_m^*\boldsymbol{x}_{i(m)}'\left\{\bar{\boldsymbol{T}}_{(m)}^{-1}\left(\hat{\bar{\boldsymbol{t}}}_{(m)} - \hat{\bar{\boldsymbol{T}}}_{(m)}\tilde{\boldsymbol{\beta}}_{(m)}\right) + \boldsymbol{r}_{(m)}\right\}$$

$$
\begin{aligned}
&= \frac{1}{N}\sum_{j\in U}\frac{I_j-\pi_j}{\pi_j}\sum_{m=1}^{M}\left\{\omega_m^*\boldsymbol{x}'_{i(m)}\bar{\boldsymbol{T}}_{(m)}^{-1}\boldsymbol{x}_{j(m)}\left(y_j-\boldsymbol{x}'_{j(m)}\tilde{\boldsymbol{\beta}}_{(m)}\right)\right\}\\
&\quad+\sum_{m=1}^{M}\left(\omega_m^*\boldsymbol{x}'_{i(m)}\boldsymbol{r}_{(m)}\right)\\
&\triangleq \frac{1}{N}\sum_{j\in U}\frac{I_j-\pi_j}{\pi_j}z_{ij}+r_i(\boldsymbol{\omega}^*).
\end{aligned}
$$

Thus

$$
\begin{aligned}
b_N^2 &= \frac{1}{N^4}\sum_{i,j,k,l\in U}\frac{(\pi_i-I_i)(\pi_j-I_j)(\pi_k-I_k)(\pi_l-I_l)z_{ik}z_{jl}}{\pi_i\pi_j\pi_k\pi_l}\\
&\quad+\frac{2}{N^3}\sum_{i,j,k\in U}\frac{(\pi_i-I_i)(\pi_j-I_j)(\pi_k-I_k)z_{ik}r_j(\boldsymbol{\omega}^*)}{\pi_i\pi_j\pi_k}\\
&\quad+\frac{1}{N^2}\sum_{i,j\in U}\frac{(\pi_i-I_i)(\pi_j-I_j)r_i(\boldsymbol{\omega}^*)r_j(\boldsymbol{\omega}^*)}{\pi_i\pi_j}\\
&\triangleq b_{1N}+b_{2N}+b_{3N}.
\end{aligned}
$$

We first consider $|E_d(b_{1N})|$. Conditions 1 and 13 imply that

$$
\begin{aligned}
|z_{ij}| &= \left|\sum_{m=1}^{M}\left\{\omega_m^*\boldsymbol{x}'_{i(m)}\bar{\boldsymbol{T}}_{(m)}^{-1}\boldsymbol{x}_{j(m)}\left(y_j-\boldsymbol{x}'_{j(m)}\tilde{\boldsymbol{\beta}}_{(m)}\right)\right\}\right|\\
&\le \sum_{m=1}^{M}\left|\omega_m^*\boldsymbol{x}'_{i(m)}\bar{\boldsymbol{T}}_{(m)}^{-1}\boldsymbol{x}_{j(m)}\left(y_j-\tilde{\mu}_{j(m)}\right)\right|\\
&\le \sum_{m=1}^{M}\omega_m^*\|\boldsymbol{x}'_{i(m)}\|\cdot\|\bar{\boldsymbol{T}}_{(m)}^{-1}\|\cdot\|x_{j(m)}\|\cdot\left|y_j-\tilde{\mu}_{j(m)}\right|\\
&= O(k_M)
\end{aligned}
$$

uniformly in $i$ and $j$. Then, by Conditions 2 and 9, we have

$$
\begin{aligned}
|E_d(b_{1N})| &= \left|\frac{1}{N^4}\sum_{i,j,k,l\in U}z_{ik}z_{jl}E_d\left\{\frac{(\pi_i-I_i)(\pi_j-I_j)(\pi_k-I_k)(\pi_l-I_l)}{\pi_i\pi_j\pi_k\pi_l}\right\}\right|\\
&\le \frac{Ck_M^2}{N^4}\sum_{i,j,k,l\in U}\left|E_d\left\{(\pi_i-I_i)(\pi_j-I_j)(\pi_k-I_k)(\pi_l-I_l)\right\}\right|\\
&\le \frac{Ck_M^2}{N^4}\left[\sum_{i,j,k,l\in\boldsymbol{D}_4}\left|E_d\left\{(I_i-\pi_i)(I_j-\pi_j)(I_k-\pi_k)(I_l-\pi_l)\right\}\right|\right.\\
&\quad\left.+\sum_{i,j,k\in\boldsymbol{D}_3}\left|E_d\left\{(I_i-\pi_i)^2(I_j-\pi_j)(I_k-\pi_k)\right\}\right|\right.
\end{aligned}
$$

$$
\begin{aligned}
&+ \sum_{i \neq j \in U} \left| E_d \left\{ (I_i - \pi_i)^2 (I_j - \pi_j)^2 + (I_i - \pi_i)(I_j - \pi_j)^3 \right\} \right| \\
&+ \sum_{i \in U} \left| E_d \left\{ (I_i - \pi_i)^4 \right\} \right| \Bigg] \\
&= O\left( n^{-2} k_M^2 \right).
\end{aligned}
$$

Next, we consider $|E_d(b_{3N})|$. Condition 1 and Lemma 14 imply that

$$
\begin{aligned}
\sum_{i \in U} E_d \left\{ r_i^2(\boldsymbol{\omega}^*) \right\} &= E_d \left( \sum_{m=1}^{M} \sum_{l=1}^{M} \omega_m^* \omega_l^* \boldsymbol{r}'_{(m)} \boldsymbol{X}'_{(m)U} \boldsymbol{X}_{(l)U} \boldsymbol{r}_{(l)} \right) \\
&\leq C \sum_{m=1}^{M} \sum_{l=1}^{M} \omega_m^* \omega_l^* \left( E_d \left\| \boldsymbol{r}_{(m)} \right\|^2 E_d \left\| \boldsymbol{r}_{(l)} \right\|^2 \right)^{1/2} \\
&= O(N n^{-2} k_M^2). \qquad \text{(S.73)}
\end{aligned}
$$

Then, from Condition 2 and Loève's $c_r$ inequality, we have

$$
\begin{aligned}
|E_d(b_{3N})| &= \left| \frac{1}{N^2} \sum_{i,j \in U} E_d \left\{ \frac{(\pi_i - I_i)(\pi_j - I_j) r_i(\boldsymbol{\omega}^*) r_j(\boldsymbol{\omega}^*)}{\pi_i \pi_j} \right\} \right| \\
&\leq E_d \left\{ \frac{1}{N} \sum_{i \in U} \frac{r_i(\boldsymbol{\omega}^*)(\pi_i - I_i)}{\pi_i} \right\}^2 \\
&\leq \frac{1}{N} \sum_{i \in U} E_d \left\{ r_i^2(\boldsymbol{\omega}^*) \right\} \\
&= O\left( n^{-2} k_M^2 \right).
\end{aligned}
$$

It remains to show that $|E_d(b_{2N})| = O(n^{-2} k_M^2)$. By the Cauchy–Schwarz inequality,

$$
\begin{aligned}
|E_d(b_{2N})| &= \left| E_d \left\{ \frac{2}{N^3} \sum_{i,j,k \in U} \frac{(\pi_i - I_i)(\pi_j - I_j)(\pi_k - I_k) z_{ik} r_j(\boldsymbol{\omega}^*)}{\pi_i \pi_j \pi_k} \right\} \right| \\
&= \frac{2}{N^3} \left| E_d \left\{ \sum_{i,k \in U} \frac{(\pi_i - I_i)(\pi_k - I_k) z_{ik}}{\pi_i \pi_k} \cdot \sum_{j \in U} \frac{(\pi_j - I_j) r_j(\boldsymbol{\omega}^*)}{\pi_j} \right\} \right| \\
&= \frac{2}{N^3} \left| E_d \left\{ \sum_{i,k \in U} \frac{(\pi_i - I_i)(\pi_k - I_k) z_{ik}}{\pi_i \pi_k} \right\}^2 \cdot E_d \left\{ \sum_{j \in U} \frac{(\pi_j - I_j) r_j(\boldsymbol{\omega}^*)}{\pi_j} \right\}^2 \right|^{1/2} \\
&= \frac{2}{N^3} \cdot \sqrt{N^4 |E_d(b_{1N})| \cdot N^2 |E_d(b_{3N})|} \\
&= O\left( n^{-2} k_M^2 \right).
\end{aligned}
$$

Therefore, $E_d(b_N^2) \leq |E_d(b_{1N})| + |E_d(b_{2N})| + |E_d(b_{3N})| = O\left(n^{-2}k_M^2\right)$.

Note that $\tilde{\mu}_{i(m)} = \tilde{\mu}_i(\boldsymbol{\omega}_m^0)$, where $\boldsymbol{\omega}_m^0$ is an $M \times 1$ vector with one in the $m$th position and zeros elsewhere. Similar to the proof of $E_d(b_N^2) = O\left(n^{-2}k_M^2\right)$, we have

$$E_d\left\{\frac{1}{N}\sum_{i\in U}\frac{I_i - \pi_i}{\pi_i}\left(\tilde{\mu}_{i(m)} - \hat{\mu}_{i(m)}\right)\right\}^2 \leq Cn^{-2}k_m^2.$$

This completes the proof of Lemma 15.

## S3.13 Proof of Theorem 8

The proof of this theorem relies on Lemmas 13–15.

**Proof.** By Equation (S.69), we have

$$\hat{\bar{t}}_y(\hat{\boldsymbol{\omega}}_0) - \bar{t}_y = a_N + b_N + c_N = \left(\hat{\bar{t}}_y - \bar{t}_y\right) + a_{2N} + b_N + c_N. \tag{S.74}$$

For $a_N$, Conditions 2 and 13 imply that

$$\begin{aligned} E_d(a_N^2) &= E_d\left[\frac{1}{N}\sum_{i\in U}\frac{(I_i - \pi_i)\left\{y_i - \tilde{\mu}_i(\boldsymbol{\omega}^*)\right\}}{\pi_i}\right]^2 \\ &= \frac{1}{N^2}\sum_{i,j\in U}\frac{\pi_{ij} - \pi_i\pi_j}{\pi_i\pi_j}\left\{y_i - \tilde{\mu}_i(\boldsymbol{\omega}^*)\right\}\left\{y_j - \tilde{\mu}_j(\boldsymbol{\omega}^*)\right\} \\ &= O\left(n^{-1}\right). \end{aligned} \tag{S.75}$$

For $a_{2N}$, it follows from Conditions 2 and 15 that

$$\begin{aligned} E_d(a_{2N}^2) &= E_d\left[\frac{1}{N}\sum_{i\in s}\frac{\mu_i - \tilde{\mu}_i(\boldsymbol{\omega}^*)}{\pi_i} - \frac{1}{N}\sum_{i\in U}\left\{\mu_i - \tilde{\mu}_i(\boldsymbol{\omega}^*)\right\}\right]^2 \\ &= \left|\frac{1}{N^2}\sum_{i,j\in U}\frac{\pi_{ij} - \pi_i\pi_j}{\pi_i\pi_j}\left\{\mu_i - \tilde{\mu}_i(\boldsymbol{\omega}^*)\right\}\left\{\mu_j - \tilde{\mu}_j(\boldsymbol{\omega}^*)\right\}\right| \\ &\leq \frac{C}{N^2}\sum_{i\in U}\left\{\mu_i - \tilde{\mu}_i(\boldsymbol{\omega}^*)\right\}^2 + \frac{C}{nN^2}\left|\sum_{i\neq j\in U}\left\{\mu_i - \tilde{\mu}_i(\boldsymbol{\omega}^*)\right\}\left\{\mu_j - \tilde{\mu}_j(\boldsymbol{\omega}^*)\right\}\right| \\ &\leq \frac{C}{N^2}\sum_{i\in U}\left\{\mu_i - \tilde{\mu}_i(\boldsymbol{\omega}^*)\right\}^2 + \frac{C}{nN}\sum_{i\in U}\left\{\mu_i - \tilde{\mu}_i(\boldsymbol{\omega}^*)\right\}^2 \end{aligned}$$

$$= O\left(\frac{\tilde{L}_N(\boldsymbol{\omega}^*)}{nN}\right)$$
$$= o\left(n^{-1}\right). \tag{S.76}$$

Next, we consider $c_N$. Using Lemmas 13 and 15, Conditions 13 and 14, Loève's $c_r$ inequality, and the Cauchy–Schwarz inequality, we have

$$\begin{aligned}
E_d(c_N^2) &= E_d\left\{\frac{1}{N}\sum_{m=1}^{M}(\hat{\omega}_{0m}-\omega_m^*)\left(\sum_{i\in U}\frac{I_i-\pi_i}{\pi_i}\hat{\mu}_{i(m)}\right)\right\}^2 \\
&= E_d\left\{\sum_{m=1}^{M}(\hat{\omega}_{0m}-\omega_m^*)\cdot\frac{1}{N}\sum_{i\in U}\frac{I_i-\pi_i}{\pi_i}\left(\hat{\mu}_{i(m)}-\tilde{\mu}_{i(m)}\right)\right. \\
&\quad \left.+\sum_{m=1}^{M}(\hat{\omega}_{0m}-\omega_m^*)\cdot\frac{1}{N}\sum_{i\in U}\frac{I_i-\pi_i}{\pi_i}\tilde{\mu}_{i(m)}\right\}^2 \\
&\le 2M\sum_{m=1}^{M}\left[E_d\left\{\frac{1}{N}\sum_{i\in U}\frac{I_i-\pi_i}{\pi_i}\left(\hat{\mu}_{i(m)}-\tilde{\mu}_{i(m)}\right)\right\}^2\right] \\
&\quad +2E_d\left\{\sum_{m=1}^{M}(\hat{\omega}_{0m}-\omega_m^*)^2\cdot\sum_{m=1}^{M}\left(\frac{1}{N}\sum_{i\in U}\frac{I_i-\pi_i}{\pi_i}\tilde{\mu}_{i(m)}\right)^2\right\} \\
&\le 2M\sum_{m=1}^{M}\left[E_d\left\{\frac{1}{N}\sum_{i\in U}\frac{I_i-\pi_i}{\pi_i}\left(\hat{\mu}_{i(m)}-\tilde{\mu}_{i(m)}\right)\right\}^2\right] \\
&\quad +C\left\{M\sum_{m=1}^{M}E_d\left(\frac{1}{N}\sum_{i\in U}\frac{I_i-\pi_i}{\pi_i}\tilde{\mu}_{i(m)}\right)^4\cdot E_d\|\hat{\boldsymbol{\omega}}_0-\boldsymbol{\omega}^*\|^4\right\}^{1/2} \\
&\le Cn^{-2}M^2k_M^2+Cn^{-1}M\tilde{\zeta}_n^{1/2} \\
&= o\left(n^{-1}\right).
\end{aligned} \tag{S.77}$$

Condition 14 and Lemma 15 imply that $E_d(b_N^2) = o\left(n^{-1}\right)$. Combining this result with (S.74), (S.76), and (S.77), we obtain that

$$\hat{\bar{t}}_y(\hat{\boldsymbol{\omega}}_0)-\bar{t}_y = \left(\hat{\bar{t}}_y-\bar{t}_y\right)+o_{\tilde{p}}\left(n^{-1/2}\right)$$

and

$$\begin{aligned}
\mathrm{MSE}_d\left(\hat{\bar{t}}_y(\hat{\boldsymbol{\omega}}_0)\right) &= E_d\left\{\hat{\bar{t}}_y(\hat{\boldsymbol{\omega}}_0)-\bar{t}_y\right\}^2 \\
&= E_d\left(\hat{\bar{t}}_y-\bar{t}_y\right)^2+o\left(n^{-1}\right)
\end{aligned}$$

$$= \mathrm{MSE}_d(\hat{\bar{t}}_y) + o\left(n^{-1}\right). \tag{S.78}$$

Theorem 2 of Berger (1996), together with the conditions $D(P\|R) \to 0$, $e_R \to 0$, and $d_R \to \infty$, implies that

$$\frac{\sqrt{n}(\hat{\bar{t}}_y - \bar{t}_y)}{\sqrt{\mathrm{MSE}_d(\hat{\bar{t}}_y)}} \xrightarrow{\tilde{d}} N(0,1).$$

Therefore, to complete the proof of this theorem, from the condition $n\mathrm{MSE}_d(\hat{\bar{t}}_y) > c_{10}$ and Equations (S.72), (S.75) and (S.78), it suffices to show that

$$\left|\widehat{\mathrm{MSE}}\left(\hat{\bar{t}}_y(\hat{\boldsymbol{\omega}}_0)\right) - \mathrm{MSE}_d\left(\hat{\bar{t}}_y(\hat{\boldsymbol{\omega}}_0)\right)\right| = o_{\tilde{p}}\left(n^{-1}\right),$$

where

$$\widehat{\mathrm{MSE}}\left(\hat{\bar{t}}_y(\hat{\boldsymbol{\omega}}_0)\right) = \frac{1}{N^2}\sum_{i,j\in s}\frac{\pi_{ij}-\pi_i\pi_j}{\pi_{ij}\pi_i\pi_j}\left\{y_i - \hat{\mu}_i(\hat{\boldsymbol{\omega}}_0)\right\}\left\{y_j - \hat{\mu}_j(\hat{\boldsymbol{\omega}}_0)\right\}$$

and

$$\mathrm{MSE}_d\left(\hat{\bar{t}}_y(\hat{\boldsymbol{\omega}}_0)\right) = \frac{1}{N^2}\sum_{i,j\in U}\frac{\pi_{ij}-\pi_i\pi_j}{\pi_i\pi_j}\left\{y_i - \tilde{\mu}_i(\boldsymbol{\omega}^*)\right\}\left\{y_j - \tilde{\mu}_j(\boldsymbol{\omega}^*)\right\} + o\left(n^{-1}\right).$$

Similar to the proof of Theorem 6, we need to verify that $|y_i - \tilde{\mu}_i(\boldsymbol{\omega}^*)| = O(1)$ and

$$\frac{1}{N}\sum_{i\in U} E_d\left\{\hat{\mu}_i(\hat{\boldsymbol{\omega}}_0) - \tilde{\mu}_i(\boldsymbol{\omega}^*)\right\}^2 = o(1).$$

The first equation follows directly from Condition 13. Moreover, by Lemmas 13–14 and Conditions 1, 2 and 13, we have

$$\begin{aligned}
E_d\left\|\hat{\boldsymbol{\beta}}_{(m)} - \tilde{\boldsymbol{\beta}}_{(m)}\right\|^2 &= E_d\left\|\bar{\boldsymbol{T}}_{(m)}^{-1}\left(\hat{\bar{\boldsymbol{t}}}_{(m)} - \hat{\bar{\boldsymbol{T}}}_{(m)}\tilde{\boldsymbol{\beta}}_{(m)}\right) + \boldsymbol{r}_{(m)}\right\|^2 \\
&\le \frac{2}{N}\sum_{i,j\in U}\frac{|E_d(I_i-\pi_i)(I_j-\pi_j)|}{\pi_i\pi_j}\left|\left\{\bar{\boldsymbol{T}}_{(m)}^{-1}\boldsymbol{x}_{i(m)}\left(y_i - \boldsymbol{x}_{i(m)}'\tilde{\boldsymbol{\beta}}_{(m)}\right)\right\}'\right. \\
&\quad \left.\left\{\bar{\boldsymbol{T}}_{(m)}^{-1}\boldsymbol{x}_{j(m)}\left(y_j - \boldsymbol{x}_{j(m)}'\tilde{\boldsymbol{\beta}}_{(m)}\right)\right\}\right| + 2E_d\left\|\boldsymbol{r}_{(m)}\right\|^2 \\
&= O(n^{-1}k_m).
\end{aligned}$$

Then, Conditions 2, 13 and 14 imply that

$$\frac{1}{N}\sum_{i\in U} E_d\left\{\hat{\mu}_i(\hat{\boldsymbol{\omega}}_0) - \tilde{\mu}_i(\boldsymbol{\omega}^*)\right\}^2$$

$$
\begin{aligned}
&= \frac{1}{N}\sum_{i\in U} E_d\left\{\hat{\mu}_i(\hat{\boldsymbol{\omega}}_0) - \tilde{\mu}_i(\hat{\boldsymbol{\omega}}_0) + \tilde{\mu}_i(\hat{\boldsymbol{\omega}}_0) - \tilde{\mu}_i(\boldsymbol{\omega}^*)\right\}^2 \\
&\le \frac{2}{N}\sum_{i\in U} E_d\left\{\sum_{m=1}^{M}\hat{\omega}_{0m}^2\sum_{m=1}^{M}\left(\boldsymbol{x}'_{i(m)}\hat{\boldsymbol{\beta}}_{(m)} - \boldsymbol{x}'_{i(m)}\tilde{\boldsymbol{\beta}}_{(m)}\right)^2\right\} \\
&\quad + \frac{2}{N}\sum_{i\in U} E_d\left\{\sum_{m=1}^{M}(\hat{\omega}_{0m} - \omega_m^*)^2\sum_{m=1}^{M}\left(\boldsymbol{x}'_{i(m)}\tilde{\boldsymbol{\beta}}_{(m)}\right)^2\right\} \\
&\le \frac{2}{N}E_d\left\{\sum_{m=1}^{M}\left(\hat{\boldsymbol{\beta}}_{(m)} - \tilde{\boldsymbol{\beta}}_{(m)}\right)'\boldsymbol{X}'_{(m)U}\boldsymbol{X}_{(m)U}\left(\hat{\boldsymbol{\beta}}_{(m)} - \tilde{\boldsymbol{\beta}}_{(m)}\right)\right\} \\
&\quad + \frac{2}{N}E_d\left\{\|\hat{\boldsymbol{\omega}}_0 - \boldsymbol{\omega}^*\|^2\cdot\sum_{m=1}^{M}\boldsymbol{y}'_U\boldsymbol{X}_{(m)U}\left(\boldsymbol{X}'_{(m)U}\boldsymbol{X}_{(m)U}\right)^{-1}\boldsymbol{X}'_{(m)U}\boldsymbol{y}_U\right\} \\
&\le Cn^{-1}Mk_M + CN^{-1}M\|\boldsymbol{y}_U\|^2\left(E_d\|\hat{\boldsymbol{\omega}}_0 - \boldsymbol{\omega}^*\|^4\right)^{1/2} \\
&= O\left(n^{-1}Mk_M + \tilde{\zeta}_n^{1/2}M\right) \\
&= o(1).
\end{aligned}
$$

This completes the proof of Theorem 8.

# S4 Relationship with High-Dimensional Regression

In this section, we explain how the number and dimensions of the candidate models relate to the number of covariates and discuss the relationship between the proposed framework and high-dimensional regression in survey sampling.

The number of candidate models $M$ and their dimensions $\{k_m\}$ depend on how the candidate models are constructed, so their relationship with the number of covariates $p$ is not fixed. For example, under the nested candidate models, where the $m$th model contains the first $m$ covariates (treating the intercept as a covariate), we have $p = M = k_M$. In this case, Conditions 3, 4, 6, and 11, which restrict the number and dimensions of candidate models, can be expressed in terms of the number of covariates by replacing both $M$ and $k_M$ with $p$. More generally, if the $m$th candidate model contains the first $k_m$ covariates, then $p = k_M$, which may be much larger than $M$. If the sets of covariates included in

the candidate models are mutually disjoint as in Ando & Li (2014), then $p = \sum_{m=1}^{M} k_m$. When the candidate models correspond to all nonempty subsets of the $p$ covariates, we have $p = \log_2(M+1)$. Consequently, we express our conditions in terms of the number and dimensions of candidate models in this paper.

Although our framework allows both $M$ and $k_M$ to diverge (so that $p$ also diverges), their growth rates are constrained by Conditions 3, 4, 6, and 11. Detailed discussions of these conditions are provided in the main manuscript. Since the theoretical properties studied in this paper are different from those in the survey sampling literature on high-dimensional regression (Ta et al. 2020, Chauvet & Goga 2022), the divergence rate of $p$ is also different. For instance, in the discussion of Condition 3, we require $p = O(n^v)$ with $0 < v < 3/28$ to establish the asymptotic optimality of the proposed model averaging estimators. By comparison, to establish asymptotic equivalence between estimators, Ta et al. (2020) require $p/n \to 0$, whereas Chauvet & Goga (2022) require $p = O(n^v)$ with $0 < v < 1/2$.

Moreover, when $p > n$, Ta et al. (2020) showed that, under some sparsity and regularity conditions, the generalized regression (GREG) estimator based on the $\ell_1$-penalized weighted least squares estimator is asymptotically equivalent to the generalized difference estimator. In contrast, the proposed model averaging method may not be directly applicable without additional dimensionality-reduction techniques. A possible solution is to incorporate a preliminary feature screening or grouping step, as is commonly adopted in high-dimensional model averaging (Ando & Li 2014, Xie et al. 2021), before constructing the candidate models. This discussion has been added to the Conclusion section in the main manuscript.

# S5 Additional Numerical Analyses

## S5.1 Simulation results for $n = 50, 150, 200$

Table S1: Values of $\text{MSE}_\mu$ for various methods in the simulation ($n = 50$)

| | $R^2$ | SCAD | ALASSO | AGG1 | AGG2 | AIC | SAIC | BIC | SBIC | Cp | SMA1 | SMA2 |
|---|---|---|---|---|---|---|---|---|---|---|---|---|
| $\rho = 0.1$ $\alpha = 0.5$ | 0.1 | 0.1708 | 0.2046 | 0.1115 | 0.1814 | 0.1561 | 0.1207 | 0.1057 | **0.0864** | 0.1457 | *0.1039* | 0.1035 |
| | 0.2 | 0.2540 | 0.2747 | 0.1627 | 0.3751 | 0.2176 | 0.1699 | 0.1815 | **0.1455** | 0.2038 | *0.1519* | 0.1516 |
| | 0.3 | 0.3188 | 0.3195 | 0.2113 | 0.5318 | 0.2687 | 0.2181 | 0.2529 | *0.2059* | 0.2614 | **0.1980** | 0.1978 |
| | 0.4 | 0.3859 | 0.3769 | *0.2643* | 0.6026 | 0.3266 | 0.2740 | 0.3342 | 0.2737 | 0.3216 | **0.2502** | 0.2501 |
| | 0.5 | 0.4257 | 0.4193 | 0.3303 | 0.6459 | 0.3753 | *0.3168* | 0.4021 | 0.3406 | 0.3721 | **0.3004** | 0.3005 |
| | 0.6 | 0.4933 | 0.4947 | 0.4164 | 0.6430 | 0.4563 | *0.4013* | 0.5001 | 0.4273 | 0.4588 | **0.3786** | 0.3787 |
| | 0.7 | 0.5798 | 0.5944 | 0.5521 | 0.7339 | 0.5719 | *0.5143* | 0.6501 | 0.5617 | 0.5755 | **0.4909** | 0.4911 |
| | 0.8 | 0.7641 | 0.7969 | 0.7963 | 0.9523 | 0.7685 | *0.7170* | 0.8907 | 0.7888 | 0.7817 | **0.6989** | 0.6991 |
| | 0.9 | 1.3004 | 1.3489 | 1.4902 | 1.6650 | 1.3354 | *1.2906* | 1.5287 | 1.4116 | 1.3555 | **1.2823** | 1.2829 |
| $\rho = 0.1$ $\alpha = 1.0$ | 0.1 | 0.1479 | 0.1966 | 0.0960 | 0.2570 | 0.1332 | 0.1020 | *0.0763* | **0.0600** | 0.1247 | 0.0862 | 0.0857 |
| | 0.2 | 0.1901 | 0.2171 | 0.1213 | 0.4521 | 0.1553 | 0.1185 | 0.1184 | **0.0895** | 0.1477 | *0.1053* | 0.1048 |
| | 0.3 | 0.2335 | 0.2404 | 0.1438 | 0.4625 | 0.1750 | 0.1376 | 0.1509 | **0.1158** | 0.1654 | *0.1241* | 0.1240 |
| | 0.4 | 0.2641 | 0.2664 | 0.1650 | 0.4133 | 0.2029 | 0.1585 | 0.1789 | **0.1390** | 0.1939 | *0.1445* | 0.1443 |
| | 0.5 | 0.2834 | 0.2762 | 0.1875 | 0.3619 | 0.2214 | 0.1780 | 0.2018 | *0.1636* | 0.2143 | **0.1628** | 0.1627 |
| | 0.6 | 0.2966 | 0.2911 | 0.2207 | 0.3674 | 0.2393 | 0.1909 | 0.2312 | *0.1873* | 0.2294 | **0.1807** | 0.1805 |
| | 0.7 | 0.3208 | 0.3192 | 0.2600 | 0.3598 | 0.2713 | *0.2254* | 0.2746 | 0.2279 | 0.2664 | **0.2137** | 0.2136 |
| | 0.8 | 0.3419 | 0.3448 | 0.3123 | 0.3907 | 0.3036 | *0.2565* | 0.3298 | 0.2733 | 0.3027 | **0.2474** | 0.2474 |
| | 0.9 | 0.3966 | 0.4123 | 0.4314 | 0.4973 | 0.3859 | *0.3433* | 0.4425 | 0.3799 | 0.3913 | **0.3369** | 0.3370 |
| $\rho = 0.6$ $\alpha = 0.5$ | 0.1 | 0.1793 | 0.2175 | 0.1125 | 0.1798 | 0.1658 | 0.1292 | *0.1081* | **0.0895** | 0.1573 | 0.1116 | 0.1109 |
| | 0.2 | 0.2578 | 0.2790 | 0.1627 | 0.3670 | 0.2220 | 0.1762 | 0.1835 | **0.1482** | 0.2150 | *0.1567* | 0.1561 |
| | 0.3 | 0.3184 | 0.3216 | 0.2081 | 0.5028 | 0.2681 | 0.2176 | 0.2519 | *0.2040* | 0.2619 | **0.1973** | 0.1971 |
| | 0.4 | 0.3801 | 0.3760 | *0.2590* | 0.5578 | 0.3257 | 0.2698 | 0.3255 | 0.2686 | 0.3181 | **0.2473** | 0.2471 |
| | 0.5 | 0.4328 | 0.4267 | 0.3292 | 0.5707 | 0.3811 | *0.3234* | 0.4116 | 0.3434 | 0.3777 | **0.3029** | 0.3028 |
| | 0.6 | 0.4916 | 0.4943 | 0.4168 | 0.6250 | 0.4622 | *0.4058* | 0.5128 | 0.4336 | 0.4616 | **0.3834** | 0.3834 |
| | 0.7 | 0.5769 | 0.5952 | 0.5445 | 0.6914 | 0.5680 | *0.5134* | 0.6468 | 0.5616 | 0.5732 | **0.4945** | 0.4946 |
| | 0.8 | 0.7662 | 0.7994 | 0.7914 | 0.9429 | 0.7823 | *0.7261* | 0.8958 | 0.7997 | 0.7910 | **0.7098** | 0.7103 |
| | 0.9 | 1.2979 | 1.3591 | 1.4641 | 1.6266 | 1.3338 | *1.2908* | 1.5332 | 1.4107 | 1.3648 | **1.2818** | 1.2821 |
| $\rho = 0.6$ $\alpha = 1.0$ | 0.1 | 0.1412 | 0.1877 | 0.0931 | 0.2521 | 0.1288 | 0.0983 | *0.0754* | **0.0596** | 0.1205 | 0.0844 | 0.0838 |
| | 0.2 | 0.1944 | 0.2237 | 0.1187 | 0.4139 | 0.1580 | 0.1211 | 0.1201 | **0.0909** | 0.1481 | *0.1087* | 0.1080 |
| | 0.3 | 0.2288 | 0.2410 | 0.1416 | 0.4110 | 0.1756 | 0.1385 | 0.1481 | **0.1156** | 0.1682 | *0.1257* | 0.1252 |
| | 0.4 | 0.2655 | 0.2632 | 0.1647 | 0.3779 | 0.2026 | 0.1592 | 0.1832 | **0.1429** | 0.1923 | *0.1456* | 0.1452 |
| | 0.5 | 0.2822 | 0.2754 | 0.1843 | 0.3325 | 0.2133 | 0.1742 | 0.2019 | *0.1636* | 0.2072 | **0.1619** | 0.1617 |
| | 0.6 | 0.3019 | 0.2991 | 0.2134 | 0.3324 | 0.2446 | 0.1974 | 0.2337 | *0.1894* | 0.2374 | **0.1856** | 0.1853 |
| | 0.7 | 0.3164 | 0.3161 | 0.2553 | 0.3408 | 0.2696 | *0.2213* | 0.2725 | 0.2233 | 0.2625 | **0.2121** | 0.2121 |
| | 0.8 | 0.3542 | 0.3522 | 0.2987 | 0.3674 | 0.3109 | *0.2635* | 0.3348 | 0.2766 | 0.3080 | **0.2534** | 0.2535 |
| | 0.9 | 0.4099 | 0.4175 | 0.4332 | 0.4885 | 0.3919 | *0.3471* | 0.4508 | 0.3862 | 0.3963 | **0.3420** | 0.3422 |

Table S2: Values of $\text{MSE}_\mu$ for various methods in the simulation ($n = 150$)

| | $R^2$ | SCAD | ALASSO | AGG1 | AGG2 | AIC | SAIC | BIC | SBIC | Cp | SMA1 | SMA2 |
|---|---|---|---|---|---|---|---|---|---|---|---|---|
| $\rho = 0.1$ $\alpha = 0.5$ | 0.1 | 0.1002 | 0.1064 | *0.0587* | 0.1826 | 0.0743 | 0.0592 | 0.0705 | 0.0592 | 0.0732 | **0.0544** | 0.0543 |
| | 0.2 | 0.1378 | 0.1375 | *0.0886* | 0.3591 | 0.1078 | 0.0889 | 0.1129 | 0.0980 | 0.1066 | **0.0818** | 0.0817 |
| | 0.3 | 0.1537 | 0.1534 | 0.1136 | 0.3611 | 0.1314 | *0.1118* | 0.1515 | 0.1320 | 0.1304 | **0.1046** | 0.1046 |
| | 0.4 | 0.1706 | 0.1745 | 0.1425 | 0.3178 | 0.1578 | *0.1373* | 0.1877 | 0.1664 | 0.1576 | **0.1285** | 0.1286 |
| | 0.5 | 0.1956 | 0.2032 | 0.1810 | 0.3092 | 0.1914 | *0.1714* | 0.2331 | 0.2097 | 0.1914 | **0.1616** | 0.1617 |
| | 0.6 | 0.2246 | 0.2368 | 0.2279 | 0.3182 | 0.2313 | *0.2118* | 0.2917 | 0.2638 | 0.2320 | **0.2037** | 0.2038 |
| | 0.7 | 0.2779 | 0.2917 | 0.2975 | 0.3660 | 0.2875 | *0.2722* | 0.3688 | 0.3359 | 0.2908 | **0.2656** | 0.2658 |
| | 0.8 | *0.3857* | 0.4014 | 0.4309 | 0.4848 | 0.4003 | 0.3890 | 0.4965 | 0.4590 | 0.4027 | **0.3835** | 0.3837 |
| | 0.9 | **0.7028** | 0.7157 | 0.7917 | 0.8374 | 0.7180 | 0.7141 | 0.8289 | 0.7880 | 0.7204 | *0.7141* | 0.7144 |
| $\rho = 0.1$ $\alpha = 1.0$ | 0.1 | 0.0780 | 0.0897 | 0.0438 | 0.2568 | 0.0513 | 0.0408 | 0.0454 | **0.0358** | 0.0505 | *0.0382* | 0.0380 |
| | 0.2 | 0.0963 | 0.0985 | 0.0537 | 0.2992 | 0.0605 | *0.0499* | 0.0614 | 0.0502 | 0.0597 | **0.0469** | 0.0468 |
| | 0.3 | 0.1079 | 0.1078 | 0.0656 | 0.2204 | 0.0723 | *0.0591* | 0.0729 | 0.0605 | 0.0709 | **0.0557** | 0.0556 |
| | 0.4 | 0.1132 | 0.1129 | 0.0737 | 0.1834 | 0.0798 | *0.0663* | 0.0854 | 0.0717 | 0.0782 | **0.0629** | 0.0628 |
| | 0.5 | 0.1178 | 0.1180 | 0.0850 | 0.1575 | 0.0877 | *0.0732* | 0.0976 | 0.0833 | 0.0866 | **0.0705** | 0.0704 |
| | 0.6 | 0.1246 | 0.1251 | 0.0967 | 0.1537 | 0.0991 | *0.0832* | 0.1101 | 0.0942 | 0.0980 | **0.0786** | 0.0786 |
| | 0.7 | 0.1340 | 0.1352 | 0.1124 | 0.1529 | 0.1130 | *0.0964* | 0.1294 | 0.1120 | 0.1122 | **0.0916** | 0.0916 |
| | 0.8 | 0.1459 | 0.1504 | 0.1349 | 0.1620 | 0.1323 | *0.1155* | 0.1581 | 0.1387 | 0.1323 | **0.1104** | 0.1105 |
| | 0.9 | 0.1602 | 0.1687 | 0.1762 | 0.1988 | 0.1620 | *0.1464* | 0.2054 | 0.1818 | 0.1632 | **0.1427** | 0.1428 |
| $\rho = 0.6$ $\alpha = 0.5$ | 0.1 | 0.1036 | 0.1113 | *0.0578* | 0.1814 | 0.0774 | 0.0616 | 0.0711 | 0.0595 | 0.0756 | **0.0562** | 0.0560 |
| | 0.2 | 0.1387 | 0.1371 | *0.0850* | 0.3304 | 0.1066 | 0.0892 | 0.1149 | 0.0988 | 0.1055 | **0.0815** | 0.0814 |
| | 0.3 | 0.1560 | 0.1557 | *0.1113* | 0.3177 | 0.1323 | 0.1127 | 0.1491 | 0.1307 | 0.1310 | **0.1049** | 0.1049 |
| | 0.4 | 0.1759 | 0.1799 | 0.1424 | 0.2822 | 0.1625 | *0.1417* | 0.1899 | 0.1686 | 0.1618 | **0.1316** | 0.1316 |
| | 0.5 | 0.1930 | 0.2018 | 0.1772 | 0.2781 | 0.1891 | *0.1695* | 0.2316 | 0.2067 | 0.1894 | **0.1607** | 0.1608 |
| | 0.6 | 0.2300 | 0.2411 | 0.2258 | 0.2987 | 0.2343 | *0.2158* | 0.2886 | 0.2615 | 0.2357 | **0.2058** | 0.2059 |
| | 0.7 | 0.2782 | 0.2946 | 0.2968 | 0.3524 | 0.2896 | *0.2750* | 0.3743 | 0.3418 | 0.2921 | **0.2680** | 0.2682 |
| | 0.8 | *0.3845* | 0.3995 | 0.4256 | 0.4673 | 0.3998 | 0.3890 | 0.4958 | 0.4594 | 0.4025 | **0.3827** | 0.3830 |
| | 0.9 | **0.7072** | 0.7213 | 0.7842 | 0.8290 | 0.7228 | 0.7180 | 0.8313 | 0.7925 | 0.7255 | *0.7176* | 0.7179 |
| $\rho = 0.6$ $\alpha = 1.0$ | 0.1 | 0.0744 | 0.0872 | 0.0411 | 0.2478 | 0.0492 | 0.0396 | 0.0444 | **0.0353** | 0.0483 | *0.0373* | 0.0370 |
| | 0.2 | 0.0959 | 0.0970 | 0.0520 | 0.2505 | 0.0598 | *0.0490* | 0.0613 | 0.0504 | 0.0590 | **0.0471** | 0.0470 |
| | 0.3 | 0.1090 | 0.1086 | 0.0631 | 0.1810 | 0.0729 | *0.0596* | 0.0715 | 0.0601 | 0.0718 | **0.0564** | 0.0563 |
| | 0.4 | 0.1135 | 0.1136 | 0.0715 | 0.1492 | 0.0809 | *0.0659* | 0.0852 | 0.0708 | 0.0794 | **0.0625** | 0.0624 |
| | 0.5 | 0.1189 | 0.1188 | 0.0806 | 0.1388 | 0.0893 | *0.0740* | 0.0969 | 0.0822 | 0.0883 | **0.0704** | 0.0704 |
| | 0.6 | 0.1285 | 0.1276 | 0.0947 | 0.1352 | 0.1020 | *0.0855* | 0.1119 | 0.0961 | 0.1007 | **0.0805** | 0.0805 |
| | 0.7 | 0.1349 | 0.1356 | 0.1092 | 0.1399 | 0.1137 | *0.0970* | 0.1304 | 0.1125 | 0.1126 | **0.0917** | 0.0917 |
| | 0.8 | 0.1421 | 0.1455 | 0.1317 | 0.1564 | 0.1294 | *0.1128* | 0.1559 | 0.1347 | 0.1289 | **0.1080** | 0.1081 |
| | 0.9 | 0.1606 | 0.1692 | 0.1773 | 0.1982 | 0.1633 | *0.1480* | 0.2079 | 0.1843 | 0.1645 | **0.1440** | 0.1441 |

Table S3: Values of $\mathrm{MSE}_\mu$ for various methods in the simulation ($n = 200$)

| | $R^2$ | SCAD | ALASSO | AGG1 | AGG2 | AIC | SAIC | BIC | SBIC | Cp | SMA1 | SMA2 |
|---|---|---|---|---|---|---|---|---|---|---|---|---|
| $\rho = 0.1$ $\alpha = 0.5$ | 0.1 | 0.0904 | 0.0933 | *0.0499* | 0.1821 | 0.0635 | 0.0517 | 0.0646 | 0.0544 | 0.0630 | **0.0469** | 0.0468 |
| | 0.2 | 0.1158 | 0.1142 | *0.0740* | 0.3347 | 0.0897 | 0.0755 | 0.0998 | 0.0873 | 0.0887 | **0.0688** | 0.0688 |
| | 0.3 | 0.1284 | 0.1296 | 0.0971 | 0.2983 | 0.1113 | *0.0956* | 0.1302 | 0.1159 | 0.1103 | **0.0889** | 0.0890 |
| | 0.4 | 0.1426 | 0.1473 | 0.1213 | 0.2631 | 0.1331 | *0.1177* | 0.1638 | 0.1475 | 0.1330 | **0.1101** | 0.1102 |
| | 0.5 | 0.1622 | 0.1691 | 0.1546 | 0.2596 | 0.1621 | *0.1468* | 0.2027 | 0.1836 | 0.1625 | **0.1376** | 0.1377 |
| | 0.6 | 0.1864 | 0.1961 | 0.1936 | 0.2649 | 0.1923 | *0.1782* | 0.2515 | 0.2295 | 0.1937 | **0.1720** | 0.1721 |
| | 0.7 | 0.2336 | 0.2458 | 0.2521 | 0.3045 | 0.2439 | *0.2331* | 0.3192 | 0.2941 | 0.2459 | **0.2271** | 0.2272 |
| | 0.8 | **0.3258** | 0.3372 | 0.3650 | 0.4064 | 0.3374 | 0.3300 | 0.4306 | 0.4003 | 0.3398 | *0.3272* | 0.3274 |
| | 0.9 | **0.6028** | 0.6109 | 0.6692 | 0.7037 | 0.6120 | *0.6106* | 0.7048 | 0.6754 | 0.6141 | 0.6110 | 0.6112 |
| $\rho = 0.1$ $\alpha = 1.0$ | 0.1 | 0.0662 | 0.0753 | 0.0359 | 0.2524 | 0.0407 | 0.0323 | 0.0391 | *0.0315* | 0.0398 | **0.0307** | 0.0306 |
| | 0.2 | 0.0809 | 0.0830 | 0.0455 | 0.2514 | 0.0507 | *0.0409* | 0.0495 | 0.0421 | 0.0497 | **0.0390** | 0.0389 |
| | 0.3 | 0.0876 | 0.0879 | 0.0523 | 0.1810 | 0.0575 | *0.0467* | 0.0604 | 0.0507 | 0.0565 | **0.0449** | 0.0448 |
| | 0.4 | 0.0935 | 0.0940 | 0.0620 | 0.1465 | 0.0657 | *0.0540* | 0.0715 | 0.0612 | 0.0650 | **0.0518** | 0.0518 |
| | 0.5 | 0.0991 | 0.0989 | 0.0681 | 0.1248 | 0.0728 | *0.0600* | 0.0807 | 0.0695 | 0.0716 | **0.0569** | 0.0569 |
| | 0.6 | 0.1043 | 0.1035 | 0.0798 | 0.1208 | 0.0815 | *0.0688* | 0.0922 | 0.0801 | 0.0808 | **0.0646** | 0.0646 |
| | 0.7 | 0.1073 | 0.1078 | 0.0885 | 0.1183 | 0.0908 | *0.0773* | 0.1075 | 0.0936 | 0.0902 | **0.0729** | 0.0729 |
| | 0.8 | 0.1179 | 0.1209 | 0.1079 | 0.1288 | 0.1071 | *0.0932* | 0.1292 | 0.1141 | 0.1067 | **0.0883** | 0.0883 |
| | 0.9 | 0.1324 | 0.1389 | 0.1467 | 0.1636 | 0.1335 | *0.1211* | 0.1731 | 0.1558 | 0.1343 | **0.1173** | 0.1174 |
| $\rho = 0.6$ $\alpha = 0.5$ | 0.1 | 0.0902 | 0.0953 | *0.0491* | 0.1817 | 0.0643 | 0.0512 | 0.0647 | 0.0547 | 0.0631 | **0.0474** | 0.0472 |
| | 0.2 | 0.1148 | 0.1130 | *0.0718* | 0.2951 | 0.0881 | 0.0748 | 0.0969 | 0.0850 | 0.0876 | **0.0687** | 0.0687 |
| | 0.3 | 0.1298 | 0.1303 | *0.0950* | 0.2556 | 0.1127 | 0.0973 | 0.1313 | 0.1161 | 0.1122 | **0.0895** | 0.0895 |
| | 0.4 | 0.1443 | 0.1485 | 0.1210 | 0.2318 | 0.1350 | *0.1197* | 0.1655 | 0.1486 | 0.1349 | **0.1119** | 0.1119 |
| | 0.5 | 0.1604 | 0.1674 | 0.1502 | 0.2239 | 0.1599 | *0.1449* | 0.2015 | 0.1831 | 0.1604 | **0.1368** | 0.1369 |
| | 0.6 | 0.1907 | 0.2019 | 0.1937 | 0.2466 | 0.1958 | *0.1819* | 0.2543 | 0.2323 | 0.1972 | **0.1749** | 0.1751 |
| | 0.7 | 0.2340 | 0.2455 | 0.2526 | 0.2929 | 0.2457 | *0.2337* | 0.3184 | 0.2942 | 0.2472 | **0.2283** | 0.2285 |
| | 0.8 | **0.3258** | 0.3376 | 0.3622 | 0.3946 | 0.3376 | 0.3307 | 0.4303 | 0.4011 | 0.3398 | *0.3274* | 0.3276 |
| | 0.9 | **0.6085** | 0.6174 | 0.6679 | 0.7007 | 0.6177 | 0.6155 | 0.6990 | 0.6731 | 0.6192 | *0.6148* | 0.6150 |
| $\rho = 0.6$ $\alpha = 1.0$ | 0.1 | 0.0662 | 0.0752 | 0.0340 | 0.2398 | 0.0406 | 0.0323 | 0.0388 | **0.0309** | 0.0387 | *0.0309* | 0.0307 |
| | 0.2 | 0.0799 | 0.0814 | 0.0419 | 0.1987 | 0.0501 | *0.0405* | 0.0501 | 0.0421 | 0.0495 | **0.0389** | 0.0388 |
| | 0.3 | 0.0884 | 0.0889 | 0.0504 | 0.1401 | 0.0578 | *0.0476* | 0.0607 | 0.0509 | 0.0570 | **0.0451** | 0.0450 |
| | 0.4 | 0.0945 | 0.0946 | 0.0590 | 0.1187 | 0.0660 | *0.0548* | 0.0705 | 0.0608 | 0.0652 | **0.0520** | 0.0519 |
| | 0.5 | 0.0984 | 0.0982 | 0.0684 | 0.1091 | 0.0726 | *0.0607* | 0.0809 | 0.0699 | 0.0715 | **0.0580** | 0.0580 |
| | 0.6 | 0.1031 | 0.1030 | 0.0753 | 0.1068 | 0.0798 | *0.0675* | 0.0924 | 0.0801 | 0.0789 | **0.0644** | 0.0643 |
| | 0.7 | 0.1118 | 0.1126 | 0.0895 | 0.1137 | 0.0936 | *0.0807* | 0.1095 | 0.0958 | 0.0931 | **0.0761** | 0.0761 |
| | 0.8 | 0.1182 | 0.1207 | 0.1073 | 0.1255 | 0.1077 | *0.0941* | 0.1312 | 0.1154 | 0.1074 | **0.0890** | 0.0891 |
| | 0.9 | 0.1335 | 0.1398 | 0.1426 | 0.1580 | 0.1341 | *0.1219* | 0.1715 | 0.1555 | 0.1346 | **0.1178** | 0.1179 |

Table S4: Values of $\text{MSE}_{\bar{y}} \times 100$ for various methods in the simulation ($n = 50$)

| | $R^2$ | SCAD | ALASSO | AGG1 | AGG2 | AIC | SAIC | BIC | SBIC | Cp | SMA1 | SMA2 |
|---|---|---|---|---|---|---|---|---|---|---|---|---|
| $\rho = 0.1$ $\alpha = 0.5$ | 0.1 | 2.3356 | 2.3532 | 2.1954 | **2.1707** | 2.2721 | 2.2149 | 2.2375 | *2.1804* | 2.2479 | 2.1916 | 2.1918 |
| | 0.2 | 2.3712 | 2.4489 | 2.2446 | 2.2381 | 2.3254 | *2.2322* | 2.3256 | 2.2324 | 2.3185 | **2.2082** | 2.2086 |
| | 0.3 | 2.6507 | 2.6019 | 2.4599 | 2.5657 | 2.5194 | *2.4403* | 2.5120 | 2.4579 | 2.5232 | **2.4072** | 2.4067 |
| | 0.4 | 2.6446 | 2.5707 | **2.4153** | 2.5575 | 2.5494 | 2.4620 | 2.5729 | 2.4771 | 2.5504 | *2.4181* | 2.4185 |
| | 0.5 | 2.5697 | 2.5846 | 2.3571 | 2.5118 | 2.4431 | *2.3565* | 2.5990 | 2.4810 | 2.4826 | **2.3404** | 2.3413 |
| | 0.6 | 2.9517 | 2.9397 | **2.7370** | 2.8836 | 2.8611 | 2.7769 | 2.9866 | 2.8411 | 2.8872 | *2.7723* | 2.7724 |
| | 0.7 | 2.7368 | 2.7214 | 2.6480 | 2.7258 | 2.7775 | *2.6199* | 2.8481 | 2.6749 | 2.7481 | **2.5827** | 2.5832 |
| | 0.8 | 3.3757 | 3.4217 | 3.3083 | 3.3530 | 3.3752 | *3.2988* | 3.5618 | 3.4139 | 3.4280 | **3.2556** | 3.2567 |
| | 0.9 | *4.3581* | 4.4543 | 4.5263 | 4.6566 | 4.3918 | 4.3608 | 4.8711 | 4.6240 | 4.4136 | **4.3113** | 4.3125 |
| $\rho = 0.1$ $\alpha = 1.0$ | 0.1 | 2.2134 | 2.2984 | 2.0239 | **1.9568** | 2.1465 | 2.0855 | 2.0056 | *1.9822* | 2.1306 | 2.0488 | 2.0478 |
| | 0.2 | 2.3335 | 2.3693 | 2.1188 | *2.0711* | 2.2238 | 2.1730 | 2.1392 | **2.0612** | 2.2287 | 2.1326 | 2.1310 |
| | 0.3 | 2.2824 | 2.3281 | 2.1578 | **2.1311** | 2.2272 | 2.1832 | 2.2001 | 2.1492 | 2.1984 | *2.1483* | 2.1474 |
| | 0.4 | 2.3816 | 2.3967 | **2.1401** | 2.1907 | 2.2909 | 2.1957 | 2.2195 | *2.1434* | 2.2738 | 2.1867 | 2.1869 |
| | 0.5 | 2.5975 | 2.5862 | 2.3989 | *2.3573* | 2.4864 | 2.3776 | 2.4212 | **2.3474** | 2.4672 | 2.3720 | 2.3720 |
| | 0.6 | 2.6270 | 2.6171 | 2.3951 | 2.3932 | 2.4695 | *2.3736* | 2.4490 | 2.3742 | 2.4314 | **2.3539** | 2.3536 |
| | 0.7 | 2.5488 | 2.5713 | 2.4537 | *2.4193* | 2.5221 | 2.4205 | 2.5119 | 2.4453 | 2.5199 | **2.4097** | 2.4098 |
| | 0.8 | 2.7294 | 2.7407 | 2.5855 | 2.5847 | 2.6551 | *2.5630* | 2.7130 | 2.5733 | 2.6243 | **2.5513** | 2.5513 |
| | 0.9 | 2.9151 | 2.9411 | 2.8385 | 2.7997 | 2.8269 | *2.7855* | 3.0396 | 2.8711 | 2.8116 | **2.7715** | 2.7704 |
| $\rho = 0.6$ $\alpha = 0.5$ | 0.1 | 2.3833 | 2.3918 | 2.1990 | **2.1049** | 2.3115 | 2.2465 | 2.1979 | *2.1636* | 2.2964 | 2.2212 | 2.2216 |
| | 0.2 | 2.5356 | 2.6353 | 2.3474 | **2.2212** | 2.5115 | 2.4172 | 2.4098 | *2.3232* | 2.4959 | 2.3649 | 2.3630 |
| | 0.3 | 2.4602 | 2.4875 | *2.2483* | **2.2088** | 2.4193 | 2.3217 | 2.3250 | 2.2493 | 2.4326 | 2.2573 | 2.2593 |
| | 0.4 | 2.5704 | 2.5375 | **2.3145** | 2.4267 | 2.4662 | 2.3542 | 2.5022 | 2.4197 | 2.4728 | *2.3263* | 2.3269 |
| | 0.5 | 2.5419 | 2.4899 | *2.2293* | 2.2915 | 2.3734 | 2.2504 | 2.3916 | 2.2918 | 2.3413 | **2.2166** | 2.2180 |
| | 0.6 | 2.7453 | 2.7203 | 2.5849 | 2.6989 | 2.6643 | *2.5467* | 2.8334 | 2.6538 | 2.6196 | **2.5309** | 2.5315 |
| | 0.7 | 3.0402 | 3.0819 | 3.0455 | 3.0805 | 3.1348 | *2.9740* | 3.3010 | 3.0651 | 3.0989 | **2.9335** | 2.9346 |
| | 0.8 | 3.4678 | 3.5087 | *3.3416* | 3.3809 | 3.5068 | 3.3614 | 3.5512 | 3.4113 | 3.4919 | **3.2858** | 3.2842 |
| | 0.9 | 4.7501 | 4.8761 | 4.8555 | 4.9126 | 4.8014 | *4.7026* | 5.2296 | 4.9295 | 4.9178 | **4.6568** | 4.6551 |
| $\rho = 0.6$ $\alpha = 1.0$ | 0.1 | 2.2071 | 2.3165 | 2.0424 | **1.8993** | 2.1742 | 2.1066 | 2.0418 | *2.0100* | 2.1662 | 2.0754 | 2.0764 |
| | 0.2 | 2.3414 | 2.4485 | 2.1667 | **2.0267** | 2.2659 | 2.1953 | 2.2108 | *2.1331* | 2.2343 | 2.1774 | 2.1772 |
| | 0.3 | 2.3544 | 2.4114 | 2.1411 | **2.0346** | 2.2429 | 2.1917 | 2.1869 | *2.1254* | 2.2476 | 2.1499 | 2.1499 |
| | 0.4 | 2.4672 | 2.4497 | *2.2347* | **2.1625** | 2.3822 | 2.2834 | 2.3404 | 2.2548 | 2.3782 | 2.2442 | 2.2450 |
| | 0.5 | 2.6687 | 2.6389 | *2.3174* | **2.2978** | 2.4845 | 2.4151 | 2.4429 | 2.3610 | 2.4720 | 2.3951 | 2.3960 |
| | 0.6 | 2.4832 | 2.4521 | *2.2534* | **2.2068** | 2.3916 | 2.3146 | 2.3542 | 2.2960 | 2.3990 | 2.2801 | 2.2809 |
| | 0.7 | 2.6330 | 2.6138 | 2.4685 | 2.4551 | 2.5186 | 2.4423 | 2.5124 | *2.4224* | 2.5249 | **2.4116** | 2.4124 |
| | 0.8 | 2.7659 | 2.7328 | 2.5022 | **2.4749** | 2.6019 | 2.5266 | 2.6024 | 2.5160 | 2.5786 | *2.4976* | 2.4978 |
| | 0.9 | 2.9262 | 2.9745 | 2.8876 | 2.8790 | 2.8731 | *2.8094* | 3.0375 | 2.8878 | 2.8851 | **2.7923** | 2.7924 |

Table S5: Values of $\text{MSE}_{\bar{y}} \times 100$ for various methods in the simulation ($n = 150$)

| | $R^2$ | SCAD | ALASSO | AGG1 | AGG2 | AIC | SAIC | BIC | SBIC | Cp | SMA1 | SMA2 |
|---|---|---|---|---|---|---|---|---|---|---|---|---|
| $\rho = 0.1$ $\alpha = 0.5$ | 0.1 | 0.6580 | 0.6529 | *0.6340* | 0.6364 | 0.6496 | 0.6356 | 0.6409 | 0.6348 | 0.6451 | **0.6313** | 0.6313 |
| | 0.2 | 0.7231 | 0.7234 | **0.6964** | 0.7374 | 0.7031 | 0.6991 | 0.7271 | 0.7139 | 0.7020 | *0.6967* | 0.6968 |
| | 0.3 | 0.7125 | 0.7188 | 0.7011 | 0.7435 | 0.7110 | *0.6932* | 0.7392 | 0.7244 | 0.7097 | **0.6892** | 0.6894 |
| | 0.4 | 0.6976 | 0.7029 | 0.6710 | 0.7204 | 0.6758 | *0.6653* | 0.7072 | 0.6928 | 0.6753 | **0.6642** | 0.6643 |
| | 0.5 | 0.7147 | 0.7189 | 0.7264 | 0.7637 | 0.7232 | *0.7088* | 0.7576 | 0.7384 | 0.7220 | **0.7039** | 0.7040 |
| | 0.6 | 0.7989 | 0.8035 | 0.8092 | 0.8297 | 0.8006 | **0.7896** | 0.8681 | 0.8448 | 0.7987 | *0.7922* | 0.7923 |
| | 0.7 | 0.7965 | 0.7995 | 0.7822 | 0.7934 | 0.7930 | *0.7793* | 0.8247 | 0.8080 | 0.7885 | **0.7750** | 0.7750 |
| | 0.8 | **0.9080** | 0.9245 | 0.9472 | 0.9606 | 0.9264 | *0.9161* | 1.0068 | 0.9911 | 0.9269 | 0.9168 | 0.9169 |
| | 0.9 | **1.0788** | 1.0909 | 1.0953 | 1.0929 | 1.0891 | 1.0834 | 1.1476 | 1.1283 | 1.0895 | *1.0797* | 1.0798 |
| $\rho = 0.1$ $\alpha = 1.0$ | 0.1 | 0.6726 | 0.6777 | 0.6589 | 0.6666 | 0.6691 | **0.6586** | 0.6719 | 0.6643 | 0.6678 | *0.6587* | 0.6587 |
| | 0.2 | 0.6506 | 0.6508 | *0.6346* | 0.6391 | 0.6389 | 0.6361 | 0.6479 | 0.6364 | 0.6389 | **0.6303** | 0.6303 |
| | 0.3 | 0.7113 | 0.7167 | *0.6843* | 0.6879 | 0.7004 | 0.6855 | 0.6974 | 0.6899 | 0.6986 | **0.6842** | 0.6843 |
| | 0.4 | 0.6774 | 0.6780 | 0.6625 | 0.6602 | 0.6570 | *0.6505* | 0.6609 | 0.6528 | 0.6574 | **0.6490** | 0.6488 |
| | 0.5 | 0.6772 | 0.6852 | 0.6637 | 0.6612 | 0.6683 | *0.6597* | 0.6815 | 0.6729 | 0.6688 | **0.6565** | 0.6563 |
| | 0.6 | 0.6530 | 0.6585 | 0.6506 | 0.6496 | 0.6530 | *0.6450* | 0.6659 | 0.6606 | 0.6524 | **0.6427** | 0.6427 |
| | 0.7 | 0.7218 | 0.7229 | 0.7063 | 0.7099 | 0.7164 | **0.7023** | 0.7314 | 0.7220 | 0.7210 | *0.7032* | 0.7033 |
| | 0.8 | 0.6701 | 0.6717 | 0.6531 | 0.6538 | 0.6655 | *0.6473* | 0.6771 | 0.6626 | 0.6582 | **0.6428** | 0.6427 |
| | 0.9 | 0.6989 | 0.7031 | 0.7154 | 0.7190 | 0.6975 | **0.6935** | 0.7411 | 0.7304 | 0.6989 | *0.6963* | 0.6964 |
| $\rho = 0.6$ $\alpha = 0.5$ | 0.1 | 0.6421 | 0.6586 | 0.6276 | **0.6035** | 0.6376 | 0.6312 | 0.6270 | *0.6201* | 0.6357 | 0.6258 | 0.6259 |
| | 0.2 | 0.7059 | 0.6916 | *0.6658* | 0.6726 | 0.6823 | 0.6666 | 0.6911 | 0.6821 | 0.6793 | **0.6619** | 0.6621 |
| | 0.3 | 0.6888 | 0.6859 | 0.6603 | 0.6685 | 0.6713 | *0.6541* | 0.6812 | 0.6670 | 0.6684 | **0.6525** | 0.6525 |
| | 0.4 | 0.7462 | 0.7445 | *0.7014* | **0.6961** | 0.7470 | 0.7275 | 0.7436 | 0.7331 | 0.7471 | 0.7126 | 0.7128 |
| | 0.5 | 0.7049 | 0.7197 | 0.6963 | 0.7082 | 0.7111 | *0.6949* | 0.7162 | 0.7139 | 0.7065 | **0.6895** | 0.6897 |
| | 0.6 | 0.8876 | 0.8975 | *0.8713* | 0.8757 | 0.8856 | 0.8731 | 0.9157 | 0.8920 | 0.8849 | **0.8652** | 0.8654 |
| | 0.7 | 0.8135 | 0.8334 | **0.7981** | 0.8015 | 0.8115 | 0.8091 | 0.8607 | 0.8448 | 0.8079 | *0.7983* | 0.7984 |
| | 0.8 | **0.8700** | 0.8750 | 0.8968 | 0.9056 | 0.8810 | *0.8724* | 0.9351 | 0.9199 | 0.8841 | 0.8732 | 0.8735 |
| | 0.9 | **1.0798** | *1.0846* | 1.1514 | 1.1652 | 1.0957 | 1.0954 | 1.1649 | 1.1537 | 1.0973 | 1.0984 | 1.0988 |
| $\rho = 0.6$ $\alpha = 1.0$ | 0.1 | 0.6852 | 0.6931 | 0.6665 | **0.6269** | 0.6768 | 0.6683 | 0.6661 | *0.6632* | 0.6766 | 0.6654 | 0.6657 |
| | 0.2 | 0.6956 | 0.6918 | 0.6501 | **0.6267** | 0.6596 | 0.6498 | 0.6570 | *0.6477* | 0.6599 | 0.6485 | 0.6487 |
| | 0.3 | 0.6885 | 0.6870 | 0.6654 | **0.6521** | 0.6707 | 0.6659 | 0.6714 | 0.6645 | 0.6709 | *0.6599* | 0.6600 |
| | 0.4 | 0.6911 | 0.6904 | 0.6452 | **0.6317** | 0.6555 | 0.6473 | 0.6573 | 0.6483 | 0.6544 | *0.6449* | 0.6450 |
| | 0.5 | 0.6793 | 0.6776 | *0.6455* | **0.6312** | 0.6666 | 0.6561 | 0.6784 | 0.6639 | 0.6652 | 0.6520 | 0.6519 |
| | 0.6 | 0.7625 | 0.7531 | 0.7345 | **0.7272** | 0.7510 | 0.7403 | 0.7566 | 0.7426 | 0.7494 | *0.7322* | 0.7325 |
| | 0.7 | 0.7361 | 0.7334 | 0.7129 | **0.7083** | 0.7150 | 0.7106 | 0.7280 | 0.7203 | 0.7143 | *0.7094* | 0.7096 |
| | 0.8 | 0.6568 | 0.6576 | 0.6306 | 0.6288 | 0.6392 | 0.6281 | 0.6363 | **0.6207** | 0.6330 | *0.6222* | 0.6223 |
| | 0.9 | *0.7632* | 0.7670 | 0.7720 | 0.7797 | 0.7656 | **0.7577** | 0.8235 | 0.8069 | 0.7695 | 0.7641 | 0.7642 |

Table S6: Values of $\text{MSE}_{\bar{y}} \times 100$ for various methods in the simulation ($n = 200$)

| | $R^2$ | SCAD | ALASSO | AGG1 | AGG2 | AIC | SAIC | BIC | SBIC | Cp | SMA1 | SMA2 |
|---|---|---|---|---|---|---|---|---|---|---|---|---|
| $\rho = 0.1$ $\alpha = 0.5$ | 0.1 | 0.4872 | 0.4785 | **0.4614** | 0.4804 | 0.4681 | 0.4635 | 0.4779 | 0.4728 | 0.4689 | *0.4630* | 0.4631 |
| | 0.2 | 0.4712 | 0.4717 | **0.4483** | 0.4750 | 0.4652 | 0.4547 | 0.4692 | 0.4604 | 0.4635 | *0.4523* | 0.4523 |
| | 0.3 | 0.5461 | 0.5508 | **0.5266** | 0.5501 | 0.5376 | 0.5321 | 0.5588 | 0.5473 | 0.5354 | *0.5317* | 0.5317 |
| | 0.4 | 0.5394 | 0.5463 | **0.5178** | 0.5264 | 0.5309 | 0.5259 | 0.5424 | 0.5379 | 0.5321 | *0.5235* | 0.5236 |
| | 0.5 | 0.5358 | 0.5366 | 0.5393 | 0.5579 | 0.5389 | *0.5321* | 0.5607 | 0.5564 | 0.5382 | **0.5291** | 0.5292 |
| | 0.6 | **0.5181** | 0.5300 | 0.5218 | 0.5347 | 0.5274 | 0.5209 | 0.5638 | 0.5486 | 0.5310 | *0.5198* | 0.5199 |
| | 0.7 | **0.5368** | 0.5412 | 0.5552 | 0.5632 | 0.5429 | *0.5391* | 0.5797 | 0.5666 | 0.5414 | 0.5416 | 0.5417 |
| | 0.8 | 0.6038 | 0.6082 | 0.6255 | 0.6354 | 0.6034 | **0.6014** | 0.6440 | 0.6357 | 0.6070 | *0.6029* | 0.6030 |
| | 0.9 | **0.6540** | *0.6567* | 0.6936 | 0.6916 | 0.6576 | 0.6592 | 0.7089 | 0.7014 | 0.6603 | 0.6639 | 0.6641 |
| $\rho = 0.1$ $\alpha = 1.0$ | 0.1 | 0.5073 | 0.5152 | 0.4942 | **0.4893** | 0.5027 | 0.4993 | 0.4937 | *0.4910* | 0.5018 | 0.4945 | 0.4944 |
| | 0.2 | 0.4916 | 0.4930 | 0.4805 | 0.4835 | 0.4817 | *0.4766* | 0.4831 | 0.4816 | 0.4824 | **0.4743** | 0.4743 |
| | 0.3 | 0.4945 | 0.4945 | 0.4759 | 0.4815 | 0.4749 | **0.4717** | 0.4880 | 0.4822 | *0.4742* | 0.4743 | 0.4742 |
| | 0.4 | 0.5534 | 0.5539 | 0.5393 | 0.5458 | 0.5412 | **0.5364** | 0.5477 | 0.5414 | 0.5403 | *0.5388* | 0.5387 |
| | 0.5 | 0.4903 | 0.4894 | 0.4737 | 0.4719 | 0.4782 | *0.4702* | 0.4727 | 0.4712 | 0.4778 | **0.4686** | 0.4686 |
| | 0.6 | 0.4743 | 0.4755 | 0.4728 | 0.4706 | 0.4766 | *0.4703* | 0.4744 | 0.4725 | 0.4763 | **0.4660** | 0.4661 |
| | 0.7 | 0.5042 | 0.5072 | 0.4975 | 0.4977 | 0.5018 | *0.4936* | 0.5028 | 0.4966 | 0.5001 | **0.4922** | 0.4923 |
| | 0.8 | 0.4861 | 0.4873 | 0.4769 | 0.4783 | 0.4854 | *0.4769* | 0.4974 | 0.4913 | 0.4845 | **0.4750** | 0.4750 |
| | 0.9 | 0.5360 | 0.5420 | 0.5293 | 0.5304 | 0.5391 | **0.5276** | 0.5452 | 0.5379 | 0.5391 | *0.5277* | 0.5278 |
| $\rho = 0.6$ $\alpha = 0.5$ | 0.1 | 0.4632 | 0.4787 | 0.4598 | **0.4556** | 0.4620 | 0.4582 | 0.4698 | 0.4653 | 0.4614 | *0.4567* | 0.4568 |
| | 0.2 | 0.5130 | 0.5228 | **0.4900** | 0.4940 | 0.5018 | 0.4949 | 0.5032 | 0.4985 | 0.5011 | *0.4927* | 0.4927 |
| | 0.3 | 0.5491 | 0.5471 | **0.5128** | 0.5202 | 0.5218 | 0.5144 | 0.5325 | 0.5255 | 0.5190 | *0.5144* | 0.5144 |
| | 0.4 | 0.5456 | 0.5441 | 0.5374 | 0.5392 | 0.5393 | **0.5338** | 0.5620 | 0.5563 | 0.5380 | *0.5344* | 0.5345 |
| | 0.5 | 0.5087 | 0.5129 | **0.4953** | 0.4994 | 0.5069 | 0.5016 | 0.5191 | 0.5148 | 0.5062 | *0.4972* | 0.4972 |
| | 0.6 | **0.5591** | 0.5669 | 0.5752 | 0.5825 | 0.5742 | *0.5650* | 0.6081 | 0.5972 | 0.5749 | 0.5650 | 0.5652 |
| | 0.7 | *0.5948* | 0.6012 | 0.6132 | 0.6191 | 0.6024 | **0.5942** | 0.6363 | 0.6258 | 0.6023 | 0.5966 | 0.5968 |
| | 0.8 | *0.5937* | 0.5991 | 0.6071 | 0.6062 | 0.6020 | 0.5969 | 0.6391 | 0.6221 | 0.6050 | **0.5910** | 0.5910 |
| | 0.9 | **0.7606** | 0.7658 | 0.7724 | 0.7794 | 0.7681 | 0.7668 | 0.8026 | 0.7900 | 0.7695 | *0.7633* | 0.7632 |
| $\rho = 0.6$ $\alpha = 1.0$ | 0.1 | 0.4864 | 0.4969 | 0.4736 | **0.4514** | 0.4768 | 0.4707 | 0.4731 | *0.4698* | 0.4743 | 0.4732 | 0.4731 |
| | 0.2 | 0.4840 | 0.4849 | 0.4611 | **0.4391** | 0.4642 | 0.4616 | 0.4571 | *0.4550* | 0.4624 | 0.4600 | 0.4599 |
| | 0.3 | 0.4848 | 0.4853 | 0.4761 | **0.4682** | 0.4749 | 0.4721 | 0.4833 | 0.4789 | 0.4753 | *0.4691* | 0.4694 |
| | 0.4 | 0.5107 | 0.5112 | 0.4944 | **0.4862** | 0.4989 | 0.4951 | 0.5022 | 0.5002 | 0.5002 | *0.4918* | 0.4918 |
| | 0.5 | 0.5383 | 0.5404 | 0.5160 | **0.5067** | 0.5232 | 0.5138 | 0.5210 | 0.5175 | 0.5204 | *0.5130* | 0.5131 |
| | 0.6 | 0.5170 | 0.5196 | 0.5060 | *0.5008* | 0.5062 | 0.5020 | 0.5125 | 0.5072 | 0.5052 | **0.4983** | 0.4983 |
| | 0.7 | 0.5073 | 0.5087 | **0.4874** | *0.4890* | 0.5033 | 0.4965 | 0.5103 | 0.5019 | 0.5037 | 0.4924 | 0.4925 |
| | 0.8 | 0.5340 | 0.5365 | 0.5327 | 0.5337 | 0.5350 | *0.5322* | 0.5539 | 0.5457 | 0.5383 | **0.5292** | 0.5292 |
| | 0.9 | 0.4881 | **0.4845** | 0.4967 | 0.4958 | 0.4879 | *0.4858* | 0.5238 | 0.5153 | 0.4882 | 0.4866 | 0.4867 |

## S5.2 Simulation results for $M = [n/3]$

Table S7: Values of $\text{MSE}_\mu$ for various methods in the simulation ($n = 50$)

| | $R^2$ | SCAD | ALASSO | AGG1 | AGG2 | AIC | SAIC | BIC | SBIC | Cp | SMA1 | SMA2 |
|---|---|---|---|---|---|---|---|---|---|---|---|---|
| $\rho = 0.1$ $\alpha = 0.5$ | 0.1 | 0.2118 | 0.3226 | 0.1211 | 0.1755 | 0.2040 | 0.1611 | *0.1115* | **0.0905** | 0.1791 | 0.1270 | 0.1263 |
| | 0.2 | 0.2865 | 0.3770 | *0.1638* | 0.3459 | 0.2563 | 0.2009 | 0.1805 | **0.1455** | 0.2321 | 0.1660 | 0.1655 |
| | 0.3 | 0.3806 | 0.4414 | *0.2188* | 0.4845 | 0.3259 | 0.2627 | 0.2592 | **0.2088** | 0.2969 | 0.2191 | 0.2186 |
| | 0.4 | 0.4646 | 0.4994 | *0.2650* | 0.5374 | 0.3823 | 0.3101 | 0.3278 | 0.2687 | 0.3508 | **0.2604** | 0.2598 |
| | 0.5 | 0.5378 | 0.5416 | *0.3332* | 0.5954 | 0.4423 | 0.3649 | 0.4081 | 0.3430 | 0.4132 | **0.3174** | 0.3171 |
| | 0.6 | 0.6403 | 0.6298 | *0.4117* | 0.6117 | 0.5358 | 0.4478 | 0.5110 | 0.4349 | 0.5082 | **0.3891** | 0.3890 |
| | 0.7 | 0.7386 | 0.7304 | *0.5372* | 0.7040 | 0.6446 | 0.5532 | 0.6522 | 0.5615 | 0.6230 | **0.4891** | 0.4890 |
| | 0.8 | 0.9041 | 0.8958 | 0.7513 | 0.9091 | 0.8306 | *0.7353* | 0.8989 | 0.7830 | 0.8266 | **0.6678** | 0.6679 |
| | 0.9 | 1.3666 | 1.3886 | 1.3426 | 1.5030 | 1.3323 | *1.2357* | 1.5064 | 1.3355 | 1.3490 | **1.1492** | 1.1496 |
| $\rho = 0.1$ $\alpha = 1.0$ | 0.1 | 0.1766 | 0.2888 | 0.1042 | 0.2420 | 0.1735 | 0.1316 | *0.0750* | **0.0599** | 0.1402 | 0.1017 | 0.1012 |
| | 0.2 | 0.2314 | 0.3332 | 0.1267 | 0.3915 | 0.2111 | 0.1595 | *0.1173* | **0.0895** | 0.1731 | 0.1267 | 0.1259 |
| | 0.3 | 0.2792 | 0.3533 | 0.1579 | 0.4272 | 0.2319 | 0.1753 | 0.1496 | **0.1161** | 0.1932 | *0.1414* | 0.1409 |
| | 0.4 | 0.3090 | 0.3669 | 0.1708 | 0.3918 | 0.2390 | 0.1867 | 0.1709 | **0.1381** | 0.2089 | *0.1563* | 0.1558 |
| | 0.5 | 0.3661 | 0.4086 | 0.2084 | 0.3822 | 0.2744 | 0.2204 | 0.2000 | **0.1628** | 0.2423 | *0.1844* | 0.1838 |
| | 0.6 | 0.3890 | 0.4162 | 0.2330 | 0.3597 | 0.3006 | 0.2393 | 0.2299 | **0.1857** | 0.2675 | *0.2007* | 0.2003 |
| | 0.7 | 0.4214 | 0.4450 | 0.2616 | 0.3726 | 0.3249 | 0.2608 | 0.2742 | **0.2237** | 0.2903 | *0.2238* | 0.2236 |
| | 0.8 | 0.4767 | 0.4869 | 0.3113 | 0.3914 | 0.3757 | 0.3088 | 0.3316 | *0.2769* | 0.3443 | **0.2697** | 0.2695 |
| | 0.9 | 0.5629 | 0.5662 | 0.4276 | 0.4886 | 0.4731 | 0.4000 | 0.4549 | *0.3813* | 0.4493 | **0.3507** | 0.3507 |
| $\rho = 0.6$ $\alpha = 0.5$ | 0.1 | 0.1954 | 0.3109 | 0.1219 | 0.1742 | 0.1991 | 0.1563 | *0.1093* | **0.0901** | 0.1720 | 0.1255 | 0.1245 |
| | 0.2 | 0.2868 | 0.3751 | *0.1653* | 0.3379 | 0.2662 | 0.2101 | 0.1797 | **0.1464** | 0.2367 | 0.1721 | 0.1714 |
| | 0.3 | 0.3816 | 0.4409 | 0.2178 | 0.4569 | 0.3238 | 0.2614 | 0.2550 | **0.2079** | 0.2967 | *0.2167* | 0.2160 |
| | 0.4 | 0.4691 | 0.5004 | **0.2594** | 0.4874 | 0.3895 | 0.3135 | 0.3257 | 0.2678 | 0.3584 | *0.2609* | 0.2605 |
| | 0.5 | 0.5500 | 0.5622 | *0.3312* | 0.5555 | 0.4578 | 0.3769 | 0.4110 | 0.3457 | 0.4249 | **0.3240** | 0.3235 |
| | 0.6 | 0.6369 | 0.6326 | *0.4087* | 0.5857 | 0.5427 | 0.4538 | 0.5105 | 0.4357 | 0.5111 | **0.3941** | 0.3940 |
| | 0.7 | 0.7482 | 0.7339 | *0.5408* | 0.6891 | 0.6580 | 0.5700 | 0.6670 | 0.5720 | 0.6363 | **0.5053** | 0.5054 |
| | 0.8 | 0.9342 | 0.9226 | 0.7552 | 0.8866 | 0.8466 | *0.7537* | 0.9187 | 0.8043 | 0.8380 | **0.6857** | 0.6858 |
| | 0.9 | 1.3727 | 1.3996 | 1.3304 | 1.4753 | 1.3302 | *1.2293* | 1.5161 | 1.3417 | 1.3505 | **1.1543** | 1.1545 |
| $\rho = 0.6$ $\alpha = 1.0$ | 0.1 | 0.1902 | 0.2997 | 0.1069 | 0.2306 | 0.1782 | 0.1374 | *0.0811* | **0.0641** | 0.1488 | 0.1060 | 0.1051 |
| | 0.2 | 0.2310 | 0.3307 | 0.1260 | 0.3586 | 0.2027 | 0.1522 | *0.1168* | **0.0894** | 0.1677 | 0.1230 | 0.1219 |
| | 0.3 | 0.2810 | 0.3483 | 0.1437 | 0.3691 | 0.2212 | 0.1731 | 0.1482 | **0.1145** | 0.1959 | *0.1420* | 0.1413 |
| | 0.4 | 0.3131 | 0.3617 | 0.1639 | 0.3353 | 0.2280 | 0.1802 | 0.1773 | **0.1401** | 0.1985 | *0.1552* | 0.1545 |
| | 0.5 | 0.3507 | 0.3935 | 0.1941 | 0.3325 | 0.2626 | 0.2070 | 0.2027 | **0.1639** | 0.2296 | *0.1783* | 0.1776 |
| | 0.6 | 0.3945 | 0.4186 | 0.2229 | 0.3381 | 0.3007 | 0.2441 | 0.2338 | **0.1904** | 0.2735 | *0.2041* | 0.2035 |
| | 0.7 | 0.4174 | 0.4426 | 0.2524 | 0.3400 | 0.3264 | 0.2602 | 0.2721 | **0.2241** | 0.2880 | *0.2250* | 0.2245 |
| | 0.8 | 0.4773 | 0.4907 | 0.3187 | 0.3921 | 0.3869 | 0.3159 | 0.3345 | *0.2785* | 0.3528 | **0.2724** | 0.2720 |
| | 0.9 | 0.5716 | 0.5651 | 0.4258 | 0.4898 | 0.4834 | 0.4066 | 0.4591 | *0.3872* | 0.4545 | **0.3567** | 0.3566 |

Table S8: Values of $\text{MSE}_\mu$ for various methods in the simulation ($n = 100$)

| | $R^2$ | SCAD | ALASSO | AGG1 | AGG2 | AIC | SAIC | BIC | SBIC | Cp | SMA1 | SMA2 |
|---|---|---|---|---|---|---|---|---|---|---|---|---|
| $\rho = 0.1$ $\alpha = 0.5$ | 0.1 | 0.1593 | 0.2725 | *0.0795* | 0.1746 | 0.1181 | 0.0928 | 0.0821 | **0.0678** | 0.1046 | 0.0830 | 0.0825 |
| | 0.2 | 0.2322 | 0.3237 | **0.1104** | 0.3165 | 0.1599 | 0.1294 | 0.1341 | 0.1131 | 0.1429 | *0.1129* | 0.1124 |
| | 0.3 | 0.3036 | 0.3791 | **0.1467** | 0.3593 | 0.2132 | 0.1752 | 0.1825 | 0.1566 | 0.1906 | *0.1474* | 0.1470 |
| | 0.4 | 0.3611 | 0.4136 | *0.1812* | 0.3602 | 0.2528 | 0.2110 | 0.2325 | 0.2003 | 0.2314 | **0.1759** | 0.1756 |
| | 0.5 | 0.4015 | 0.4411 | *0.2207* | 0.3639 | 0.3062 | 0.2558 | 0.2847 | 0.2513 | 0.2757 | **0.2122** | 0.2118 |
| | 0.6 | 0.4647 | 0.4951 | *0.2775* | 0.3945 | 0.3696 | 0.3162 | 0.3544 | 0.3154 | 0.3397 | **0.2601** | 0.2598 |
| | 0.7 | 0.5486 | 0.5650 | *0.3503* | 0.4423 | 0.4504 | 0.3923 | 0.4515 | 0.4076 | 0.4198 | **0.3235** | 0.3234 |
| | 0.8 | 0.6636 | 0.6619 | *0.4752* | 0.5467 | 0.5726 | 0.5140 | 0.6144 | 0.5539 | 0.5480 | **0.4291** | 0.4290 |
| | 0.9 | 0.9137 | 0.9210 | 0.7994 | 0.8672 | 0.8561 | *0.7913* | 1.0057 | 0.9198 | 0.8467 | **0.6925** | 0.6926 |
| $\rho = 0.1$ $\alpha = 1.0$ | 0.1 | 0.1257 | 0.2460 | 0.0645 | 0.2304 | 0.0899 | 0.0713 | *0.0544* | **0.0424** | 0.0781 | 0.0653 | 0.0647 |
| | 0.2 | 0.1718 | 0.2712 | 0.0821 | 0.2982 | 0.1004 | 0.0803 | 0.0788 | **0.0633** | 0.0891 | *0.0759* | 0.0754 |
| | 0.3 | 0.2090 | 0.2963 | 0.0944 | 0.2648 | 0.1206 | 0.0960 | 0.0948 | **0.0779** | 0.1076 | *0.0889* | 0.0884 |
| | 0.4 | 0.2363 | 0.3139 | 0.1068 | 0.2315 | 0.1357 | 0.1084 | 0.1087 | **0.0900** | 0.1201 | *0.0983* | 0.0978 |
| | 0.5 | 0.2561 | 0.3326 | 0.1182 | 0.2100 | 0.1532 | 0.1234 | 0.1263 | **0.1048** | 0.1322 | *0.1096* | 0.1091 |
| | 0.6 | 0.2764 | 0.3409 | 0.1341 | 0.2051 | 0.1694 | 0.1382 | 0.1460 | *0.1221* | 0.1508 | **0.1210** | 0.1206 |
| | 0.7 | 0.3022 | 0.3632 | 0.1625 | 0.2205 | 0.1870 | 0.1536 | 0.1687 | *0.1439* | 0.1645 | **0.1385** | 0.1381 |
| | 0.8 | 0.3285 | 0.3829 | 0.1856 | 0.2306 | 0.2190 | 0.1822 | 0.2016 | *0.1728* | 0.1951 | **0.1569** | 0.1566 |
| | 0.9 | 0.3968 | 0.4291 | 0.2423 | 0.2748 | 0.2903 | 0.2462 | 0.2715 | *0.2366* | 0.2620 | **0.2057** | 0.2055 |
| $\rho = 0.6$ $\alpha = 0.5$ | 0.1 | 0.1603 | 0.2706 | *0.0791* | 0.1696 | 0.1191 | 0.0947 | 0.0827 | **0.0678** | 0.1028 | 0.0851 | 0.0842 |
| | 0.2 | 0.2400 | 0.3369 | **0.1111** | 0.2874 | 0.1670 | 0.1352 | 0.1379 | *0.1148* | 0.1509 | 0.1182 | 0.1174 |
| | 0.3 | 0.3057 | 0.3782 | **0.1423** | 0.3184 | 0.2120 | 0.1719 | 0.1841 | 0.1574 | 0.1896 | *0.1466* | 0.1460 |
| | 0.4 | 0.3534 | 0.4055 | **0.1771** | 0.3161 | 0.2500 | 0.2104 | 0.2279 | 0.1994 | 0.2297 | *0.1780* | 0.1775 |
| | 0.5 | 0.4059 | 0.4481 | *0.2196* | 0.3270 | 0.2956 | 0.2497 | 0.2838 | 0.2502 | 0.2710 | **0.2113** | 0.2109 |
| | 0.6 | 0.4638 | 0.4931 | *0.2663* | 0.3520 | 0.3623 | 0.3104 | 0.3545 | 0.3141 | 0.3332 | **0.2564** | 0.2561 |
| | 0.7 | 0.5492 | 0.5610 | *0.3460* | 0.4227 | 0.4492 | 0.3894 | 0.4543 | 0.4050 | 0.4151 | **0.3199** | 0.3197 |
| | 0.8 | 0.6578 | 0.6610 | *0.4674* | 0.5299 | 0.5755 | 0.5113 | 0.6157 | 0.5538 | 0.5477 | **0.4277** | 0.4276 |
| | 0.9 | 0.9145 | 0.9229 | *0.7854* | 0.8468 | 0.8520 | 0.7886 | 1.0115 | 0.9272 | 0.8445 | **0.6937** | 0.6939 |
| $\rho = 0.6$ $\alpha = 1.0$ | 0.1 | 0.1349 | 0.2584 | 0.0640 | 0.2153 | 0.0883 | 0.0724 | *0.0549* | **0.0434** | 0.0799 | 0.0669 | 0.0660 |
| | 0.2 | 0.1778 | 0.2764 | *0.0790* | 0.2525 | 0.1092 | 0.0872 | 0.0808 | **0.0640** | 0.0961 | 0.0794 | 0.0786 |
| | 0.3 | 0.2048 | 0.2920 | 0.0888 | 0.2195 | 0.1176 | 0.0925 | 0.0914 | **0.0753** | 0.1020 | *0.0865* | 0.0857 |
| | 0.4 | 0.2363 | 0.3088 | 0.1010 | 0.1921 | 0.1383 | 0.1109 | 0.1099 | **0.0911** | 0.1208 | *0.1004* | 0.0998 |
| | 0.5 | 0.2587 | 0.3262 | 0.1144 | 0.1792 | 0.1482 | 0.1199 | 0.1236 | **0.1031** | 0.1305 | *0.1093* | 0.1087 |
| | 0.6 | 0.2782 | 0.3343 | 0.1337 | 0.1878 | 0.1705 | 0.1417 | 0.1466 | **0.1238** | 0.1548 | *0.1239* | 0.1234 |
| | 0.7 | 0.2979 | 0.3604 | 0.1522 | 0.1938 | 0.1904 | 0.1593 | 0.1686 | *0.1435* | 0.1714 | **0.1385** | 0.1380 |
| | 0.8 | 0.3387 | 0.3902 | 0.1858 | 0.2181 | 0.2272 | 0.1898 | 0.2050 | *0.1767* | 0.2038 | **0.1650** | 0.1644 |
| | 0.9 | 0.3996 | 0.4332 | 0.2455 | 0.2786 | 0.2871 | 0.2420 | 0.2747 | *0.2407* | 0.2560 | **0.2071** | 0.2068 |

Table S9: Values of $\text{MSE}_\mu$ for various methods in the simulation ($n = 150$)

| | $R^2$ | SCAD | ALASSO | AGG1 | AGG2 | AIC | SAIC | BIC | SBIC | Cp | SMA1 | SMA2 |
|---|---|---|---|---|---|---|---|---|---|---|---|---|
| $\rho = 0.1$ $\alpha = 0.5$ | 0.1 | 0.1480 | 0.2713 | *0.0653* | 0.1718 | 0.0890 | 0.0729 | 0.0722 | **0.0605** | 0.0838 | 0.0690 | 0.0685 |
| | 0.2 | 0.2169 | 0.3209 | **0.0920** | 0.2730 | 0.1249 | 0.1013 | 0.1144 | 0.0988 | 0.1137 | *0.0948* | 0.0942 |
| | 0.3 | 0.2665 | 0.3489 | *0.1169* | 0.2802 | 0.1561 | 0.1317 | 0.1499 | 0.1307 | 0.1446 | **0.1161** | 0.1157 |
| | 0.4 | 0.3075 | 0.3818 | *0.1459* | 0.2762 | 0.1971 | 0.1661 | 0.1896 | 0.1686 | 0.1800 | **0.1415** | 0.1412 |
| | 0.5 | 0.3437 | 0.4101 | *0.1761* | 0.2837 | 0.2314 | 0.1983 | 0.2331 | 0.2088 | 0.2128 | **0.1684** | 0.1681 |
| | 0.6 | 0.3952 | 0.4469 | *0.2170* | 0.2990 | 0.2889 | 0.2499 | 0.2928 | 0.2639 | 0.2638 | **0.2041** | 0.2039 |
| | 0.7 | 0.4597 | 0.5009 | *0.2721* | 0.3358 | 0.3604 | 0.3126 | 0.3682 | 0.3338 | 0.3260 | **0.2527** | 0.2525 |
| | 0.8 | 0.5615 | 0.5767 | *0.3668* | 0.4187 | 0.4579 | 0.4095 | 0.4964 | 0.4564 | 0.4233 | **0.3319** | 0.3318 |
| | 0.9 | 0.7595 | 0.7596 | *0.5869* | 0.6279 | 0.6833 | 0.6313 | 0.7950 | 0.7383 | 0.6581 | **0.5192** | 0.5192 |
| $\rho = 0.1$ $\alpha = 1.0$ | 0.1 | 0.1174 | 0.2332 | 0.0494 | 0.2135 | 0.0572 | 0.0459 | *0.0449* | **0.0357** | 0.0523 | 0.0495 | 0.0490 |
| | 0.2 | 0.1563 | 0.2627 | 0.0602 | 0.2198 | 0.0755 | 0.0600 | 0.0599 | **0.0496** | 0.0683 | *0.0594* | 0.0589 |
| | 0.3 | 0.1909 | 0.2844 | 0.0715 | 0.1954 | 0.0894 | 0.0712 | 0.0718 | **0.0600** | 0.0782 | *0.0686* | 0.0682 |
| | 0.4 | 0.2024 | 0.2871 | 0.0812 | 0.1748 | 0.0910 | 0.0746 | 0.0844 | **0.0714** | 0.0827 | *0.0742* | 0.0737 |
| | 0.5 | 0.2132 | 0.2918 | 0.0899 | 0.1588 | 0.1068 | 0.0870 | 0.0969 | **0.0820** | 0.0962 | *0.0826* | 0.0822 |
| | 0.6 | 0.2391 | 0.3191 | 0.1039 | 0.1604 | 0.1221 | 0.1017 | 0.1120 | *0.0966* | 0.1097 | **0.0946** | 0.0942 |
| | 0.7 | 0.2560 | 0.3335 | 0.1170 | 0.1571 | 0.1393 | 0.1161 | 0.1293 | *0.1110* | 0.1256 | **0.1044** | 0.1040 |
| | 0.8 | 0.2866 | 0.3615 | 0.1379 | 0.1662 | 0.1602 | *0.1362* | 0.1582 | 0.1385 | 0.1470 | **0.1234** | 0.1230 |
| | 0.9 | 0.3313 | 0.3909 | 0.1798 | 0.2030 | 0.2105 | *0.1795* | 0.2059 | 0.1839 | 0.1869 | **0.1546** | 0.1543 |
| $\rho = 0.6$ $\alpha = 0.5$ | 0.1 | 0.1475 | 0.2705 | *0.0633* | 0.1637 | 0.0866 | 0.0690 | 0.0706 | **0.0592** | 0.0796 | 0.0676 | 0.0668 |
| | 0.2 | 0.2184 | 0.3192 | **0.0902** | 0.2346 | 0.1266 | 0.1032 | 0.1153 | 0.0985 | 0.1141 | *0.0948* | 0.0941 |
| | 0.3 | 0.2675 | 0.3489 | **0.1129** | 0.2376 | 0.1541 | 0.1296 | 0.1513 | 0.1327 | 0.1414 | *0.1163* | 0.1157 |
| | 0.4 | 0.3143 | 0.3882 | **0.1417** | 0.2388 | 0.1947 | 0.1656 | 0.1894 | 0.1671 | 0.1794 | *0.1421* | 0.1416 |
| | 0.5 | 0.3484 | 0.4125 | *0.1734* | 0.2494 | 0.2300 | 0.1965 | 0.2314 | 0.2077 | 0.2096 | **0.1688** | 0.1683 |
| | 0.6 | 0.3957 | 0.4477 | *0.2160* | 0.2736 | 0.2826 | 0.2455 | 0.2908 | 0.2614 | 0.2586 | **0.2044** | 0.2041 |
| | 0.7 | 0.4730 | 0.5097 | *0.2727* | 0.3241 | 0.3667 | 0.3215 | 0.3670 | 0.3347 | 0.3328 | **0.2579** | 0.2575 |
| | 0.8 | 0.5553 | 0.5717 | *0.3656* | 0.4108 | 0.4556 | 0.4036 | 0.4977 | 0.4578 | 0.4219 | **0.3314** | 0.3313 |
| | 0.9 | 0.7576 | 0.7531 | *0.5885* | 0.6307 | 0.6801 | 0.6270 | 0.7899 | 0.7369 | 0.6544 | **0.5191** | 0.5191 |
| $\rho = 0.6$ $\alpha = 1.0$ | 0.1 | 0.1186 | 0.2378 | 0.0484 | 0.1866 | 0.0584 | 0.0461 | *0.0454* | **0.0359** | 0.0533 | 0.0503 | 0.0495 |
| | 0.2 | 0.1587 | 0.2707 | *0.0598* | 0.1850 | 0.0756 | 0.0601 | 0.0610 | **0.0500** | 0.0671 | 0.0606 | 0.0598 |
| | 0.3 | 0.1818 | 0.2722 | 0.0685 | 0.1595 | 0.0796 | *0.0663* | 0.0715 | **0.0603** | 0.0751 | 0.0680 | 0.0673 |
| | 0.4 | 0.2065 | 0.2910 | *0.0776* | 0.1360 | 0.0978 | 0.0804 | 0.0852 | **0.0721** | 0.0882 | 0.0777 | 0.0771 |
| | 0.5 | 0.2187 | 0.3059 | 0.0864 | 0.1324 | 0.1044 | 0.0851 | 0.0968 | **0.0824** | 0.0939 | *0.0831* | 0.0825 |
| | 0.6 | 0.2387 | 0.3167 | 0.1018 | 0.1403 | 0.1229 | 0.1001 | 0.1113 | *0.0957* | 0.1074 | **0.0933** | 0.0927 |
| | 0.7 | 0.2542 | 0.3238 | 0.1150 | 0.1450 | 0.1369 | 0.1136 | 0.1314 | *0.1132* | 0.1234 | **0.1046** | 0.1042 |
| | 0.8 | 0.2832 | 0.3503 | 0.1338 | 0.1562 | 0.1563 | *0.1323* | 0.1533 | 0.1340 | 0.1416 | **0.1211** | 0.1207 |
| | 0.9 | 0.3314 | 0.3902 | 0.1826 | 0.2009 | 0.2070 | *0.1765* | 0.2085 | 0.1858 | 0.1859 | **0.1561** | 0.1558 |

Table S10: Values of $\text{MSE}_\mu$ for various methods in the simulation ($n = 200$)

| | $R^2$ | SCAD | ALASSO | AGG1 | AGG2 | AIC | SAIC | BIC | SBIC | Cp | SMA1 | SMA2 |
|---|---|---|---|---|---|---|---|---|---|---|---|---|
| $\rho = 0.1$ $\alpha = 0.5$ | 0.1 | 0.1397 | 0.2638 | *0.0540* | 0.1667 | 0.0723 | 0.0590 | 0.0642 | **0.0540** | 0.0684 | 0.0573 | 0.0569 |
| | 0.2 | 0.2056 | 0.3066 | **0.0783** | 0.2353 | 0.0992 | 0.0827 | 0.0998 | 0.0874 | 0.0942 | *0.0798* | 0.0794 |
| | 0.3 | 0.2459 | 0.3360 | **0.0991** | 0.2261 | 0.1299 | 0.1087 | 0.1311 | 0.1170 | 0.1196 | *0.0995* | 0.0992 |
| | 0.4 | 0.2782 | 0.3621 | *0.1198* | 0.2198 | 0.1578 | 0.1330 | 0.1621 | 0.1452 | 0.1438 | **0.1179** | 0.1175 |
| | 0.5 | 0.3165 | 0.3910 | *0.1471* | 0.2280 | 0.1906 | 0.1636 | 0.2028 | 0.1840 | 0.1759 | **0.1427** | 0.1424 |
| | 0.6 | 0.3576 | 0.4243 | *0.1828* | 0.2482 | 0.2339 | 0.2040 | 0.2519 | 0.2295 | 0.2133 | **0.1730** | 0.1728 |
| | 0.7 | 0.4161 | 0.4721 | *0.2260* | 0.2730 | 0.2986 | 0.2615 | 0.3140 | 0.2889 | 0.2678 | **0.2111** | 0.2109 |
| | 0.8 | 0.4939 | 0.5260 | *0.3035* | 0.3420 | 0.3822 | 0.3416 | 0.4263 | 0.3967 | 0.3522 | **0.2757** | 0.2756 |
| | 0.9 | 0.6704 | 0.6705 | *0.4844* | 0.5150 | 0.5778 | 0.5332 | 0.6781 | 0.6343 | 0.5461 | **0.4271** | 0.4270 |
| $\rho = 0.1$ $\alpha = 1.0$ | 0.1 | 0.1114 | 0.2336 | 0.0408 | 0.1941 | 0.0439 | *0.0354* | 0.0393 | **0.0316** | 0.0414 | 0.0414 | 0.0408 |
| | 0.2 | 0.1428 | 0.2404 | 0.0491 | 0.1800 | 0.0521 | *0.0424* | 0.0503 | **0.0422** | 0.0496 | 0.0470 | 0.0466 |
| | 0.3 | 0.1688 | 0.2645 | 0.0586 | 0.1519 | 0.0651 | *0.0529* | 0.0614 | **0.0517** | 0.0608 | 0.0558 | 0.0553 |
| | 0.4 | 0.1821 | 0.2752 | 0.0665 | 0.1372 | 0.0704 | **0.0583** | 0.0708 | *0.0603* | 0.0665 | 0.0620 | 0.0616 |
| | 0.5 | 0.1946 | 0.2849 | 0.0752 | 0.1274 | 0.0781 | **0.0655** | 0.0814 | 0.0704 | 0.0761 | *0.0680* | 0.0676 |
| | 0.6 | 0.2097 | 0.2987 | 0.0828 | 0.1242 | 0.0889 | *0.0750* | 0.0916 | 0.0802 | 0.0843 | **0.0748** | 0.0743 |
| | 0.7 | 0.2265 | 0.3081 | 0.0956 | 0.1288 | 0.1002 | *0.0849* | 0.1064 | 0.0930 | 0.0955 | **0.0835** | 0.0831 |
| | 0.8 | 0.2551 | 0.3320 | 0.1137 | 0.1365 | 0.1234 | *0.1050* | 0.1294 | 0.1147 | 0.1134 | **0.0993** | 0.0989 |
| | 0.9 | 0.2907 | 0.3611 | 0.1465 | 0.1628 | 0.1547 | *0.1355* | 0.1719 | 0.1548 | 0.1473 | **0.1246** | 0.1244 |
| $\rho = 0.6$ $\alpha = 0.5$ | 0.1 | 0.1394 | 0.2648 | **0.0533** | 0.1567 | 0.0696 | 0.0564 | 0.0637 | *0.0539* | 0.0658 | 0.0575 | 0.0568 |
| | 0.2 | 0.2033 | 0.3107 | **0.0757** | 0.1979 | 0.1018 | 0.0840 | 0.1004 | 0.0876 | 0.0938 | *0.0800* | 0.0793 |
| | 0.3 | 0.2476 | 0.3362 | **0.0965** | 0.1899 | 0.1250 | 0.1058 | 0.1304 | 0.1159 | 0.1182 | *0.0986* | 0.0980 |
| | 0.4 | 0.2815 | 0.3663 | *0.1198* | 0.1922 | 0.1568 | 0.1344 | 0.1634 | 0.1470 | 0.1469 | **0.1197** | 0.1192 |
| | 0.5 | 0.3171 | 0.3911 | *0.1465* | 0.2033 | 0.1958 | 0.1692 | 0.2020 | 0.1830 | 0.1796 | **0.1440** | 0.1436 |
| | 0.6 | 0.3618 | 0.4285 | *0.1830* | 0.2324 | 0.2407 | 0.2115 | 0.2524 | 0.2312 | 0.2209 | **0.1753** | 0.1749 |
| | 0.7 | 0.4211 | 0.4686 | *0.2275* | 0.2650 | 0.2980 | 0.2628 | 0.3186 | 0.2942 | 0.2707 | **0.2134** | 0.2130 |
| | 0.8 | 0.4954 | 0.5267 | *0.3041* | 0.3355 | 0.3845 | 0.3453 | 0.4264 | 0.3958 | 0.3532 | **0.2780** | 0.2778 |
| | 0.9 | 0.6659 | 0.6639 | *0.4790* | 0.5084 | 0.5737 | 0.5257 | 0.6730 | 0.6323 | 0.5358 | **0.4232** | 0.4231 |
| $\rho = 0.6$ $\alpha = 1.0$ | 0.1 | 0.1089 | 0.2313 | 0.0391 | 0.1630 | 0.0452 | *0.0375* | 0.0394 | **0.0321** | 0.0434 | 0.0413 | 0.0407 |
| | 0.2 | 0.1417 | 0.2490 | 0.0468 | 0.1376 | 0.0518 | *0.0427* | 0.0492 | **0.0413** | 0.0492 | 0.0479 | 0.0472 |
| | 0.3 | 0.1719 | 0.2687 | 0.0563 | 0.1217 | 0.0657 | *0.0540* | 0.0610 | **0.0516** | 0.0622 | 0.0565 | 0.0558 |
| | 0.4 | 0.1832 | 0.2675 | 0.0634 | 0.1144 | 0.0697 | **0.0577** | 0.0697 | *0.0592* | 0.0652 | 0.0614 | 0.0608 |
| | 0.5 | 0.1929 | 0.2801 | 0.0695 | 0.1060 | 0.0784 | **0.0649** | 0.0801 | 0.0694 | 0.0734 | *0.0670* | 0.0665 |
| | 0.6 | 0.2136 | 0.3007 | 0.0816 | 0.1080 | 0.0911 | *0.0766* | 0.0926 | 0.0803 | 0.0851 | **0.0760** | 0.0755 |
| | 0.7 | 0.2300 | 0.3117 | 0.0925 | 0.1129 | 0.1040 | *0.0861* | 0.1088 | 0.0951 | 0.0969 | **0.0856** | 0.0852 |
| | 0.8 | 0.2504 | 0.3285 | 0.1092 | 0.1277 | 0.1201 | *0.1040* | 0.1299 | 0.1147 | 0.1148 | **0.0988** | 0.0984 |
| | 0.9 | 0.2928 | 0.3606 | 0.1440 | 0.1604 | 0.1603 | *0.1384* | 0.1719 | 0.1533 | 0.1483 | **0.1253** | 0.1250 |

Table S11: Values of $\text{MSE}_{\bar{y}} \times 100$ for various methods in the simulation ($n = 50$)

| | $R^2$ | SCAD | ALASSO | AGG1 | AGG2 | AIC | SAIC | BIC | SBIC | Cp | SMA1 | SMA2 |
|---|---|---|---|---|---|---|---|---|---|---|---|---|
| $\rho = 0.1$ $\alpha = 0.5$ | 0.1 | 2.4335 | 2.6698 | 2.2872 | *2.2529* | 2.4883 | 2.3826 | 2.2562 | **2.2215** | 2.4473 | 2.3093 | 2.3066 |
| | 0.2 | 2.5462 | 2.7367 | *2.2280* | 2.2324 | 2.4956 | 2.3416 | 2.2761 | **2.1931** | 2.3939 | 2.2766 | 2.2749 |
| | 0.3 | 2.7850 | 2.9158 | **2.4568** | 2.5798 | 2.7053 | 2.6003 | 2.6502 | 2.5397 | 2.6956 | *2.5163* | 2.5157 |
| | 0.4 | 2.8109 | 2.8980 | **2.3587** | 2.4326 | 2.6075 | 2.4991 | 2.4833 | 2.3868 | 2.5770 | *2.3817* | 2.3785 |
| | 0.5 | 3.1337 | 3.1756 | 2.7745 | 2.8770 | 3.0406 | 2.8365 | 2.8972 | **2.7497** | 2.8613 | *2.7604* | 2.7603 |
| | 0.6 | 3.0808 | 3.0271 | *2.7801* | 2.8871 | 2.9715 | 2.7908 | 3.0086 | 2.8380 | 2.9290 | **2.7090** | 2.7093 |
| | 0.7 | 3.2647 | 3.2200 | *2.8178* | 2.8943 | 3.1346 | 2.9061 | 3.1669 | 2.9946 | 3.0606 | **2.7753** | 2.7768 |
| | 0.8 | 3.5069 | 3.4795 | 3.3299 | 3.4625 | 3.4576 | *3.2683* | 3.6524 | 3.4328 | 3.5165 | **3.1462** | 3.1495 |
| | 0.9 | 4.5089 | 4.6381 | 4.3010 | 4.3834 | 4.4075 | *4.1759* | 4.7571 | 4.3817 | 4.3751 | **4.0078** | 4.0076 |
| $\rho = 0.1$ $\alpha = 1.0$ | 0.1 | 2.2717 | 2.4755 | 2.0929 | **2.0175** | 2.2255 | 2.1799 | 2.0376 | *2.0196* | 2.1691 | 2.1167 | 2.1166 |
| | 0.2 | 2.1411 | 2.3163 | **1.9566** | *1.9630* | 2.1769 | 2.0560 | 2.0343 | 1.9740 | 2.1344 | 2.0081 | 2.0073 |
| | 0.3 | 2.4549 | 2.5737 | 2.2355 | *2.2288* | 2.4234 | 2.3095 | 2.3137 | **2.2249** | 2.3545 | 2.2544 | 2.2539 |
| | 0.4 | 2.3531 | 2.5393 | 2.1304 | *2.1210* | 2.3863 | 2.2351 | 2.1554 | **2.1011** | 2.2898 | 2.1703 | 2.1693 |
| | 0.5 | 2.8121 | 2.8316 | **2.4098** | 2.4611 | 2.6258 | 2.5113 | 2.5259 | *2.4402* | 2.5954 | 2.4444 | 2.4454 |
| | 0.6 | 2.5731 | 2.5782 | 2.3298 | 2.3158 | 2.4105 | *2.3020* | 2.3955 | 2.3374 | 2.3844 | **2.2755** | 2.2756 |
| | 0.7 | 2.8484 | 2.9165 | *2.4373* | 2.4497 | 2.6163 | 2.4829 | 2.5951 | 2.4746 | 2.5796 | **2.4307** | 2.4306 |
| | 0.8 | 3.0285 | 3.0600 | 2.6190 | *2.6167* | 2.8403 | 2.6839 | 2.7659 | 2.6272 | 2.7258 | **2.6109** | 2.6108 |
| | 0.9 | 2.8477 | 2.9770 | 2.5190 | 2.5390 | 2.7025 | 2.5491 | 2.6752 | *2.5030* | 2.6529 | **2.4666** | 2.4673 |
| $\rho = 0.6$ $\alpha = 0.5$ | 0.1 | 2.3915 | 2.7468 | 2.1752 | **2.0003** | 2.5194 | 2.3964 | 2.1726 | *2.1245* | 2.4618 | 2.2663 | 2.2695 |
| | 0.2 | 2.6483 | 2.8265 | 2.3662 | **2.2163** | 2.5846 | 2.4821 | 2.3923 | *2.3053* | 2.4834 | 2.3930 | 2.3942 |
| | 0.3 | 2.5746 | 2.6589 | **2.1500** | 2.2033 | 2.3811 | 2.2557 | 2.3014 | *2.1883* | 2.3297 | 2.1914 | 2.1893 |
| | 0.4 | 2.6165 | 2.7429 | 2.2873 | **2.2646** | 2.5805 | 2.4101 | 2.3743 | *2.2729* | 2.4860 | 2.3115 | 2.3098 |
| | 0.5 | 3.0528 | 3.1191 | 2.6769 | 2.7578 | 2.8583 | 2.7153 | 2.7741 | **2.6373** | 2.7766 | *2.6466* | 2.6462 |
| | 0.6 | 3.2916 | 3.2010 | *2.6825* | 2.6999 | 2.9900 | 2.8066 | 2.8824 | 2.7381 | 2.9402 | **2.6675** | 2.6697 |
| | 0.7 | 3.5306 | 3.4177 | *3.0254* | 3.0348 | 3.3227 | 3.1476 | 3.4166 | 3.2129 | 3.3193 | **3.0136** | 3.0150 |
| | 0.8 | 4.1697 | 4.1563 | *3.7728* | 3.8221 | 3.9589 | 3.8055 | 4.1484 | 3.9393 | 4.0017 | **3.6674** | 3.6656 |
| | 0.9 | 4.8924 | 4.9501 | 4.7133 | 4.8579 | 4.8560 | *4.6316* | 5.2348 | 4.8921 | 4.8839 | **4.5143** | 4.5148 |
| $\rho = 0.6$ $\alpha = 1.0$ | 0.1 | 2.5971 | 2.8473 | 2.3655 | **2.1549** | 2.5384 | 2.4800 | 2.3442 | *2.3190* | 2.5090 | 2.3989 | 2.4012 |
| | 0.2 | 2.3051 | 2.4699 | 2.0473 | **1.9967** | 2.2119 | 2.1026 | 2.1028 | *2.0429* | 2.1347 | 2.0589 | 2.0616 |
| | 0.3 | 2.4713 | 2.7001 | *2.1174* | **2.0364** | 2.4100 | 2.2722 | 2.1914 | 2.1241 | 2.3570 | 2.2103 | 2.2104 |
| | 0.4 | 2.5871 | 2.6565 | 2.2429 | **2.1631** | 2.4033 | 2.3059 | 2.3032 | *2.2325* | 2.3623 | 2.2574 | 2.2551 |
| | 0.5 | 2.7205 | 2.7829 | *2.2233* | **2.1594** | 2.4618 | 2.3513 | 2.3028 | 2.2453 | 2.4060 | 2.3076 | 2.3067 |
| | 0.6 | 2.7210 | 2.8243 | **2.2686** | *2.2765* | 2.5189 | 2.4042 | 2.3905 | 2.3226 | 2.4634 | 2.3379 | 2.3380 |
| | 0.7 | 2.6130 | 2.7251 | 2.3460 | 2.3361 | 2.4467 | 2.3705 | 2.3761 | **2.3026** | 2.3825 | *2.3042* | 2.3039 |
| | 0.8 | 2.5794 | 2.7216 | 2.4288 | 2.3991 | 2.5618 | 2.4384 | 2.4856 | *2.3600* | 2.5366 | **2.3493** | 2.3500 |
| | 0.9 | 2.8513 | 2.8443 | 2.5372 | *2.5237* | 2.7227 | 2.5484 | 2.7370 | 2.5873 | 2.7062 | **2.4689** | 2.4688 |

Table S12: Values of $\text{MSE}_{\bar{y}} \times 100$ for various methods in the simulation ($n = 100$)

| | $R^2$ | SCAD | ALASSO | AGG1 | AGG2 | AIC | SAIC | BIC | SBIC | Cp | SMA1 | SMA2 |
|---|---|---|---|---|---|---|---|---|---|---|---|---|
| $\rho = 0.1$ $\alpha = 0.5$ | 0.1 | 1.1503 | 1.2286 | **1.0401** | 1.0535 | 1.0557 | *1.0443* | 1.0671 | 1.0543 | 1.0597 | 1.0455 | 1.0458 |
| | 0.2 | 1.1663 | 1.2341 | **1.0238** | 1.0561 | 1.0984 | 1.0676 | 1.0767 | *1.0517* | 1.0787 | 1.0578 | 1.0574 |
| | 0.3 | 1.2952 | 1.3857 | *1.1501* | 1.1697 | 1.2335 | 1.1925 | 1.1704 | 1.1514 | 1.1904 | **1.1456** | 1.1452 |
| | 0.4 | 1.2769 | 1.3042 | *1.0862* | 1.1117 | 1.1772 | 1.1243 | 1.1434 | 1.0990 | 1.1319 | **1.0753** | 1.0755 |
| | 0.5 | 1.2736 | 1.3206 | **1.0852** | 1.1212 | 1.1820 | 1.1286 | 1.1929 | 1.1430 | 1.1528 | *1.0870* | 1.0870 |
| | 0.6 | 1.3150 | 1.3639 | **1.1723** | 1.2172 | 1.2959 | 1.2307 | 1.2749 | 1.2440 | 1.2731 | *1.1836* | 1.1837 |
| | 0.7 | 1.4016 | 1.4723 | *1.2106* | 1.2270 | 1.3302 | 1.2814 | 1.3098 | 1.2560 | 1.3115 | **1.2014** | 1.2006 |
| | 0.8 | 1.5240 | 1.5060 | 1.3355 | 1.3640 | 1.3833 | *1.3307* | 1.4530 | 1.4056 | 1.3627 | **1.2707** | 1.2706 |
| | 0.9 | 1.8465 | 1.8643 | 1.7636 | 1.7632 | 1.8144 | *1.7475* | 1.9104 | 1.8426 | 1.8086 | **1.6475** | 1.6474 |
| $\rho = 0.1$ $\alpha = 1.0$ | 0.1 | 1.0433 | 1.2084 | 1.0113 | *1.0018* | 1.0368 | 1.0231 | 1.0042 | **0.9895** | 1.0281 | 1.0202 | 1.0193 |
| | 0.2 | 1.1088 | 1.1492 | 0.9932 | 1.0038 | 1.0112 | 0.9988 | 1.0099 | **0.9892** | 1.0027 | *0.9927* | 0.9922 |
| | 0.3 | 1.1625 | 1.2161 | *1.0463* | **1.0389** | 1.1025 | 1.0764 | 1.0796 | 1.0581 | 1.0942 | 1.0510 | 1.0507 |
| | 0.4 | 1.1691 | 1.2236 | **1.0262** | 1.0318 | 1.0594 | 1.0343 | 1.0549 | *1.0313* | 1.0334 | 1.0399 | 1.0392 |
| | 0.5 | 1.2006 | 1.2565 | **1.0432** | 1.0497 | 1.0955 | 1.0683 | 1.0688 | *1.0464* | 1.0650 | 1.0508 | 1.0503 |
| | 0.6 | 1.2217 | 1.3084 | *1.0671* | **1.0567** | 1.1155 | 1.0978 | 1.1051 | 1.0864 | 1.0960 | 1.0830 | 1.0826 |
| | 0.7 | 1.2261 | 1.3515 | 1.1022 | 1.0941 | 1.1461 | 1.1076 | *1.0753* | **1.0588** | 1.1020 | 1.0854 | 1.0847 |
| | 0.8 | 1.2019 | 1.2517 | 1.0374 | **1.0251** | 1.0726 | 1.0567 | 1.0794 | 1.0484 | 1.0880 | *1.0321* | 1.0321 |
| | 0.9 | 1.2460 | 1.2632 | *1.1143* | 1.1209 | 1.1986 | 1.1475 | 1.1543 | 1.1229 | 1.1804 | **1.1000** | 1.0998 |
| $\rho = 0.6$ $\alpha = 0.5$ | 0.1 | 1.1666 | 1.2565 | 1.0842 | **1.0410** | 1.1466 | 1.1109 | 1.1011 | 1.0803 | 1.1286 | *1.0787* | 1.0802 |
| | 0.2 | 1.2106 | 1.3123 | 1.0418 | **1.0237** | 1.1090 | 1.0931 | 1.0508 | *1.0324* | 1.0969 | 1.0659 | 1.0667 |
| | 0.3 | 1.2996 | 1.3339 | **1.0908** | 1.1166 | 1.1935 | 1.1484 | 1.1591 | 1.1467 | 1.1821 | *1.1152* | 1.1166 |
| | 0.4 | 1.4131 | 1.4888 | *1.1718* | **1.1503** | 1.2647 | 1.2327 | 1.2109 | 1.1898 | 1.2508 | 1.1998 | 1.1998 |
| | 0.5 | 1.3471 | 1.3981 | **1.1509** | *1.1538* | 1.2440 | 1.1993 | 1.2012 | 1.1864 | 1.1975 | 1.1638 | 1.1645 |
| | 0.6 | 1.4055 | 1.4426 | **1.1699** | *1.1709* | 1.2816 | 1.2292 | 1.2820 | 1.2443 | 1.2541 | 1.1826 | 1.1822 |
| | 0.7 | 1.4719 | 1.4760 | **1.1579** | *1.1599* | 1.3390 | 1.2881 | 1.2715 | 1.2317 | 1.3088 | 1.2021 | 1.2019 |
| | 0.8 | 1.5626 | 1.5659 | *1.4194* | 1.4373 | 1.4986 | 1.4256 | 1.5566 | 1.5029 | 1.4644 | **1.3732** | 1.3736 |
| | 0.9 | 1.9126 | 1.9239 | 1.8319 | 1.8329 | 1.8808 | *1.7986* | 2.1261 | 1.9962 | 1.8433 | **1.7092** | 1.7088 |
| $\rho = 0.6$ $\alpha = 1.0$ | 0.1 | 1.2033 | 1.3168 | 1.0721 | **1.0053** | 1.0927 | 1.0851 | 1.0687 | *1.0622* | 1.0920 | 1.0860 | 1.0858 |
| | 0.2 | 1.1221 | 1.2565 | 1.0349 | **0.9896** | 1.0841 | 1.0648 | 1.0392 | *1.0279* | 1.0592 | 1.0503 | 1.0504 |
| | 0.3 | 1.2143 | 1.2894 | 1.0244 | **0.9909** | 1.0588 | 1.0319 | 1.0205 | *1.0065* | 1.0424 | 1.0331 | 1.0332 |
| | 0.4 | 1.3490 | 1.4410 | 1.1794 | **1.1415** | 1.2368 | 1.2062 | 1.1687 | *1.1561* | 1.2239 | 1.1934 | 1.1932 |
| | 0.5 | 1.2044 | 1.2667 | 1.0242 | **0.9938** | 1.0912 | 1.0541 | 1.0242 | *1.0173* | 1.0602 | 1.0338 | 1.0334 |
| | 0.6 | 1.2468 | 1.2843 | *1.0818* | **1.0710** | 1.1082 | 1.0865 | 1.1035 | 1.0849 | 1.0940 | 1.0854 | 1.0856 |
| | 0.7 | 1.2422 | 1.3343 | *1.1069* | **1.0956** | 1.1786 | 1.1449 | 1.1604 | 1.1254 | 1.1678 | 1.1181 | 1.1184 |
| | 0.8 | 1.3047 | 1.3528 | 1.1122 | **1.1026** | 1.1880 | 1.1403 | 1.1364 | 1.1157 | 1.1547 | *1.1099* | 1.1101 |
| | 0.9 | 1.3890 | 1.3790 | *1.1977* | 1.2078 | 1.2524 | 1.2062 | 1.2353 | 1.2116 | 1.2282 | **1.1855** | 1.1856 |

Table S13: Values of $\text{MSE}_{\bar{y}} \times 100$ for various methods in the simulation ($n = 150$)

| | $R^2$ | SCAD | ALASSO | AGG1 | AGG2 | AIC | SAIC | BIC | SBIC | Cp | SMA1 | SMA2 |
|---|---|---|---|---|---|---|---|---|---|---|---|---|
| $\rho = 0.1$ $\alpha = 0.5$ | 0.1 | 0.7126 | 0.7911 | *0.6431* | **0.6369** | 0.6677 | 0.6591 | 0.6543 | 0.6452 | 0.6657 | 0.6488 | 0.6486 |
| | 0.2 | 0.7631 | 0.8120 | *0.7283* | 0.7481 | 0.7471 | 0.7361 | 0.7525 | 0.7410 | 0.7454 | **0.7232** | 0.7231 |
| | 0.3 | 0.7906 | 0.8286 | 0.7027 | 0.7207 | 0.7254 | 0.7173 | *0.6996* | **0.6951** | 0.7227 | 0.7012 | 0.7010 |
| | 0.4 | 0.8756 | 0.9505 | **0.7780** | 0.8076 | 0.8064 | 0.7890 | 0.8236 | 0.8034 | 0.7921 | *0.7832* | 0.7830 |
| | 0.5 | 0.8218 | 0.8269 | **0.7172** | 0.7414 | 0.7520 | 0.7437 | 0.7755 | 0.7601 | 0.7465 | *0.7174* | 0.7174 |
| | 0.6 | 0.8479 | 0.8583 | *0.7332* | 0.7466 | 0.7767 | 0.7488 | 0.7533 | 0.7410 | 0.7603 | **0.7129** | 0.7126 |
| | 0.7 | 0.9543 | 0.9810 | *0.8254* | 0.8376 | 0.8679 | 0.8378 | 0.8598 | 0.8535 | 0.8373 | **0.8010** | 0.8006 |
| | 0.8 | 1.0274 | 1.0483 | *0.8910* | 0.9025 | 0.9535 | 0.9140 | 0.9920 | 0.9573 | 0.9220 | **0.8732** | 0.8732 |
| | 0.9 | 1.0842 | 1.0888 | *0.9519* | 0.9572 | 1.0161 | 0.9876 | 1.1176 | 1.0706 | 1.0128 | **0.9181** | 0.9178 |
| $\rho = 0.1$ $\alpha = 1.0$ | 0.1 | 0.6702 | 0.7389 | **0.6344** | 0.6435 | 0.6450 | 0.6405 | 0.6421 | *0.6359* | 0.6435 | 0.6390 | 0.6387 |
| | 0.2 | 0.7025 | 0.7512 | 0.6285 | 0.6368 | 0.6332 | *0.6260* | 0.6293 | **0.6221** | 0.6277 | 0.6285 | 0.6282 |
| | 0.3 | 0.7565 | 0.8503 | 0.6705 | 0.6794 | 0.6765 | 0.6676 | *0.6646* | **0.6569** | 0.6805 | 0.6714 | 0.6710 |
| | 0.4 | 0.8063 | 0.8304 | **0.7088** | *0.7150* | 0.7233 | 0.7193 | 0.7340 | 0.7214 | 0.7226 | 0.7201 | 0.7197 |
| | 0.5 | 0.7214 | 0.7550 | *0.6075* | **0.6040** | 0.6293 | 0.6192 | 0.6206 | 0.6103 | 0.6260 | 0.6169 | 0.6165 |
| | 0.6 | 0.7718 | 0.8040 | **0.6825** | *0.6825* | 0.7158 | 0.6994 | 0.7060 | 0.7005 | 0.6997 | 0.6918 | 0.6918 |
| | 0.7 | 0.7463 | 0.8012 | *0.6837* | 0.6837 | 0.7144 | 0.6936 | 0.7021 | 0.6937 | 0.7031 | **0.6815** | 0.6815 |
| | 0.8 | 0.8044 | 0.8684 | 0.7185 | *0.7155* | 0.7286 | 0.7180 | 0.7317 | 0.7183 | 0.7221 | **0.7112** | 0.7108 |
| | 0.9 | 0.8369 | 0.8771 | **0.7500** | *0.7530* | 0.7827 | 0.7692 | 0.8085 | 0.7901 | 0.7645 | 0.7567 | 0.7568 |
| $\rho = 0.6$ $\alpha = 0.5$ | 0.1 | 0.7156 | 0.7957 | 0.6448 | **0.6310** | 0.6586 | 0.6540 | 0.6575 | *0.6446* | 0.6577 | 0.6486 | 0.6495 |
| | 0.2 | 0.7860 | 0.8692 | *0.6795* | **0.6727** | 0.7180 | 0.7041 | 0.7005 | 0.6864 | 0.7102 | 0.6929 | 0.6931 |
| | 0.3 | 0.7841 | 0.8566 | **0.6742** | 0.6766 | 0.6940 | 0.6877 | 0.7349 | 0.7163 | 0.7005 | *0.6765* | 0.6769 |
| | 0.4 | 0.8497 | 0.9167 | **0.7000** | 0.7101 | 0.7426 | 0.7190 | 0.7396 | 0.7255 | 0.7221 | *0.7075* | 0.7078 |
| | 0.5 | 0.8547 | 0.9146 | 0.7339 | **0.7306** | 0.7648 | 0.7416 | 0.7717 | 0.7523 | 0.7435 | *0.7316* | 0.7310 |
| | 0.6 | 0.8357 | 0.8618 | 0.7445 | *0.7444* | 0.7898 | 0.7713 | 0.7877 | 0.7737 | 0.7876 | **0.7376** | 0.7379 |
| | 0.7 | 0.9575 | 0.9768 | *0.8115* | 0.8158 | 0.8674 | 0.8434 | 0.8501 | 0.8285 | 0.8404 | **0.8057** | 0.8056 |
| | 0.8 | 0.9735 | 0.9651 | 0.8483 | *0.8469* | 0.9254 | 0.8830 | 0.9167 | 0.8945 | 0.8973 | **0.8303** | 0.8303 |
| | 0.9 | 1.0859 | 1.0980 | *0.9783* | 0.9877 | 1.0321 | 1.0039 | 1.0900 | 1.0658 | 1.0465 | **0.9402** | 0.9403 |
| $\rho = 0.6$ $\alpha = 1.0$ | 0.1 | 0.7284 | 0.7795 | 0.6821 | **0.6466** | 0.6959 | 0.6891 | 0.6744 | *0.6697* | 0.6902 | 0.6839 | 0.6848 |
| | 0.2 | 0.6988 | 0.8081 | 0.6431 | **0.6275** | 0.6731 | 0.6580 | *0.6414* | 0.6417 | 0.6602 | 0.6524 | 0.6532 |
| | 0.3 | 0.8009 | 0.8583 | 0.6883 | **0.6657** | 0.7056 | 0.7009 | 0.6885 | *0.6814* | 0.7013 | 0.6984 | 0.6983 |
| | 0.4 | 0.7307 | 0.7942 | *0.6571* | **0.6482** | 0.6689 | 0.6618 | 0.6631 | 0.6596 | 0.6695 | 0.6606 | 0.6608 |
| | 0.5 | 0.7506 | 0.8288 | *0.6607* | **0.6528** | 0.6833 | 0.6660 | 0.6722 | 0.6665 | 0.6733 | 0.6616 | 0.6612 |
| | 0.6 | 0.7336 | 0.7966 | 0.6571 | **0.6479** | 0.6677 | 0.6574 | 0.6795 | 0.6652 | *0.6550* | 0.6633 | 0.6632 |
| | 0.7 | 0.8401 | 0.8887 | 0.7215 | 0.7093 | 0.7467 | 0.7302 | *0.7033* | **0.6949** | 0.7263 | 0.7131 | 0.7128 |
| | 0.8 | 0.8308 | 0.8775 | *0.7116* | **0.7019** | 0.7368 | 0.7185 | 0.7319 | 0.7213 | 0.7325 | 0.7136 | 0.7138 |
| | 0.9 | 0.7991 | 0.8223 | 0.6738 | 0.6753 | 0.6904 | 0.6733 | 0.6864 | 0.6771 | **0.6646** | *0.6665* | 0.6664 |

Table S14: Values of $\mathrm{MSE}_{\bar{y}} \times 100$ for various methods in the simulation ($n = 200$)

| | $R^2$ | SCAD | ALASSO | AGG1 | AGG2 | AIC | SAIC | BIC | SBIC | Cp | SMA1 | SMA2 |
|---|---|---|---|---|---|---|---|---|---|---|---|---|
| $\rho = 0.1$ $\alpha = 0.5$ | 0.1 | 0.4882 | 0.5472 | 0.4526 | 0.4644 | 0.4642 | 0.4530 | 0.4543 | *0.4522* | 0.4585 | **0.4513** | 0.4512 |
| | 0.2 | 0.5389 | 0.5798 | *0.4774* | 0.4906 | 0.4854 | 0.4784 | 0.4865 | 0.4802 | 0.4828 | **0.4765** | 0.4762 |
| | 0.3 | 0.5851 | 0.6105 | **0.5140** | 0.5289 | 0.5466 | 0.5272 | 0.5292 | 0.5234 | 0.5367 | *0.5197* | 0.5196 |
| | 0.4 | 0.5197 | 0.5822 | **0.4470** | 0.4593 | 0.4802 | 0.4649 | 0.4599 | *0.4539* | 0.4671 | 0.4561 | 0.4556 |
| | 0.5 | 0.6224 | 0.6426 | **0.5585** | *0.5647* | 0.5936 | 0.5807 | 0.5813 | 0.5740 | 0.5802 | 0.5649 | 0.5650 |
| | 0.6 | 0.6400 | 0.6768 | **0.5595** | 0.5676 | 0.6000 | 0.5806 | 0.6035 | 0.5902 | 0.5815 | *0.5596* | 0.5595 |
| | 0.7 | 0.6064 | 0.6371 | *0.4925* | 0.5002 | 0.5444 | 0.5260 | 0.5071 | 0.4977 | 0.5213 | **0.4920** | 0.4919 |
| | 0.8 | 0.6620 | 0.6778 | *0.5797* | 0.5847 | 0.6120 | 0.5910 | 0.6358 | 0.6238 | 0.5935 | **0.5676** | 0.5676 |
| | 0.9 | 0.7398 | 0.7437 | *0.6909* | 0.6978 | 0.7157 | 0.6997 | 0.7833 | 0.7617 | 0.7165 | **0.6496** | 0.6497 |
| $\rho = 0.1$ $\alpha = 1.0$ | 0.1 | 0.4892 | 0.5524 | 0.4611 | 0.4661 | 0.4618 | **0.4584** | 0.4611 | *0.4589* | 0.4598 | 0.4607 | 0.4605 |
| | 0.2 | 0.5438 | 0.5692 | 0.4844 | 0.4850 | 0.4833 | **0.4795** | 0.4908 | 0.4873 | *0.4819* | 0.4821 | 0.4820 |
| | 0.3 | 0.5327 | 0.5719 | **0.4915** | 0.4944 | 0.4978 | *0.4938* | 0.5050 | 0.5002 | 0.4973 | 0.4957 | 0.4954 |
| | 0.4 | 0.5184 | 0.5745 | 0.4817 | 0.4824 | 0.4812 | *0.4759* | 0.4844 | 0.4791 | **0.4734** | 0.4818 | 0.4813 |
| | 0.5 | 0.5721 | 0.6206 | 0.5182 | **0.5108** | 0.5294 | 0.5231 | 0.5266 | *0.5180* | 0.5291 | 0.5202 | 0.5201 |
| | 0.6 | 0.5215 | 0.5624 | 0.4612 | **0.4605** | 0.4696 | 0.4611 | 0.4814 | 0.4754 | 0.4632 | *0.4605* | 0.4606 |
| | 0.7 | 0.5267 | 0.5693 | 0.4778 | 0.4760 | 0.4743 | **0.4656** | 0.4743 | *0.4683* | 0.4730 | 0.4688 | 0.4686 |
| | 0.8 | 0.5977 | 0.6413 | 0.5190 | **0.5157** | 0.5376 | 0.5245 | 0.5197 | *0.5173* | 0.5227 | 0.5180 | 0.5177 |
| | 0.9 | 0.5844 | 0.6105 | 0.5151 | 0.5128 | **0.5036** | *0.5045* | 0.5348 | 0.5288 | 0.5086 | 0.5072 | 0.5071 |
| $\rho = 0.6$ $\alpha = 0.5$ | 0.1 | 0.4808 | 0.5451 | 0.4411 | **0.4267** | 0.4525 | 0.4484 | 0.4433 | *0.4394* | 0.4482 | 0.4480 | 0.4481 |
| | 0.2 | 0.5351 | 0.6023 | *0.4718* | **0.4702** | 0.4899 | 0.4764 | 0.4849 | 0.4783 | 0.4809 | 0.4751 | 0.4755 |
| | 0.3 | 0.5567 | 0.5957 | *0.4827* | **0.4825** | 0.4987 | 0.4888 | 0.4975 | 0.4916 | 0.4933 | 0.4870 | 0.4873 |
| | 0.4 | 0.6026 | 0.6387 | *0.5155* | **0.5106** | 0.5256 | 0.5166 | 0.5353 | 0.5306 | 0.5225 | 0.5184 | 0.5184 |
| | 0.5 | 0.5791 | 0.6045 | **0.5029** | *0.5075* | 0.5189 | 0.5112 | 0.5326 | 0.5267 | 0.5093 | 0.5079 | 0.5080 |
| | 0.6 | 0.6844 | 0.7046 | **0.5696** | *0.5738* | 0.6162 | 0.5997 | 0.5907 | 0.5823 | 0.6102 | 0.5751 | 0.5751 |
| | 0.7 | 0.6650 | 0.6849 | **0.5417** | *0.5428* | 0.5901 | 0.5727 | 0.5915 | 0.5792 | 0.5813 | 0.5474 | 0.5474 |
| | 0.8 | 0.7305 | 0.7341 | *0.6061* | 0.6084 | 0.6473 | 0.6328 | 0.6691 | 0.6563 | 0.6345 | **0.6021** | 0.6019 |
| | 0.9 | 0.7556 | 0.7520 | *0.6461* | 0.6520 | 0.7087 | 0.6848 | 0.7295 | 0.7109 | 0.6923 | **0.6396** | 0.6394 |
| $\rho = 0.6$ $\alpha = 1.0$ | 0.1 | 0.5408 | 0.6048 | 0.5066 | **0.4835** | 0.5118 | 0.5056 | 0.5015 | *0.4997* | 0.5068 | 0.5062 | 0.5068 |
| | 0.2 | 0.4905 | 0.5548 | 0.4455 | **0.4248** | 0.4574 | 0.4531 | 0.4443 | *0.4407* | 0.4515 | 0.4488 | 0.4491 |
| | 0.3 | 0.5334 | 0.5714 | 0.4871 | **0.4786** | 0.4914 | 0.4875 | 0.4875 | *0.4840* | 0.4919 | 0.4844 | 0.4847 |
| | 0.4 | 0.6008 | 0.6310 | *0.5101* | **0.4985** | 0.5262 | 0.5136 | 0.5231 | 0.5173 | 0.5203 | 0.5150 | 0.5151 |
| | 0.5 | 0.5770 | 0.6271 | 0.5013 | **0.4906** | 0.5035 | *0.4965* | 0.5056 | 0.4988 | 0.4977 | 0.5024 | 0.5024 |
| | 0.6 | 0.5823 | 0.6122 | *0.5182* | **0.5122** | 0.5253 | 0.5219 | 0.5325 | 0.5266 | 0.5186 | 0.5239 | 0.5242 |
| | 0.7 | 0.5608 | 0.5960 | 0.5055 | **0.5017** | 0.5171 | 0.5055 | 0.5197 | 0.5116 | 0.5208 | *0.5048* | 0.5051 |
| | 0.8 | 0.5569 | 0.6051 | 0.4862 | *0.4847* | 0.4955 | **0.4843** | 0.4934 | 0.4898 | 0.4929 | 0.4873 | 0.4869 |
| | 0.9 | 0.5874 | 0.6251 | *0.5251* | 0.5254 | 0.5430 | 0.5319 | 0.5386 | 0.5308 | 0.5401 | **0.5221** | 0.5223 |

## S5.3 Simulation results for $\sigma^2 = 1$

Table S15: Values of $\text{MSE}_\mu$ for various methods in the simulation ($n = 50$)

| | $R^2$ | SCAD | ALASSO | AGG1 | AGG2 | AIC | SAIC | BIC | SBIC | Cp | SMA1 | SMA2 |
|---|---|---|---|---|---|---|---|---|---|---|---|---|
| $\rho = 0.1$ $\alpha = 0.5$ | 0.1 | 0.1797 | 0.2224 | 0.1130 | 0.1801 | 0.1686 | 0.1289 | 0.1076 | **0.0880** | 0.1587 | *0.1074* | 0.1070 |
| | 0.2 | 0.2530 | 0.2734 | 0.1629 | 0.3857 | 0.2222 | 0.1760 | 0.1807 | **0.1480** | 0.2149 | *0.1527* | 0.1525 |
| | 0.3 | 0.3240 | 0.3236 | 0.2092 | 0.5471 | 0.2662 | 0.2173 | 0.2537 | *0.2052* | 0.2578 | **0.1959** | 0.1957 |
| | 0.4 | 0.3769 | 0.3684 | *0.2664* | 0.6058 | 0.3223 | 0.2666 | 0.3268 | 0.2708 | 0.3158 | **0.2437** | 0.2437 |
| | 0.5 | 0.4243 | 0.4153 | 0.3317 | 0.6081 | 0.3736 | *0.3150* | 0.4001 | 0.3384 | 0.3702 | **0.2979** | 0.2979 |
| | 0.6 | 0.4886 | 0.4917 | 0.4227 | 0.6297 | 0.4564 | *0.4017* | 0.5064 | 0.4362 | 0.4575 | **0.3767** | 0.3767 |
| | 0.7 | 0.5806 | 0.5937 | 0.5570 | 0.7131 | 0.5698 | *0.5112* | 0.6542 | 0.5683 | 0.5740 | **0.4888** | 0.4887 |
| | 0.8 | 0.7693 | 0.8021 | 0.8056 | 0.9370 | 0.7715 | *0.7211* | 0.9043 | 0.7991 | 0.7845 | **0.6953** | 0.6953 |
| | 0.9 | 1.3112 | 1.3711 | 1.4672 | 1.6090 | 1.3388 | *1.2965* | 1.5358 | 1.4167 | 1.3633 | **1.2613** | 1.2614 |
| $\rho = 0.1$ $\alpha = 1.0$ | 0.1 | 0.1431 | 0.1896 | 0.0929 | 0.2576 | 0.1233 | 0.0967 | *0.0764* | **0.0614** | 0.1134 | 0.0814 | 0.0810 |
| | 0.2 | 0.1863 | 0.2156 | 0.1208 | 0.4636 | 0.1525 | 0.1166 | 0.1154 | **0.0889** | 0.1412 | *0.1006* | 0.1003 |
| | 0.3 | 0.2348 | 0.2470 | 0.1475 | 0.4768 | 0.1815 | 0.1433 | 0.1517 | **0.1180** | 0.1735 | *0.1251* | 0.1248 |
| | 0.4 | 0.2590 | 0.2568 | 0.1644 | 0.4125 | 0.1909 | 0.1515 | 0.1721 | **0.1380** | 0.1853 | *0.1382* | 0.1381 |
| | 0.5 | 0.2778 | 0.2719 | 0.1852 | 0.3587 | 0.2111 | 0.1691 | 0.1995 | *0.1630* | 0.2029 | **0.1558** | 0.1557 |
| | 0.6 | 0.2924 | 0.2890 | 0.2149 | 0.3457 | 0.2405 | 0.1950 | 0.2269 | *0.1862* | 0.2337 | **0.1791** | 0.1789 |
| | 0.7 | 0.3149 | 0.3127 | 0.2492 | 0.3484 | 0.2643 | *0.2177* | 0.2667 | 0.2183 | 0.2564 | **0.2047** | 0.2046 |
| | 0.8 | 0.3411 | 0.3429 | 0.3144 | 0.3907 | 0.3102 | *0.2620* | 0.3309 | 0.2783 | 0.3090 | **0.2511** | 0.2511 |
| | 0.9 | 0.3943 | 0.4066 | 0.4318 | 0.4970 | 0.3833 | *0.3399* | 0.4390 | 0.3718 | 0.3865 | **0.3279** | 0.3280 |
| $\rho = 0.6$ $\alpha = 0.5$ | 0.1 | 0.1831 | 0.2194 | 0.1147 | 0.1811 | 0.1627 | 0.1273 | 0.1081 | **0.0870** | 0.1515 | *0.1072* | 0.1065 |
| | 0.2 | 0.2487 | 0.2703 | 0.1616 | 0.3753 | 0.2163 | 0.1711 | 0.1836 | **0.1500** | 0.2097 | *0.1527* | 0.1522 |
| | 0.3 | 0.3283 | 0.3325 | *0.2046* | 0.5017 | 0.2796 | 0.2218 | 0.2537 | 0.2050 | 0.2696 | **0.1987** | 0.1982 |
| | 0.4 | 0.3775 | 0.3715 | *0.2557* | 0.5344 | 0.3233 | 0.2655 | 0.3251 | 0.2688 | 0.3181 | **0.2439** | 0.2436 |
| | 0.5 | 0.4286 | 0.4213 | 0.3354 | 0.5495 | 0.3820 | *0.3240* | 0.4091 | 0.3440 | 0.3781 | **0.3029** | 0.3027 |
| | 0.6 | 0.4951 | 0.4940 | 0.4197 | 0.5880 | 0.4556 | *0.4005* | 0.5106 | 0.4377 | 0.4561 | **0.3801** | 0.3800 |
| | 0.7 | 0.5901 | 0.6053 | 0.5567 | 0.6852 | 0.5767 | *0.5188* | 0.6610 | 0.5784 | 0.5811 | **0.4976** | 0.4976 |
| | 0.8 | 0.7762 | 0.8056 | 0.8019 | 0.9181 | 0.7819 | *0.7283* | 0.9109 | 0.8074 | 0.7949 | **0.6999** | 0.6999 |
| | 0.9 | 1.3052 | 1.3616 | 1.4714 | 1.6188 | 1.3395 | *1.2962* | 1.5494 | 1.4255 | 1.3667 | **1.2601** | 1.2602 |
| $\rho = 0.6$ $\alpha = 1.0$ | 0.1 | 0.1419 | 0.1886 | 0.0920 | 0.2571 | 0.1240 | 0.0949 | *0.0752* | **0.0590** | 0.1152 | 0.0797 | 0.0790 |
| | 0.2 | 0.1921 | 0.2273 | 0.1170 | 0.4247 | 0.1569 | 0.1220 | 0.1193 | **0.0919** | 0.1490 | *0.1057* | 0.1051 |
| | 0.3 | 0.2268 | 0.2372 | 0.1387 | 0.4159 | 0.1724 | 0.1361 | 0.1508 | **0.1193** | 0.1633 | *0.1232* | 0.1228 |
| | 0.4 | 0.2567 | 0.2579 | 0.1523 | 0.3488 | 0.1884 | 0.1490 | 0.1733 | *0.1352* | 0.1819 | **0.1350** | 0.1347 |
| | 0.5 | 0.2796 | 0.2786 | 0.1812 | 0.3246 | 0.2139 | 0.1696 | 0.1979 | *0.1630* | 0.2067 | **0.1590** | 0.1586 |
| | 0.6 | 0.2880 | 0.2858 | 0.2114 | 0.3287 | 0.2339 | 0.1892 | 0.2288 | *0.1861* | 0.2267 | **0.1770** | 0.1768 |
| | 0.7 | 0.3108 | 0.3096 | 0.2462 | 0.3272 | 0.2642 | *0.2164* | 0.2706 | 0.2213 | 0.2582 | **0.2046** | 0.2044 |
| | 0.8 | 0.3419 | 0.3450 | 0.3082 | 0.3830 | 0.3045 | *0.2592* | 0.3276 | 0.2726 | 0.3027 | **0.2465** | 0.2465 |
| | 0.9 | 0.4098 | 0.4196 | 0.4287 | 0.4921 | 0.3992 | *0.3561* | 0.4549 | 0.3909 | 0.4030 | **0.3442** | 0.3443 |

Table S16: Values of $\text{MSE}_\mu$ for various methods in the simulation ($n = 100$)

| | $R^2$ | SCAD | ALASSO | AGG1 | AGG2 | AIC | SAIC | BIC | SBIC | Cp | SMA1 | SMA2 |
|---|---|---|---|---|---|---|---|---|---|---|---|---|
| $\rho = 0.1$ $\alpha = 0.5$ | 0.1 | 0.1204 | 0.1384 | 0.0750 | 0.1823 | 0.0966 | 0.0757 | 0.0802 | **0.0664** | 0.0937 | *0.0680* | 0.0678 |
| | 0.2 | 0.1769 | 0.1759 | *0.1074* | 0.3759 | 0.1360 | 0.1103 | 0.1363 | 0.1143 | 0.1325 | **0.1015** | 0.1014 |
| | 0.3 | 0.2054 | 0.2010 | 0.1420 | 0.4403 | 0.1671 | *0.1409* | 0.1818 | 0.1543 | 0.1657 | **0.1293** | 0.1292 |
| | 0.4 | 0.2320 | 0.2334 | 0.1813 | 0.4188 | 0.2055 | *0.1764* | 0.2284 | 0.1982 | 0.2041 | **0.1630** | 0.1630 |
| | 0.5 | 0.2556 | 0.2615 | 0.2215 | 0.3864 | 0.2433 | *0.2132* | 0.2894 | 0.2529 | 0.2436 | **0.2010** | 0.2010 |
| | 0.6 | 0.2939 | 0.3087 | 0.2837 | 0.3985 | 0.2914 | *0.2631* | 0.3550 | 0.3148 | 0.2933 | **0.2508** | 0.2508 |
| | 0.7 | 0.3572 | 0.3765 | 0.3787 | 0.4710 | 0.3681 | *0.3419* | 0.4543 | 0.4080 | 0.3713 | **0.3294** | 0.3295 |
| | 0.8 | 0.4829 | 0.5037 | 0.5349 | 0.6043 | 0.5025 | *0.4826* | 0.6161 | 0.5615 | 0.5070 | **0.4693** | 0.4694 |
| | 0.9 | *0.8687* | 0.8933 | 0.9846 | 1.0514 | 0.8902 | 0.8810 | 1.0309 | 0.9744 | 0.8993 | **0.8649** | 0.8650 |
| $\rho = 0.1$ $\alpha = 1.0$ | 0.1 | 0.0984 | 0.1184 | 0.0588 | 0.2608 | 0.0711 | 0.0550 | 0.0548 | **0.0423** | 0.0681 | *0.0491* | 0.0488 |
| | 0.2 | 0.1289 | 0.1350 | 0.0740 | 0.3850 | 0.0875 | 0.0688 | 0.0777 | **0.0625** | 0.0845 | *0.0629* | 0.0627 |
| | 0.3 | 0.1451 | 0.1453 | 0.0866 | 0.2992 | 0.0985 | 0.0776 | 0.0923 | *0.0753* | 0.0947 | **0.0727** | 0.0725 |
| | 0.4 | 0.1536 | 0.1514 | 0.0998 | 0.2435 | 0.1112 | *0.0894* | 0.1088 | 0.0901 | 0.1086 | **0.0838** | 0.0837 |
| | 0.5 | 0.1662 | 0.1651 | 0.1139 | 0.2158 | 0.1251 | *0.1016* | 0.1261 | 0.1046 | 0.1227 | **0.0937** | 0.0936 |
| | 0.6 | 0.1690 | 0.1696 | 0.1311 | 0.2079 | 0.1334 | *0.1113* | 0.1452 | 0.1221 | 0.1323 | **0.1062** | 0.1062 |
| | 0.7 | 0.1838 | 0.1837 | 0.1507 | 0.2057 | 0.1546 | *0.1307* | 0.1674 | 0.1437 | 0.1523 | **0.1235** | 0.1235 |
| | 0.8 | 0.1970 | 0.2009 | 0.1843 | 0.2270 | 0.1777 | *0.1531* | 0.2034 | 0.1768 | 0.1772 | **0.1474** | 0.1475 |
| | 0.9 | 0.2263 | 0.2375 | 0.2446 | 0.2790 | 0.2254 | *0.2024* | 0.2737 | 0.2402 | 0.2272 | **0.1962** | 0.1963 |
| $\rho = 0.6$ $\alpha = 0.5$ | 0.1 | 0.1211 | 0.1389 | 0.0727 | 0.1815 | 0.0971 | 0.0769 | 0.0811 | **0.0669** | 0.0945 | *0.0691* | 0.0688 |
| | 0.2 | 0.1741 | 0.1741 | *0.1056* | 0.3587 | 0.1341 | 0.1105 | 0.1367 | 0.1150 | 0.1316 | **0.1014** | 0.1012 |
| | 0.3 | 0.2052 | 0.2026 | 0.1395 | 0.3950 | 0.1662 | *0.1394* | 0.1842 | 0.1569 | 0.1646 | **0.1307** | 0.1306 |
| | 0.4 | 0.2303 | 0.2319 | *0.1763* | 0.3563 | 0.2055 | 0.1768 | 0.2290 | 0.2003 | 0.2045 | **0.1645** | 0.1644 |
| | 0.5 | 0.2568 | 0.2631 | 0.2191 | 0.3459 | 0.2438 | *0.2131* | 0.2876 | 0.2520 | 0.2437 | **0.2013** | 0.2014 |
| | 0.6 | 0.2972 | 0.3131 | 0.2818 | 0.3764 | 0.2970 | *0.2683* | 0.3568 | 0.3182 | 0.2989 | **0.2549** | 0.2550 |
| | 0.7 | 0.3624 | 0.3798 | 0.3711 | 0.4392 | 0.3730 | *0.3469* | 0.4588 | 0.4107 | 0.3762 | **0.3331** | 0.3331 |
| | 0.8 | 0.4809 | 0.5041 | 0.5345 | 0.5927 | 0.4990 | *0.4788* | 0.6165 | 0.5617 | 0.5033 | **0.4661** | 0.4662 |
| | 0.9 | *0.8677* | 0.8957 | 0.9808 | 1.0456 | 0.8936 | 0.8820 | 1.0334 | 0.9720 | 0.9009 | **0.8659** | 0.8661 |
| $\rho = 0.6$ $\alpha = 1.0$ | 0.1 | 0.0964 | 0.1186 | 0.0560 | 0.2547 | 0.0717 | 0.0550 | 0.0549 | **0.0424** | 0.0684 | *0.0498* | 0.0494 |
| | 0.2 | 0.1288 | 0.1364 | 0.0709 | 0.3321 | 0.0879 | 0.0689 | 0.0801 | **0.0636** | 0.0845 | *0.0641* | 0.0638 |
| | 0.3 | 0.1418 | 0.1418 | 0.0842 | 0.2567 | 0.0956 | 0.0769 | 0.0926 | *0.0759* | 0.0932 | **0.0724** | 0.0722 |
| | 0.4 | 0.1519 | 0.1504 | 0.0948 | 0.1990 | 0.1073 | *0.0875* | 0.1066 | 0.0895 | 0.1054 | **0.0825** | 0.0824 |
| | 0.5 | 0.1619 | 0.1631 | 0.1107 | 0.1876 | 0.1239 | *0.1013* | 0.1241 | 0.1044 | 0.1211 | **0.0952** | 0.0950 |
| | 0.6 | 0.1721 | 0.1733 | 0.1275 | 0.1853 | 0.1389 | *0.1141* | 0.1473 | 0.1228 | 0.1361 | **0.1081** | 0.1081 |
| | 0.7 | 0.1827 | 0.1818 | 0.1490 | 0.1915 | 0.1546 | *0.1297* | 0.1689 | 0.1451 | 0.1533 | **0.1233** | 0.1233 |
| | 0.8 | 0.1931 | 0.1969 | 0.1761 | 0.2154 | 0.1773 | *0.1523* | 0.2011 | 0.1727 | 0.1765 | **0.1448** | 0.1448 |
| | 0.9 | 0.2263 | 0.2366 | 0.2429 | 0.2690 | 0.2271 | *0.2041* | 0.2758 | 0.2418 | 0.2292 | **0.1972** | 0.1973 |

Table S17: Values of $\mathrm{MSE}_\mu$ for various methods in the simulation ($n = 150$)

| | $R^2$ | SCAD | ALASSO | AGG1 | AGG2 | AIC | SAIC | BIC | SBIC | Cp | SMA1 | SMA2 |
|---|---|---|---|---|---|---|---|---|---|---|---|---|
| $\rho = 0.1$ $\alpha = 0.5$ | 0.1 | 0.1054 | 0.1139 | *0.0599* | 0.1824 | 0.0788 | 0.0632 | 0.0713 | 0.0600 | 0.0775 | **0.0568** | 0.0567 |
| | 0.2 | 0.1371 | 0.1359 | *0.0871* | 0.3586 | 0.1055 | 0.0878 | 0.1128 | 0.0972 | 0.1041 | **0.0800** | 0.0799 |
| | 0.3 | 0.1543 | 0.1535 | 0.1151 | 0.3666 | 0.1310 | *0.1113* | 0.1502 | 0.1313 | 0.1300 | **0.1034** | 0.1035 |
| | 0.4 | 0.1734 | 0.1781 | 0.1448 | 0.3149 | 0.1597 | *0.1399* | 0.1897 | 0.1678 | 0.1591 | **0.1302** | 0.1302 |
| | 0.5 | 0.1924 | 0.2006 | 0.1804 | 0.2999 | 0.1893 | *0.1704* | 0.2341 | 0.2085 | 0.1898 | **0.1600** | 0.1601 |
| | 0.6 | 0.2274 | 0.2403 | 0.2298 | 0.3172 | 0.2328 | *0.2144* | 0.2929 | 0.2645 | 0.2346 | **0.2039** | 0.2040 |
| | 0.7 | 0.2763 | 0.2901 | 0.2928 | 0.3518 | 0.2881 | *0.2723* | 0.3664 | 0.3345 | 0.2905 | **0.2626** | 0.2627 |
| | 0.8 | *0.3822* | 0.3974 | 0.4275 | 0.4721 | 0.3969 | 0.3860 | 0.4961 | 0.4609 | 0.3993 | **0.3776** | 0.3777 |
| | 0.9 | **0.7059** | 0.7196 | 0.7854 | 0.8260 | 0.7197 | 0.7163 | 0.8307 | 0.7914 | 0.7234 | *0.7077* | 0.7078 |
| $\rho = 0.1$ $\alpha = 1.0$ | 0.1 | 0.0780 | 0.0904 | 0.0440 | 0.2595 | 0.0504 | 0.0404 | 0.0447 | **0.0351** | 0.0501 | *0.0375* | 0.0373 |
| | 0.2 | 0.0975 | 0.0996 | 0.0547 | 0.3035 | 0.0631 | 0.0504 | 0.0604 | *0.0498* | 0.0616 | **0.0470** | 0.0469 |
| | 0.3 | 0.1067 | 0.1071 | 0.0650 | 0.2201 | 0.0706 | *0.0574* | 0.0728 | 0.0608 | 0.0692 | **0.0547** | 0.0546 |
| | 0.4 | 0.1133 | 0.1127 | 0.0749 | 0.1818 | 0.0801 | *0.0661* | 0.0847 | 0.0711 | 0.0792 | **0.0623** | 0.0622 |
| | 0.5 | 0.1189 | 0.1191 | 0.0861 | 0.1577 | 0.0880 | *0.0735* | 0.0978 | 0.0831 | 0.0872 | **0.0705** | 0.0705 |
| | 0.6 | 0.1272 | 0.1273 | 0.0974 | 0.1525 | 0.1010 | *0.0849* | 0.1115 | 0.0962 | 0.0994 | **0.0805** | 0.0805 |
| | 0.7 | 0.1332 | 0.1335 | 0.1092 | 0.1471 | 0.1118 | *0.0944* | 0.1277 | 0.1102 | 0.1104 | **0.0891** | 0.0890 |
| | 0.8 | 0.1398 | 0.1439 | 0.1342 | 0.1619 | 0.1286 | *0.1112* | 0.1545 | 0.1355 | 0.1284 | **0.1062** | 0.1063 |
| | 0.9 | 0.1592 | 0.1676 | 0.1774 | 0.1980 | 0.1603 | *0.1449* | 0.2040 | 0.1802 | 0.1616 | **0.1401** | 0.1402 |
| $\rho = 0.6$ $\alpha = 0.5$ | 0.1 | 0.1029 | 0.1115 | *0.0584* | 0.1819 | 0.0754 | 0.0611 | 0.0708 | 0.0592 | 0.0741 | **0.0557** | 0.0554 |
| | 0.2 | 0.1378 | 0.1350 | *0.0867* | 0.3273 | 0.1051 | 0.0876 | 0.1148 | 0.0985 | 0.1040 | **0.0808** | 0.0807 |
| | 0.3 | 0.1558 | 0.1550 | *0.1124* | 0.3077 | 0.1319 | 0.1126 | 0.1508 | 0.1325 | 0.1308 | **0.1044** | 0.1044 |
| | 0.4 | 0.1743 | 0.1786 | 0.1418 | 0.2708 | 0.1608 | *0.1405* | 0.1892 | 0.1678 | 0.1605 | **0.1301** | 0.1301 |
| | 0.5 | 0.1912 | 0.2010 | 0.1747 | 0.2643 | 0.1895 | *0.1694* | 0.2303 | 0.2066 | 0.1897 | **0.1590** | 0.1591 |
| | 0.6 | 0.2234 | 0.2358 | 0.2252 | 0.2892 | 0.2276 | *0.2094* | 0.2918 | 0.2634 | 0.2292 | **0.2006** | 0.2007 |
| | 0.7 | 0.2776 | 0.2926 | 0.2959 | 0.3461 | 0.2885 | *0.2737* | 0.3670 | 0.3364 | 0.2912 | **0.2641** | 0.2643 |
| | 0.8 | *0.3818* | 0.3966 | 0.4259 | 0.4634 | 0.3967 | 0.3856 | 0.5019 | 0.4641 | 0.4000 | **0.3770** | 0.3771 |
| | 0.9 | **0.7072** | 0.7193 | 0.7831 | 0.8257 | 0.7214 | 0.7178 | 0.8262 | 0.7881 | 0.7254 | *0.7091* | 0.7092 |
| $\rho = 0.6$ $\alpha = 1.0$ | 0.1 | 0.0756 | 0.0882 | 0.0418 | 0.2485 | 0.0511 | 0.0403 | 0.0447 | **0.0357** | 0.0498 | *0.0382* | 0.0379 |
| | 0.2 | 0.1008 | 0.1035 | 0.0540 | 0.2511 | 0.0637 | *0.0507* | 0.0626 | 0.0513 | 0.0621 | **0.0483** | 0.0481 |
| | 0.3 | 0.1089 | 0.1080 | 0.0626 | 0.1841 | 0.0732 | *0.0591* | 0.0727 | 0.0613 | 0.0717 | **0.0557** | 0.0555 |
| | 0.4 | 0.1157 | 0.1157 | 0.0727 | 0.1502 | 0.0822 | *0.0669* | 0.0853 | 0.0718 | 0.0808 | **0.0637** | 0.0635 |
| | 0.5 | 0.1181 | 0.1181 | 0.0802 | 0.1321 | 0.0879 | *0.0724* | 0.0969 | 0.0816 | 0.0867 | **0.0690** | 0.0689 |
| | 0.6 | 0.1272 | 0.1269 | 0.0937 | 0.1341 | 0.1003 | *0.0838* | 0.1104 | 0.0947 | 0.0995 | **0.0795** | 0.0795 |
| | 0.7 | 0.1326 | 0.1327 | 0.1081 | 0.1388 | 0.1120 | *0.0943* | 0.1304 | 0.1135 | 0.1113 | **0.0908** | 0.0908 |
| | 0.8 | 0.1436 | 0.1461 | 0.1319 | 0.1568 | 0.1316 | *0.1146* | 0.1568 | 0.1369 | 0.1315 | **0.1088** | 0.1089 |
| | 0.9 | 0.1640 | 0.1722 | 0.1788 | 0.2006 | 0.1656 | *0.1501* | 0.2070 | 0.1846 | 0.1667 | **0.1445** | 0.1446 |

Table S18: Values of $\mathrm{MSE}_\mu$ for various methods in the simulation ($n = 200$)

| | $R^2$ | SCAD | ALASSO | AGG1 | AGG2 | AIC | SAIC | BIC | SBIC | Cp | SMA1 | SMA2 |
|---|---|---|---|---|---|---|---|---|---|---|---|---|
| $\rho = 0.1$ $\alpha = 0.5$ | 0.1 | 0.0899 | 0.0946 | *0.0495* | 0.1822 | 0.0639 | 0.0521 | 0.0639 | 0.0537 | 0.0630 | **0.0469** | 0.0468 |
| | 0.2 | 0.1139 | 0.1116 | 0.0742 | 0.3383 | 0.0880 | *0.0741* | 0.0984 | 0.0859 | 0.0874 | **0.0681** | 0.0681 |
| | 0.3 | 0.1295 | 0.1315 | 0.0979 | 0.2982 | 0.1124 | *0.0971* | 0.1295 | 0.1151 | 0.1119 | **0.0894** | 0.0893 |
| | 0.4 | 0.1420 | 0.1472 | 0.1218 | 0.2596 | 0.1339 | *0.1178* | 0.1630 | 0.1465 | 0.1336 | **0.1095** | 0.1095 |
| | 0.5 | 0.1608 | 0.1680 | 0.1542 | 0.2485 | 0.1607 | *0.1462* | 0.2024 | 0.1824 | 0.1611 | **0.1373** | 0.1373 |
| | 0.6 | 0.1895 | 0.2003 | 0.1928 | 0.2604 | 0.1947 | *0.1805* | 0.2524 | 0.2309 | 0.1964 | **0.1723** | 0.1724 |
| | 0.7 | 0.2347 | 0.2472 | 0.2567 | 0.3071 | 0.2461 | *0.2344* | 0.3176 | 0.2931 | 0.2479 | **0.2264** | 0.2265 |
| | 0.8 | *0.3265* | 0.3375 | 0.3663 | 0.4040 | 0.3379 | 0.3312 | 0.4314 | 0.4021 | 0.3402 | **0.3249** | 0.3250 |
| | 0.9 | **0.6074** | 0.6154 | 0.6704 | 0.6994 | 0.6160 | 0.6150 | 0.6989 | 0.6724 | 0.6186 | *0.6091* | 0.6091 |
| $\rho = 0.1$ $\alpha = 1.0$ | 0.1 | 0.0661 | 0.0752 | 0.0361 | 0.2541 | 0.0401 | 0.0323 | 0.0389 | *0.0311* | 0.0394 | **0.0306** | 0.0305 |
| | 0.2 | 0.0835 | 0.0853 | 0.0452 | 0.2429 | 0.0517 | *0.0419* | 0.0513 | 0.0429 | 0.0508 | **0.0394** | 0.0393 |
| | 0.3 | 0.0885 | 0.0884 | 0.0533 | 0.1809 | 0.0579 | *0.0469* | 0.0599 | 0.0506 | 0.0567 | **0.0446** | 0.0445 |
| | 0.4 | 0.0900 | 0.0907 | 0.0591 | 0.1440 | 0.0624 | *0.0518* | 0.0707 | 0.0602 | 0.0617 | **0.0497** | 0.0497 |
| | 0.5 | 0.0994 | 0.0991 | 0.0687 | 0.1235 | 0.0729 | *0.0617* | 0.0814 | 0.0699 | 0.0720 | **0.0576** | 0.0576 |
| | 0.6 | 0.1042 | 0.1040 | 0.0785 | 0.1202 | 0.0807 | *0.0678* | 0.0913 | 0.0793 | 0.0794 | **0.0641** | 0.0641 |
| | 0.7 | 0.1096 | 0.1099 | 0.0922 | 0.1237 | 0.0922 | *0.0784* | 0.1082 | 0.0945 | 0.0914 | **0.0738** | 0.0738 |
| | 0.8 | 0.1172 | 0.1206 | 0.1098 | 0.1332 | 0.1074 | *0.0934* | 0.1302 | 0.1152 | 0.1072 | **0.0886** | 0.0886 |
| | 0.9 | 0.1304 | 0.1380 | 0.1435 | 0.1592 | 0.1322 | *0.1195* | 0.1704 | 0.1529 | 0.1329 | **0.1150** | 0.1151 |
| $\rho = 0.6$ $\alpha = 0.5$ | 0.1 | 0.0906 | 0.0943 | *0.0495* | 0.1815 | 0.0644 | 0.0525 | 0.0650 | 0.0547 | 0.0637 | **0.0477** | 0.0476 |
| | 0.2 | 0.1169 | 0.1142 | *0.0730* | 0.3003 | 0.0901 | 0.0751 | 0.1008 | 0.0879 | 0.0891 | **0.0689** | 0.0688 |
| | 0.3 | 0.1296 | 0.1306 | *0.0962* | 0.2468 | 0.1124 | 0.0974 | 0.1310 | 0.1166 | 0.1120 | **0.0889** | 0.0889 |
| | 0.4 | 0.1433 | 0.1465 | 0.1207 | 0.2183 | 0.1343 | *0.1186* | 0.1649 | 0.1481 | 0.1340 | **0.1106** | 0.1106 |
| | 0.5 | 0.1637 | 0.1712 | 0.1533 | 0.2266 | 0.1628 | *0.1478* | 0.2026 | 0.1838 | 0.1632 | **0.1383** | 0.1384 |
| | 0.6 | 0.1903 | 0.2011 | 0.1928 | 0.2450 | 0.1962 | *0.1822* | 0.2510 | 0.2301 | 0.1972 | **0.1738** | 0.1739 |
| | 0.7 | 0.2339 | 0.2460 | 0.2501 | 0.2850 | 0.2438 | *0.2327* | 0.3157 | 0.2904 | 0.2456 | **0.2251** | 0.2253 |
| | 0.8 | *0.3265* | 0.3368 | 0.3593 | 0.3894 | 0.3375 | 0.3302 | 0.4290 | 0.4007 | 0.3396 | **0.3242** | 0.3243 |
| | 0.9 | **0.6049** | 0.6126 | 0.6651 | 0.6945 | 0.6145 | 0.6127 | 0.7025 | 0.6769 | 0.6161 | *0.6077* | 0.6078 |
| $\rho = 0.6$ $\alpha = 1.0$ | 0.1 | 0.0680 | 0.0752 | 0.0350 | 0.2429 | 0.0406 | 0.0327 | 0.0398 | *0.0322* | 0.0399 | **0.0315** | 0.0313 |
| | 0.2 | 0.0824 | 0.0838 | 0.0436 | 0.2025 | 0.0505 | *0.0412* | 0.0514 | 0.0428 | 0.0496 | **0.0394** | 0.0392 |
| | 0.3 | 0.0880 | 0.0881 | 0.0498 | 0.1413 | 0.0576 | *0.0466* | 0.0610 | 0.0516 | 0.0563 | **0.0452** | 0.0451 |
| | 0.4 | 0.0924 | 0.0929 | 0.0585 | 0.1204 | 0.0651 | *0.0538* | 0.0703 | 0.0597 | 0.0644 | **0.0507** | 0.0506 |
| | 0.5 | 0.0978 | 0.0971 | 0.0660 | 0.1093 | 0.0706 | *0.0595* | 0.0795 | 0.0690 | 0.0696 | **0.0565** | 0.0564 |
| | 0.6 | 0.1041 | 0.1031 | 0.0772 | 0.1066 | 0.0815 | *0.0692* | 0.0931 | 0.0803 | 0.0807 | **0.0648** | 0.0647 |
| | 0.7 | 0.1090 | 0.1097 | 0.0876 | 0.1069 | 0.0906 | *0.0772* | 0.1077 | 0.0946 | 0.0900 | **0.0737** | 0.0736 |
| | 0.8 | 0.1176 | 0.1205 | 0.1045 | 0.1212 | 0.1067 | *0.0937* | 0.1306 | 0.1151 | 0.1063 | **0.0885** | 0.0886 |
| | 0.9 | 0.1316 | 0.1386 | 0.1424 | 0.1599 | 0.1332 | *0.1206* | 0.1707 | 0.1535 | 0.1336 | **0.1156** | 0.1157 |

Table S19: Values of $\text{MSE}_{\bar{y}} \times 100$ for various methods in the simulation ($n = 50$)

| | $R^2$ | SCAD | ALASSO | AGG1 | AGG2 | AIC | SAIC | BIC | SBIC | Cp | SMA1 | SMA2 |
|---|---|---|---|---|---|---|---|---|---|---|---|---|
| $\rho = 0.1$ $\alpha = 0.5$ | 0.1 | 2.3463 | 2.4610 | 2.2018 | *2.1788* | 2.2888 | 2.2336 | 2.1974 | **2.1643** | 2.2935 | 2.2120 | 2.2114 |
| | 0.2 | 2.4075 | 2.3861 | *2.2233* | 2.3180 | 2.3382 | 2.2433 | 2.2787 | **2.2229** | 2.3303 | 2.2344 | 2.2347 |
| | 0.3 | 2.5834 | 2.5546 | **2.2903** | 2.4632 | 2.4963 | 2.3481 | 2.4657 | 2.3516 | 2.4435 | *2.3132* | 2.3122 |
| | 0.4 | 2.7951 | 2.8021 | **2.5596** | 2.6984 | 2.7322 | 2.6352 | 2.6766 | 2.5703 | 2.7446 | *2.5633* | 2.5645 |
| | 0.5 | 2.5418 | 2.5485 | 2.3489 | 2.4797 | 2.4923 | 2.3318 | 2.4233 | *2.3205* | 2.4438 | **2.3056** | 2.3058 |
| | 0.6 | 2.7379 | 2.7849 | 2.6708 | 2.7353 | 2.6875 | *2.6393* | 2.9173 | 2.7766 | 2.7206 | **2.5685** | 2.5661 |
| | 0.7 | 3.1503 | 3.1009 | 3.0130 | 3.1035 | 3.0902 | *2.9883* | 3.2207 | 3.0610 | 3.1026 | **2.9271** | 2.9273 |
| | 0.8 | 3.3510 | 3.4394 | 3.5328 | 3.6437 | 3.3425 | *3.2757* | 3.7971 | 3.5518 | 3.3690 | **3.2254** | 3.2259 |
| | 0.9 | 4.7502 | 4.8426 | 4.7626 | 4.7972 | 4.8175 | *4.7076* | 5.0328 | 4.8091 | 4.8446 | **4.6144** | 4.6146 |
| $\rho = 0.1$ $\alpha = 1.0$ | 0.1 | 2.2736 | 2.4017 | 2.1263 | *2.0819* | 2.2304 | 2.1606 | 2.0932 | **2.0647** | 2.1834 | 2.1407 | 2.1401 |
| | 0.2 | 2.2244 | 2.2873 | *2.0493* | **2.0405** | 2.1541 | 2.0778 | 2.1125 | 2.0507 | 2.1262 | 2.0540 | 2.0543 |
| | 0.3 | 2.3795 | 2.4302 | *2.1814* | **2.1790** | 2.2879 | 2.2230 | 2.2675 | 2.2030 | 2.3074 | 2.1850 | 2.1842 |
| | 0.4 | 2.4431 | 2.4556 | 2.1867 | 2.2203 | 2.2753 | 2.1928 | 2.2210 | **2.1601** | 2.2812 | *2.1800* | 2.1804 |
| | 0.5 | 2.6505 | 2.6520 | *2.4077* | 2.4657 | 2.4984 | 2.4110 | 2.4890 | 2.4130 | 2.4695 | **2.3912** | 2.3910 |
| | 0.6 | 2.5058 | 2.5452 | 2.3702 | 2.3411 | 2.4117 | *2.3327* | 2.4398 | 2.3412 | 2.3909 | **2.3240** | 2.3245 |
| | 0.7 | 2.5256 | 2.4971 | 2.3792 | 2.3904 | 2.3997 | *2.3337* | 2.4572 | 2.3633 | 2.4006 | **2.2890** | 2.2899 |
| | 0.8 | 2.5817 | 2.6132 | *2.4187* | **2.3849** | 2.5510 | 2.4776 | 2.6016 | 2.4990 | 2.5481 | 2.4316 | 2.4306 |
| | 0.9 | 2.6891 | 2.7155 | 2.7261 | 2.6992 | 2.7300 | *2.6177* | 2.8287 | 2.6837 | 2.7332 | **2.6006** | 2.6011 |
| $\rho = 0.6$ $\alpha = 0.5$ | 0.1 | 2.2377 | 2.2454 | 2.0700 | **2.0060** | 2.1102 | 2.0603 | 2.0905 | 2.0519 | 2.0974 | *2.0305* | 2.0326 |
| | 0.2 | 2.5192 | 2.5855 | 2.3126 | **2.2208** | 2.4359 | 2.3552 | 2.3642 | *2.3036* | 2.4119 | 2.3070 | 2.3074 |
| | 0.3 | 2.5584 | 2.5939 | *2.3258* | **2.3038** | 2.4823 | 2.3915 | 2.5026 | 2.3892 | 2.4602 | 2.3558 | 2.3558 |
| | 0.4 | 2.6686 | 2.6349 | *2.3490* | **2.2927** | 2.5354 | 2.4177 | 2.4833 | 2.3785 | 2.5142 | 2.3550 | 2.3551 |
| | 0.5 | 2.7819 | 2.7973 | 2.5378 | **2.5259** | 2.7398 | 2.5932 | 2.7155 | 2.6232 | 2.6956 | *2.5370* | 2.5374 |
| | 0.6 | 2.9476 | 2.9681 | 2.8076 | *2.8031* | 2.8860 | 2.8063 | 3.0177 | 2.9019 | 2.8942 | **2.7588** | 2.7584 |
| | 0.7 | 3.1788 | 3.2082 | 3.1347 | 3.1962 | 3.1604 | *3.0636* | 3.3902 | 3.2205 | 3.1879 | **3.0263** | 3.0269 |
| | 0.8 | 3.5468 | 3.5933 | 3.4926 | 3.5135 | 3.4814 | *3.4276* | 3.8639 | 3.5852 | 3.5571 | **3.3667** | 3.3656 |
| | 0.9 | 4.4869 | 4.5859 | 4.6678 | 4.7624 | 4.5195 | *4.4597* | 4.9739 | 4.7192 | 4.6163 | **4.3910** | 4.3902 |
| $\rho = 0.6$ $\alpha = 1.0$ | 0.1 | 2.1098 | 2.1914 | 1.9922 | **1.8354** | 2.0997 | 2.0341 | 2.0123 | *1.9633* | 2.0790 | 2.0055 | 2.0044 |
| | 0.2 | 2.4161 | 2.4544 | *2.1944* | **2.1073** | 2.3015 | 2.2334 | 2.2796 | 2.2092 | 2.2854 | 2.1994 | 2.2003 |
| | 0.3 | 2.4777 | 2.4803 | 2.2394 | **2.1311** | 2.2935 | 2.2378 | 2.3366 | 2.2498 | 2.2722 | *2.2259* | 2.2252 |
| | 0.4 | 2.2441 | 2.2749 | *1.9828* | **1.9238** | 2.1124 | 2.0512 | 2.1137 | 2.0426 | 2.1140 | 2.0165 | 2.0187 |
| | 0.5 | 2.4083 | 2.4222 | 2.2004 | **2.1572** | 2.2846 | *2.1924* | 2.2601 | 2.2116 | 2.2711 | 2.1985 | 2.1966 |
| | 0.6 | 2.3980 | 2.4236 | 2.2699 | **2.2227** | 2.3723 | 2.2898 | 2.3513 | 2.2848 | 2.3716 | *2.2454* | 2.2461 |
| | 0.7 | 2.6576 | 2.6556 | 2.4506 | **2.4129** | 2.5668 | 2.4847 | 2.5668 | 2.4667 | 2.5613 | *2.4449* | 2.4463 |
| | 0.8 | 2.7509 | 2.7635 | 2.5796 | 2.5802 | 2.6272 | 2.5539 | 2.5754 | **2.5132** | 2.6198 | *2.5172* | 2.5176 |
| | 0.9 | 2.8173 | 2.8408 | 2.7409 | 2.7563 | 2.8260 | *2.6961* | 2.9525 | 2.7825 | 2.7849 | **2.6721** | 2.6722 |

Table S20: Values of $\text{MSE}_{\bar{y}} \times 100$ for various methods in the simulation ($n = 100$)

| | $R^2$ | SCAD | ALASSO | AGG1 | AGG2 | AIC | SAIC | BIC | SBIC | Cp | SMA1 | SMA2 |
|---|---|---|---|---|---|---|---|---|---|---|---|---|
| $\rho = 0.1$ $\alpha = 0.5$ | 0.1 | 1.0198 | 1.0444 | 0.9903 | *0.9871* | 1.0322 | 1.0039 | 0.9928 | **0.9826** | 1.0242 | 0.9922 | 0.9917 |
| | 0.2 | 1.1157 | 1.1002 | *1.0713* | 1.1273 | 1.0914 | 1.0763 | 1.1112 | 1.0922 | 1.0847 | **1.0698** | 1.0701 |
| | 0.3 | 1.1077 | 1.0901 | **1.0168** | 1.0907 | 1.0687 | 1.0339 | 1.0850 | 1.0589 | 1.0633 | *1.0276* | 1.0278 |
| | 0.4 | 1.1996 | 1.1913 | **1.1313** | 1.1942 | 1.1591 | 1.1362 | 1.2006 | 1.1701 | 1.1575 | *1.1324* | 1.1324 |
| | 0.5 | 1.2138 | 1.2082 | *1.1495* | 1.1670 | 1.1829 | 1.1563 | 1.2332 | 1.1966 | 1.1817 | **1.1401** | 1.1403 |
| | 0.6 | 1.2765 | 1.2892 | 1.2445 | 1.2662 | 1.2734 | *1.2340* | 1.3023 | 1.2725 | 1.2795 | **1.2182** | 1.2180 |
| | 0.7 | 1.3674 | 1.3955 | 1.3414 | 1.3629 | 1.3591 | *1.3322* | 1.4372 | 1.3898 | 1.3663 | **1.3252** | 1.3252 |
| | 0.8 | 1.5895 | 1.6103 | 1.5882 | 1.5939 | 1.5772 | *1.5649* | 1.6476 | 1.5979 | 1.5845 | **1.5567** | 1.5567 |
| | 0.9 | **1.6718** | 1.7046 | 1.8144 | 1.8567 | 1.6940 | 1.6891 | 1.8689 | 1.8265 | 1.6982 | *1.6847* | 1.6850 |
| $\rho = 0.1$ $\alpha = 1.0$ | 0.1 | 1.0158 | 1.0436 | 0.9797 | **0.9485** | 1.0078 | 0.9897 | 0.9765 | *0.9630* | 1.0046 | 0.9770 | 0.9769 |
| | 0.2 | 1.0623 | 1.0714 | **0.9925** | 1.0298 | 1.0239 | *0.9979* | 1.0251 | 1.0115 | 1.0233 | 1.0058 | 1.0058 |
| | 0.3 | 1.0053 | 1.0075 | 0.9690 | 0.9664 | 0.9663 | **0.9468** | 0.9789 | 0.9590 | 0.9695 | *0.9487* | 0.9484 |
| | 0.4 | 1.1623 | 1.1596 | *1.1071* | 1.1112 | 1.1398 | 1.1188 | 1.1226 | **1.1069** | 1.1353 | 1.1084 | 1.1084 |
| | 0.5 | 1.1756 | 1.1654 | 1.1219 | 1.1185 | 1.1236 | *1.1115* | 1.1439 | 1.1178 | 1.1304 | **1.0988** | 1.0986 |
| | 0.6 | 1.0670 | 1.0782 | 1.0253 | 1.0207 | 1.0366 | **1.0108** | 1.0517 | 1.0333 | 1.0338 | *1.0144* | 1.0144 |
| | 0.7 | 1.1486 | 1.1438 | 1.0936 | 1.0975 | 1.1236 | 1.0912 | 1.0990 | **1.0701** | 1.1242 | *1.0759* | 1.0758 |
| | 0.8 | 1.1610 | 1.1514 | 1.1672 | 1.1806 | 1.1460 | *1.1191* | 1.1815 | 1.1543 | 1.1347 | **1.1191** | 1.1193 |
| | 0.9 | 1.0956 | 1.1111 | 1.0865 | 1.0925 | 1.1012 | *1.0793* | 1.1493 | 1.1078 | 1.1030 | **1.0679** | 1.0678 |
| $\rho = 0.6$ $\alpha = 0.5$ | 0.1 | 1.0572 | 1.0666 | 0.9958 | **0.9454** | 1.0299 | 1.0101 | 1.0029 | *0.9918* | 1.0301 | 0.9977 | 0.9975 |
| | 0.2 | 1.1100 | 1.1076 | *1.0139* | **1.0028** | 1.0576 | 1.0359 | 1.0628 | 1.0421 | 1.0502 | 1.0201 | 1.0202 |
| | 0.3 | 1.1865 | 1.1812 | **1.1209** | 1.1745 | 1.1600 | 1.1345 | 1.1891 | 1.1707 | 1.1598 | *1.1287* | 1.1293 |
| | 0.4 | 1.1525 | 1.1591 | *1.1024* | 1.1265 | 1.1189 | 1.1048 | 1.1436 | 1.1193 | 1.1136 | **1.0893** | 1.0894 |
| | 0.5 | 1.2241 | 1.2227 | **1.1423** | *1.1502* | 1.1919 | 1.1636 | 1.2318 | 1.1912 | 1.2007 | 1.1525 | 1.1522 |
| | 0.6 | 1.2627 | 1.2885 | 1.2610 | 1.2670 | 1.2405 | *1.2217* | 1.2801 | 1.2512 | 1.2541 | **1.2076** | 1.2082 |
| | 0.7 | 1.2928 | 1.3231 | 1.2992 | 1.3184 | 1.3206 | *1.2862* | 1.4167 | 1.3675 | 1.3219 | **1.2764** | 1.2760 |
| | 0.8 | 1.3070 | 1.3328 | 1.3660 | 1.3807 | 1.3315 | *1.3040* | 1.4341 | 1.3819 | 1.3314 | **1.2861** | 1.2868 |
| | 0.9 | **1.8133** | 1.8473 | 1.9681 | 1.9726 | 1.8362 | 1.8331 | 2.0049 | 1.9462 | 1.8417 | *1.8188* | 1.8194 |
| $\rho = 0.6$ $\alpha = 1.0$ | 0.1 | 1.0934 | 1.0958 | 1.0339 | **0.9736** | 1.0775 | 1.0505 | 1.0320 | *1.0222* | 1.0710 | 1.0402 | 1.0405 |
| | 0.2 | 1.0808 | 1.0815 | 1.0056 | **0.9629** | 1.0208 | 1.0071 | 1.0154 | *1.0039* | 1.0142 | 1.0116 | 1.0118 |
| | 0.3 | 1.1395 | 1.1468 | 1.0734 | **1.0304** | 1.0892 | 1.0654 | 1.0885 | *1.0636* | 1.0817 | 1.0668 | 1.0667 |
| | 0.4 | 1.0636 | 1.0782 | 1.0087 | **0.9783** | 1.0189 | *1.0049* | 1.0246 | 1.0122 | 1.0176 | 1.0062 | 1.0064 |
| | 0.5 | 1.1618 | 1.1598 | 1.0920 | **1.0656** | 1.1088 | 1.0893 | 1.1053 | 1.0838 | 1.1078 | *1.0835* | 1.0841 |
| | 0.6 | 1.2247 | 1.2242 | 1.1619 | **1.1419** | 1.2046 | 1.1628 | 1.1660 | 1.1492 | 1.2012 | *1.1461* | 1.1461 |
| | 0.7 | 1.1785 | 1.1786 | *1.1052* | **1.0959** | 1.1412 | 1.1188 | 1.1352 | 1.1186 | 1.1479 | 1.1077 | 1.1073 |
| | 0.8 | 1.1511 | 1.1639 | 1.1003 | **1.0884** | 1.1481 | 1.1122 | 1.1370 | 1.1241 | 1.1488 | *1.0985* | 1.0989 |
| | 0.9 | 1.1900 | 1.1904 | *1.1421* | **1.1412** | 1.1788 | 1.1577 | 1.2208 | 1.1812 | 1.1872 | 1.1496 | 1.1495 |

Table S21: Values of $\text{MSE}_{\bar{y}} \times 100$ for various methods in the simulation ($n = 150$)

| | $R^2$ | SCAD | ALASSO | AGG1 | AGG2 | AIC | SAIC | BIC | SBIC | Cp | SMA1 | SMA2 |
|---|---|---|---|---|---|---|---|---|---|---|---|---|
| $\rho = 0.1$ $\alpha = 0.5$ | 0.1 | 0.6899 | 0.6849 | **0.6530** | 0.6648 | 0.6622 | 0.6562 | 0.6687 | 0.6611 | 0.6600 | *0.6543* | 0.6542 |
| | 0.2 | 0.7044 | 0.6983 | 0.6845 | 0.7045 | 0.6915 | *0.6800* | 0.6935 | 0.6873 | 0.6888 | **0.6750** | 0.6752 |
| | 0.3 | 0.7573 | 0.7635 | *0.7274* | 0.7625 | 0.7339 | 0.7277 | 0.7511 | 0.7412 | 0.7353 | **0.7233** | 0.7233 |
| | 0.4 | 0.7760 | 0.7748 | **0.7431** | 0.7638 | 0.7715 | 0.7547 | 0.7655 | 0.7576 | 0.7678 | *0.7463* | 0.7462 |
| | 0.5 | 0.7240 | 0.7235 | 0.7147 | 0.7361 | 0.7263 | 0.7119 | 0.7237 | *0.7101* | 0.7271 | **0.7002** | 0.7002 |
| | 0.6 | **0.7538** | 0.7633 | 0.7734 | 0.7959 | 0.7727 | 0.7574 | 0.8191 | 0.8011 | 0.7803 | *0.7562* | 0.7564 |
| | 0.7 | 0.7898 | 0.7990 | *0.7741* | 0.7832 | 0.7908 | 0.7748 | 0.8038 | 0.7931 | 0.7907 | **0.7685** | 0.7685 |
| | 0.8 | 0.8483 | 0.8480 | 0.8386 | 0.8434 | 0.8415 | *0.8372* | 0.8894 | 0.8663 | 0.8439 | **0.8317** | 0.8316 |
| | 0.9 | **1.1486** | 1.1649 | 1.2037 | 1.2285 | *1.1559* | 1.1594 | 1.2398 | 1.2154 | 1.1580 | 1.1562 | 1.1563 |
| $\rho = 0.1$ $\alpha = 1.0$ | 0.1 | 0.6600 | 0.6701 | *0.6382* | 0.6446 | 0.6532 | 0.6446 | 0.6443 | **0.6370** | 0.6532 | 0.6403 | 0.6403 |
| | 0.2 | 0.7145 | 0.7114 | *0.6752* | 0.6772 | 0.6891 | 0.6861 | 0.6814 | **0.6742** | 0.6854 | 0.6809 | 0.6806 |
| | 0.3 | 0.7067 | 0.6995 | *0.6757* | 0.6813 | 0.6846 | 0.6762 | 0.6904 | 0.6794 | 0.6864 | **0.6743** | 0.6744 |
| | 0.4 | 0.7332 | 0.7285 | 0.6991 | 0.7042 | 0.6988 | **0.6936** | 0.7031 | 0.6970 | 0.6964 | *0.6940* | 0.6938 |
| | 0.5 | 0.6920 | 0.6956 | 0.6842 | 0.6787 | 0.6732 | **0.6697** | 0.6971 | 0.6933 | 0.6744 | *0.6717* | 0.6717 |
| | 0.6 | 0.7114 | 0.7090 | **0.6712** | 0.6779 | 0.6856 | 0.6735 | 0.6992 | 0.6870 | 0.6810 | *0.6731* | 0.6731 |
| | 0.7 | 0.6640 | 0.6603 | *0.6261* | 0.6267 | 0.6442 | 0.6318 | 0.6336 | 0.6274 | 0.6407 | **0.6242** | 0.6241 |
| | 0.8 | 0.6937 | 0.6935 | 0.6960 | 0.6963 | 0.6869 | **0.6757** | 0.7164 | 0.7019 | 0.6904 | *0.6774* | 0.6774 |
| | 0.9 | 0.6250 | 0.6290 | 0.6430 | 0.6448 | *0.6205* | **0.6149** | 0.6702 | 0.6563 | 0.6242 | 0.6230 | 0.6231 |
| $\rho = 0.6$ $\alpha = 0.5$ | 0.1 | 0.7068 | 0.7085 | 0.6823 | **0.6464** | 0.6864 | 0.6761 | 0.6863 | 0.6771 | 0.6860 | *0.6711* | 0.6710 |
| | 0.2 | 0.7515 | 0.7400 | *0.7236* | 0.7321 | 0.7445 | 0.7275 | 0.7327 | 0.7253 | 0.7438 | **0.7216** | 0.7219 |
| | 0.3 | 0.7334 | 0.7285 | **0.7041** | 0.7131 | 0.7250 | 0.7088 | 0.7321 | 0.7284 | 0.7239 | *0.7047* | 0.7050 |
| | 0.4 | 0.7304 | 0.7328 | *0.6991* | 0.7009 | 0.7152 | 0.7050 | 0.7104 | 0.7033 | 0.7167 | **0.6902** | 0.6904 |
| | 0.5 | 0.6565 | 0.6640 | 0.6528 | 0.6615 | 0.6531 | *0.6406* | 0.6981 | 0.6801 | 0.6558 | **0.6395** | 0.6396 |
| | 0.6 | 0.7892 | 0.7976 | 0.7759 | 0.7803 | 0.7807 | **0.7691** | 0.8381 | 0.8144 | 0.7770 | *0.7694* | 0.7696 |
| | 0.7 | *0.7392* | 0.7475 | 0.7733 | 0.7809 | 0.7454 | **0.7346** | 0.8273 | 0.8148 | 0.7481 | 0.7415 | 0.7416 |
| | 0.8 | *0.8201* | 0.8306 | 0.8382 | 0.8439 | 0.8305 | 0.8252 | 0.9014 | 0.8710 | 0.8364 | **0.8141** | 0.8139 |
| | 0.9 | *1.0542* | 1.0621 | 1.0900 | 1.0997 | 1.0678 | 1.0629 | 1.1580 | 1.1145 | 1.0703 | **1.0534** | 1.0534 |
| $\rho = 0.6$ $\alpha = 1.0$ | 0.1 | 0.6649 | 0.6728 | 0.6510 | **0.6279** | 0.6621 | 0.6523 | 0.6523 | *0.6466* | 0.6599 | 0.6490 | 0.6491 |
| | 0.2 | 0.7382 | 0.7433 | 0.6904 | **0.6607** | 0.6932 | 0.6877 | 0.6928 | *0.6870* | 0.6931 | 0.6886 | 0.6885 |
| | 0.3 | 0.7115 | 0.6999 | 0.6758 | **0.6596** | 0.6864 | *0.6741* | 0.6918 | 0.6841 | 0.6842 | 0.6744 | 0.6746 |
| | 0.4 | 0.6975 | 0.6920 | 0.6582 | **0.6414** | 0.6779 | 0.6620 | 0.6640 | *0.6570* | 0.6793 | 0.6579 | 0.6580 |
| | 0.5 | 0.6773 | 0.6834 | *0.6441* | **0.6346** | 0.6724 | 0.6591 | 0.6643 | 0.6553 | 0.6708 | 0.6506 | 0.6506 |
| | 0.6 | 0.7027 | 0.7110 | 0.6822 | **0.6739** | 0.6895 | 0.6777 | 0.6974 | 0.6908 | 0.6904 | *0.6757* | 0.6757 |
| | 0.7 | 0.7375 | 0.7372 | 0.7156 | 0.7117 | 0.7186 | *0.7112* | 0.7439 | 0.7303 | 0.7171 | **0.7102** | 0.7103 |
| | 0.8 | 0.7411 | 0.7471 | 0.7257 | 0.7234 | 0.7339 | *0.7202* | 0.7519 | 0.7444 | 0.7373 | **0.7197** | 0.7199 |
| | 0.9 | 0.7039 | 0.7121 | 0.6955 | *0.6913* | 0.7029 | 0.6925 | 0.7246 | 0.7102 | 0.6992 | **0.6880** | 0.6882 |

Table S22: Values of $\text{MSE}_{\bar{y}} \times 100$ for various methods in the simulation ($n = 200$)

| | $R^2$ | SCAD | ALASSO | AGG1 | AGG2 | AIC | SAIC | BIC | SBIC | Cp | SMA1 | SMA2 |
|---|---|---|---|---|---|---|---|---|---|---|---|---|
| $\rho = 0.1$ $\alpha = 0.5$ | 0.1 | 0.4662 | 0.4675 | 0.4540 | 0.4663 | 0.4572 | *0.4525* | 0.4649 | 0.4581 | 0.4592 | **0.4508** | 0.4508 |
| | 0.2 | 0.4928 | 0.4920 | *0.4713* | 0.5011 | 0.4771 | 0.4721 | 0.4884 | 0.4826 | 0.4790 | **0.4692** | 0.4692 |
| | 0.3 | 0.4769 | 0.4771 | 0.4638 | 0.4899 | 0.4644 | *0.4563* | 0.4706 | 0.4619 | 0.4596 | **0.4546** | 0.4545 |
| | 0.4 | 0.5189 | 0.5196 | *0.5090* | 0.5215 | 0.5152 | 0.5137 | 0.5379 | 0.5274 | 0.5187 | **0.5050** | 0.5052 |
| | 0.5 | 0.5199 | 0.5274 | 0.5202 | 0.5374 | 0.5137 | *0.5084* | 0.5409 | 0.5337 | 0.5146 | **0.5065** | 0.5065 |
| | 0.6 | 0.5562 | 0.5580 | 0.5509 | 0.5560 | 0.5565 | *0.5488* | 0.5682 | 0.5585 | 0.5575 | **0.5440** | 0.5440 |
| | 0.7 | 0.5644 | 0.5701 | *0.5598* | 0.5662 | 0.5673 | 0.5641 | 0.5991 | 0.5881 | 0.5692 | **0.5563** | 0.5562 |
| | 0.8 | **0.6174** | 0.6229 | 0.6562 | 0.6626 | 0.6266 | 0.6214 | 0.6734 | 0.6583 | 0.6255 | *0.6214* | 0.6215 |
| | 0.9 | **0.7385** | 0.7443 | 0.7714 | 0.7764 | 0.7477 | 0.7450 | 0.7717 | 0.7669 | 0.7511 | *0.7424* | 0.7424 |
| $\rho = 0.1$ $\alpha = 1.0$ | 0.1 | 0.5232 | 0.5243 | **0.4925** | 0.5023 | 0.4971 | 0.4964 | 0.5023 | 0.4984 | *0.4958* | 0.4991 | 0.4989 |
| | 0.2 | 0.4883 | 0.4951 | 0.4721 | **0.4708** | 0.4785 | 0.4747 | 0.4752 | *0.4718* | 0.4784 | 0.4725 | 0.4725 |
| | 0.3 | 0.5029 | 0.5089 | 0.4918 | 0.4935 | 0.4957 | 0.4884 | 0.4934 | *0.4883* | 0.4954 | **0.4868** | 0.4868 |
| | 0.4 | 0.5013 | 0.5024 | **0.4845** | 0.4866 | 0.4877 | 0.4870 | 0.4951 | 0.4900 | 0.4883 | *0.4865* | 0.4866 |
| | 0.5 | 0.4714 | 0.4682 | *0.4425* | **0.4398** | 0.4531 | 0.4515 | 0.4497 | 0.4433 | 0.4509 | 0.4459 | 0.4457 |
| | 0.6 | 0.4716 | 0.4682 | 0.4540 | 0.4580 | 0.4530 | **0.4483** | 0.4668 | 0.4638 | *0.4509* | 0.4531 | 0.4531 |
| | 0.7 | 0.5076 | 0.5082 | 0.4956 | 0.4975 | 0.4897 | **0.4861** | 0.5022 | 0.4974 | 0.4886 | *0.4864* | 0.4864 |
| | 0.8 | 0.4954 | 0.4969 | 0.4953 | 0.4944 | 0.4962 | *0.4899* | 0.5012 | 0.4962 | 0.4947 | **0.4891** | 0.4891 |
| | 0.9 | 0.5015 | 0.5030 | *0.4902* | 0.4923 | 0.4994 | 0.4912 | 0.5046 | 0.4979 | 0.4984 | **0.4870** | 0.4870 |
| $\rho = 0.6$ $\alpha = 0.5$ | 0.1 | 0.4966 | 0.4948 | 0.4793 | **0.4761** | 0.4841 | 0.4793 | 0.4884 | 0.4841 | 0.4831 | *0.4774* | 0.4776 |
| | 0.2 | 0.5485 | 0.5418 | *0.5278* | 0.5441 | 0.5339 | 0.5289 | 0.5432 | 0.5370 | 0.5348 | **0.5257** | 0.5257 |
| | 0.3 | 0.5508 | 0.5531 | 0.5093 | **0.5031** | 0.5322 | 0.5225 | 0.5128 | *0.5086* | 0.5326 | 0.5147 | 0.5145 |
| | 0.4 | 0.5526 | 0.5539 | *0.5217* | **0.5161** | 0.5422 | 0.5315 | 0.5431 | 0.5314 | 0.5426 | 0.5242 | 0.5241 |
| | 0.5 | 0.5640 | 0.5629 | 0.5523 | 0.5519 | 0.5588 | *0.5506* | 0.5705 | 0.5558 | 0.5566 | **0.5434** | 0.5434 |
| | 0.6 | 0.6095 | 0.6160 | 0.6040 | 0.6072 | 0.6113 | *0.6034* | 0.6231 | 0.6202 | 0.6143 | **0.5967** | 0.5968 |
| | 0.7 | **0.5672** | 0.5779 | 0.5824 | 0.5836 | 0.5782 | 0.5729 | 0.6195 | 0.6123 | 0.5785 | *0.5704* | 0.5706 |
| | 0.8 | *0.6281* | 0.6355 | 0.6351 | 0.6376 | 0.6367 | 0.6307 | 0.6746 | 0.6645 | 0.6379 | **0.6270** | 0.6270 |
| | 0.9 | *0.7422* | 0.7468 | 0.7483 | 0.7557 | 0.7463 | 0.7449 | 0.7793 | 0.7708 | 0.7495 | **0.7402** | 0.7402 |
| $\rho = 0.6$ $\alpha = 1.0$ | 0.1 | 0.5302 | 0.5290 | 0.5108 | **0.4839** | 0.5118 | *0.5060* | 0.5103 | 0.5067 | 0.5092 | 0.5072 | 0.5072 |
| | 0.2 | 0.5050 | 0.5063 | *0.4827* | **0.4642** | 0.4897 | 0.4849 | 0.4862 | 0.4832 | 0.4893 | 0.4831 | 0.4832 |
| | 0.3 | 0.4964 | 0.4938 | *0.4701* | **0.4603** | 0.4791 | 0.4727 | 0.4821 | 0.4765 | 0.4797 | 0.4732 | 0.4732 |
| | 0.4 | 0.5005 | 0.5015 | 0.4770 | **0.4705** | 0.4844 | 0.4774 | 0.4770 | *0.4728* | 0.4849 | 0.4770 | 0.4771 |
| | 0.5 | 0.4925 | 0.4964 | 0.4782 | **0.4688** | 0.4836 | 0.4771 | 0.4883 | 0.4851 | 0.4821 | *0.4755* | 0.4755 |
| | 0.6 | 0.5137 | 0.5092 | 0.4978 | **0.4903** | 0.4988 | 0.4950 | 0.5178 | 0.5092 | 0.4978 | *0.4920* | 0.4921 |
| | 0.7 | 0.4857 | 0.4854 | 0.4768 | 0.4759 | 0.4741 | **0.4675** | 0.4859 | 0.4806 | 0.4746 | *0.4697* | 0.4697 |
| | 0.8 | 0.5296 | 0.5345 | 0.5220 | **0.5187** | 0.5252 | 0.5215 | 0.5438 | 0.5357 | 0.5255 | *0.5198* | 0.5199 |
| | 0.9 | 0.5108 | 0.5134 | 0.5105 | 0.5109 | 0.5127 | **0.5069** | 0.5367 | 0.5279 | 0.5148 | *0.5081* | 0.5082 |

## S5.4 Simulation under varying model misspecification

Following the simulation setup in Section 5.1, we generate a finite population according to

$$y_i = \sum_{j=1}^{101} \beta_j x_{ij} + e_i, \qquad i = 1, \ldots, 2000,$$

where $\beta_j = \sqrt{1.5} j^{-1}$ for $j = 1, \ldots, 51$ and $\beta_j = 0$ for $j = 52, \ldots, 101$. We set $x_{i1} = 1$, while the remaining covariates $\{x_{ij}\}$and errors $\{e_i\}$ are independent and identically distributed (i.i.d) as $N(0, 1)$. We again consider the nested candidate models, with the $m$th model containing the first $m$ regressors. We vary the number of candidate models $M$ across $\{11, 21, 31, 41, 51, 61, 71, 81, 91, 101\}$ to examine different degrees of model misspecification. When $M < 51$, all candidate models are misspecified because they omit relevant regressors. When $M = 51$, the largest candidate model coincides with the true model. We draw a sample of size $n = 300$ from the finite population using $\pi$PS sampling, with inclusion probabilities specified as in Section 5.1. The procedure is repeated $D = 1000$ times to compare the model selection and averaging methods (a)–(k). We also report the results for the largest candidate model (LARGE).

Tables S23–S24 show that the proposed methods (SMA1 and SMA2) yield similar results and perform best overall. When the candidate models are severely underfitted ($M = 11$), the largest-model estimator performs well because it incorporates the most information. As $M$ increases, the proposed methods improve and achieve the best performance among the methods considered. When the largest candidate model is overfitted ($M > 51$), its performance gradually deteriorates as $M$ increases, possibly because including irrelevant covariates increases the estimation variance. However, the proposed methods continue to perform best. Moreover, the model averaging methods (SAIC, SBIC, and SMA1) generally outperform their model selection counterparts (AIC, BIC, and $C_p$). As $M$ increases, the performance of AGG1 improves, ranking second only to the proposed methods in terms of $\text{MSE}_\mu$.

Table S23: Values of $\text{MSE}_\mu$ for various methods in the simulation

| | $M$ | SCAD | ALASSO | AGG1 | AGG2 | AIC | SAIC | BIC | SBIC | Cp | LARGE | SMA1 | SMA2 |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| $\rho = 0.1$ | 11 | *0.1432* | 0.1472 | 0.1631 | 0.2111 | 0.1479 | 0.1466 | 0.1814 | 0.1704 | 0.1483 | **0.1426** | 0.1472 | 0.1473 |
| | 21 | 0.1203 | 0.1279 | 0.1278 | 0.1761 | 0.1262 | *0.1172* | 0.1752 | 0.1610 | 0.1271 | 0.1181 | **0.1113** | 0.1114 |
| | 31 | 0.1410 | 0.1470 | 0.1206 | 0.1707 | 0.1330 | *0.1190* | 0.1773 | 0.1631 | 0.1324 | 0.1355 | **0.1054** | 0.1054 |
| | 41 | 0.1672 | 0.1700 | *0.1164* | 0.1651 | 0.1369 | 0.1205 | 0.1771 | 0.1621 | 0.1349 | 0.1626 | **0.1028** | 0.1028 |
| | 51 | 0.1983 | 0.2023 | *0.1176* | 0.1646 | 0.1448 | 0.1269 | 0.1797 | 0.1639 | 0.1410 | 0.2052 | **0.1059** | 0.1058 |
| | 61 | 0.2210 | 0.2314 | *0.1162* | 0.1646 | 0.1434 | 0.1251 | 0.1763 | 0.1616 | 0.1378 | 0.2507 | **0.1045** | 0.1044 |
| | 71 | 0.2377 | 0.2629 | *0.1187* | 0.1656 | 0.1431 | 0.1260 | 0.1785 | 0.1636 | 0.1389 | 0.3031 | **0.1074** | 0.1073 |
| | 81 | 0.2570 | 0.2974 | *0.1184* | 0.1650 | 0.1477 | 0.1297 | 0.1789 | 0.1639 | 0.1417 | 0.3636 | **0.1088** | 0.1086 |
| | 91 | 0.2736 | 0.3311 | *0.1180* | 0.1644 | 0.1465 | 0.1274 | 0.1768 | 0.1617 | 0.1400 | 0.4232 | **0.1087** | 0.1084 |
| | 101 | 0.2858 | 0.3589 | *0.1182* | 0.1624 | 0.1497 | 0.1303 | 0.1790 | 0.1636 | 0.1402 | 0.4879 | **0.1099** | 0.1097 |
| $\rho = 0.6$ | 11 | *0.1442* | 0.1471 | 0.1634 | 0.2016 | 0.1487 | 0.1476 | 0.1804 | 0.1700 | 0.1488 | **0.1434** | 0.1480 | 0.1481 |
| | 21 | 0.1194 | 0.1268 | 0.1249 | 0.1607 | 0.1255 | *0.1163* | 0.1773 | 0.1626 | 0.1263 | 0.1168 | **0.1106** | 0.1108 |
| | 31 | 0.1422 | 0.1475 | 0.1202 | 0.1573 | 0.1338 | *0.1191* | 0.1771 | 0.1626 | 0.1336 | 0.1358 | **0.1055** | 0.1056 |
| | 41 | 0.1691 | 0.1717 | *0.1166* | 0.1528 | 0.1389 | 0.1217 | 0.1790 | 0.1637 | 0.1367 | 0.1652 | **0.1041** | 0.1041 |
| | 51 | 0.1995 | 0.2032 | *0.1167* | 0.1508 | 0.1454 | 0.1279 | 0.1765 | 0.1613 | 0.1416 | 0.2051 | **0.1056** | 0.1055 |
| | 61 | 0.2194 | 0.2307 | *0.1162* | 0.1517 | 0.1443 | 0.1260 | 0.1776 | 0.1625 | 0.1389 | 0.2491 | **0.1064** | 0.1063 |
| | 71 | 0.2384 | 0.2622 | *0.1156* | 0.1511 | 0.1457 | 0.1268 | 0.1778 | 0.1619 | 0.1389 | 0.3036 | **0.1063** | 0.1061 |
| | 81 | 0.2547 | 0.2957 | *0.1157* | 0.1490 | 0.1447 | 0.1270 | 0.1772 | 0.1633 | 0.1390 | 0.3605 | **0.1077** | 0.1074 |
| | 91 | 0.2727 | 0.3282 | *0.1158* | 0.1485 | 0.1454 | 0.1261 | 0.1768 | 0.1615 | 0.1384 | 0.4233 | **0.1083** | 0.1080 |
| | 101 | 0.2814 | 0.3586 | *0.1165* | 0.1499 | 0.1469 | 0.1285 | 0.1780 | 0.1622 | 0.1391 | 0.4876 | **0.1098** | 0.1094 |

Table S24: Values of $\text{MSE}_{\bar{y}} \times 100$ for various methods in the simulation

| | $M$ | SCAD | ALASSO | AGG1 | AGG2 | AIC | SAIC | BIC | SBIC | Cp | LARGE | SMA1 | SMA2 |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| $\rho = 0.1$ | 11 | *0.3354* | 0.3364 | 0.3454 | 0.3515 | 0.3363 | 0.3370 | 0.3501 | 0.3483 | 0.3366 | **0.3352** | 0.3382 | 0.3383 |
| | 21 | *0.2981* | 0.3006 | 0.3051 | 0.3088 | 0.3029 | 0.2992 | 0.3236 | 0.3196 | 0.3049 | **0.2976** | 0.3005 | 0.3005 |
| | 31 | 0.3565 | 0.3567 | *0.3447* | 0.3487 | 0.3487 | 0.3450 | 0.3618 | 0.3570 | 0.3471 | 0.3541 | **0.3414** | 0.3414 |
| | 41 | 0.3426 | 0.3403 | *0.3210* | 0.3223 | 0.3293 | 0.3234 | 0.3387 | 0.3331 | 0.3260 | 0.3407 | **0.3206** | 0.3205 |
| | 51 | 0.3287 | 0.3344 | 0.3220 | 0.3272 | 0.3266 | *0.3204* | 0.3431 | 0.3368 | 0.3214 | 0.3346 | **0.3153** | 0.3152 |
| | 61 | 0.3173 | 0.3262 | 0.3073 | 0.3179 | 0.3093 | *0.3048* | 0.3249 | 0.3202 | 0.3051 | 0.3318 | **0.2999** | 0.2999 |
| | 71 | 0.3520 | 0.3584 | *0.3094* | 0.3114 | 0.3197 | 0.3118 | 0.3266 | 0.3233 | 0.3137 | 0.3624 | **0.3056** | 0.3056 |
| | 81 | 0.3524 | 0.3647 | *0.3015* | 0.3048 | 0.3061 | 0.3021 | 0.3187 | 0.3153 | 0.3063 | 0.3873 | **0.2983** | 0.2983 |
| | 91 | 0.3775 | 0.4071 | 0.3348 | 0.3400 | 0.3366 | *0.3333* | 0.3534 | 0.3493 | 0.3381 | 0.4249 | **0.3314** | 0.3313 |
| | 101 | 0.3768 | 0.3966 | 0.3236 | 0.3250 | 0.3290 | *0.3229* | 0.3362 | 0.3329 | 0.3264 | 0.4446 | **0.3196** | 0.3194 |
| $\rho = 0.6$ | 11 | *0.3459* | 0.3481 | 0.3525 | 0.3508 | 0.3472 | 0.3468 | 0.3576 | 0.3541 | 0.3473 | **0.3456** | 0.3473 | 0.3473 |
| | 21 | 0.3129 | 0.3147 | 0.3134 | 0.3161 | 0.3115 | **0.3092** | 0.3278 | 0.3222 | 0.3119 | 0.3124 | *0.3099* | 0.3099 |
| | 31 | 0.3216 | 0.3219 | *0.3122* | 0.3139 | 0.3173 | 0.3126 | 0.3287 | 0.3250 | 0.3155 | 0.3163 | **0.3081** | 0.3081 |
| | 41 | 0.3384 | 0.3414 | 0.3328 | 0.3322 | 0.3388 | *0.3305* | 0.3554 | 0.3489 | 0.3380 | 0.3407 | **0.3230** | 0.3231 |
| | 51 | 0.3546 | 0.3575 | 0.3299 | *0.3282* | 0.3429 | 0.3357 | 0.3473 | 0.3441 | 0.3402 | 0.3581 | **0.3254** | 0.3255 |
| | 61 | 0.3544 | 0.3527 | *0.3236* | 0.3244 | 0.3332 | 0.3243 | 0.3428 | 0.3404 | 0.3325 | 0.3656 | **0.3202** | 0.3204 |
| | 71 | 0.3593 | 0.3656 | 0.3118 | *0.3104* | 0.3158 | **0.3101** | 0.3381 | 0.3329 | 0.3134 | 0.3837 | 0.3106 | 0.3106 |
| | 81 | 0.3836 | 0.4004 | 0.3333 | *0.3330* | 0.3447 | 0.3383 | 0.3461 | 0.3441 | 0.3411 | 0.4178 | **0.3330** | 0.3329 |
| | 91 | 0.3752 | 0.3936 | 0.3237 | 0.3292 | 0.3281 | 0.3235 | 0.3373 | 0.3350 | *0.3204* | 0.4299 | **0.3194** | 0.3195 |
| | 101 | 0.3826 | 0.4056 | 0.3241 | *0.3231* | 0.3349 | 0.3271 | 0.3454 | 0.3408 | 0.3321 | 0.4561 | **0.3211** | 0.3212 |

## S5.5 Simulation of weight selection consistency

Following the simulation setup in Section 5.1, we generate a finite population according to

$$y_i = \sum_{j=1}^{11} \beta_j x_{ij} + e_i, \qquad i = 1, \ldots, 2000,$$

where $\beta_j = cj^{-1}$ for $j = 1, \ldots, 6$ and $\beta_j = 0$ for $j = 7, \ldots, 11$. We set $x_{i1} = 1$, and the remaining covariates $\{x_{ij}\}$ and errors $\{e_i\}$ are i.i.d. as $N(0, 1)$. The parameter $c$ controls the $R^2$ of the model. We again consider nested candidate models, with the $m$th model containing the first $m$ regressors. Thus, the 6th through 11th candidate models are correct models, as they contain the true model. We draw a sample from the finite population using $\pi$PS sampling, with inclusion probabilities specified as in Section 5.1, and repeat the procedure $D = 1000$ times. Figure S1 shows the sum of the estimated weights assigned by SMA1 to the correct models across different sample sizes. This sum approaches one as the sample size increases, numerically verifying the selection consistency of the weights established in Theorem 4. Convergence becomes faster as $R^2$ increases.

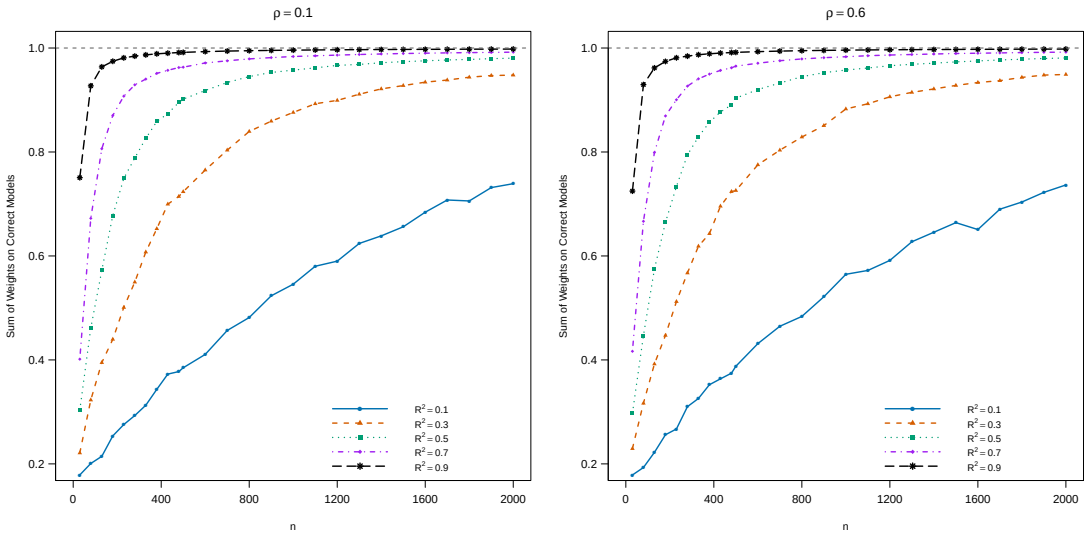



Figure S1: Sum of the estimated weights assigned by SMA1 to correct models across different sample sizes ($\rho = 0.1$, left; $\rho = 0.6$, right).

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